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Audit and feedback: effects on professional practice (Review)

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ABSTRACT

Background

Audit and feedback (A&F) is a widely used strategy to improve professional practice. This is supported by prior Cochrane reviews and behavioural theories describing how healthcare professionals are prompted to modify their practice when given data showing that their clinical practice is inconsistent with a desirable target. Yet there remains uncertainty regarding the effects of A&F on improving healthcare practice and the characteristics of A&F that lead to a greater impact.

Objectives

To assess the effects of A&F on the practice of healthcare professionals and to examine factors that may explain variation in the effectiveness of A&F.

Search methods

With the Cochrane Effective Practice and Organisation of Care (EPOC) group information scientist, we updated our search strategy to include studies published from 2010 to June 2020. Search updates were performed on 28 February 2019 and 11 June 2020. We searched MEDLINE (Ovid), Embase (Ovid), CINAHL (EBSCO), the Cochrane Library, clinicaltrials.gov (all dates to June 2020), WHO ICTRP (all dates to February Week 3 2019, no information available in 2020 due to COVID-19 pandemic). An updated search and duplicate screen was completed on February 14, 2022; studies that met inclusion criteria are included in the 'Studies awaiting classification' section.

Selection criteria

Randomised trials, including cluster-trials and cross-over and factorial designs, featuring A&F (defined as measurement of clinical performance over a specified period of time (*audit*) and provision of the resulting data to clinicians or clinical teams (*feedback)*) in any trial arm that reported objectively measured health professional practice outcomes.

Data collection and analysis

For this updated review, we re-extracted data for each study arm, including theory-informed variables regarding how the A&F was conducted and behaviour change techniques for each intervention, as well as study-level characteristics including risk of bias. For each study, we extracted outcome data for every healthcare professional practice targeted by A&F. All data were extracted by a minimum of two independent review authors.

For studies with dichotomous outcomes that included arms with and without A&F, we calculated risk differences (RDs) (absolute difference between arms in proportion of desired practice completed) and also odds ratios (ORs). We synthesised the median RDs and interquartile ranges (IQRs) across all trials. We then conducted meta-analyses, accounting for multiple outcomes from a given study and weighted by effective sample size, using reported (or imputed, when necessary) intra-cluster correlation coefficients. Next, we explored the role of baseline performance, co-interventions, targeted behaviour, and study design factors on the estimated effects of A&F. Finally, we conducted exploratory meta-regressions to test preselected variables that might be associated with A&F effect size: characteristics of the *audit* (number of indicators, aggregation of data); delivery of the *feedback* (multi-modal format, local champion, nature of comparator, repeated delivery); and components supporting *action* (facilitation, provision of specific plans for improvement, co-development of action plans).

Main results

We included 292 studies with 678 arms; 133 (46%) had a low risk of bias, 41 (14%) unclear, and 113 (39%) had a high risk of bias. There were 26 (9%) studies conducted in low- or middle-income countries. In most studies (237, 81%), the recipients of A&F were physicians. Professional practices most commonly targeted in the studies were prescribing (138 studies, 47%) and test-ordering (103 studies, 35%). Most studies featured multifaceted interventions: the most common co-interventions were clinician education (377 study arms, 56%) and reminders (100 study arms, 15%). Forty-eight unique behaviour change techniques were identified within the study arms (mean 5.2, standard deviation 2.8, range 1 to 29).

Synthesis of 558 dichotomous outcomes measuring professional practices from 177 studies testing A&F versus control revealed a median absolute improvement in desired practice of 2.7%, with an IQR of 0.0 to 8.6. Meta-analyses of these studies, accounting for multiple outcomes from the same study and weighting by effective sample size accounting for clustering, found a mean absolute increase in desired practice of 6.2% (95% confidence interval (CI) 4.1 to 8.2; moderate-certainty evidence) and an OR of 1.47 (95% CI 1.31 to 1.64; moderate-certainty evidence). Effects were similar for pre-planned subgroup analyses focused on prescribing and test-ordering outcomes. Lower baseline performance and increased number of co-interventions were both associated with larger intervention effects.

Meta-regressions comparing the presence versus absence of specific A&F components to explore heterogeneity, accounting for baseline performance and number of co-interventions, suggested that A&F effects were greater with individual-recipient-level data rather than team-level data, comparing performance to top-peers or a benchmark, involving a local champion with whom the recipient had a relationship, using interactive modalities rather than just didactic or just written format, and with facilitation to support engagement, and action plans to improve performance. The meta-regressions did not find significant effects with the number of indicators in the audit, comparison to average performance of all peers, or co-development of action plans. Contrary to expectations, repeated delivery was associated with lower effect size. Direct comparisons from head-to-head trials support the use of peer-comparisons versus no comparison at all and the use of design elements in feedback that facilitate the identification and action of high-priority clinical items.

Authors' conclusions

A&F can be effective in improving professional practice, but effects vary in size. A&F is most often delivered along with co-interventions which can contribute additive effects. A&F may be most effective when designed to help recipients prioritise and take action on high-priority clinical issues and with the following characteristics:

1. targets important performance metrics where health professionals have substantial room for improvement (*audit*);

2. measures the individual recipient's practice, rather than their team or organisation (*audit*);

3. involves a local champion with an existing relationship with the recipient (*feedback*);

4. includes multiple, interactive modalities such as verbal and written (*feedback*);

5. compares performance to top peers or a benchmark (*feedback*);

6. facilitates engagement with the feedback (*action*);

7. features an actionable plan with specific advice for improvement (*action*).

These conclusions require further confirmatory research; future research should focus on discerning ways to optimise the effectiveness of A&F interventions.

PLAIN LANGUAGE SUMMARY

Audit and feedback: effects on professional practice

Key messages

- Audit and feedback in healthcare is when a health professional's performance is evaluated and compared to professional standards (audit). Then the health professionals are given the results of the comparison (feedback), with the hope that it might help them improve their performance.

- Audit and feedback helps to improve health professional performance a little to a moderate amount. It works best when it shows health professionals how they compare to top performers, focuses on important areas for improvement, and includes tips for making changes. Audit and feedback can be even more helpful when combined with other supports like reminders or extra training.

- Future research should focus on finding the best ways to improve audit and feedback interventions.

What is meant by audit and feedback in healthcare?

Audit and feedback is often used in healthcare organisations to improve healthcare professionals’ performance. In an audit and feedback process, an individual’s professional practice or performance is measured and then compared to professional standards or targets. In other words, their professional performance is “audited”. During the "feedback", the results of this comparison are then provided to health professionals. The aim of this process is to encourage healthcare professionals to take action or make changes to follow standards.

What did we do?

We searched for all the studies in which healthcare professionals were randomised to receive audit and feedback and in which the results on professional practice were measured.

What did we find?

We found 292 studies that met the requirements. We found that audit and feedback is often used together with other strategies to improve quality of care, such as educational meetings or reminders. Most studies measured the effect of audit and feedback on doctors, although some studies measured the effect on nurses or pharmacists. Audit and feedback was used to influence their performance in different areas, including the proper use of prescription treatments or test-ordering.

The exact way that audit and feedback was delivered varied widely across the studies. Sometimes health professionals were given feedback verbally, other times in writing, on an electronic dashboard or through multiple modes. In some studies, this feedback was given to them by the researchers responsible for the study, while in other studies, feedback was given by supervisors or colleagues. In some studies, health professionals were given feedback only once, while others were given feedback monthly. Sometimes, they were also given or supported to create an action plan with suggestions or advice about how to improve their performance.

Main results: What happens when health professionals are audited and provided with feedback?

The effect of using audit and feedback varied widely across the included studies, but most often it achieves small-to-moderate improvements in quality of care.

Audit and feedback may be most effective when recipients can see how their own performance compares to their high-performing peers, when it helps the health professional to identify and take action on high-priority clinical issues, and when it focuses on areas where health professionals have substantial room for improvement. Other features of audit and feedback that are associated with greater effects are when it involves measurement of the individual recipient's practice (rather than their team or organisation); comes from a respected peer with an existing relationship to the recipient; includes multiple modalities (e.g., verbal and written); and features an action plan with advice for improvement.

In addition, the effect of audit and feedback may change when combined with other strategies that support improved quality of care, such as education or reminders.

What are the limitations of the evidence?

The quality of the evidence is moderate and further research is needed to confirm the features of audit and feedback that are most likely to achieve the greatest effects in different situations.

How up to date is this evidence?

This review updates our previous review. The evidence analysed is up-to-date to June 2020.

SUMMARY OF FINDINGS

Summary of findings 1. Audit and feedback for health professionals

Patient or population: Healthcare professionals

Settings: Primary and secondary care

Intervention: Audit and feedback with or without other interventions[1]

Comparison: Usual care

Outcomes	Effect size[2]	No. of health professionals (no. of studies)	Certainty of the evidence (GRADE)	Comments
Compliance with any desired practice in the audit and feedback intervention	6.2% increase (95% confidence interval (CI) 4.1 to 8.2) Odds ratio of 1.47 (95% CI 1.31 to 1.64)	18,789 clusters of health professionals for 177 studies	Moderate[3]	Lower baseline performance and increased number of co-interventions are both associated with larger effects. Low-certainty evidence[4] from meta-regressions suggests that A&F effects were greater with: • individual-recipient-level rather than team-level data; • feedback delivered from a local champion with whom the recipient had a relationship; • interactive modalities rather than just didactic or just written; • facilitation to support engagement with the feedback; and • action plans provided to improve performance.
Compliance with desired practice for prescribing	6.1% increase (95% confidence interval (CI) 3.0 to 9.2) Odds ratio of 1.43 (95% CI 1.21 to 1.70)	10,891 clusters of health professionals for 85 studies	Moderate[3]	-
Compliance with desired practice for test ordering	4.9% increase (95% confidence interval (CI) 2.4 to 7.3) Odds ratio of 1.37 (95% CI 1.19 to 1.57)	4036 clusters of health professionals for 65 studies	Moderate[3]	-

GRADE Working Group grades of evidence:
High certainty: We are very confident that the true effect lies close to that of the estimate of the effect.
Moderate certainty: We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low certainty: Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect.
Very low certainty: We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect.

1 - The effect of audit and feedback *alone* on professional practice was similar to audit and feedback as the core, essential feature in *multi-faceted* interventions.
2 - The effect is based on meta-analyses that accounted for multiple outcomes from the same study and weighted by effective sample size accounting for clustering and adjusted for baseline performance and number of EPOC co-interventions.
3 - Downgraded to moderate certainty because of inconsistency in the results that could not be fully explained.
4 - Downgraded to low certainty because of limited head-to-head trials testing these comparisons (i.e. indirectness).

BACKGROUND

This review updates a previous Cochrane review of the effects of audit and feedback (A&F) (Ivers 2012), defined as measurement of clinical performance over a specified period of time (*audit*) and provision of the resulting data to clinicians or clinical teams (*feedback)*.

Earlier versions of this review found that A&F generally has a small-to-moderate effect on professional practice, but the effects of audit and feedback are highly variable (Ivers 2012; Jamtvedt 2003; Jamtvedt 2006; Thomson O'Brien 1997a; Thomson O'Brien 1997b).

The most recent version of this review (Ivers 2012) identified a series of factors that seemed to explain heterogeneity, proposing that A&F was most effective when the health professionals are not performing well to begin with; the person delivering the A&F is a supervisor or colleague; it is provided more than once; it is given both verbally and in writing; and it includes clear targets and an action plan.

Description of the condition

Research has demonstrated persistent overuse, underuse, and misuse of healthcare services across the full range of healthcare professionals (e.g. physicians, nurses, pharmacists), settings (e.g. primary care, secondary care), and clinical domains (e.g. prescribing, test-ordering) (Braithwaite 2018; McGlynn 2003; Seddon 2001; Squires 2022). In response, health professionals, organisations, and systems expend substantial effort and resources on quality improvement initiatives to improve system performance. These initiatives, whether they occur in inpatient or outpatient settings, and regardless of which health professionals they involve, all share the broad goal of increasing compliance with desired professional practice. To improve compliance with desired practice, it is common to start by measuring gaps between existing and ideal performance (i.e. conducting an audit), and ensuring the relevant health professionals are aware of such gaps (i.e. providing feedback).

Description of the intervention

A&F involves providing healthcare professionals and/or organisations with a summary of clinical performance over time on objectively measured quality indicators and is a foundational component of many quality improvement activities. The audit component involves measuring quality of care against standards of practice using chart audits, electronic health records, patient surveys and/or routine administrative data. The feedback component involves providing a summary report on performance to relevant healthcare professionals, administrators or organisations with the purpose of prompting change to align with desired practice. Terminologies sometimes used as synonyms for A&F include benchmarking, practice profiles, report cards, and dashboards. High-performing health systems tend to feature A&F as an evidence-based, scalable, and relatively inexpensive strategy to encourage uptake of best practices (Brown 2012; Hysong 2006).

How the intervention might work

The observation that the effects of A&F are greatest if baseline compliance is low supports the idea that A&F is felt to be effective as a tool to improve practice because it may overcome healthcare providers' limited ability to accurately self-assess (Davis 2006). Under this assumption, providers are thought to be inherently motivated to improve care, but may not act to change their current practices because they are unaware of their suboptimal performance. In turn, they may be prompted to modify their practice if given feedback that their clinical practice was inconsistent with their peers or with accepted guidelines.

In addition to individual behaviour change theory, organisational theories focused on quality improvement offer clues regarding potential important effect modifiers, including organisational culture with respect to quality improvement, and the 'actionability' of feedback reports (Hysong 2006). Van der Veer and colleagues (Van der Veer 2010) conducted a systematic review of the impact on quality of care of using medical registries to produce feedback reports, usually as part of multifaceted interventions, for healthcare professionals. They noted that important effect modifiers seemed to be the quality of the data provided to recipients, the motivation and interest of recipients, and the organisational support for quality improvement.

A strong theoretical foundation explains in principle how A&F works, drawing primarily from Control Theory (Carver 1982) and Feedback Intervention Theory (FIT) (Kluger 1996). Both Control Theory and FIT have been empirically substantiated in A&F applied to healthcare in secondary analyses of our prior review (Gardner 2010; Hysong 2009)). Brehaut and colleagues conducted a review of existing theories relevant to A&F and interviews with interdisciplinary leaders in fields relevant to A&F, ultimately identifying "15 recommendations for optimising feedback" (Brehaut 2016). The Brown study conducted a meta-synthesis of A&F-related qualitative studies, which conceptualised a range of potential mediators and moderators for A&F, and developed the "Clinical Performance-Feedback Intervention Theory (CP-FIT)" (Brown 2019). In synthesising this work, we can say that to be effective, recipients of A&F must: 1) engage with and understand the data; 2) validate and decide to act upon the data; and 3) be able to improve quality of care in ways that are measured in the data (Desveaux 2021).

Nevertheless, there remains uncertainty about how to optimise the effects of A&F interventions. For example, there is conflicting evidence regarding the role of peer comparison in feedback (Kiefe 2001; Søndergaard 2002; Wones 1987). Further, the role of participant involvement in either target-setting or in feedback interventions seems promising (Anonymous 1992) but remains uncertain (Nasser 2008).

Why it is important to do this review

A&F is increasingly common, but there remains uncertainty about how to optimise the effects of this intervention. The aim of the current update is to investigate the effectiveness of A&F to improve processes of care and to examine a range of factors that could influence the effectiveness of this intervention informed by current theory. The answer to whether A&F can improve quality of care has been established; herein, we hope to update knowledge regarding estimated effects, but (arguably more importantly) also advance knowledge regarding how to best design and implement feedback to improve quality of care. We focused on processes of care (compliance with desired practice) but not clinical outcomes in this review because effects on patient outcomes will depend to a great extent on the strength of association between actions under the control of the health professional (the proximal behaviour focus of

A&F), and the downstream effects of those clinical actions, which will vary widely.

OBJECTIVES

Our main objectives were to explore the average effect of A&F across trials with diverse targeted behaviours and co-interventions, and which intervention components or co-interventions are linked to stronger effects.

We addressed two primary questions in this review:

1. What is the average effect of A&F, across trials with diverse targeted behaviours and a range of co-interventions?
2. Which intervention components or co-interventions are associated with larger effects?

For question 1, we considered the following comparisons.

- *Comparison A:* A&F alone or within a multifaceted intervention compared with usual care.
- *Comparison B:* A&F alone compared with usual care.
- *Comparison C:* A&F within a multifaceted intervention compared with usual care.

Comparisons B and C involve unique subsets of the studies in Comparison A, with the goal of validating the main effect estimate and exploring whether A&F effects vary in the presence of a multi-faceted intervention.

For question 2, we conducted *meta-regressions* using the Comparison A sample of studies *(*studies comparing A&F with or without co-interventions to usual care, using study/contrast level results), which adjusted for the number of co-interventions and baseline performance. Specifically, we examine three hierarchical groupings of variables because these align with the theoretical understanding of intervention mechanisms(Brown 2019) and also because they offer face validity in terms of the steps that intervention developers must consider for implementation: 1) *audit (i.e. measuring quality of care*, 2) *feedback (i.e. presenting results to health professionals)*, and 3) *action (i.e. directing the changes needed to improve quality of care).*

In addition, we also qualitatively described *Comparison D*: trials comparing head-to-head different ways of implementing A&F and then examined *Comparison E*: A&F plus co-intervention(s) compared with A&F alone, as well as *Comparison F:* trials comparing multifaceted interventions that include audit and feedback.

METHODS

Criteria for considering studies for this review

Types of studies

Randomised controlled trials (RCTs), including cluster-randomised trials, stepped-wedge trials, cross-over trials, and factorial trials.

Types of participants

Healthcare professionals who are responsible for patient care.

Types of interventions

A&F, defined as measurement of clinical performance over a specified period of time (*audit*) and provision of the resulting data to clinicians or clinical teams (*feedback)*. The audit component involves measuring quality of care against standards of practice using chart audits, electronic health records, patient surveys, and/or routine administrative data. The feedback component involves providing a summary report on performance to relevant healthcare professionals, administrators, or organisations with the purpose of prompting improved quality of care and may be delivered in a written, electronic or verbal format.

For this update, we included any intervention that included A&F. This is distinct from the prior version of the review when we only included studies in which the A&F was used alone or was qualitatively deemed a 'core' aspect of a multi-faceted intervention.

We used the EPOC classification (EPOC 2002) scheme to identify the other components of the multifaceted interventions. As with all other abstracted variables, this process was completed independently by two abstractors and discrepancies were resolved through discussion, including other authors as needed.

Since the prior review, our team has developed and applied novel methods to extract behaviour change techniques as components of quality improvement strategies such as A&F from trial reports (Presseau 2015). Behaviour change techniques are the elemental components of intervention content (Michie 2013). Each A&F intervention typically has multiple behaviour change techniques (e.g. feedback on behaviours, feedback on outcomes, social comparison, information on health consequences, credible sources, instruction on how to perform a behaviour, action planning, reviewing goals, problem-solving) and A&F interventions are often implemented alongside co-interventions such as clinical decision-support or incentives (Stanworth 2022). Examining interventions at this level can help identify which components (and which theory-informed combinations) can help test and develop theory to further advance our ability to optimise A&F interventions (Michie 2018). We used the behaviour change techniques taxonomy (Michie 2013), which describes 93 different techniques for changing behaviour. We have previously applied this taxonomy to reliably identify and classify components of diabetes quality improvement (QI) interventions (Presseau 2015). Using this established taxonomy may help support replication of effective A&F interventions (Lawrenson 2018).

Studies that focus on real-time feedback for procedural skills were excluded, as were studies in which the feedback focused on performance in tests or simulated patient interactions. Studies that featured facilitated relay of communication regarding patient status or symptoms, but that did not provide a summary of physician performance were also excluded. Even if the term 'feedback' was used in the manuscript, the study was excluded if the so-called feedback intervention was best classified as a reminder (especially when the intervention was at the point of care), or another category in the EPOC (EPOC 2015) classification of quality improvement interventions other than 'A&F' (see also: Shojania 2006).

Types of outcome measures

We focused on objectively measured health professional performance (e.g. prescribing, test-ordering, counselling, referring, diagnosing, screening, immunising and making treatment decisions). For this update, we abstracted outcomes at the first time point available after the intervention. We did not abstract additional data from the primary trial report nor from separate articles or companion reports wherein longer-term follow-up was reassessed; sustainability of effects is of great interest, but beyond the scope of this review. Studies that provided data only on patient outcomes (rather than professional practice targeted by the A&F) or only on cost of care were excluded, as were studies that measured knowledge or performance in a test situation only. As patient outcomes depend entirely on the strength of association between the professional behaviour(s) targeted by A&F and subsequent changes in patient behaviour(s), and because A&F is typically designed to directly change health professional behaviours, we focused on these outcomes in this review.

Primary outcomes

The primary outcome was the effect on compliance with desired professional practice. We assessed any objective measure of desired professional practice targeted by the A&F intervention (e.g. whether prescribing, test-ordering, referring, counselling, or other action of the health professional was guideline-concordant). Increases in outcomes should be interpreted as an “increase in desired practice” or an improvement in the professional behaviour(s) targeted by the A&F.

Search methods for identification of studies

The search strategy from the previous review was revised and updated by the Cochrane EPOC group information scientist, Paul Miller. The updated search strategy contained subject headings and free-text keyword searches for our key concepts, and we applied modified methodological filters for study design as appropriate (Lefebvre 2008; Lefebvre 2023). The searches for each database were restricted to studies published from 2010 to June 2020, except for trial registries which were searched for all dates, due to not being included in the prior review. However, the WHO registry was not searched in 2020 as it was made unavailable by WHO due to an inability to cope with increased usage during the COVID-19 pandemic. A top-up search was run up to February 2022; studies that met inclusion criteria at the title and abstract level are available in Studies awaiting classification.

Electronic searches

We searched the following electronic databases:

- Cochrane Central Register of Controlled Trials (CENTRAL 2020, Issue 6), including the Cochrane Effective Practice and Organisation of Care (EPOC) Group Specialised Register;
- MEDLINE (Ovid MEDLINE® Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE® Daily and Ovid MEDLINE®) (2010 to 11 June 2020);
- Embase (Ovid) (2010 to 11 June 2020);
- CINAHL (EBSCOhost) (2010 to 11 June 2020);
- EPOC Trials Register (2010 to 31 December 2014).

The search strategies for the current version of the review are available in Appendix 1, search strategies for the 2022 top-up search are available in Appendix 2.

Searching other resources

We searched the following trial registers for ongoing studies:

- ClinicalTrials.gov (www.clinicaltrials.gov) (11 June 2020);
- WHO International Clinical Trials Registry Platform (ICTRP) (www.who.int/trialsearch) (2 February 2019).

We also searched for full reports published up until July 2021 (one year after the literature search date) linked to relevant trial registries and protocols identified in the literature search.

Data collection and analysis

Selection of studies

Two review authors independently screened the titles and abstracts in Covidence (Covidence 2024) and applied inclusion criteria to the updated searches; complete manuscripts were sought in the case of uncertainty and differences of opinion resolved through consensus. Conference abstracts were included if they provided sufficient data, a full report could be found, or missing data could be obtained from the investigators. We reassessed whether each study from the previous review met the inclusion criteria. Data from relevant trial registries and protocols were also included when available. In this update, given the large number of available trials, we required cluster trials to have at least four clusters per arm as trials with fewer than this have substantial analytical challenges (Donner 1994).

Data extraction and management

Two review authors independently abstracted all data from the included studies in Excel using detailed abstraction sheets. Data from studies included in the previous review were re-abstracted due to changes in the data abstraction form for this updated review.

A revised version of the EPOC data collection checklist (EPOC 2002) was used to collect information on study design, type of interventions compared, type of targeted behaviour, participants, setting, methods, outcomes, and results. We used the EPOC taxonomy (EPOC 2015) to describe the interventions present in each trial arm and then extracted further details about the A&F intervention itself using an extraction guide informed by Clinical Performance Feedback Intervention Theory (CP-FIT) (Brown 2019) regarding components of interventions likely to be both important and frequently reported. Discrepancies between authors were resolved through discussion, with final decisions made by the lead author when necessary. A coding manual is available upon request.

We also abstracted data on behaviour change techniques to provide an alternative, more detailed way to describe ‘active’ intervention components (e.g. specific strategies designed to change health professional behaviours). Trained extractors used a behaviour change technique codebook to code behaviour change techniques present in each study arm for A&F and co-interventions, as well as the behaviours and actors/groups targeted by those BCTs (Crawshaw 2023). We used the 93-item behaviour change techniques taxonomy version 1 (Michie 2013) to describe interventions for each study arm, acknowledging that even arms with "*just*" A&F might be considered complex, multi-faceted interventions when examined at the elemental level. Two additional behaviour-change techniques - 'Education (unspecified)' and 'Feedback (unspecified)' - were added during the coding process, such that 95 behaviour-change techniques were considered for coding (Crawshaw 2023). In the context of extracting behaviour-change techniques, we considered study arms as treatment or control/comparator based primarily on how relative intervention content was reported in the source materials (manuscripts, supplementary materials, and study protocols). For studies without explicit control/comparator arms (e.g. head-to-head trials of A&F), we assigned the lowest intensity treatment arm as a control/comparator arm.

Assessment of risk of bias in included studies

Two review authors independently assessed the risk of bias for each study and extracted data for newly identified studies using a revised data collection form; discrepancies were resolved by consensus with a third author as needed. The risk of bias for each main outcome in all studies included in the review was assessed according to the criteria outlined in the Cochrane Handbook for Systematic Reviews of Interventions (Higgins 2011a; Higgins 2011b) and Cochrane EPOC criteria (EPOC 2017). The risk of bias assessment included the following domains:

1. Random sequence generation;
2. Allocation concealment;
3. Blinding of outcome assessment;
4. Incomplete outcome data;
5. Selective outcome reporting;
6. Other bias: recruitment bias, baseline imbalance, protection against contamination, incorrect analysis.

In this update, recruitment bias and incorrect analysis were added as cluster trial design-specific other risks of bias (Higgins 2011b). Blinding of participants was not included as it is typically not possible to blind healthcare providers to receive audit and feedback. We graded each potential source of bias as high, low or unclear risk with justification for our judgement in the 'Risk of bias' table. We summarised the risk of bias judgements across different studies for each domain in a 'Risk of bias graph'.

An overall assessment of the risk of bias (high, unclear or low risk of bias) was assigned to each of the included studies using the Cochrane Collaboration's tool for assessing risk of bias (Higgins 2011b). Studies with a low risk of bias for all key domains representing selection bias (random sequence allocation and allocation concealment) or detection bias (blinding of outcome assessment) were considered to have a 'low' risk of bias. Studies where the risk of bias in at least one of these domains was unclear or judged to have some bias that could plausibly raise doubts about the conclusions were considered to have an 'unclear' risk of bias. Studies with a high risk of bias for at least one of these domains or with other features that seriously limited confidence in the results that we were able to extract were considered to have a 'high' risk of bias.

For the studies included in the previous review, we reassessed 10% to ensure consistency, and found there to be little inconsistency between the most recently added and previous studies. Any discrepancies between the conclusions regarding risk of bias using the new and previous approaches were resolved through consensus.

Measures of treatment effect

All outcomes were expressed as the degree of compliance with desired practice because we considered this the shared latent target for all A&F interventions, expressed as a proportion. In this update, when several outcomes were reported in a trial, we extracted all those relevant to the A&F intervention - that is, those outcomes that could have been affected by the feedback. We did not extract summary or composite performance measures if individual measures were available. For example, rather than extract a composite measure of adherence to guideline-concordant care for diabetes, we would extract each available metric of adherence to guideline recommendations, whether for prescribing, test-ordering, or other processes of care.

When outcomes were assessed multiple times, we extracted the outcomes closest to the time after the intervention ended, rather than the longest follow-up available, as this best represents the proximal effects of A&F. For cross-over studies, we took data from the first period; for stepped-wedge trials, we took the overall effect estimate. The numerator and denominator for the compliance proportion were recorded when available. When only the proportion was reported, the underlying population size giving rise to the proportion was estimated based on the number of visits or participants, in order to determine the numerator and denominator.

For interventions in which the goal was to reduce targeted behaviour to be more concordant with guidelines (e.g. inappropriate x-rays), we 'inverted' the proportion of outcome data to allow for consistent interpretation. For example, if the study reported that professionals in the intervention arm reduced inappropriate x-rays from 25% to 15%, we would extract that compliance with guideline-recommended x-ray utilisation improved in that arm from 75% to 85%; such changes were extracted in duplicate and then rechecked by the lead author.

Across studies, effects were summarised on both the absolute and relative scale, by reporting both the mean risk difference (RD) and the odds ratio (OR) - see data synthesis section below.

Unit of analysis issues

Cluster-randomised trials

Trials of A&F are typically randomised at the cluster level, either at the level of the health professional or at the level of groups of health professionals. Meta-analyses of such trials must ensure proper accounting for clustering to avoid underestimating the standard error (SE) of the estimate of effect (Higgins 2024). As in the previous versions of this review, we have not abstracted the observed SEs, P values, or confidence intervals to inform our quantitative analyses, as the literature is fraught with variable and oft-inappropriate analyses of clustered data. We did extract intra-cluster correlations (ICCs) when reported. When not reported, we imputed ICCs based on the median of the extracted values in order to estimate an effective sample size for each trial.

Studies with more than two arms

Many trials had multiple arms and also contributed data across multiple outcomes. Our analyses that were based on random-effects models included all such data, accounting for the study identifier as a random intercept in the model. We conducted stratified analyses to focus on specific arms and also sensitivity analyses, selecting only one outcome per arm to examine the consistency of findings.

Dealing with missing data

Missing data regarding results, the characteristics of the studies or of the A&F intervention were not imputed. When missing numerators were apparent, we attempted to extract these based on the reported proportions, and the most appropriate denominator reported, such as the number of patients or visits. As noted above, missing data regarding ICCs were imputed based on the median of the extracted ICCs.

Assessment of heterogeneity

We explored heterogeneity by measuring Higgins I^2 values (Higgins 2003) for the meta-analysis of absolute risk differences and the odds ratios and by visually examining plots that allowed us to explore the range of effect sizes for comparisons of interest. We repeated this exercise for studies targeting prescribing and test-ordering.

Assessment of reporting biases

We produced funnel plots for asymmetry to look for publication bias (Cumpston 2019; Page 2021). We then conducted a modified Egger test (Lin 2018), based on our primary log-odds ratio-based meta-analysis approaches (see below). Specifically, for each of these two meta-analysis models, we included a covariate for the standard error of the estimate. The coefficient of the regression model is reported, which represents the increase in effect size per 1-point increase in the standard error of the study estimate. We further repeated this analysis using the odds ratio analysis that was based on the median reported effect size for each study.

Data synthesis

Quantitative data synthesis focused on trials with dichotomous outcomes measuring compliance with desired professional practice. For studies that did not provide such data (i.e. continuous outcome data only), we summarised effects narratively.

For Comparisons A, B, and C, for the studies with dichotomous outcomes, we summarised descriptively the distribution of all estimates observed across studies that included a comparison between arms with A&F and arms without. We then conducted a meta-analysis using a contrast-based 2-step framework for trials comparing any intervention with A&F, either alone or as part of a multi-faceted intervention, to interventions without A&F. In the first stage, we calculated study-outcome specific effect estimates for A&F versus no A&F, and their standard error, for each study-outcome. In the second stage, we ran a random-effects meta-analysis model on the effect estimates, with two separate levels of random intercepts, one at the study level (to account for multiple outcomes) and one at the study-outcome level.

We conducted the first step (extraction of A&F effect estimates) in two different ways: first, using logistic count regression with an identity link, to obtain a risk difference effect estimate, and second, using logistic count regression with the logit link, to obtain log odds ratio estimates. Logistic count regressions for each study used a data format that generally had two rows for each study: a row representing the post-no A&F data, and a row representing the post-A&F data. When there were multiple arms with or without A&F, these were collapsed based on the count so that there were just two rows of data. Numerator and denominator counts for the first-stage regressions were adjusted to account for the estimated ICC by calculating the design effect and estimating the effective sample size for events and total sample (Higgins 2024). The first-stage logistic regression models were conducted using the glm function in R.

In the second stage, we ran separate random-effects meta-analysis models for risk differences and log-odds ratios, meaning that all models were run in duplicate. All the second-stage meta-analyses included two separate levels of random intercepts, one at the study level (to account for the dependence between multiple outcomes) and one at the study-outcome level (to allow within-study effect estimates to vary). The second stage meta-analysis was conducted using the rma.mv function in the metafor package of R. Second-stage meta-analyses were conducted first, for all studies with A&F versus no- A&F comparisons within them, regardless of the presence of co-interventions ("Comparison A" studies). Also, amongst the Comparison A studies, we conducted a meta-regression, with a variable added for the average number of co-interventions in the A&F arms, and the average number of co-interventions in the 'no A&F' arms. This was an approach to make unconfounded inferences about the average effects of A&F adjusted for the number of co-interventions. Next, we conducted meta-analyses for specific strata defined by the number of co-interventions in the A&F and 'no A&F' groups:

- 'Comparison B' studies = 0 EPOC co-interventions in the A&F and no- A&F arms;
- 'Comparison C' studies = 0 EPOC co-interventions in no-A&F arms vs. > 0 EPOC co-interventions in A&F arms.

These subgroup analyses for Comparison B and Comparison C were conducted to check the model-based results from Comparison A, examining the effects of A&F with and without co-interventions, with the goal of providing answers to two commonly posed questions:

1. What is the effect of A&F when used alone (to estimate the isolated effect of A&F) - Comparison B;
2. What is the effect of multi-faceted interventions featuring A&F - Comparison C.

The latter question represents the more common situation observed in real-world scenarios and in trials.

Finally, we narratively summarised trials that contributed data as seen below:

- Comparison D (head-to-head trials comparing different ways of conducting A&F);
- Comparison E (trials testing A&F combined with co-interventions vs. A&F alone);
- Comparison F (trials testing A&F combined with co-interventions vs. A&F combined with different co-interventions).

When possible, we reported the risk difference to describe the effects observed in studies summarised within these comparisons to facilitate comparison to the quantitative analyses.

Subgroup analysis and investigation of heterogeneity

We examined heterogeneity through a variety of approaches.

First, we stratified the Comparison A studies based on characteristics of the study design, the targeted behaviour, and the intervention and re-ran meta-analyses in these subsets of studies.

Next, to examine the role of A&F features in heterogeneity, we then conducted a one-step logistic meta-regression model to explore the association of specific intervention components with the outcome (Jackson 2018). Analyses were conducted without adjustment and with adjustment for the number of non-A&F EPOC co-interventions (linear predictor) and the baseline performance (linear predictor). Analyses included a random intercept for the study, to account for correlated outcomes within studies. For each A&F feature, separate unadjusted and adjusted logistic meta-regression models were fitted.

In the meta-regressions, we examined A&F features for which a clear, theory-informed hypothesis exists. These were informed by the behavioural theories described above, from the recommendations for clinical performance feedback (Brehaut 2016) and also from Colquhoun and colleagues' theory-informed hypotheses (Colquhoun 2017). We conceptually grouped the features of A&F. We explored as follows for ease of understanding:

- *Audit* (number of indicators, aggregation of data);
- *Feedback* (type of format, involvement of local champion, type of comparator, repeated delivery);
- *Action* (engagement facilitated, plans for improvement provided, co-developed action plans).

We acknowledge that some of these A&F features may have overlapping mechanisms of action related to the steps required for recipients to turn data into improved clinical performance. Furthermore, we recognise that a wide range of additional variables may mediate or moderate the effects of A&F, but the above variables were felt to be most readily extractable from trial reports.

We hypothesised that greater effects would be observed with:

- Fewer targeted behaviours (e.g. prescribing, test-ordering, etc.) in the audit because it limits cognitive load and enables focused individual and organisational action;
- Disaggregated data (recipient's own performance, rather than team) in the audit because it is easier to validate by the recipient and enables the recipient to feel accountable for the results;
- Multi-modal feedback format (verbal plus visual) because the data may become more memorable this way;
- Delivery of the feedback by a local champion because this may improve the credibility and acceptability of the data and may increase engagement;
- Comparison to top peers or a benchmark rather than the average in the feedback as this sets a target for performance;
- Repeated delivery of the feedback as this reinforces the feedback loop compared to once-only feedback;
- Facilitation to enable engagement with feedback and subsequent *action;*
- Action plans provided so that recipients know specifically what action to do to achieve higher scores;
- Co-development of action plans because this may help recipients feel the desired action is feasible and social construction of the feedback loop may enable a culture of data-driven quality improvement.

Sensitivity analysis

Primary models included all outcomes from each study. As a sensitivity analysis, we re-ran the models using the median result to represent the outcome for each study from which multiple outcomes were extracted.

As a further sensitivity analysis, we imputed the 25th and 75th percentile of ICCs rather than the median for those studies where an ICC was not reported.

Summary of findings and assessment of the certainty of the evidence

Per EPOC Guidance (EPOC 2017), we created 'Summary of findings' tables to highlight the degree of confidence in the results using the GRADE approach (Schünemann 2024). Two authors independently assessed GRADE and disagreements were resolved through discussion until a consensus was reached. The Summary of findings table compares A&F (alone or with co-interventions) with usual care. We reported the following outcomes:

- Compliance with any desired practice in the audit and feedback intervention;
- Compliance with desired practice for prescribing;
- Compliance with desired practice for test ordering.

RESULTS

Description of studies

Results of the search

For this update, we screened 15,702 new records and assessed the full texts of 559. We included 292 studies in 276 full-text articles. Of note, 182 new studies were added to this review since the previous update (Ivers 2012), and 30 were removed from this previous version of the review as they no longer met our updated inclusion criteria. See study flow diagram for details (Figure 1).

Figure 1. PRISMA study flow diagram † A 'top-up' search was conducted in February 2022; records were screened at the title and abstract level only and none were included in the analysis. These records are listed under studies awaiting classification. ‡ Details of 61 studies excluded during data extraction are summarised in Characteristics of

excluded studies; details of the remaining 362 excluded studies are available on request. * CRT (Cluster Randomised Trial) required at least 4 units per arm for inclusion.

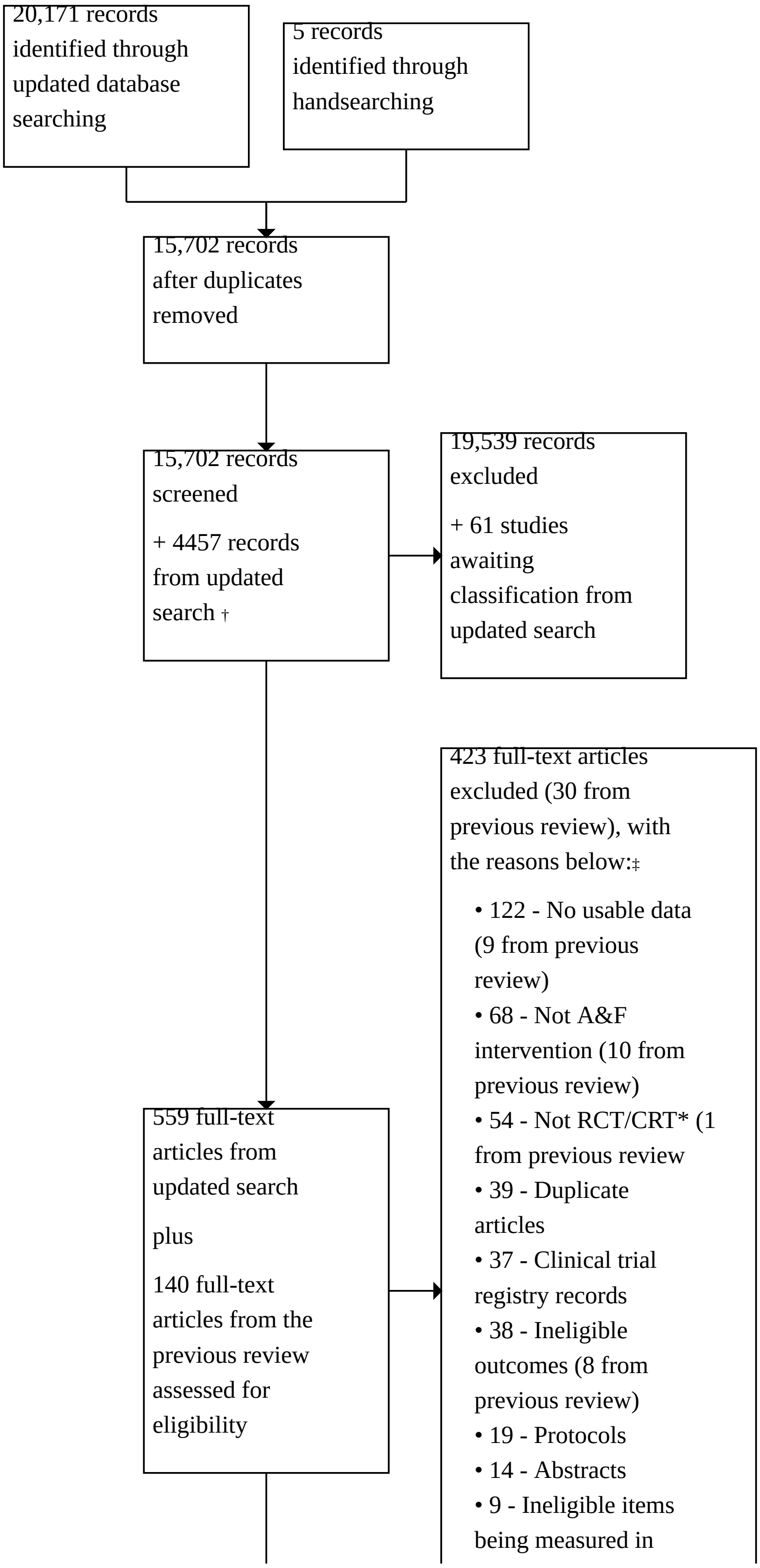
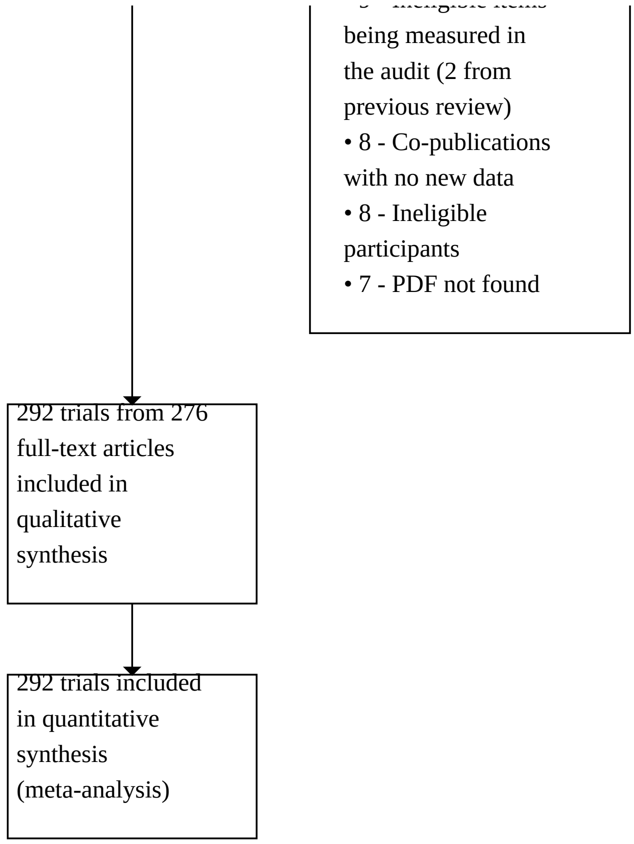


Figure 1. (Continued)



All extracted data are available upon request; the general characteristics of the included studies are described in Table 1.

We found no ongoing studies. An additional 61 studies meeting inclusion criteria were identified in a top-up search performed in February 2022 and are briefly described in Characteristics of studies awaiting classification.

Included studies

The unit of allocation was groups of clusters of healthcare professionals (e.g. clinics, wards, hospitals) in 221 studies, a single healthcare provider in 65 studies; this was unclear in five studies (2698 communities/patient groups) (Diamantouros 2017; Fabbri 2019; Hocking 2018; Navarro 2012; O'Connell 1999), and patients were the unit of allocation in one study (Belcher 1990). Twenty-five studies had four arms, 39 studies had three, 224 had two and the remaining four had over four arms.

Characteristics of setting and professionals

One hundred and forty-nine studies were based in North America (120 in the USA, 29 in Canada), 37 in the UK or Ireland, 18 in Australia, 17 based in Asia, nine based in Africa, four based in South America, and 61 elsewhere in Europe. Twenty-six studies were from low- and middle-income countries.

Medical doctors were the targeted health professionals in 237 trials. Eight studies explicitly targeted pharmacists, 36 studies targeted nurses, and nine studies targeted long-term care staff. General or family practice was the most common clinical speciality area, targeted in 170 trials. One hundred and sixty-three trials were solely in a primary care clinic or office setting, 18 trials were in a community care setting (including nursing homes), 25 trials were in other outpatient settings (including specialist clinics), 65 were in inpatient settings, and the clinical setting was mixed in 21 studies.

Targeted behaviours

There were 138 studies aiming to improve appropriate prescribing and 103 targeting test utilisation. Fifty-nine trials focused on treatment decisions or actions and three focused on improving screening rates. The remaining trials varied widely across conditions and targeted behaviours. In 225 (77%) studies, recipients were required to increase their current behaviour, while 65 (22%) studies required a decrease in current behaviour to better align with the desired standard.

Description of co-interventions

Using the EPOC coding scheme (EPOC 2015) to categorise components of each intervention arm, we identified 17 out of 106 EPOC-taxonomy-co-interventions in the study arms. *The EPOC taxonomy* coding framework is organised into four categories: Implementation Strategies, Delivery Arrangements, Financial Arrangements and Governance Arrangements. For the full dataset (including trial arms with and without A&F), see Table 2.

Implementation Strategies from the EPOC taxonomy included:

- Clinician Education (377 study arms (55.6%));
- Reminders (100 study arms (14.7%));
- Patient Mediated Interventions (60 study arms (8.8%));
- Managerial Supervision (14 study arms (2%));
- Organisational Culture (3 study arms (0.4%)).

Delivery Arrangements included:

- Information and Communication Technology (62 study arms (9.1%));
- Teamwork (20 study arms (2.9%));
- Coordination of Patients (9 study arms (1.3%));
- Where care is provided (9 study arms (1.3%));
- Who provides care (6 study arms (0.9%));
- Enabling Access (3 study arms (0.4%));
- How/When care is delivered (3 study arms (0.4%)).

Financial Arrangements included:

- Pathways (49 study arms (7.2%));
- Financial Interventions (20 study arms (2.9%));
- Public Reporting (2 study arms (0.3%)).

Governance Arrangements included:

- Authority and Governance (13 study arms (1.9%)).

The majority of study arms, 372/678 (55%), featured multi-faceted interventions. A&F as the **only** EPOC intervention component was found in 86 study arms (12.7%).

The median number of EPOC components across all study arms was two per arm (range 1-8). There were 376 (55%) study arms that included only one EPOC component category (Implementation strategies, Delivery arrangements, Financial arrangements and Governance arrangements), 124 (18%) arms with two categories, 25 (4%) arms with three categories, only one (0.1%) arm with four categories and 152 (22%) arms that did not include any EPOC component categories. These 152 study arms represent "control" arms for the purpose of our analyses.

When examining EPOC taxonomy intervention categories used **in combination with A&F** (n = 435), we identified 327 (75%) study arms with additional Implementation Strategies: 'Clinician Education' (n = 303, 70% study arms) was the most common EPOC Implementation Strategy used with A&F, followed by 'Reminders' (n = 87, 20% study arms) and 'Patient Mediated Interventions' (n = 52, 12% study arms). The most common components used with A&F from other EPOC categories were 'Information and Communication Technology' (Delivery Arrangements), found in 52 (12%) study arms, 'Pathways' (Financial Arrangements), found in 43 (10%) study arms, and 'Authority and Governance' (Governance Arrangements), found in 11 (3%) study arms.

We extracted *behaviour change techniques* in all study arms, including control/comparator arms. Our analysis of behaviour change techniques included 369 'treatment' arms and 292 'control/comparator' arms (some control/comparator arms included A&F components).

Approximately half (48/95; 51%) of the behaviour change techniques were identified across 369 treatment arms at least once (mean = 5.2 per arm, SD = 2.8 behaviour change techniques per arm, range = 1 to 29 behaviour change techniques). The ten most common behaviour change techniques were:

1. 'Feedback on behaviour' (326/369, 88%; e.g. providing feedback on drug prescribing);
2. 'Instruction on how to perform the behaviour' (258/369, 70%; e.g. issuing clinical guidelines to staff);
3. 'Social comparison' (195/369, 53%; e.g. providing feedback on the performance of peers);
4. 'Credible source' (154/369, 42%; e.g. endorsements from a respected professional body);
5. 'Education (unspecified)' (112/369, 30%; e.g. giving a lecture to staff);
6. 'Social support (practical)' (89/369, 24%; e.g. providing support to staff to achieve the target behaviour or outcome);
7. 'Prompts/cues' (81/369, 22%; e.g. a computer prompt when ordering a routine test);
8. 'Discrepancy between current behaviour and goal' (77/369, 21%; e.g. contrasting healthcare performance with a standard);
9. 'Problem-solving' (76/369, 21%; e.g. identifying barriers and generating solutions towards achieving an audit standard); and
10. 'Restructuring the social environment' (68/369, 18%; e.g. team members taking on an additional role to support practice change).

In summary, a total of 134/292 (46%) control/comparator arms included at least one behaviour change technique. A total of 35/95 behaviour change techniques (37%) were identified across 292 control/comparator arms at least once. Excluding control/comparator arms with no behaviour-change technique, the mean number of behaviour change techniques in 134/292 control/comparator arms was 3.0 (SD = 2.3, range = 1 to 15 behaviour-change techniques). The five most common behaviour change techniques reported in control/comparator arms were identical to those listed above in the treatment arms: 1) 'Instruction on how to perform the behaviour' (79/292; 27%); 2) 'Feedback on behaviour' (68/292, 23%); 3) 'Social comparison' (40/292; 14%); 4) 'Credible source' (35/292; 12%); and 5) 'Education (unspecified)' (29/292; 10%).

Characteristics of A&F interventions

Detailed information regarding the characteristics of A&F interventions can be found in Table 3.

Audit

The total number of study arms with A&F was 435.

The majority of arms with A&F reported only one measure or quality indicator in the audit (296/435, 71%), 99 (24%) arms reported two indicators, 48 (12%) and 52 (13%) arms reported three and four indicators respectively, and 160 (37%) arms reported more than five indicators. There were 269 (62%) arms which aggregated data at the organisational or team level, and 195 (45%) arms at the individual health professional level. In the majority of arms, the source of the data was provided by manual chart review (177 arms, 41%) or computerised search of charts (164 arms, 38%).

Feedback

Feedback was delivered repeatedly in 264 (39%) arms. The frequency of the feedback was weekly in six arms, monthly in 88 arms, repeated but less than monthly in 118 arms, and only once in 171 arms.

The format of the feedback was clearly reported in 281 (96%) studies and in 407 (60%) arms. Feedback was delivered in multiple formats in 164 (24%) arms. One hundred and nine arms had verbal didactic feedback (16%), 157 (23%) had verbal interactive feedback, 346 (51%) had written feedback, and 31 (5%) had electronic dashboard feedback.

In 72 (11%) arms, feedback was delivered by a local champion while 108 (16%) arms had feedback delivered by an external champion.

Comparators were used in the feedback in a variety of ways, with many interventions featuring multiple approaches. In total, 332 (49%) arms with A&F provided a comparator in the feedback intervention: 122 (18%) arms used feedback that included historical personal trend data; 96 (14%) arms provided a target or information about the performance of top peers; and 235 (35%) provided an average performance of all those audited and/or of peers.

Action

Plans for improvement were provided to recipients in 428 (63%) arms, with 327 (48%) arms receiving general suggestions and 101 (15%) arms receiving feedback with explicit, detailed suggestions. Ninety-one (13%) arms involved plans for improvement that were co-developed between recipients and feedback providers.

The target of change or action plans was at the organisational or team level for 250 (57%) arms and provider-level for 331 (76%) arms.

Outcome measures

As this was an inclusion criterion for this update, all trials measured professional practice, such as prescribing or test-ordering. Some trials reported both practice and patient outcomes such as smoking status or blood pressure, but our focus in this update was on professional practice, since this is the proximal target of A&F. There was a mixture of dichotomous outcomes, measured at different levels of analysis (for example, the proportion compliance with guidelines or the proportion of patients with appropriate management or proportion of patient-visits when the desired professional behaviour occurred) and continuous outcome measures (for example, total number of laboratory tests, or total number of prescriptions) across and within studies.

ICC

Across the 872 dichotomous outcome measures, 203 (23%) reported an ICC value for at least a single outcome. The median ICC value reported was 0.096 (P25 = 0.048, P75 = 0.15). In linear models examining correlates of reported ICC values, we observed that values did not differ between reporting behaviour (P = 0.35), but larger studies reported smaller ICC values (P = 0.003). Missing ICC values were imputed as the median (0.096) in the primary analyses.

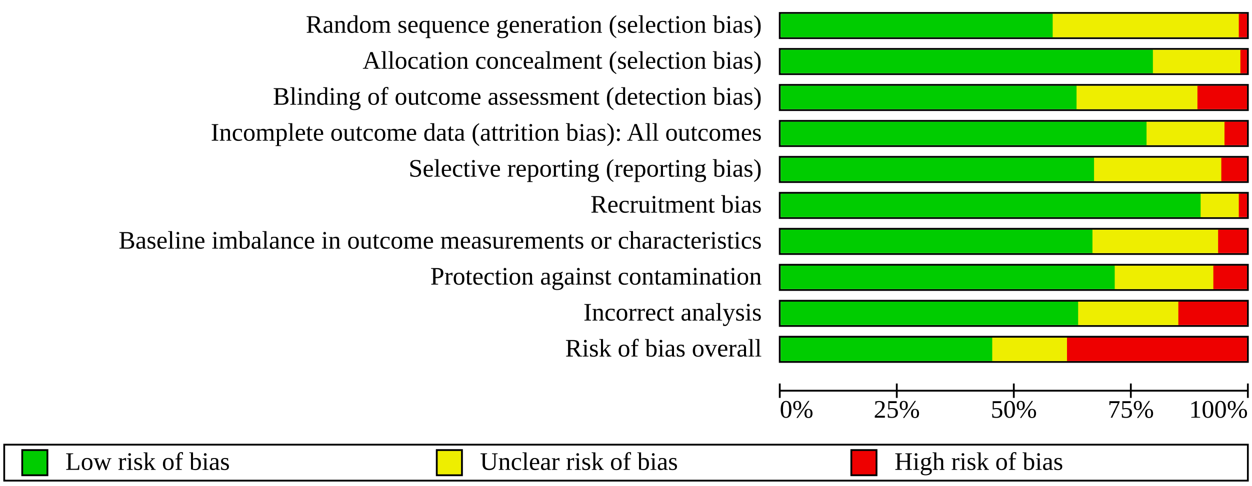
Excluded studies

We assessed 699 full texts for eligibility (559 studies from the updated search in addition to the 140 included studies from the previous review, Ivers 2012) and excluded 423 studies for not meeting inclusion criteria - of these, we have listed 61 studies in Characteristics of excluded studies that were included for data extraction, but upon closer review, were excluded for not meeting eligibility criteria. Details of the remainder of the excluded studies may be available upon request. The most common reasons for exclusion were: no reporting of eligible outcomes, not truly meeting criteria as an A&F intervention, and ineligible study design.

Risk of bias in included studies

See Figure 2.

Figure 2. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.



Of the 292 studies, 133 (46%) had a low risk of bias, 41 (14%) had an unclear risk of bias, and 113 (39%) had a high risk of bias. The most common sources of a high risk of bias related to: incorrect analysis methods that appropriately adjusted for clustering (39 trials, 14%), lack of outcome assessor blinding (e.g. when outcomes were reported by participating healthcare professionals) (29 trials, 10%), and due to selective outcome reporting (15 trials, 5%). Clarity of reporting regarding the risk of bias variables was frequently inadequate. For example, the nature of the randomisation sequence was unclear in 112 trials (39%), outcome assessor blinding was unclear in 76 trials (27%), similarity at baseline was unclear in 78 trials (27%), and risk of contamination was unclear in 63 trials (22%). Randomisation was clearly concealed (or there was cluster randomisation) in 234 trials (82%). There was adequate follow-up data in 225 trials (79%).

Allocation

We judged 169 studies out of 292 (58%) to have been randomised using appropriate methods. Only four studies (1%) were judged as at high risk for using non-random or inappropriate random sequence allocation methods. One hundred and seventeen studies (40%) did not report or did not provide enough information to adequately judge the sequence allocation methods.

We assessed allocation concealment as adequate in 235 studies (80%). Three studies (1%) were judged as being at high risk and 54 studies (18%) did not provide sufficient information to judge whether allocation concealment was adequate.

Blinding

Blinding of intervention recipients was not assessed. We rated 186 studies (64%) as appropriately blinding outcome assessment using objective outcome measures. We judged 29 (10%) studies as being at high risk and 77 (26%) as having unclear risk because they did not provide enough information to assess the appropriateness of blinding.

Incomplete outcome data

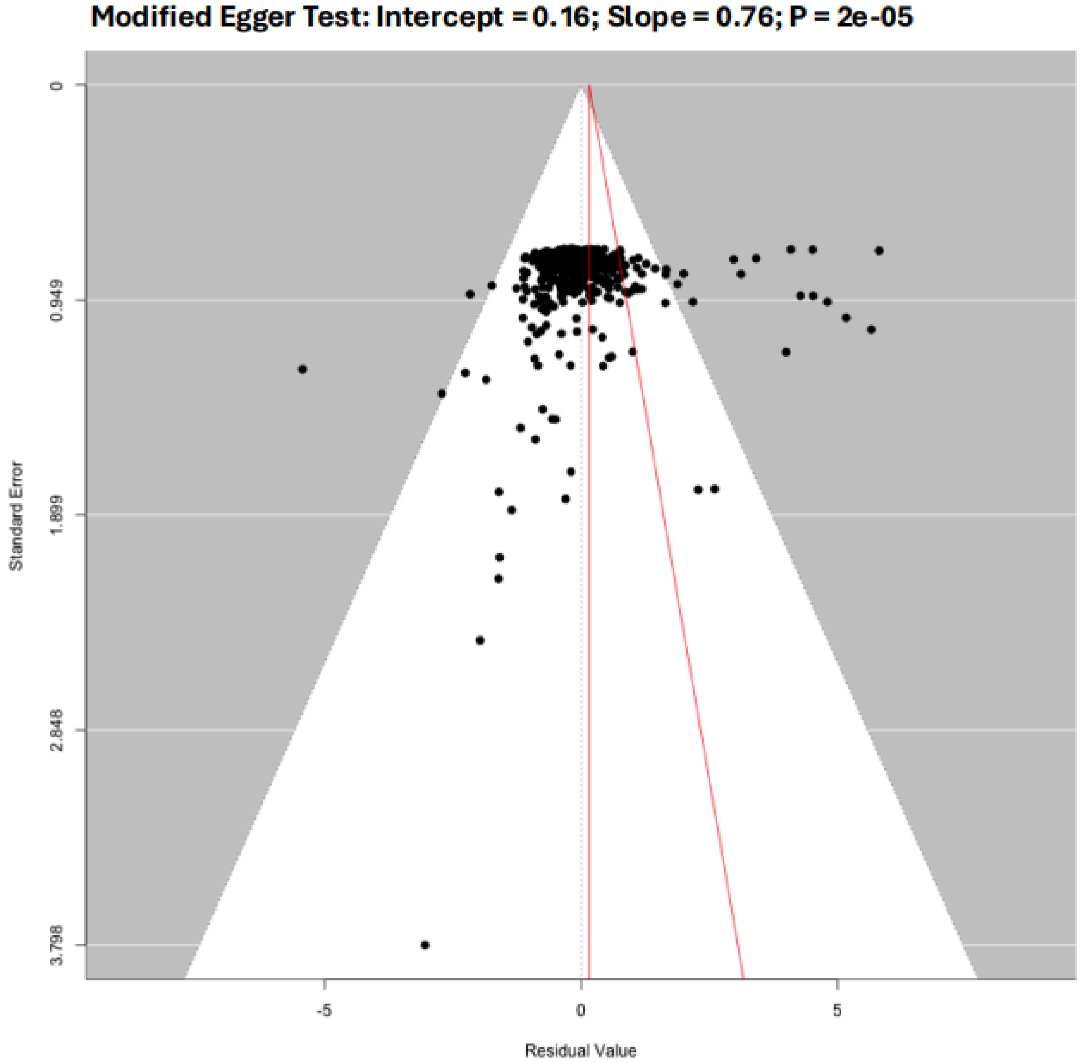
We assessed the majority of studies as being at low risk (231 studies, 79%) for incomplete outcome data. We judged 13 studies (4%) as being at high risk for incomplete outcome data and 49 studies (17%) as unclear for not providing adequate information.

Selective reporting

We assessed 197 studies (67%) as having low risk for selective outcome reporting. We judged 15 studies (5%) as being at high risk for not reporting all outcomes presented in the protocol and 81 studies (28%) as being unclear as no protocol was available.

Figure 3 includes the results of the modified Egger test and the funnel plot to examine for reporting bias. The analysis considering all outcomes from every trial comparing A&F versus no-A&F was significant, but the analysis using the median outcome for each trial only was not statistically significant.

Figure 3. Funnel Plot taking all outcomes across all trials on odds ratio scale and modified Egger test indicating reporting bias



Other potential sources of bias

Baseline performance was not reported in 11 studies (Buffington 1991; Diamantouros 2017; Gonzales 2013; Huis 2013; Kaufmann-Kolle 2011; Leviton 1999; Maddocks 2011; Mertens 2015; Robling 2002; Wathne 2018; Zafar 2019). As baseline imbalance in cluster trials is more likely than in patient randomised trials, effects from such studies must be viewed critically.

Effects of interventions

See: **Summary of findings 1** Audit and feedback for health professionals

See Summary of findings 1.

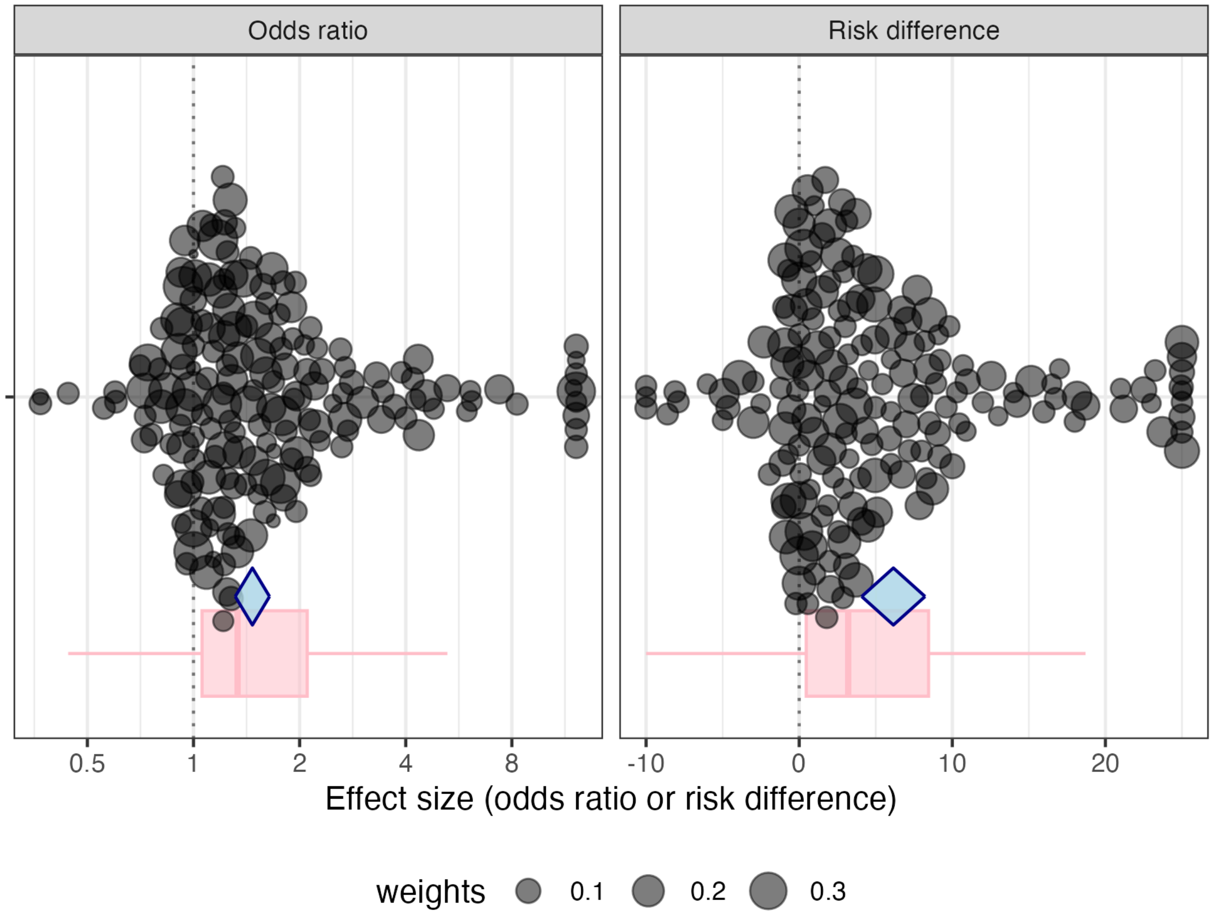
The primary outcome for quantitative analyses was the effect on compliance with desired professional practice. Therefore, quantitative analyses focused on those trial arms for which we had dichotomous data measuring processes of care targeted by the A&F with a suitable control (absence of A&F). Other studies are summarised qualitatively below.

Comparison A. Any intervention with audit and feedback, either alone or as part of a multi-faceted intervention, versus interventions without audit and feedback

In this quantitative analysis, we included data from 177 studies with 558 dichotomous outcomes of professional practice. These studies included 18,789 clusters/groups of health providers. Of note, 22 studies included in the quantitative synthesis for this comparison were at high risk of bias (12.4%). These analyses are summarised in **Table A**, which illustrates that effect sizes were consistent, whether considering all outcomes targeted by A&F or focusing specifically on the prespecified subset of outcomes measuring prescribing or test-ordering.

The range of effects observed across these studies is illustrated using an "orchard plot" in Figure 4, on both the odds ratio scale and risk difference scale. This highlights the substantial variation of findings in these studies and the presence of numerous trials with outlier effects. Figure 4 also provides summary results in the form of a "box plot" for median and interquartile range, as well as a meta-analytic estimate. A decline in performance (i.e. risk difference < 0% or odds ratio < 1) occurred in 27% of the studies. We observed absolute improvements in practice greater than 10% in 17 (9.6%) studies or an odds ratio greater than 2.0 in 28 (15.8%) studies.

Figure 4. Effects of Audit and Feedback versus Usual Care: Orchard Plot, Box Plot, and Meta-Analysis on Risk Difference and Odds Ratio Scale The black dots (Orchard Plot) represent every outcome assessed, including multiple outcomes per trial and should be considered a descriptive summary of findings. The pink line output (Box Plot) shows the median and interquartile range of effects across trials. The blue diamond (Meta-Analysis) presents the model-based effect estimate and 95% confidence interval.



The median risk difference (RD), considering all outcomes without adjustment, was 2.7%, with interquartile range (IQR) 0.0 to 8.6. The risk difference via meta-analysis was 6.2% (95% confidence interval [CI] 4.1 to 8.2; moderate-certainty evidence). The heterogeneity was extreme (I^2 = 98.8). The meta-analytic odds ratio (OR) was 1.47 (95% CI 1.31 to 1.64, I^2 = 92.2). Sensitivity analyses using the 25th percentile and 75th percentile for ICC produced did not change the results. Sensitivity analyses using the median to represent each study with multiple outcomes produced slightly lower effect sizes: RD 4.3% (95% CI 2.1 to 6.6; I^2 = 98.0) and OR 1.34 (95% CI 1.19 to 1.50; I^2 = 91.1).

There were 169 prescribing outcomes measured across 85 studies testing A&F versus usual care, which randomised 10,891 clusters. The median RD for prescribing was 2.0% (IQR -0.2 to 7.0). The RD for prescribing estimated through meta-analysis was 6.1% (95% CI 3.0 to 9.2), I^2 = 99.5; the estimated OR for prescribing was 1.43 (95% CI 1.21 to 1.70), I^2 = 94.4.

We also identified 166 test-ordering outcomes from 65 studies, which randomised 4036 clusters. The median RD for test-ordering was 2.5% (IQR -0.8 to 8.4). The estimated RD for test-ordering was 4.9% (95% CI 2.4 to 7.3), I^2 = 96.4; the estimated OR was 1.37 (95% CI 1.19 to 1.57), I^2 = 84.9.

Table A. Observed effect sizes estimated via meta-analysis for all outcomes targeted by A&F and for prespecified outcomes (prescribing and test-ordering)

Outcomes	No. of studies	No. of study outcomes	No. of clusters	Median risk difference (IQR)	Estimated risk difference (95% CI)	Estimated odds ratio (95% CI)
All	177	558	18,789	2.7% (0.0, 8.6)	6.2% (4.1, 8.2)	1.47 (1.31, 1.64)
Prescribing	85	169	10,891	2.0% (-0.2, 7.0)	6.1% (3.0, 9.2)	1.43 (1.21, 1.70)
Test-ordering	65	166	4036	2.5% (-0.8, 8.4)	4.9% (2.4, 7.3)	1.37 (1.19, 1.57)

Investigation of heterogeneity

We explored the contribution of study characteristics: year of publication before or after inclusion of studies from the prior Cochrane review (Ivers 2012), duration from randomisation greater or less than 12 months, risk of bias, unit of randomisation, and unit of analysis. Mostly, small differences in effects were noted across these strata, as summarised in **Table B** below, but it does appear that newer trials may be achieving larger effects. Heterogeneity remained high in all subgroups.

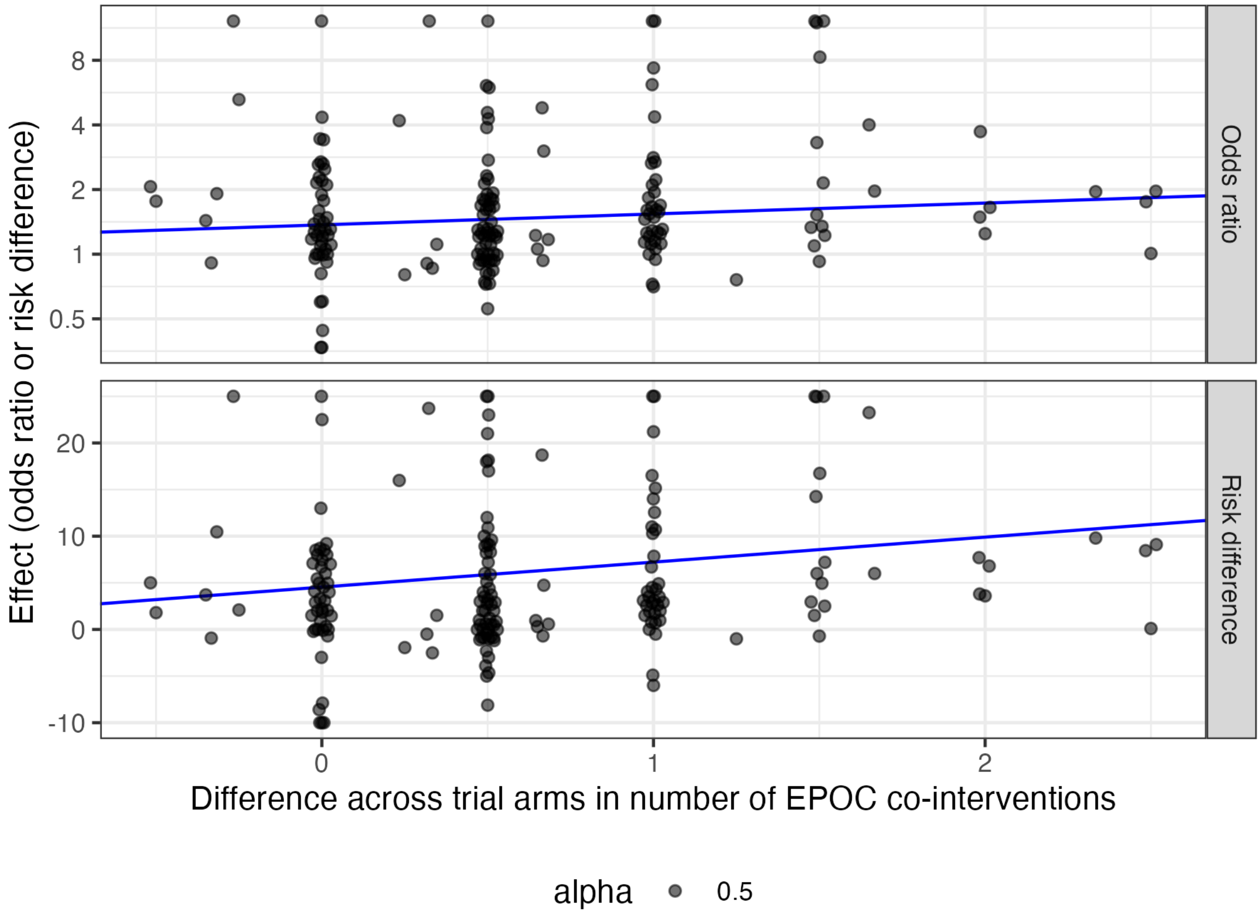
Table B. Effect size variation across study characteristics

Outcomes	No. of studies	No. of study outcomes	No. of clusters	Median risk difference (IQR)	Estimated risk difference (95% CI)	Estimated odds ratio (95% CI)
Study year < 2010	102	337	9105	2.2% (-0.3, 7.6)	4.2% (2.3, 6.0)	1.28 (1.18, 1.39)
Study year ≥ 2010	75	221	9684	3.5% (0.0, 9.3)	9.0% (5.2, 12.9)	1.79 (1.43, 2.25)
Duration < 12 months	54	120	5020	3.1% (-0.2, 9.5)	8.3% (3.7, 12.9)	1.65 (1.26, 1.56)
Duration ≥ 12 months	122	432	13,805	2.6% (0.0, 8.2)	5.5% (3.3, 7.6)	1.40 (1.26, 1.56)
Risk of bias - high	22	76	3798	2.5% (-0.1, 7.0)	2.9% (0.2, 5.6)	1.22 (1.02, 1.46)
Risk of bias - unclear	40	121	3452	2.9% (0.4, 8.1)	7.4% (1.3, 13.6)	1.53 (1.07, 2.18)
Risk of bias - low	115	361	11,539	2.6% (-0.4, 9.0)	6.1% (4.0, 8.1)	1.47 (1.33, 1.62)
Randomised teams	139	479	12,243	2.6% (0.0, 8.3)	5.9% (4.1, 7.8)	1.43 (1.30, 1.58)

Randomised providers	37	74	4106	3.1% (-0.4, 9.4)	6.7% (-0.8, 14.1)	1.53 (1.04, 2.26)
Unit of analysis - encounters	40	96	9303	1.9% (0.0, 6.0)	6.1% (1.9, 10.3)	1.43 (1.14, 1.81)
Unit of analysis - patients	113	412	6485	2.8% (-0.2, 8.7)	5.6% (3.2, 7.9)	1.40 (1.25, 1.56)
Unit of analysis - providers	26	50	3070	3.9% (0.8, 16.5)	10.1% (2.4, 17.8)	2.00 (1.24, 3.23)

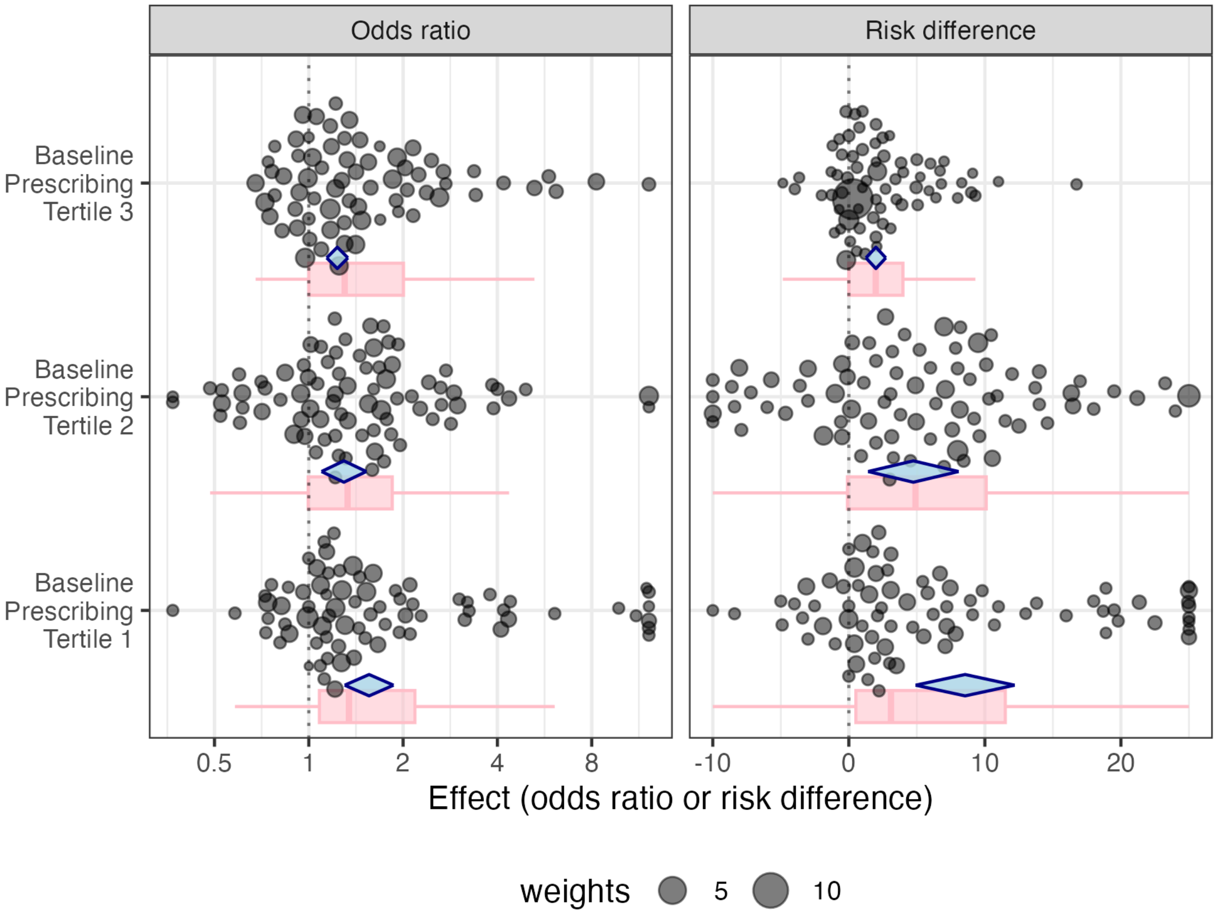
We examined the role of co-interventions by producing a scatter plot and regression line in which the A&F effect size (on both the risk difference and odds ratio scales) was estimated for those study contrasts with the same number of co-interventions in each arm (e.g. study comparing A&F plus reminders versus reminders), and when there was a difference in the number of co-interventions (e.g. a study comparing A&F plus reminders versus control). Figure 5 demonstrates that effect estimates were greater with increased difference in the co-interventions present in the A&F arm compared to the control arm.

Figure 5. Effect of A&F (in risk difference and odds ratio scale) varies by EPOC-defined co-interventions in the trial arms. X-axis denotes the difference in the number of EPOC interventions within trials, subtracting those in arms without A&F from those with A&F. Note that some trials had more EPOC interventions in the non-A&F arms. In general, trials with more EPOC interventions along with A&F had greater effect sizes.



We also tested the role of room for improvement (i.e. baseline performance), similarly producing orchard plots that stratified results by tertile of baseline performance in the outcome. Figure 6 illustrates that those studies with the greatest room for improvement (i.e. those studies in the lowest tertile for baseline performance) generally had greater effects. The lowest tertile estimated RD was 8.6% (95% CI 4.9 to 12.2) and the estimated OR was 1.56 (95% CI 1.30 to 1.86); the highest tertile estimated RD was 2.0% (95% CI 1.3 to 2.7) and estimated OR was 1.23 (95% CI 1.14 to 1.33).

Figure 6. Effects of A&F by tertiles of baseline performance: orchard plot and box plot on risk difference and odds ratio scales



Finally, we explored the role of characteristics of the *audit* (number of quality indicators, level of aggregation of data), *feedback* (multiple formats, local champion, type of comparator, repeated delivery), and supports for *action* (facilitation, provision of specific action plans, co-development of action plans). Each characteristic was tested in an independent model, adjusting for the count of EPOC co-interventions, baseline performance, and the overall A&F effect, such that the resulting output indicates the multiplicative effect of these components beyond the average A&F effects. Results from these meta-regressions are summarised in **Table C**.

Table C. Estimated effect size with and without specific components of the A&F interventions

A&F intervention component	Arms with component (N = 579)	Odds ratio multiplier (95% CI)
Audit		
Less than 5 quality indicators in the A&F	218 (38%)	0.84 (0.69, 1.02)
Individual (rather than team-level) data	166 (29%)	1.15 (1.02, 1.29)
Feedback		
Comparison to average or all peers	109 (19%)	1.03 (0.90, 1.18)

Comparison to top peers or benchmark	35 (6%)	1.18 (1.02, 1.36)
Multiple episodes of feedback (rather than one-time-only)	224 (39%)	0.90 (0.81, 0.99)
Multiple formats (e.g. verbal, written, interactive)	163 (28%)	1.33 (1.27, 1.39)
Delivery from a local champion	69 (12%)	1.10 (1.00, 1.21)
Action		
Facilitation to engage with the feedback	131 (23%)	1.11 (1.05, 1.18)
Specific action plans for improvement given	87 (15%)	1.16 (1.10, 1.23)
Specific action plans co-developed with recipients	80 (14%)	1.06 (0.96, 1.17)

We found that A&F effects were enhanced when using individual-recipient-level data rather than team-level data, comparing to top peers or a benchmark, when delivered from a local champion with whom the recipient had a relationship, through multiple modalities rather than just didactic or just written, when there were facilitation supports to engage with the feedback and develop a response, and when action plans to improve were provided. The meta-regressions found no significant effects for a number of quality indicators in the audit, comparing to average or all peers, or co-development of action plans. Multiple episodes rather than one-time-only feedback was associated with a lower effect size.

These findings are held with a low level of confidence due to indirectness of the evidence. It is possible these meta-regression findings are confounded by the complexity of the targeted behaviour and the intervention, and thus must be interpreted cautiously.

Comparison B. Subgroup analysis of trials comparing interventions with audit and feedback without co-interventions to no active intervention

We identified a subset of 14 trials (with 26 comparisons) testing the effects of A&F in trials without any other active interventions in either arm of the comparison with dichotomous measures of compliance with desired practice. These trials randomised a total of 3568 clusters of health professionals. The median RD observed in these comparisons was 4.2% (IQR 0.1 to 8.0). The meta-analytic estimate for RD was 2.6% (95% CI -0.4 to 5.7; I^2 = 78.1). The meta-analytic estimate for OR was 1.13 (95% CI 0.97 to 1.31; I^2 = 74.6).

Comparison C. Subgroup analysis of studies comparing audit and feedback with co-interventions compared to no active intervention

We identified a subset of 95 trials (with 300 comparisons) testing the effects of A&F with one or more co-interventions to no active intervention with dichotomous measures of compliance with desired practice. These trials randomised a total of 8144 clusters of health professionals. The median RD observed in these comparisons was 2.6% (IQR -0.3 to 10.1). The meta-analytic estimate for RD was 6.7% (95% CI 4.1 to 9.2; I^2 = 97.6). The meta-analytic estimate for OR was 1.47 (1.29 to 1.68; I^2 = 88.7).

Studies for comparisons A, B, C with outcomes ineligible for meta-analysis: Stepped-wedge trials

See Table 4.

There were 12 stepped-wedge trials of multi-faceted interventions featuring A&F that could not be included in the quantitative analysis (Amanyire 2016; Brown 2018; Diamantouros 2017; Dreischulte 2016; Fuller 2012; Huffman 2018; Morrison 2015; Ralph 2018; Rodriguez 2015; Scales 2016; Trent 2019; Wu 2019). Seven of the stepped-wedge trials targeted prescribing: Huffman 2018 and Wu 2019 both used A&F plus training to implement pathways for acute coronary syndromes, leading to increased prescribing of guideline-recommended medication classes at discharge in India and China, respectively. Amanyire 2016 used A&F plus education and managerial supervision to increase anti-retroviral prescribing in Uganda (relative risk [RR] 2.11, 95% CI 2.03 to 2.20); Diamantouros 2017 used a multicomponent intervention featuring A&F to improve thromboprophylaxis prescribing in hospital with mixed effects across types of hospital wards; Dreischulte 2016 used A&F plus education, decision-support, and financial incentives to reduce high-risk prescriptions in Scotland (OR 0.63, 95% CI 0.57 to 0.68); and Ralph 2018 used A&F as a key part of a multi-faceted quality improvement approach in remote areas of Australia but did not result in a significant change in adherence to penicillin injections for preventing rheumatic fever (OR 0.78, 95% CI 0.54 to 1.11).

Two trials focused on changing the practice of specialist physicians: in the USA, Trent 2019 showed that monthly emailed feedback to emergency room physicians improved compliance with guideline-recommended management of pneumonia and sepsis (OR 1.7; 95% CI 1.2, 2.5; 4-22% absolute increase in compliance); in Australia, Brown 2018 found that A&F plus opinion leader education led to no changes in referrals by urologists to radiation oncologists (RR 1.06; 95% CI 0.74 to 1.51).

Two trials focused on hand hygiene: in Argentina, Rodriguez 2015 showed that leadership engagement with A&F led to a small improvement in hand-hygiene (OR 1.08, 95% CI 1.03 to 1.14); in the UK, Fuller 2012 showed that frequent observation, feedback and personalised action planning resulted in better hand hygiene for intensive care wards (OR 1.44; 95% CI 1.18, 1.76; 7–9% absolute increase in compliance), but not in other wards.

Finally, two stepped-wedge trials focused on the management of cardiac arrest in Canada: Morrison 2015 compared education and training of local champions with and without A&F and educational outreach, finding no change in temperature management; Scales 2016 compared the same interventions to usual care, finding improvement in neurological assessments (OR 1.79, 95% CI 1.01 to 3.19).

Studies for comparisons A, B, C with outcomes ineligible for meta-analysis: Continuous outcome data only

See Table 5.

There were 26 trials which evaluated A&F interventions against usual care but with outcomes only reported as a continuous measure. One trial used feedback to successfully improve infectious disease management in hospitals in Laos (Wahlström 2003). Another trial focused on improving chart documentation for children (Socolar 1998), without effect. A third trial used feedback of various processes of primary care based on patient survey data (Vingerhoets 2001), also finding no effect. Two other trials used feedback to increase the number of referrals to smoking cessation programmes, one showing effectiveness (Wadland 2007) and one not (Houston 2015). Nine additional trials with only continuous outcomes focused on prescribing; six with a positive effect (Bregnhøj 2009; Dormuth 2012; McConnell 1982; Navarro 2012; Nejad 2016; Smith 1998), and three with no effects compared to usual care (Hemkens 2017; Holm 1990; Wattal 2015). Twelve others focused on test-ordering: three showing desirable changes (Everett 1983; Kerry 2000; Martin 1980); four with no effects (Marton 1985; Ryskina 2018; Van der Weijden 1999; Wones 1987); and the rest with mixed effects (Ruangkanchanasetr 1993a; Ruangkanchanasetr 1993b; Ruangkanchanasetr 1993c; Verstappen 2003a; Verstappen 2003b). For example, Verstappen 2003a and Verstappen 2003b showed significant improvement in appropriate use of some tests but not others using the same intervention in the same context. It seems likely that the complexity of the targeted behaviour to be changed (i.e. the specific test targeted) plays an important role.

Comparison D. Different ways of providing audit and feedback (head-to-head comparisons)

Seventeen trials included head-to-head comparisons of different ways of conducting A&F, which we organised below according to studies directly testing different approaches to: i) data audited, ii) peer comparisons, iii) information provided, and iv) method of delivery. Some of the 17 trials included multiple comparisons across these categories. As summarised in **Table D**, the most consistent findings from these trials were that peer comparisons seem to amplify effects (versus no peer comparison), and design elements in the feedback to help recipients identify and act upon high-priority clinical items seem helpful.

Table D. Summary of findings of head-to-head trials comparing different ways of providing A&F

Variation in A&F	No. of studies	Findings
Type and amount of data audited	2	1 found effects of involvement in self-audit and 1 found no difference between feedback from administrative data versus electronic records.
Type of peer comparison	9	4/6 found statistically significant improvement with peer comparison versus without. 3 trials found benefit when comparing top-peers versus all peers, with 1/3 statistically significant.
Information provided in feedback	8	6 trials found benefit with carefully designed feedback that directed recipient attention to high-priority action; 5/6 statistically significant. 2 trials testing action planning addendums; 1/2 found statistically significant benefit.
Type of delivery	4	1 trial found verbal feedback significantly more effective than written; 1 found dashboard plus verbal significantly more effective than just verbal. 1 trial found that text messages were no more effective than paper and 1 trial found that increasing frequency from 2 to 3 times over 9 months was not more effective.

Type and amount of data audited

Two studies directly tested the type and amount of data audited and used for the feedback reports.

Brady 1988a found that when resident physicians conducted a self-audit at baseline, it led to improvements compared with simply receiving the data for mammographic screening rates (adjusted RD 8%; P reported as < 0.05) but a non-statistically significant improvement for influenza vaccination rates (adjusted RD 1.5%). Herrin 2006 compared feedback based on administrative data alone to the addition of patient-specific clinical data from medical records. They did not find any differences statistically in processes of care for diabetes management.

Type of peer comparisons

Nine trials tested the role of peer comparison data; six of these directly tested feedback with peer comparison to feedback without. Wones 1987 found no difference in the number of daily inpatient laboratory tests when adding peer comparison data to the A&F. Søndergaard 2002 tested the addition of peer comparison for outpatient asthma management, but this test was confounded by also testing aggregate-only versus patient-specific information in the feedback. Elouafkaoui 2016 was a factorial trial testing A&F for dentists; one factor tested the provision of local comparator data to no comparator data and found a 0.36% absolute reduction in antibiotic prescribing (P = 0.057). Patel M 2018 found in their adjusted models a 5.8% (P = 0.008) improvement in statin prescribing in primary care when the feedback included peer comparison compared to no feedback at all and 4.1% (P = 0.11) improvement when the feedback did not have peer comparison. Boet 2018 found that A&F with peer ranking to anaesthesiologists led to a non-significant improvement in the use of warmers (adjusted RD 0.9%). Navathe 2020 found in their adjusted analysis that feedback with peer comparisons had a 3.1-percentage-point higher composite quality score measuring adherence to a range of primary care guideline recommendations than those receiving feedback without peer comparisons (P = 0.048).

Three studies directly tested whether comparisons were more effective when focused on top-performing peers. Kiefe 2001 compared feedback featuring a mean score of peers with feedback that featured an "achievable benchmark" (the mean score of the top 10% of peers). They found that the achievable benchmark group improved quality of care for diabetes (adjusted RD 3%, IQR 2% to 4%). In particular, statistically significant increases were observed for influenza vaccination (OR 1.54, 95% CI 1.26 to 1.96), foot examination, (OR 1.33, 95% CI 1.05 to 1.69) and haemoglobin A1C measurement (OR 1.33, 95% CI 1.04 to 1.69), while cholesterol measurement (OR 1.20, 95% CI 0.95 to 1.51) and triglyceride measurement (OR 1.15, 95% CI 0.92 to 1.44) had non-statistically significant increases. Schneider 2008 found that identifying top performers in feedback presented in a quality circle (i.e. learning collaborative) led to non-statistically-significant improvements in the management of asthma (adjusted RD 1.6%). Similarly, Kaufmann-Kolle 2011 found that open benchmarking and identification of the top performers in quality circles led to small, non-statistically-significant improvement in prescribing of interacting medications (adjusted RD 0.6%) and asthma care (adjusted RD 0.5%).

Information provided in the feedback

Six studies tested approaches to focus the feedback recipients' attention on high-priority actions. Søndergaard 2002 tested the provision of patient-specific or just aggregate prescribing data for outpatient asthma management, but this test was confounded by also testing peer comparison information in the feedback, as noted above. Mitchell 2005 found that feedback that emphasised patients at greatest risk led to a non-statistically significant increase in treatment of blood pressure in primary care (adjusted RD 4.6%). Elouafkaoui 2016 was a factorial trial testing A&F for dentists; one factor tested the provision of behavioural messages to no added message and found a 0.51% absolute reduction in antibiotic prescribing (P = 0.005). Nejad 2016 found that a redesigned feedback letter focused on one topic (prescribing of steroids) resulted in a greater (desired) reduction in that targeted behaviour than with usual multi-topic prescribing feedback (P = 0.016). Similarly, Soleymani 2019 found that a redesigned feedback letter focusing on inappropriate steroid and antibiotic prescribing led to statistically significant 1% absolute reductions in the proportion of prescriptions for each of these (P < 0.01). Whidden 2018 found that adding a dashboard which visually summarised key areas of strength or needing improvement with detailed data to monthly feedback meetings with a supervisor led to improvements in number of visits (P = 0.03) but not quality of care (P = 0.15) delivered by community health workers in Mali.

Two studies tested whether adding action-planning sections to feedback would improve results. Ivers 2013 found that adding an action-planning section to feedback sent to family physicians led to no significant changes in diabetes care, with the power limited due to poor fidelity. Roos-Blom 2019 found that an action toolkit appended to a quality improvement dashboard in the ICU led to improvements in adequate pain management (adjusted RD 9.9%).

Type of delivery

Batty 2001 compared modes of A&F interventions for in-hospital benzodiazepine prescriptions. The verbal presentation was more effective than the written feedback (adjusted RD 24%) in this trial. In Whidden 2018, community health workers received verbal feedback with or without access to a dashboard. Those with access to the dashboard and verbal feedback had a 4% absolute improvement in both timeliness and quality of care. Nejad 2016 found no differences in prescribing between those receiving feedback via traditional paper letters and via text messages. Elouafkaoui 2016 was a factorial trial testing A&F for dentists; one factor tested the provision of feedback three times over nine months to two times over nine months and found no difference in antibiotic prescribing.

Comparison E. Audit and feedback combined with complementary interventions versus audit and feedback alone

Sixty-five studies compared co-interventions alongside A&F to A&F alone. Below, the results of these studies are summarised within categories related to the co-intervention. While some of the multi-faceted interventions may fit into multiple categories, we only described the findings from each comparison in a given trial once. However, multi-arm studies may be described in multiple sections corresponding to the type of comparison.

The findings of these studies below are summarised qualitatively, within each type of intervention-comparison (in bold text), organised narratively by targeted behaviour change and setting. Outcomes targeted are italicised/underlined to facilitate identification of relevant trials by readers with specific interests.

Table E. Summary of findings of head-to-head trials comparing A&F with co-intervention(s) to A&F alone

Comparison	No. of comparisons	Findings

A&F with outreach/ facilitation/ detailing versus A&F alone	26	2 of 3 studies focused on high-risk prescribing showed statistically significant improvement, with the other conducted in nursing homes. 2 of 3 studies focused on antibiotic prescribing appeared to have additional benefit from detailing. Only 1 of 5 studies found improved management of patients with high cardiovascular risk and/or diabetes; that study facilitated a patient recall system in the clinics. 8 studies focusing on processes related to prevention in primary care such as cancer screening and immunisation found mixed effects and 7 studies in specialist or inpatient settings found mixed effects.
A&F with other education strategies versus A&F alone	19	6 studies evaluated a seminar: only 2, focused on antibiotic prescribing in lower-income countries, appeared effective. 3 studies tested the addition of small group learning and each appeared to be effective, while only 2 of 8 studies testing larger group education resulted in statistically significant benefit. 2 others testing written educational materials found limited benefit.
A&F with reminders/ decision-support versus A&F alone	11	7 studies were in primary care; 2 studies found that reminders add benefit for test-ordering in primary care; 3 studies found mixed effects on prescribing; 2 found benefit for a range of prevention activities. 4 studies in other settings found limited or no effects.
Other	13	5 trials compared A&F plus patient educational materials or reminders with A&F alone and only 1, focusing on diabetes, found benefit. 5 trials compared A&F with case management-type interventions to A&F alone; 2 focused on cardiovascular risk management revealed benefit, but trials focused on vaccinations or diabetes care did not. 4 trials compared A&F with financial incentives to A&F alone with mixed effects.

Audit and feedback with educational outreach, practice facilitation, and/or academic detailing versus audit and feedback alone

Twenty-six studies compared A&F alone to the combination of A&F and practice facilitation, educational outreach or academic detailing. Such interventions involved experts providing support at the workplace of health professionals receiving A&F to increase knowledge, usually with a logistical focus on putting the knowledge into practice.

Four studies focused on *high-risk prescribing*, looking at inappropriate prescribing, high-risk medications, and/or medications at risk of interactions. The PINCER trial (Avery 2012) found that educational outreach by a pharmacist reduced unsafe prescribing practices by general practitioners compared to feedback alone for the primary outcomes of NSAID (nonsteroidal anti-inflammatory drug) use without PPI (proton-pump-inhibitor) (OR 0.58, 95% CI 0.38 to 0.89), beta-blocker use in asthmatics (OR 0.73, 95% CI 0.58 to 0.91), and ACE (angiotensin-converting enzyme) or diuretic use without electrolyte measurements (OR 0.51, 95% CI 0.34 to 0.78). Clyne 2015 showed that adding detailing by a pharmacist with patient-specific advice was more effective than feedback alone, reducing the percentage of patients in primary care with inappropriate prescriptions (adjusted OR 0.32; 95% CI, 0.15 to 0.70; P = 0.02). Lopez-Picazo 2011 tested pharmacist detailing, focused on potential drug interactions compared to a report alone in primary care and found a non-statistically significant improvement (adjusted RD 0.9%). However, Tadrous 2020 found that detailing in nursing homes did not further decrease antipsychotic prescribing compared to feedback alone (OR 1.06, 95% CI 0.93 to 1.20; P = 0.49).

Two factorial studies in Sudan both found reductions in the number of inappropriate *antibiotic prescriptions* with academic detailing compared to A&F alone (Awad 2006 - reduction in mean of 1.9 for feedback alone, reduction in mean of 5.9 for feedback plus detailing; P value not reported for this comparison; Eltayeb 2005 - reduction in mean of 3.8 for feedback alone, reduction in mean of 14.2 for feedback plus detailing; P value not reported for this comparison). However, in Ireland, Naughton 2009 found that detailing added no further benefit to antibiotic prescribing in primary care compared to feedback alone (reduction of 2% in both arms; P = 0.26).

Five studies focused on primary care management of patients with *high cardiovascular risk and/or diabetes*. The educational outreach in Moher 2001 focused on showing primary care providers how to utilise the feedback reports to develop and implement systematic patient recall systems. This resulted in an improvement (adjusted RD 22%; P = 0.002) in the proportion of patients with adequate assessment of cardiovascular risk factors, but no differences in actual treatment. Ornstein 2004 found statistically significant improvement in only two of 21 outcomes related to preventive cardiovascular care in the primary care setting and a difference in overall improvement that was not statistically significant (adjusted RD 5.5%, P > 0.2). Similarly, Naughton 2007 found that academic detailing plus feedback led to no additional improvement compared to feedback alone for prescribing statins and anticoagulants for high cardiovascular risk patients in primary care (median adjusted RD -0.5%; P value not reported). Ward 1996 found that outreach led to small, but statistically significant improvements in mean diabetes care scores compared with postal feedback alone (mean score 1.82 versus 0.37; P < 0.001). Rask 2001 found that outreach improved diabetes care for only one of six professional outcomes (median adjusted RD 9.5%; P value not reported).

Eight studies focused on *preventative activities in primary care*. Siriwardena 2002 found that only two of seven outcomes related to immunisation rates improved (median adjusted RD 5%, P value not reported). However, Quinley 2004 showed that an educational outreach phone call led to increased pneumococcal vaccination rates compared to feedback alone. Mold 2008 found that academic detailing and practice facilitation led to increased implementation in primary care for immunisations (median adjusted RD 8%; P value not reported) and cancer screening (median adjusted RD 10%; P value not reported). Kinsinger 1998 combined A&F with educational outreach, aiming to help primary care providers improve office systems to increase breast cancer screening rates. The intervention did improve the office systems and found an increase in the proportion of patients discussing mammograms (adjusted RD 4.75%; P = 0.01), but not a statistically significant difference in actual mammography rates (P = 0.56) compared with feedback alone. Ornstein 2010 found that academic detailing plus practice reports led to increased colon cancer screening compared to feedback alone (adjusted RD 4.9%; 95% CI 3.8% to 6.1%). However, Price-Haywood 2014 tested detailing visits focused on communication techniques in addition to feedback compared to feedback alone, which led to no changes in cancer screening rates in primary care (P values not reported). Similarly, Borgiel 1999 found that educational outreach plus a comprehensive primary care practice assessment report did not lead to improvements across nine quality of care indicators (adjusted RD for global quality of care score = -0.2%; P value not reported). Yano 2008 had a more specific focus on smoking cessation, but also found no effects on counselling or referrals, due to limited uptake of implementation strategies in participating clinics.

Seven studies focused on *specialist or inpatient care settings*, finding limited intervention effectiveness. Soumerai 1998 found that opinion-leader-led outreach led to improvements in two of four outcomes related to the management of acute myocardial infarction (median adjusted RD 8.5; P value not reported). Lakshminarayan 2010 found significant improvement after opinion-leader outreach in two of 10 outcomes related to management of acute ischaemic strokes in hospital, but no overall effect (median adjusted RD 4%; P value reported as non-significant). Williams 2016 found that quality improvement facilitation in addition to quality indicator feedback improved some processes for inpatient stroke care. Patel S 2018 added a refined dashboard and repeated meetings to routine feedback and showed that it improved the content of discharge summaries (unadjusted RD 16%; P < 0.0001), but not the timeliness of summaries (unadjusted RD 3%; P = 0.111) or the completion of medication reconciliation (unadjusted RD 2%; P = 0.416). McClellan 2004 found that adding quality improvement facilitation led to differences in two of six outcomes related to the management of dialysis. Guadagnoli 2000 found no differences for breast cancer treatment (adjusted RD -2%; P value not reported). One other study, Cheater 2006 focused on another complex condition and found that educational outreach to primary care nurses did not lead to improvements in assessment or management of urinary incontinence compared to A&F alone.

Audit and feedback plus other educational interventions versus audit and feedback alone

Nineteen studies tested educational interventions other than outreach or detailing or facilitation.

The effect of adding a **seminar** to A&F was tested in six studies. Both Eltayeb 2005 (reduction in mean*inappropriate prescribing* of 3.8 for feedback alone, reduction in mean of 11.6 for feedback plus seminar; P value not reported for this comparison) and Awad 2006 (reduction in mean of 1.9 for feedback alone, reduction in mean of 5.1 for feedback plus seminar; P value not reported for this comparison) found that adding seminars to A&F reduced *inappropriate prescribing* of antibiotics in Sudan. Robling 2002 found minimal difference in compliance by primary care with guidelines for Magnetic Resonance Imaging of the lumbar spine or knee when adding a seminar to feedback (unadjusted RD 4%, high risk of bias; P value not reported). Sauaia 2000 found non-statistically-significant improvement from the addition of verbal feedback in a group setting by an expert cardiologist mainly to nurses and managers with feedback reports alone for improving eight quality of care outcomes related to acute *management of myocardial infarction* (median adjusted RD 7%; P value for each outcome > 0.05). Pavese 2014 found that a seminar for hospital departments led by a local infectious disease leader did not lead to further improvements compared to feedback alone for appropriateness of *blood cultures* (median adjusted RD across four outcomes = -4%; P all > 0.1). Lopez-Picazo 2011 tested a series of seven pharmacist-led group seminars focused on potential *drug interactions* along with provision of a prescribing interaction feedback report compared to a report alone in primary care and found a non-statistically significant improvement (adjusted RD 0.8%).

Three studies tested the combination of **small groups or personalised learning** with A&F, compared to A&F alone. Herbert 2004 compared combining feedback with problem-based learning groups in primary care to feedback alone and found the combination had a greater effect on appropriate use of *antihypertensives* (adjusted RD 7.4%; P value not reported). Also, in primary care, Verstappen 2004 compared groups that focused on identifying gaps and developing quality improvement plans for decreasing total *laboratory tests* ordered to feedback alone and found the groups to be more effective (adjusted change relative to baseline performance in the A&F alone arm = 9%, P = 0.005). Kaminski 2016 found that a personalised training course for those conducting *colonoscopy* in addition to feedback was better than feedback alone for adenoma detection rate (OR 1.61; 95% CI 1.29 to 2.01).

Eight studies involved **educational programmes in larger groups or teams**. Rubin 2001 compared written feedback delivered only to the hospital administration with the addition of verbal feedback at staff meetings. They did not find a difference in the appropriateness of red blood cell *transfusions* (adjusted RD -2%; no statistical test reported for this comparison). Hayes 2001 compared written feedback alone with feedback enhanced by the participation of a trained physician champion who carried out staff meetings, quality improvement tools, and a project liaison for *anticoagulant management* of venous thrombosis, finding no statistically significant effect on the quality of care for venous thrombosis (P > 0.2 for each of five process outcomes). Hayes 2002 was a very similar trial targeting *heart failure,* again finding no statistically significant effect (median adjusted RD -1%; P values not reported). Spoorenberg 2015 also tested a similar intervention, this time focusing on *inpatient antibiotic prescribing* for complicated urinary tract infections, and also found no differences (RD for quality score = -0.6; P-value not reported). Also, in the hospital setting, Kritchevsky 2008 found that adding a quality improvement collaborative to feedback alone did not improve the utilisation of antibiotics within one hour prior to surgery (adjusted RD -3.6%; absolute risk reduction (ARR) -3.8, 95% CI -13.9 to 6.2). Likewise, Filardo 2009 found that education regarding continuous quality improvement had no statistically significant impact on hospital-based *quality indicators for pneumonia or heart failure* compared to feedback alone (median adjusted RD -1.7%; P = 0.47), although the authors reported that this finding may be due to poor participation in the intervention group. However, Horbar 2004 found that feedback plus a quality improvement collaborative against feedback alone led to improved *care for preterm infants* (OR 5.38; 95% CI 2.84 to 10.20). Likewise, Smith-Bindman 2020 found that adding a quality improvement collaborative to feedback led to greater reductions in *radiation dose during imaging*.

Finally, two studies tested **written educational materials**. Marton 1985 also found that offering a manual outlining *laboratory costs* slightly reduced laboratory test utilisation compared to feedback alone (mean reduction of 0.32 versus 0.14 tests per patient-visit; P value not reported). Finally, Anderson 1994 found little or no difference when they compared feedback given to large groups as part of a continuing medical education programme, with and without sending individualised feedback reports to participants for *prophylaxis of venous thromboembolism* (adjusted RD 0%; P value not reported for this comparison).

Audit and feedback with clinician reminders or decision-support versus audit and feedback alone

Eleven studies evaluated adding point-of-care clinician reminders or decision-support to A&F.

Two of these studies aimed to reduce outpatient test-ordering in primary care. In a 2x2 factorial trial, Eccles 2001 found that adding reminders to A&F reduced *x-ray utilisation* compared to A&F alone (adjusted change relative to baseline control = 46%; P value not reported). In another 2x2 factorial trial, Thomas 2006 found that feedback and reminders both significantly reduced *blood test utilisation* and that the effect seemed to be additive, but not synergistic (adjusted change relative to baseline performance in the A&F alone arm -2%; OR 0.78, 95% CI 0.71 to 0.85 for both versus OR 0.87, 95% CI 0.81 to 0.94 for reminders alone, P value not reported).

Three studies focused on *prescribing in primary care*. Baker 1997 found improvement by adding reminders to feedback in the management of chronic benzodiazepine prescriptions (median adjusted RD 1.7%, high risk of bias; no overall statistical test conducted). Ziemer 2006 assessed clinical inertia in diabetes and found that the combination of reminders and feedback had a greater effect on treatment intensification than feedback alone (adjusted RD 7.25%; P value not reported). Meeker 2016 showed that reminding clinicians to provide a justification, or prompting them to reconsider the prescription, plus monthly peer comparison emails was not more effective at reducing inappropriate antibiotic prescriptions than the emails alone.

Two additional trials, also in primary care, tried to improve quality for a wide range of outcomes. Tierney 1986a in a complex factorial trial found that reminders together with A&F were more effective than feedback alone for provision of *preventive services* by internal medicine trainees working in a primary care clinic (unadjusted RD 8.0%; P value not reported). Coma 2019 examined the addition of point-of-care alerts and recommendations to monthly feedback for a range of 10 primary care guideline recommendations relative to cardiovascular risk reduction or vaccinations, finding that the combination led to more frequent completion of the relevant processes of care (OR 1.29; 95% CI 1.25 to 1.34).

Four trials tested this comparison within specialist or inpatient care settings. Boekeloo 1990 found mixed results across various quality indicators for *cholesterol management in hospital*when reminders (red tabs in paper charts) were combined with feedback compared with feedback alone. One dentistry trial, Bahrami 2004, compared A&F with a computer decision-support system to A&F alone to improve the management of impacted molars by *dentists*; neither intervention produced a statistically significant improvement (adjusted RD 6%; P value not reported for this comparison). Working with *rheumatology clinics*, Lesuis 2018 found that adding decision-support to feedback alone found no changes in adherence to relevant guideline recommendations (adjusted RD 2; P = 0.6). Luitjes 2018 tested a decision-support system and team meeting to discuss A&F data compared to feedback reports alone for improving care of *hypertensive pregnant patients* admitted to hospital, finding no differences in guideline-concordant treatment, counselling or investigations (P value not reported).

Audit and feedback with patient-mediated or organisational interventions versus audit and feedback alone

Five trials compared A&F plus **patient educational materials or reminders** with A&F alone. Buffington 1991 found that mailed patient reminders resulted in little or no difference from weekly feedback alone for *influenza vaccination* rates (adjusted RD 1%, high risk of bias; P value reported as 'not significant'). O'Connor 2009 found that mailed information with reminders to patients with *diabetes*did not increase the effectiveness of a feedback intervention for up-to-date investigations for these patients (P value not reported for this comparison). Weitzman 2009 found that the addition of patient reminders to feedback using both a letter and phonecall to urge comprehensive follow-up resulted in improved control of diabetes based on achieving glucose, cholesterol and blood pressure targets (median adjusted RD 4.4%; OR 2.4, P < 0.01). Hallsworth 2016 tested the addition of posters and leaflets in primary care offices to clinician feedback but found no effects from those patient-oriented interventions on *antibiotic prescribing*. Fabbri 2019 compared A&F with and without public reporting to improve *maternal care* in India, finding no differences in maternal care.

Five trials compared A&F with **case management-type interventions** to A&F alone. Moher 2001 compared mailed feedback to feedback plus a nurse recall system in a three-arm study. The nurse recall system improved the proportion of patients with adequate *assessment of cardiovascular risk factors* compared to feedback alone (adjusted RD 33%; ARR = 33, 95% CI 19 to 46). However, Herrin 2006 found that adding a diabetes resource nurse resulted in little or no change in *diabetes management*across six process outcomes; all comparisons reported with P values > 0.1). Likewise, Harris 2013 found that additional diabetes education from experts plus access to a community pharmacist did not increase *insulin initiation* for patients with diabetes in primary care compared to feedback alone (adjusted RD -1%; P = 0.23). Pape 2011 tested whether adding pharmacist case management to software providing quality reporting and benchmarking improved *cholesterol management* in primary care, finding that patients in the intervention arm were also 15% more likely to receive a prescription for a lipid-lowering medication (P = 0.008). One study added a telephone follow-up to A&F targeting *pneumococcal vaccine* coverage (Quinley 2004). This was an administrative task that encouraged the use of the feedback reports and required no clinical expertise and the intervention resulted in little or no difference in vaccine use across the two subgroups of physicians analysed (median adjusted RD 0.97%; P = 0.07, 0.09).

Four studies examined **financial incentives**. Fairbrother 1999 had three arms that compared A&F alone to A&F plus a one-off "financial bonus" based on up-to-date coverage for four *immunisations*, and A&F plus "enhanced fee for service" (five dollars for each vaccine administered within 30 days of its due date). Rates of immunisation improved from 29% to 54% coverage in the bonus group after eight months (adjusted RD 12.7%; no P value comparing bonus group to feedback alone). The enhanced fee-for-service group decreased performance relative to feedback alone (adjusted RD -8.3%; P value not reported for this comparison). Petersen 2013 used a factorial trial design to test feedback plus incentives for physicians, or practices, or both, compared to feedback alone for care of *hypertension in primary care*. Use of guideline recommended medications increased non-significantly for the physician-incentive group by 4.7% (95% CI −1.4, 10.9), for the practice-incentive group by 0.6% (95% CI −5.3, 7.3) and for the group receiving both 2.9% (95% CI −2.8, 9.6). Ryan 2014 explored whether A&F plus financial incentives and supports for advanced use of electronic medical records could improve *cardiovascular quality metrics in primary care*, with improvement only amongst the subset of metrics that had financial incentives. Navathe 2020 tested the addition to feedback an incentive available to both the provider and the patient for improving *diabetes control*. Unfortunately, recruitment to the incentive trial arm was inadequate to allow for analysis.

Comparison F. Comparing multi-faceted interventions that both include audit and feedback

Nineteen studies directly compared two different multi-faceted interventions featuring A&F. They are summarised here based on the interventions featured in the multi-faceted interventions (in bold text). Outcomes targeted are italicised/underlined to facilitate identification of relevant trials by readers with specific interests.

Table F. Summary of findings of head-to-head trials comparing A&F with co-intervention(s) to A&F with co-intervention(s)

Co-intervention	No. of comparisons	Findings
Outreach/ facilitation/ detailing	8	3 of 4 studies testing A&F plus educational outreach/detailing/facilitation versus educational seminars found benefit. 2 studies tested different forms of facilitation, finding minimal differences. 1 study found no benefit from adding continuous quality improvement support to A&F plus detailing. 1 study found A&F plus detailing was enhanced by practice facilitation.
Other education strategies	4	2 studies comparing in-person versus virtual education alongside A&F found no differences. 2 studies showed that more repeated A&F alongside education was effective, especially if it featured action planning.
Reminders/ Decision-support	3	1 study tested different forms of reminders alongside A&F with no effects. 2 studies involved adding patient reminders to clinician reminders and education and A&F with mixed effects.
Other	4	1 study tested adding a nurse specialist to A&F plus education in nursing homes without benefit. 1 study tested specialist support in addition to A&F plus education for diabetes management in primary care with benefit on some indicators but not others. 1 study testing financial incentives plus A&F and reminders showed mixed effects. 1 study testing different types of financial incentives showed mixed effects.

Eight studies involved testing **interventions featuring educational outreach or academic detailing or practice facilitation** alongside A&F against other multi-faceted interventions featuring A&F. Two of these tested A&F plus detailing versus A&F and seminars for inappropriate *prescribing of antibiotics* as part of factorial trials conducted in Sudan: Eltayeb 2005 (reduction in mean of 11.6 for feedback plus seminar, reduction in mean of 14.2 for feedback plus detailing; P value not reported for this comparison) and Awad 2006 (reduction in mean of 5.1 for feedback plus seminar, reduction in mean of 5.9 for feedback plus detailing; P value not reported for this comparison). Lopez-Picazo 2011 was another factorial trial that tested pharmacist-led seminars plus feedback versus detailing plus feedback for potential *drug interactions* and found similar effects (adjusted RD 0.1%). Ayieko 2011 added outreach and facilitation to feedback and education in hospitals in Kenya, finding significant improvement across nearly all 16 *paediatric inpatient process outcomes*.

In four of the seven studies, outreach or detailing or facilitation was in both trial arms. For example, Ward 1996 compared A&F

with detailing delivered by a physician-peer with A&F delivered by a nurse. They found that peer-physician detailing plus feedback led to no change in the *management of diabetes* (0.5% difference in mean compliance score across 21 process measures). They also noted that the physician interviews were longer (25 minutes versus 14 minutes; P < 0.001) and that there was a significant variation in effect size across the different physicians providing the outreach. Goldberg 1998 was a three-arm study in which two arms received feedback plus detailing and one of these was also supported through a continuous quality improvement strategy. No consistent differences in the targeted process outcomes related to *hypertension or depression management* were observed, which was attributed to poor implementation strategy fidelity. Mold 2014 was a factorial trial in which all four arms received feedback and detailing. The trial tested the addition of one or both strategies of practice facilitation and learning collaboratives for the implementation of *asthma* guidelines. Across six process measures, improvement was noted in each arm. In multivariable models, the addition of practice facilitation, but not learning collaboratives, resulted in statistically significant improvement in two outcomes (assessment of severity: OR 2.5; 95% CI 1.7, 3.8 and assessment of control: OR 2.3; 95% CI 1.5, 3.5). Also, in primary care, Persell 2020 tested two forms of practice facilitation on *cardiovascular risk reduction*, both involving feedback, finding improvements of 2% for aspirin (P = 0.337), 3% for statins (P = 0.055), and 2% for management of smoking (P = 0.385), with suboptimal fidelity of the implementation strategies.

Four additional studies involved some **other forms of educational programmes** with A&F in each arm. Two of these compared in-person education alongside A&F to virtual education alongside A&F. Brunette 2015 involved a seminar and presentation of prescribing data from a local psychiatry expert about *smoking cessation* delivered via videoconference or in-person, finding both effective but no differences in outcomes between groups. Gilkey 2014 was a three-arm trial in which two arms received education including feedback of immunisation data, either via a webinar or in-person, finding no differences in *vaccination rates* between those arms. Two others tested whether enhancing the combination of feedback and education could lead to greater effects. In Bond 2011, one arm received feedback and educational materials while the other received repeated feedback and seminars, leading to an increase of 8.9% (P = 0.04) in *influenza vaccination*. Ayieko 2019 randomised Kenyan hospitals to feedback plus educational meetings or the same plus enhanced feedback that was more frequent and featuring action planning prompts; performance in classifying and treating *childhood pneumonia* was better in the enhanced feedback arm (OR 3.6; 95% CI 1.17, 11.1).

Three studies involved testing **different ways of combining reminders** (including a range of forms of clinical decision-support) along with A&F. Gonzales 2013 involved education and feedback for the management of *bronchitis* plus either patient education provided at check-in or patient education prompted through computerised decision-support. There were no differences in antibiotic prescriptions between those arms (P = 0.67). Fiks 2013 compared education, feedback, and decision-support with and without patient reminders, finding that the latter helped increase completion of the HPV *vaccination* series (adjusted RD 9%; 95% CI 5, 13). Meeker 2016 tested monthly peer comparison emails about *antibiotic prescribing* plus different types of prompts within related decision-support: asking clinicians to provide a justification for their prescription, or prompting them to reconsider the prescription through a list of alternative approaches. The reminder to provide a justification led to slightly fewer antibiotic prescriptions (adjusted RD 2%, P value not reported).

Two studies involved testing whether the addition of **organisational interventions** to support patient management increased the effects of multi-faceted interventions featuring A&F. Rantz 2001 tested feedback plus education with or without support from a geriatric nurse specialist in *nursing homes* and found no differences, perhaps due to limited uptake of the clinical support. Goderis 2010 involved family physicians in Belgium and compared a usual quality improvement intervention with feedback and education to a more intensive intervention with more frequent feedback plus greater support from specialists for management of patients with *diabetes* not achieving targets. The intensive intervention was estimated to cost $50 more annually per patient and resulted in a 9% increase in the use of antiplatelets (P < 0.001) and a 3% increase in statins (P = 0.168).

Two studies featured **financial incentives**. Ryan 2014 tested feedback plus decision-support with or without financial incentives in primary care patients with *vascular risk*, finding a 19% improvement in management of smoking (P value not reported) and a 2% improvement in provision of anti-thrombotic prescriptions (P value not reported). They also found that when financially incentivised quality indicators improved, other non-incentivized indicators got worse. In the Petersen 2013 factorial trial, feedback plus incentives for physicians, or practices, or both, were tested against feedback alone for care of *hypertension* in primary care. Use of guideline-recommended medications increased non-significantly for the physician-incentive group by 4.7% (95% CI −1.4, 10.9), for the practice-incentive group by 0.6% (95% CI −5.3, 7.3) and for the group receiving both 2.9% (95% CI −2.8, 9.6).

DISCUSSION

Summary of main results

This synthesis of 292 RCTs demonstrates that A&F can be a useful intervention to improve health professionals' compliance with desired practice. The main statistical analysis focused on 177 trials in which 558 dichotomous outcomes relevant to the A&F intervention were synthesised. The studies had wide-ranging effects for A&F, with a median risk difference of 2.7% and an interquartile range for all outcomes of 0.0% to 8.6%. The meta-analyses found that A&F was associated with an adjusted risk difference in compliance with desired practice of 6.2% (95% CI 4.1 to 8.2) or an adjusted odds ratio of 1.47 (95% CI 1.31 to 1.64).

Interestingly, although the overall range of effect estimates are similar to those observed in prior versions of this review, newer trials published after 2010 seem to be achieving slightly larger effect sizes, which may indicate incremental advances in the science and practice of A&F. Coupled with evidence suggesting cost-effectiveness of A&F and increasing availability of data for audit (Moore 2022), the results of this review provide evidence for routine implementation of A&F in circumstances where accurate measurement of high priority professional practices with room for improvement can be done at limited cost and where health professionals have capacity to engage with and act upon feedback.

Similar to the prior version of this review, approximately one-quarter of outcomes related to A&F were either unchanged or worsened, while one-quarter of outcomes across the trials had fairly large improvements. As in the prior version of the review, a wide range of effects was observed, and especially the observation that many outcomes experienced a decline in performance post-exposure to A&F should serve to remind readers that A&F is not benign. It further suggests the importance of understanding more about when, why, and how best to use this intervention. In other words, just as a generally effective medication should not necessarily be provided to patients without monitoring for side effects, the implementation of A&F as a quality improvement intervention should be done thoughtfully.

The substantial statistical heterogeneity observed within each of the meta-analyses highlights that A&F in these trials was implemented, and its effects were measured in a wide variety of ways. Effects did not vary significantly when narrowing the outcomes for prescribing or test-ordering, but low baseline performance and the addition of co-interventions were both associated with increased effect sizes. Recognising that prescribing and test-ordering are very broad categories, it is possible that subdividing the studies into more similar subgroups could be a useful way to address heterogeneity. Study design characteristics, including risk of bias, were also not associated with estimated effects; in fact, low risk of bias studies typically achieved larger effects.

The evidence summarised in this review suggests that characteristics of the ***audit***(e.g. level of data aggregation, room for improvement), the ***feedback*** delivery (e.g. interactive modalities, involving a local champion), and the way in which ***action*** in response to the feedback is actively supported (e.g. facilitation, action planning) may be associated with the effectiveness of A&F interventions.

While the meta-regression models must be interpreted with caution, there are some findings that warrant emphasis. For example, the models indicate that feedback interventions may be more effective when the feedback involves a local champion with a relationship to the feedback recipients. This may be understood in part as an approach to advancing the perception amongst recipients of feedback that the data are credible and valid. As highlighted in the CP-FIT (Brown 2019), when health professionals engage with feedback, they will not generate an intention to change practice unless they first accept the key messages. Messages that performance has not been as good as expected or desired can generate cognitive dissonance, which might be resolved in part through the involvement of trusted colleagues (i.e. a local champion with whom the recipient has a positive relationship might improve the perceived validity of the data).

We also found that feedback interventions appear to be more effective when delivered in a multi-modal, interactive fashion (e.g. written and verbal). While this fits with our hypothesis, it is possible that multi-modal feedback occurs more often in trials when the desired clinical behaviours are more complex and/or when there is active facilitation to enable a response to the feedback (i.e. potential confounding in the meta-regression models with outcomes, context, and/or intervention design). It is also worth noting that while our analyses found individual-level data to be associated with greater effects than team-level data, it is plausible that some complex change efforts can only be achieved at the level of the team or organisation. Future analyses are needed to consider potential interactions between A&F characteristics and the complexity of desired behaviour change.

The number of quality indicators in A&F was not statistically significantly associated with effect size in the meta-regression, which was contrary to expectations based on a hypothesis related to cognitive burden. However, direct comparisons from head-to-head trials (Comparison D) suggest the use of design elements in feedback that facilitate the identification and action of high-priority clinical items may be effective (Mitchell 2005; Nejad 2016; Soleymani 2019). Again, this aligns with CP-FIT, which emphasises that feedback interventions must account for the potentially limited capacity of the recipients to engage. With this in mind, the regression models suggesting the potential benefits of facilitation to help recipients engage with the feedback and of action planning to help recipients respond effectively is not surprising. Interestingly, the meta-regressions do not support prior recommendations that active participation in goal-setting plays an important role in response to feedback (Anonymous 1992), which may reflect a lack of statistical power in this comparison. Another potential explanation is that this may be true only when health professionals know how to change the behaviours needed to achieve such goals. Building self-efficacy in specific actions needed to improve performance metrics may be one mechanism by which provision of action plans and facilitation seem to enhance the effects of A&F.

Direct comparisons from head-to-head trials (Comparison D) suggest that the use of peer-comparisons (versus no comparison) can be effective. As highlighted by the findings of Navathe 2020 and Elouafkaoui 2016, peer comparison can amplify the effects of feedback compared to A&F without a comparator. The need for an explicit comparator fits with the theory about the mechanism of A&F, as it is the newfound awareness of a gap between intended and actual practice that motivates action. Meta-regression results indicate that comparisons which feature a target and/or performance of top peers are more effective, aligned with the findings of Kiefe 2001, which demonstrated how comparing feedback recipients to the top 10% of peers leads to greater improvements over comparing to the mean performance of peers. However, Schneider 2008 and Kaufmann-Kolle 2011 both found that identifying top performers in the context of a quality circle did not increase the effectiveness of feedback. The difference in these findings may reflect the role of explicit goal-/target-setting in determining the reaction to feedback (Carver 1982; Locke 2002) since high-performing peers may act as a goal or target. Another explanation may stem from Feedback Intervention Theory (Kluger 1996), which emphasises that feedback which diverts attention to the 'self' rather than the 'task' can be counter-productive. In other words, there is a risk that some peer-groups may distract from the desired behaviour change rather than enable it. The concepts of 'growth mindset' and 'psychological safety' may also help explain why some individuals or groups are more or less likely to respond to feedback (Desveaux 2024).

A&F is commonly used in the context of multi-faceted quality improvement interventions. We found that trial arms had a median of two co-interventions, as defined using the EPOC taxonomy. The co-interventions most commonly observed in trials were from the EPOC categories of Clinician Education, Reminders, and Patient-Mediated Interventions. When using the behaviour change technique taxonomy, we found that treatment arms had five behaviour change techniques on average, but as many as 29, highlighting the complexity of interventions in trials featuring A&F.

The narrative synthesis for Comparison E highlights inconsistent effects for common combinations of A&F with co-interventions, lending uncertainty to the best combinations of A&F and EPOC strategy co-interventions. The regression models evaluating EPOC taxonomy co-interventions suggest that these would supplement the effects of A&F in an additive fashion. It is likely that additional co-interventions eventually provide diminishing returns. It is also likely that some co-interventions are synergistic with A&F and that this may vary by context (i.e. context-specific determinants of the target behaviour) or targeted behaviour. Financial or social incentives in the workplace context that may affect health professional behaviour are likely to interact with A&F effects, as highlighted in CP-FIT; future work should explore these and other healthcare organisational features to understand where, when, and how to best use A&F.

In addition to the design of the A&F intervention and co-interventions, it is likely that the characteristics of the context and the recipients might influence the effectiveness of feedback. The meta-analyses did not find differences in effect sizes across broadly defined targeted behaviours, but it remains possible that A&F might also be best suited for changing specific types of behaviours. In some trials, the intention was to increase prescribing, test-ordering or referrals (addressing under-use), while in others the goal was to reduce utilisation (addressing over-use). Some trials sought to improve all of these - and more. Trials often focused their efforts on a smaller number of quality indicators, while in the real-world, A&F might be used as a component of large-scale health system or organisational change efforts.

Ultimately, a single 'best' way to conduct A&F may not exist and the 'ideal' design of the intervention will depend on the target behaviour, the target recipients, and the context in which they work; future research should focus on how to inform decisions about the best 'fit' between A&F intervention design, the targeted behaviour change(s), and recipient abilities/constraints.

Overall completeness and applicability of evidence

Although the variation in effect size is noteworthy and requires further study, the consistency of meta-analysed effect sizes found in this review compared with the previous review, despite changes in methodology, is reassuring. While the best way to design and deliver feedback remains uncertain, this review provides greater certainty about its likely effect compared with usual care across a variety of clinical situations.

Given the large number of RCTs included in this review and the stability in effect size observed over time, we believe it is unlikely that missing or new trials of A&F versus usual care would substantially alter the estimated effect of A&F on professional practice. Thus, future trials should aim to determine the best way to deliver A&F in head-to-head trials rather than comparing A&F to usual care.

Quality of the evidence

Overall, we judged only 66% of the included studies to have a low risk of bias and 12% to have a high risk of bias. As with the previous review, we found no association between the overall risk of bias (low versus unclear) and the estimate of effect.

Although we included almost 300 randomised trials, we downgraded the overall estimate of effect to moderate due to substantial unexplained heterogeneity, despite our exploration of study characteristics, intervention characteristics, and targeted behaviours. Our examination of reporting bias suggested some risk of unreported outcomes. In particular, noting the significant Egger tests when considering all outcomes from the trials and the non-significant test when considering the median result from each trial suggests a risk that some trial authors were not reporting all measured outcomes in their trials, although differences in size of study and/or intensity of intervention components could also explain some or all of these results.

We did not assess the funding source or declarations of interest of each individual study included in this review. The prevalence of financial conflicts of interest in quality improvement trials and its empirical association with effect size is not clear; we would invite collaborators to help us conduct original research on this topic using our dataset, possibly by using the approaches described at https://tacit.one/.

Potential biases in the review process

Unlike the prior version of the review, our inclusion criteria did not require that at least one arm of the trial use A&F as the core, essential feature of the intervention. This may have resulted in the inclusion of trials of multi-faceted interventions where the main effects of the intervention were unlikely to be due to feedback. Our analyses sought to explore the role of co-interventions and suggest, as expected, that in the presence of multiple co-interventions, the effects attributable to the A&F component may be proportionally smaller. We also found that the overall effects of multi-faceted A&F interventions increased as the number of co-interventions increased from one to two to three and four or more. We acknowledge that the number of non-A&F co-interventions is unlikely to be truly linear, that different co-interventions will have different effects, and the estimated effect size for A&F may be inflated by the routine presence of co-interventions. Unfortunately, statistical power was insufficient to separately adjust for each co-intervention. We also did not examine interactions between A&F and specific EPOC strategy co-interventions; understanding co-interventions with the potential for synergy (or dysynergy) is an important area for future research.

In many studies, educational materials such as guidelines were distributed to all groups, thus meeting a pragmatic definition of usual care. Previous reviews (Farmer 2008; Grimshaw 2004) found that printed educational materials have a small (but potentially important) effect. By considering the delivery of printed materials or guidelines as usual care rather than co-interventions for some studies, we may have created a conservative bias in studies comparing feedback to printed materials, but an overestimation of the effect attributed to A&F in studies where feedback plus printed materials are compared to no intervention. We also note that it can be difficult to delineate the distinction between a component of the A&F and a co-intervention. For example, verbal delivery of the feedback by a local champion, with facilitation and action planning is similar to the co-intervention known as practice facilitation. Further analyses of behaviour change techniques may be useful for unpacking the effects of active ingredients, given the potential for overlap when using the EPOC taxonomy. Likewise, if any co-interventions that may have been synergistic with A&F were not described or not captured by our extractions, we may have over-estimated the effects of A&F.

Unlike the prior version of this review, we did not extract or summarise effects on patient outcomes and did not include studies that only examined patient outcomes without measuring professional processes of care. Effects on patient outcomes depend on the combined effectiveness of feedback on professional practice and the effectiveness of the clinical intervention (delivered as a result of the change in professional practice). Since the effectiveness of feedback is typically small or moderate and the effectiveness of targeted processes of care on patient outcomes is typically small or moderate, feedback can only be expected to have a small effect on patient outcomes when measured over a short time in a trial. However, if focused on a topic where there is large room for improvement and where the connection between professional practice and measurable patient outcomes is clear, effects on patient outcomes through A&F interventions have been demonstrated (e.g. Dreischulte 2016).

As highlighted in recent theoretical developments in the field (Brown 2019), there are many possible factors that could potentially alter the effectiveness of A&F. We were unable to abstract some important factors, especially organisational and contextual characteristics. We limited the exploration of factors in this review for pragmatic reasons (based on the feasibility of abstraction) and to limit the risk of spurious findings. It is entirely possible that variables we did not analyse could be more important than those we included. For example, we know that the effects of A&F are dependent on engagement with the feedback (akin to the effects of medication being dependent on actually swallowing a tablet), but in many trials, this interaction was not measured; further research is needed to understand how to increase engagement with A&F (Daneman 2021). Changes in professional performance in a given trial are dependent on many factors we could not quantitatively examine, including, but not limited to, the complexity of the targeted behaviour, readiness to change of the recipients, and intensity of co-interventions. We explored some of these by examining the consistency of effects observed across a range of targeted behavioural categories and adjusting for the number of co-interventions. Further research is needed to unpack the extent to which intervention components address underlying barriers to desired professional behaviour.

In our quantitative analyses, we focused on trials featuring contrasts of A&F versus usual care where it was possible to calculate a difference in completion of a desired practice, extracted from each trial as a dichotomous outcome. Those trials measuring outcomes on the continuous scale may be more likely to show statistically significant effects and, if so, excluding these from the meta-analyses may produce a conservative bias. Accounting for sample size in the meta-analyses required knowledge of cluster size and the intra-cluster correlation coefficient, which were not always reported. While many studies reported outcomes across multiple time points, our extraction and analyses focused on the time point nearest to the end of the completion of A&F intervention, meaning we were unable to examine learning or decay effects.

In general, the outputs of our meta-analyses and regressions should be interpreted with caution. We sought to synthesise and describe a very large, heterogeneous dataset, which we felt could not be summarised in a useful way through a narrative approach alone. Given the substantial heterogeneity, with I^2 estimates well above 80%, the quantitative analyses should be seen as a quantitative aid to the narrative synthesis. Relatedly, the lack of statistically significant effects in the meta-regressions should not be considered akin to evidence in favour of the null.

We plan to close this over-arching review, and develop future reviews of A&F that are focused on more specific questions, which will be easier to update, analyse, and interpret. We invite collaborations for future analyses - especially those focused on a subset of outcomes that may have less heterogeneity.

Agreements and disagreements with other studies or reviews

A systematic review of electronic A&F included seven studies (Tuti 2017), finding a weighted pooled OR of compliance with desired practice of 1.93 (95% CI 1.36 to 2.73), which is comparable but somewhat higher than the findings from this review. The previous update of this Cochrane review found similar estimates of the effect of A&F on professional practices based upon evidence from 70 studies reporting dichotomous outcomes on professional practice. It also tested five specific characteristics of feedback design in a meta-regression in an attempt to identify important active ingredients of A&F.

Compared to the prior version, ***this update includes 10 more years of trials and over twice the number of trials in the quantitative analyses***. It aligns with the prior version regarding the importance of the source of feedback, and the inclusion of targets and action planning. In this review, we looked at whether having a local champion involved with a known relationship to the recipients was associated with increased effectiveness as a proxy for data source credibility. The sources of feedback in the prior review associated with the lowest effect size were 'professional standards review organisation' and 'representative of the employer or purchaser'. This fits well with previous qualitative work comparing high- and low-performing hospitals, finding that feedback with a punitive tone seems to be less effective (Hysong 2006). It also fits with the finding from the meta-regression models in the current review that feedback may be more effective when delivered via trusted, credible sources (i.e. local champions). However, we acknowledge that the source of the data used in the audit is likely to affect perceived credibility and acceptability of the feedback and invite collaborators to explore this domain further.

Our results regarding action-planning are also consistent with independent re-analyses of the previous version of this review. In this update, we further unpacked different approaches to enabling a desired response to feedback, including facilitating engagement with the feedback and co-creating action plans. While these approaches are promising, the scalability and cost-effectiveness of these more intensive approaches to A&F interventions is uncertain. A recent review suggests that A&F is, in general, cost-effective, but did not compare approaches to delivering the intervention (Moore 2022).

One noteworthy difference was that the meta-regression results in this review found that repeated feedback was not superior to one-time-only feedback. It is possible that this reflects wide variation in the targeted behaviour, in the duration of the trial, and in the timing of feedback. For example, the effects of repeating feedback after a year may not be noticeable in a trial. It is noteworthy that Elouafkaoui 2016 found that feedback three times a year was no different from feedback two times a year. The meta-regression analysis would have classified both these as multiple rather than one-time-only, highlighting the need for further analyses that unpack frequency and timing.

The prior version of this review found that feedback aiming to change prescribing habits may be more effective than feedback targeting other behaviours, positing that prescribing may often be easier to change than other clinical targets of A&F. However, this update of the review did not replicate that finding; effects for prescribing, test-ordering, and other targeted behaviours were similar. A recent review of dashboards for test-ordering and/or prescribing included 11 trials and found mixed effects (Xie 2022). The prior review also highlighted that many studies focused on de-implementation (reducing undesirable target behaviours), but we did not extract or analyse such data in this review and would welcome collaborators to conduct analyses such as these in future.

Increasingly, the risks of A&F are being recognised in terms of the potential for gaming and for enacting "Goodhart's Law" through which measures stop accurately reflecting the underlying goal once they are used as a target (Duncan 2022). This review adds to the knowledge of such risks by highlighting that approximately 25% of outcomes measured in the trials actually declined in performance. Further work is needed to understand how to prevent perverse or unintended consequences of A&F, which may occur even when other targeted outcomes improve.

AUTHORS' CONCLUSIONS

Implications for practice

The body of evidence supporting the use of A&F to improve professional practice is substantial. In general, A&F leads to moderate effects that vary based on the way the feedback is designed and delivered. A systematic review of economic evaluations suggested that A&F was likely to be cost-effective (Moore 2022). The increasing availability of electronic health records and routine administrative data increases the feasibility and reduces the costs of audits. Given this, we believe that there is a strong rationale for broad implementation of A&F as a foundational element of organisational quality improvement efforts in circumstances where accurate measurement of high-priority professional practices with room for improvement can be done at limited cost and where health professionals have the capacity to engage with and act upon feedback.

The results of this review also provide signals about factors influencing the effectiveness of A&F. Feedback appears more effective when it features data specific for the recipient, comparing performance to top-peers or a benchmark, delivered by a local champion with an existing relationship to the recipient, through interactive modalities, rather than just didactic or just written, with active facilitation to support interaction with the data, and when action plans to improve are provided to improve the targeted professional practices. Effects are also generally larger when there is more room for improvement.

This suggests that healthcare systems and organisations planning to implement A&F should audit data at the level of aggregation where recipients are more likely to accept responsibility for the observed performance and try to provide a measurable target with a suggested action plan for improvement in the feedback. These design options are unlikely to entail substantial costs. Other design features associated with larger effects (facilitation and use of local champions and/or other co-interventions requiring ongoing human resources) are likely to incur additional costs and so healthcare systems and organisations will have to consider the local feasibility and affordability of including these features. As with any quality improvement strategy, efforts to change provider practice should be targeted at behaviours for which there is direct evidence that changes in health professional processes will improve patient outcomes and/or health system performance.

Despite the substantial body of evidence showing improvements in performance following A&F, some studies observed decreased performance. It is important that healthcare systems and organisations wishing to implement A&F ensure that the A&F addresses clinical practices which are salient and actionable to the targeted healthcare professionals and consider potentially negative consequences (Duncan 2022), and develop mitigation strategies and monitor for harms. Additionally, those planning A&F as a quality improvement strategy should consider the extent to which the A&F addresses the known barriers to achieving the targeted professional practice, and try to address potential barriers to fidelity and actionability of the intervention.

Implications for research

There are two main research audiences for this review: those who wish to implement and rigorously evaluate the effectiveness of a local audit and feedback intervention and those who wish to examine the underlying cognition and behavioural control mechanisms to advance generalisable knowledge on how to best design and deliver these interventions. Rather than separate streams of work, we believe that each new A&F intervention may provide an opportunity to incorporate research on underlying mechanisms and/or evaluations of different ways of designing and/or delivering the feedback to explore how to optimise this intervention in routine practice settings. Therefore, as in the prior version of this review, we call for collaboration between 'applied' scientists aiming to improve local quality of care and 'basic' scientists aiming to produce generalisable knowledge.

This update provides an evidence base that can be used to inform future interventions in terms of components associated with greater effects. The review also provides an evidence base to inform the sample size calculations for future trials. Those planning A&F-based interventions for specific clinical targets might use the review to identify relevant studies and purposefully replicate effective strategies.

We are pleased to note that newer A&F trials seem to be achieving slightly larger effect sizes. However, if, in another decade, the scientific community has once again doubled the number of trials in the A&F literature, we hope that it will come with even more meaningful improvements in the effects seen from the interventions. This is necessary to avoid research waste and address concerns about stagnant science previously observed (Ivers 2014a). Achieving that will mean systematically building upon and advancing the current evidence base. Business as usual will not suffice (Ivers 2014b). To do this, the field would benefit from more attention to four areas: improved reporting and methods; explicit use of theory, empirical evidence, and logic to develop hypotheses and to design the intervention and comparison arms; a focus on professional practices with substantial room for improvement and for which there is compelling evidence of potential patient benefits from changes to health professional behaviour that can be audited; and more head-to-head trials (e.g. comparing different ways of providing feedback).

At a minimum, to contribute to the literature, trials need to be well-designed and clearly reported (Simera 2010). Better reporting of study methods, targeted behaviours, characteristics of participants, and the context are needed (Davidoff 2009). A clear, thorough description of the intervention, ideally with illustrative examples, would be useful (Hoffman 2014). Given uncertain fidelity in many of the trials summarised here, we urge researchers in future to report on implementation outcomes, especially the engagement of recipients with feedback reports. Likewise, it is important that future trials carefully consider the intended target of the intervention and precisely describe the targeted behaviours, ideally including an assessment of their complexity and perceived importance.

Researchers should examine prespecified hypotheses regarding how their audit and feedback intervention will be acted upon in practice and how intervention components might enhance the effects of A&F. Well-designed, mixed-methods process evaluations embedded within trials can be useful to explore and provide insights into the complex dynamics underlying the variable effectiveness of audit and feedback. This is especially important given the paucity of data currently available on A&F intervention fidelity. Research related to personalising feedback to specific user needs and understanding how competing priorities play a role in recipient action could help advance the field. In addition to some of the psychology literature referred to in the background section, the education and the organisational/management literature suggest how the design and delivery of feedback might be optimised to improve performance (see, for example, Shute 2008). We encourage creative and interdisciplinary approaches to advancing our understanding of A&F.

Finally, although there have been more trials over time directly comparing different ways of conducting feedback interventions, there is a continued need to emphasise this type of head-to-head trial. The cumulated evidence suggests that further two-arm trials comparing feedback with usual care are likely to be of limited value unless conducted in settings where research is relatively scant (e.g. low-resource settings, non-physician health professionals, etc.). Therefore, the focus of future researchers interested in improving professional practice should shift from whether audit and feedback works better than usual care to discerning ways to optimise the effectiveness of audit and feedback interventions for particular contexts or clinical practices.

In brief, understanding how to best 'fit' the A&F intervention for a given purpose and how to 'tailor' for a given recipient are key scientific challenges going forward. It is possible that advances in artificial intelligence could be leveraged toward these goals. Such work may be facilitated through implementation laboratories and meta-laboratories (Grimshaw 2019). This would involve sequential trials testing ways to improve the effects of A&F in a given context and systematic efforts to share emergent best practices and/or replicate findings in new contexts. Such efforts are currently being coordinated by an international collaboration known as the A&F MetaLab.

The utility of future updates of this review will depend on the availability of new, well-designed (and well-reported) trials and on our ability to recognise, abstract, and analyse important explanatory factors. With that in mind, we recommend that this over-arching review be closed, with future reviews of A&F focused on more specific questions, which will be easier to update, analyse, and interpret. For example, future reviews might focus on advancing understanding of the effects of A&F in specific settings, recipients, clinical topics, and/or intervention components. In addition, with electronic data enabling routine audit and feedback becoming increasingly ubiquitous, it will be important to monitor A&F interventions for unintended consequences (Duncan 2022) and to develop 'stopping rules' to 'retire' certain metrics from A&F interventions.

Our collated set of trials can be a rich source of additional hypothesis testing. We invite collaborations and would be pleased to share our dataset with groups looking to build upon this work.

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- Sign-off Editor (final editorial decision): Lisa Bero, University of Colorado Anschutz Medical Campus
- Managing Editor (selected peer reviewers, collated peer reviewer comments, provided editorial guidance to authors, edited the article): Joey Kwong, Cochrane Central Editorial Service
- Editorial Assistant (conducted editorial policy checks and supported editorial team): Jacob Hester, Cochrane Central Editorial Service
- Copy Editor (copy-editing and production): Anne Lethaby, Cochrane Central Production Service
- Peer reviewers (provided comments and recommended an editorial decision): Eva Pagano, Clinical Epidemiology, Città della Salute e della Scienza Hospital, Turin, Italy (clinical/content review), Anders Anell, Lund University, School of Economics and Management (clinical/content review), Brian Duncan (consumer review), Jennifer Hilgart, Cochrane Evidence Production and Methods Directorate (methods review), Irma Klerings, Cochrane Austria, Department for Evidence-based Medicine and Evaluation, University for Continuing Education, Krems, Austria (search review).

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* Indicates the major publication for the study

CHARACTERISTICS OF STUDIES

Characteristics of included studies *[ordered by study ID]*

Allison 2005

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family/internal medicine N health professionals: 209 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: traditional continuing medical education + delayed access to multicomponent continuing medical education (intervention); Intervention: feedback on chlamydia screening rates for the physician's office
Outcomes	Clinical behaviour category: screening, prescribing Format of Audit & Feedback: written Source: administrative/billing data Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	First physician in a practice to log on was randomised and others in the same practice were subsequently allocated the same arm. Allocation likely concealed

Allison 2005 *(Continued)*

Blinding of outcome assessment (detection bias)	Low risk	Routine electronic data collected
Incomplete outcome data (attrition bias) All outcomes	Low risk	No loss of outcome data
Selective reporting (reporting bias)	Low risk	Adequately reported as described in methods, no protocol available
Recruitment bias	High risk	Randomisation was performed when the first physician from an office recruited. Others may have been recruited from the same office after that and may have been aware of group allocation at that point.
Baseline imbalance in outcome measurements or characteristics	Low risk	No differences in baseline characteristics. Although there were few differences between groups in baseline outcomes, these were adjusted in the analysis.
Protection against contamination	Low risk	All physicians from the same office allocated to the same group
Incorrect analysis	Unclear risk	GEE used which can adjust for clustering, but adjustment for clusters not clearly reported
Risk of bias overall	High risk	Recruitment bias judged as high risk; incorrect analysis judged as unclear

Althabe 2008

Study characteristics	
Methods	Design: parallelCluster-RCT
Participants	Country: Argentina, Uruguay Setting: hospital inpatient Speciality: obstetrics N health professionals: 490 N patients: 5466
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: monthly A&F reports
Outcomes	Clinical behaviour category: prescribing, treatment decision/action Format of Audit & Feedback: written Source: computerised search of charts Frequency: monthly Instructions for improvement: not reported
Notes	

Althabe 2008 *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence (see supp file)
Allocation concealment (selection bias)	Low risk	The article stated that the allocation sequence was randomly selected. The Protocol stated that 'the allocation procedure will be done by an independent data center (Research Triangle Institute, North Carolina, USA) and their results communicated to the study coordinating unit. Thus, there will be a clear separation between the generator of the intervention allocation and the study coordination'.
Blinding of outcome assessment (detection bias)	High risk	Unlikely that all outcome assessors were blind, and some outcome measures (e.g. measuring amount of blood and self-administered questionnaire) could be subject to bias.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No hospitals lost to follow up. Outcomes for all hospitals reported as planned. Small proportions of missing data in questionnaires about readiness to change (secondary outcome) in intervention group (5.3%) and control group (2.1%). ITT analysis was used.
Selective reporting (reporting bias)	Low risk	All outcomes between the protocol and the study appear to be consistent.
Recruitment bias	Low risk	Recruitment did not occur after randomisation.
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline was reasonably balanced.
Protection against contamination	Low risk	Due to the cluster design, contamination is unlikely.
Incorrect analysis	Unclear risk	Adjustments to take clustering into account is not clearly described in methods.
Risk of bias overall	High risk	Due to blinding

Althabe 2019

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Congo/Zambia Setting: obstetrics and gynaecology clinic or ward Speciality: obstetrics N health professionals: NR N patients: 35,350
Interventions	Number of arms: 2

Althabe 2019 *(Continued)*

	Description of groups: Control: no intervention; Intervention: monthly A&F reports
Outcomes	Clinical behaviour category: screening, prescribing Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: monthly Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	It is likely an adequate randomisation method was used. Quote: assigned clinics to groups using a covariate-constrained randomisation procedure
Allocation concealment (selection bias)	Low risk	Allocation was performed centrally at one time.
Blinding of outcome assessment (detection bias)	High risk	Blinding of outcome assessors not clearly reported. Some outcomes self-reported by patients, leading to potential for detection bias
Incomplete outcome data (attrition bias) All outcomes	Low risk	No loss of clusters or missing outcome data
Selective reporting (reporting bias)	Low risk	Primary outcomes reported are consistent with registered protocol NCT02353117.
Recruitment bias	Low risk	Unlikely individuals recruited to the trial after clusters randomised
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline appears balanced between groups.
Protection against contamination	Low risk	Due to cluster design, contamination is unlikely.
Incorrect analysis	Low risk	Analysis took into account clusters.
Risk of bias overall	High risk	Due to blinding

Amanyire 2016

Study characteristics

Methods	Design: step wedge
Participants	Country: Uganda Setting: primary care

Amanyire 2016 *(Continued)*

	Speciality: GP/infectious disease N health professionals: NR N patients: 12,024
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: CME only
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: verbal (didactic) Source: computerised search of charts Frequency: bi-annually Instructions for improvement: not reported
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Likely that allocation was performed by statistician and concealed to participants and other personnel
Blinding of outcome assessment (detection bias)	Low risk	Although knowledge of allocation was known, outcome data was collected on a programme (medical records) and a random sample verified for accuracy by research personnel. It is unlikely that the outcome data would be biased.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No clusters lost to follow-up. No patient data were excluded (some outcomes were measured in a random sample of patients).
Selective reporting (reporting bias)	Unclear risk	All prespecified outcomes in the clinical trial registry are reported here. But there are also added outcomes. It is unclear why/when these outcomes were added.
Recruitment bias	Unclear risk	As this is a stepped-wedge design, there is an unclear risk of recruitment bias as participants were recruited in waves.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unclear as only one baseline measure is provided.
Protection against contamination	High risk	Although this study used a cluster design, there is a high risk of contamination bias as confirmed by the authors.
Incorrect analysis	High risk	No evidence that clustering was taken into account during analysis.
Risk of bias overall	High risk	Due to protection against contamination and incorrect analyses

Anderson 1994

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital inpatient Speciality: internal medicine N health professionals: 646 N patients: 3158
Interventions	Number of arms: 3 Description of groups: Control: no intervention; Intervention 1: group A&F (during lecture) + education; Intervention 2: group A&F (during lecture) + individual A&F + education + quality assurance
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: verbal (didactic) Source: manual chart review Frequency: monthly/variable Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Described as "drawing lots"
Allocation concealment (selection bias)	Low risk	Cluster trial, allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	All hospitals followed up
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess

Anderson 1994 *(Continued)*

Protection against contamination	Low risk	No evidence of contamination
Incorrect analysis	Unclear risk	Logistic regression with adjustment for relevant risk factors, selected patient characteristics, and the dual sampling rates employed in large and small hospitals. Not sure if this is taking into account clustering
Risk of bias overall	Low risk	Low risk across all important domains

Aspy 2008

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: community care Speciality: GP/family N health professionals: 16 N patients: 1366
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + academic detailing +practice facilitator + IT support
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: not reported/not applicable Source: manual chart review Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Sequence generated using an internet randomisation programme
Allocation concealment (selection bias)	Unclear risk	It is unclear if all clusters were allocated at the same time and there is no study flow diagram.
Blinding of outcome assessment (detection bias)	Low risk	Data were collected using chart audit even though it is not clear if the research assistant who audited all the practices at baseline and post-intervention was blind to the group allocation of the practice. Outcome as objectively assessed
Incomplete outcome data (attrition bias)	Low risk	All 16 practices were audited at baseline and 9 months post-intervention - there was no loss to follow up.

Aspy 2008 *(Continued)*

All outcomes		
Selective reporting (reporting bias)	Unclear risk	Authors presented percentage (of eligible) patients who were recommended mammography (the outcome of the study). There was nothing else presented with regard to the outcome data (e.g. denominator, variance, per practice data) and it is unclear if this was pre-planned (as there was no trial registration or protocol published).
Recruitment bias	Low risk	Chart audit of eligible cases done by research assistant so risk of recruitment bias is low
Baseline imbalance in outcome measurements or characteristics	Low risk	There were no differences in participant characteristics or mammography rates between the two groups at baseline.
Protection against contamination	Low risk	There is no apparent risk for contamination between intervention and control sites. "There was no overlap between intervention and usual care practices because there were 16 clinicians from 16 different practices."
Incorrect analysis	Low risk	Analysis was performed "using the Rao-Scott x2 analysis to account for clustering".
Risk of bias overall	Low risk	Low risk across all important domains

Avery 2012

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: NR N patients: 480,942
Interventions	Number of arms: 2 Description of groups: Intervention 1: simple computerised feedback + brief educational materials; Intervention 2: pharmacist-led intervention + simple feedback
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: once/variable Instructions for improvement: co-developed plans
Notes	
Risk of bias	

Avery 2012 *(Continued)*

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Practices allocated by an independent web-based randomisation service
Allocation concealment (selection bias)	Low risk	Access to the allocation sequence was restricted to the clinical trials unit data manager.
Blinding of outcome assessment (detection bias)	Low risk	"All outcome data were extracted from patient records by pre-specified electronic searches and could not be altered before leaving the practice. Data were then sent electronically and automatically entered into a database on the trial manager's computer. Researchers who cleaned the data and the trial statisticians were masked to allocation." (p. 2)
Incomplete outcome data (attrition bias) All outcomes	Low risk	No practices lost to follow-up. "Repeated cross-sectional design accounts for no loss to follow-up of patients." Both intervention and control groups increased by approx 8% (Fig 1, p. 4)
Selective reporting (reporting bias)	Low risk	All quantitative outcomes prespecified in the published protocol and trial registration were reported. Qualitative outcomes were reported in a separate publication.
Recruitment bias	Low risk	Patients identified from electronic medical records system. Individual GPs already working in practices (p. 2)
Baseline imbalance in outcome measurements or characteristics	Low risk	Stratification and block randomisation
Protection against contamination	Low risk	Control practices received simple feedback but not pharmacist intervention..
Incorrect analysis	Low risk	Clustering taken into account in analyses
Risk of bias overall	Low risk	Low risk across all important domains

Awad 2006

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Sudan Setting: primary care Speciality: GP/family N health professionals: NR N patients: NR
Interventions	Number of arms: 4 Description of groups: Control: no intervention; Intervention 1: A&F + written recommendations for improvements + reminder two weeks later; Intervention 2: A&F + written recommendations for improve-

Awad 2006 *(Continued)*

	ments + reminder two weeks later + 4-hour seminar; Intervention 3: A&F + written recommendations for improvements + reminder two weeks later + academic detailing
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: monthly Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomisation method not described
Allocation concealment (selection bias)	Unclear risk	It is unclear if all clusters were allocated at the same time and there is no study flow diagram.
Blinding of outcome assessment (detection bias)	Low risk	Retrospective collection from records. Patient records unlikely to be recorded with bias
Incomplete outcome data (attrition bias) All outcomes	Low risk	No clusters missing. Retrospective data collection, appears to be no missing data
Selective reporting (reporting bias)	Low risk	Reporting appears complete but no protocol or trial registration to check
Recruitment bias	Unclear risk	Unclear when randomisation occurred or if new practitioners recruited during study period
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline values for key outcomes similar between groups
Protection against contamination	Low risk	Clustering performed by the health centre - unlikely contamination
Incorrect analysis	Low risk	Analysis adjusted for clusters
Risk of bias overall	Unclear risk	Unclear risk across important domains

Ayieko 2011

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Kenya

Ayieko 2011 *(Continued)*

	Setting: hospital inpatient Speciality: internal medicine N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: evidence-based guidelines + training + job aides + local facilitation + supervision + face-to-face feedback; Intervention 2: evidence-based guidelines + didactic training + job aides + written feedback
Outcomes	Clinical behaviour category: screening, prescribing, counselling, immunisation Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: semi-annually Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not enough information to determine sequence generation
Allocation concealment (selection bias)	Low risk	Allocation appears adequately concealed.
Blinding of outcome assessment (detection bias)	Low risk	Although data collectors could not be blinded, independent re-evaluations showed high internal consistency and structured forms and SOPs were used, so unlikely that outcome assessment was biased
Incomplete outcome data (attrition bias) All outcomes	Low risk	No hospital lost to follow-up in either treatment or control group (Figure 2 trial profile). Median number of children per hospital excluded from analysis reasonably similar: 158 (treatment), 133 (control)
Selective reporting (reporting bias)	Low risk	The study protocol includes 'surveys' at 6-month intervals (up to 33 months). Results included in research article report data at baseline (survey 1) and 18 months (survey 4). The authors explain the difference (see quote) and this was judged to be low risk of bias.
Recruitment bias	Low risk	No participants were recruited to the study post-randomisation, hence there is a low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Some baseline variables were adjusted for, and the baseline was mostly balanced with exceptions noted. Authors note limitations in adjusting for baseline imbalances.
Protection against contamination	Low risk	Restricted randomisation was used for allocation, which was not concealed. The geographical distance between participating hospitals was considerable, however, so reduced chance of contamination

Ayieko 2011 *(Continued)*

Incorrect analysis	Low risk	Appropriate analytical methods accounting for clustering were used.
Risk of bias overall	Low risk	Low risk across all important domains

Ayieko 2019

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Kenya Setting: hospital inpatient Speciality: internal medicine N health professionals: NR N patients: 2299
Interventions	Number of arms: 2 Description of groups: Intervention 1: enhanced A&F + standard A&F + pneumonia guidelines + network support + half-day training on new guidelines; Intervention 2: standard A&F + pneumonia guidelines + network support + half-day training on new guidelines
Outcomes	Clinical behaviour category: diagnosis Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: monthly/bi-monthly Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	The trial statistician concealed the sequence from investigators until training on the new policy was conducted in all hospitals and the intervention was assigned (information taken from protocol).
Blinding of outcome assessment (detection bias)	Low risk	Participants unlikely to record outcomes with bias. Data abstractors were blinded to allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	All 12 clusters randomised were analysed.

Ayieko 2019 *(Continued)*

Selective reporting (reporting bias)	Low risk	Outcomes consistent with published protocol and trial registration
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Low risk	Similar characteristics across groups
Protection against contamination	Unclear risk	Authors note the potential for contamination in the control group
Incorrect analysis	Low risk	Clustering taken into account in analysis
Risk of bias overall	Low risk	Low risk across all important domains

Bahrami 2004

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: dental clinic Speciality: NA N health professionals: 51 N patients: 5276
Interventions	Number of arms: 4 Description of groups: Control 1: guideline + education course; Control 2: computer-aided learning with decision support; Intervention 1: guideline + education course + A&F; Intervention 2: computer-aided learning with decision support + A&F
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: not reported/not applicable Source: manual chart review Frequency: not reported/not applicable Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer random number table

Bahrami 2004 *(Continued)*

Allocation concealment (selection bias)	Low risk	Cluster trial, allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Data extractor was blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Less than 10% dropout, spread among groups
Selective reporting (reporting bias)	Unclear risk	Unable to assess
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 1
Protection against contamination	Low risk	Separate practices
Incorrect analysis	Low risk	Clustering taken into account in analysis
Risk of bias overall	Low risk	Low risk across all important domains

Baker 1997

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: NR N patients: 1731
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F only; Intervention 2: A&F + reminders
Outcomes	Clinical behaviour category: treatment decision/action, screening, counselling Format of Audit & Feedback: written Source: manual chart review, computerised search of charts Frequency: monthly Instructions for improvement: general

Baker 1997 *(Continued)*

Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random-number table
Allocation concealment (selection bias)	Low risk	Cluster trial, allocation after recruitment completed
Blinding of outcome assessment (detection bias)	High risk	Unable to blind second abstractor
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data from all 18 practices
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 1
Protection against contamination	Low risk	Separate practices
Incorrect analysis	Low risk	Logistic regression with random-effects in the linear predictor
Risk of bias overall	High risk	Due to blinding

Baker 2003a

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: NR N patients: 3049
Interventions	Number of arms: 3 Description of groups: Control 1: evidenced-based guidelines; Control 2: review criteria; Intervention: review criteria + A&F

Baker 2003a *(Continued)*

Outcomes	Clinical behaviour category: prescribing, counselling, testing/exams, diagnosis Format of Audit & Feedback: written Source: manual chart review Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random number table
Allocation concealment (selection bias)	Low risk	Cluster trial, allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	No sites lost to follow-up
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	Recruitment did not occur after randomisation.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Low risk	GPs work separately
Incorrect analysis	Low risk	Generalised estimating equation
Risk of bias overall	Low risk	Low risk across all important domains

Baker 2003b

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family

Baker 2003b *(Continued)*

	N health professionals: NR N patients: 3043
Interventions	Number of arms: 3 Description of groups: Control 1: evidenced-based guidelines; Control 2: review criteria; Intervention: review criteria + A&F
Outcomes	Clinical behaviour category: diagnosis, treatment decision/action, screening, prescribing, counselling Format of Audit & Feedback: written Source: manual chart review Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random number table
Allocation concealment (selection bias)	Low risk	Cluster trial, allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	No sites lost to follow-up
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	Recruitment did not occur after randomisation.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Low risk	GPs work separately
Incorrect analysis	Low risk	Generalised estimating equation
Risk of bias overall	Low risk	Low risk across all important domains

Baker 2003c

Study characteristics

Baker 2003c *(Continued)*

Methods	Design: arallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: 96 N patients: NA
Interventions	Number of arms: 2 Description of groups: Intervention: A&F for serum lipids and viscosity + guideline; Control: A&F for thyroid function, rheumatic factor and urine culture + guidelines
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: both written and verbal Source: administrative/billing data Frequency: quarterly Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated for allocation
Allocation concealment (selection bias)	Low risk	Cluster trial, allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Data collectors blinded
Incomplete outcome data (attrition bias) All outcomes	Low risk	All practices completed study.
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	Recruitment did not occur after randomisation.
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 2
Protection against contamination	Low risk	Separate practices
Incorrect analysis	High risk	Two-sample t-test

Baker 2003c *(Continued)*

Risk of bias overall	High risk	Due to incorrect analysis

Baker 2003d

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: 96 N patients: NA
Interventions	Number of arms: 2 Description of groups: Intervention: A&F for thyroid function, rheumatic factor and urine culture + guidelines; Control: A&F for serum lipids and viscosity + guideline
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: both written and verbal Source: administrative/billing data Frequency: quarterly Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated for allocation
Allocation concealment (selection bias)	Low risk	Cluster trial, allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Data collectors blinded
Incomplete outcome data (attrition bias) All outcomes	Low risk	All practices completed study.
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported.
Recruitment bias	Low risk	Recruitment did not occur after randomisation.

Baker 2003d *(Continued)*

Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 2
Protection against contamination	Low risk	Separate practices
Incorrect analysis	High risk	Two-sample t-test
Risk of bias overall	High risk	Due to incorrect analysis

Balas 1998

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: dialysis centres Speciality: nephrology N health professionals: 10 N patients: 152
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: 12 monthly reports combining centre-specific information or practice patterns with the latest published evidence
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: written Source: administrative/billing data Frequency: monthly Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation
Allocation concealment (selection bias)	Low risk	Cluster trial, allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Data collectors blinded

Balas 1998 *(Continued)*

Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unable to assess
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	Recruitment did not occur after randomisation.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	High risk	Two-sample Student's t test
Risk of bias overall	High risk	Due to incorrect analysis

Baldwin 2010

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Ireland Setting: nursing homes Speciality: long-term care N health professionals: 338 N patients: 793
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: written A&F report and verbal feedback + 2-hour infection control training + designated infection control workers (received additional 5-hour training)
Outcomes	Clinical behaviour category: screening Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: quarterly Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**

Baldwin 2010 *(Continued)*

Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Not specifically reported but based on study flow diagram; it appears all clusters allocated centrally at one time.
Blinding of outcome assessment (detection bias)	High risk	Data were collected at all time points by a researcher (NB) who was not blinded to treatment allocation. Although authors stated that an independent nurse blinded to treatment allocation also collected data, this was done only in 2 randomly selected houses at each time point.
Incomplete outcome data (attrition bias) All outcomes	Low risk	The number of participant who dropped out from the study were similar across groups and reasons for withdrawal were similar across groups.
Selective reporting (reporting bias)	Low risk	The trial was not registered and no study protocol was found; however all outcomes specified in the methods were analysed and reported.
Recruitment bias	Low risk	Study investigators stated that new participants were added to the study at each data collection point; however, results were presented separately for those recruited at baseline and those added later.
Baseline imbalance in outcome measurements or characteristics	Low risk	Pair-matched randomisation done, no baseline imbalances between groups
Protection against contamination	Low risk	There was no evidence of any contamination between the intervention and control homes.
Incorrect analysis	Low risk	Investigators accounted for clustering in the analysis.
Risk of bias overall	High risk	Due to blinding

Barkun 2013

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Canada Setting: hospital inpatient Speciality: internal medicine N health professionals: NR N patients: 826
Interventions	Number of arms: 2 Description of groups: Control: guidelines only; Intervention: guidelines + benchmarked profiles of adherence + interactive workshops/education sessions + written reminders
Outcomes	Clinical behaviour category: screening Format of Audit & Feedback: both written and verbal

Barkun 2013 *(Continued)*

	Source: manual chart review Frequency: once Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Central allocation
Blinding of outcome assessment (detection bias)	Low risk	It's not clear whether assessors were blinded, but they collected chart data of objective measures, so it's low risk if they were not.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Dropouts appear balanced with reasons reported. ITT analysis was used.
Selective reporting (reporting bias)	Low risk	Both primary and secondary outcomes match trial registration data (https://clinicaltrials.gov/ct2/show/NCT00840008).
Recruitment bias	Low risk	No evidence of recruitment after randomisation and allocation
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline is reasonably balanced and well reported. Analyses adjusted for some baseline variables
Protection against contamination	Low risk	Due to cluster design, contamination is unlikely.
Incorrect analysis	Unclear risk	Authors investigated cluster-level differences, but unclear if analyses presented were adjusted for clustering.
Risk of bias overall	Low risk	Low risk across all important domains

Batty 2001

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: hospital inpatient Speciality: internal medicine N health professionals: 313 N patients: 2805

Batty 2001 *(Continued)*

Interventions	Number of arms: 3 Description of groups: Control: verbal and bulletin intervention on other topic; Intervention 1: interactive verbal A&F; Intervention 2: written A&F
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: once Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial, allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data from all 17 hospitals
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	High risk	Figure 2, variability at baseline between groups
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Low risk	Logistic regression and estimates of standard errors of odds ratios in regression model adjusted for clustering
Risk of bias overall	High risk	Due to baseline imbalance

Beck 2005

Study characteristics

Beck 2005 *(Continued)*

Methods	Design: parallel cluster-RCT
Participants	Country: Canada Setting: hospital inpatient Speciality: internal medicine N health professionals: NR N patients: 11,402
Interventions	Number of arms: 2 Description of groups: Intervention 1: report cards (rapid feedback); Intervention 2: report cards (delayed feedback)
Outcomes	Clinical behaviour category: prescribing, counselling, other Format of Audit & Feedback: written Source: computerised search of charts, administrative/billing data Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation
Allocation concealment (selection bias)	Low risk	Cluster trial, allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Data collected from database
Incomplete outcome data (attrition bias) All outcomes	Low risk	No evidence of attrition
Selective reporting (reporting bias)	Unclear risk	Unable to assess
Recruitment bias	Low risk	Recruitment did not occur after randomisation.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Low risk	No evidence of contamination
Incorrect analysis	Low risk	Adjusted odds ratios were calculated using generalised estimating equation extensions of logistic regression procedures for cluster-randomised trials.

Beck 2005 *(Continued)*

Risk of bias overall	Low risk	Low risk across all important domains

Beeckman 2013

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Belgium Setting: nursing homes Speciality: long-term care N health professionals: 118 N patients: 464
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: electronic decision-support system + multi-faceted tailored implementation including education + A&F + reminders + leadership (introduction of key nurse role)
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: written Source: manual chart review Frequency: monthly Instructions for improvement: not reported
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Not specifically reported but based on study flow diagram, it appears all clusters were allocated at the same time.
Blinding of outcome assessment (detection bias)	High risk	All outcome assessors (health professionals and research personnel) were aware of allocations and outcomes have a subjective element.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Overall, flow of participants (patients and health professionals) was not reported or discussed. Patients: insufficient reporting of missing outcome data and not discussed. Health professionals: some loss to follow-up evident from the data reported for knowledge and attitude. Attrition not described or discussed
Selective reporting (reporting bias)	Low risk	All prespecified outcomes were reported at prespecified time points (according to Methods).

Beeckman 2013 *(Continued)*

		*No trial registration or protocol linked
Recruitment bias	Low risk	When wards were allocated to intervention/control, all health professionals involved in pressure ulcer prevention and all residents in that ward were included. There was unlikely to be recruitment made for the study, and no physicians refused the eligible residents be contacted; thus, there is unlikely to have been any bias introduced into participant recruitment.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Some baseline imbalances in gender (nursing home residents and health professionals) but not discussed so it is unclear if this reflects different populations in some wards.
Protection against contamination	High risk	11 wards in 4 nursing homes randomised. The trial explicitly reports that: "clustering took place at ward level to prevent possible cross-contamination of the study groups", however, this means that a nursing home could have both experimental and control wards within it. There is a high potential for bias from contamination as there are both experimental and control arms in the same nursing home.
Incorrect analysis	Unclear risk	Authors described that analysis was performed at the ward level, but this is not described enough to assess whether appropriate analysis was performed.
Risk of bias overall	High risk	Due to blinding and protection against contamination

Belcher 1990

Study characteristics	
Methods	Design: other
Participants	Country: USA Setting: tertiary-care centre Speciality: internal medicine N health professionals: NR N patients: 1224
Interventions	Number of arms: 4 Description of groups: Control 1: no intervention; Control 2: mailed invitation to patients to self-refer to Health Promotion Clinic; Control 3: information packet mailed to patients; Intervention: chart flowsheet (with recommended prevention activities, frequency, and follow-up guidelines) + training sessions + A&F
Outcomes	Clinical behaviour category: screening, treatment decision/action, immunisation, testing/exams Format of Audit & Feedback: written Source: manual chart review Frequency: annually Instructions for improvement: general
Notes	

Belcher 1990 *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomisation method not described
Allocation concealment (selection bias)	Unclear risk	It is unclear if all clusters were assigned at the same time, and there was no study flow diagram.
Blinding of outcome assessment (detection bias)	Unclear risk	Not enough information about outcome assessment processes
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition by year 5 appeared balanced between groups (28-35%); however, reasons for attrition not reported so unable to determine if missing data were balanced as the numbers missing were not insubstantial.
Selective reporting (reporting bias)	Low risk	No protocol or trial registration available, however, all outcomes in methods were reported in the results.
Recruitment bias	Low risk	No evidence of enrolment post-randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	No significant differences in baseline measures
Protection against contamination	Low risk	Subjects assigned to non-HPC groups were not accepted by the HPC staff.
Incorrect analysis	Unclear risk	Unable to assess
Risk of bias overall	Unclear risk	Unclear risk across important domains

Bentz 2007

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 279 N patients: 93,809
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: monthly A&F reports
Outcomes	Clinical behaviour category: treatment decision/action, counselling Format of Audit & Feedback: both written and verbal

Bentz 2007 *(Continued)*

	Source: computerised search of charts Frequency: monthly Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote "clinics matched and randomized"
Allocation concealment (selection bias)	Low risk	Cluster trial, recruitment not influenced by allocation
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	No evidence of attrition
Selective reporting (reporting bias)	Low risk	Appropriate outcomes accounted for
Recruitment bias	Low risk	Recruitment did not occur after randomisation.
Baseline imbalance in outcome measurements or characteristics	Low risk	Clinics comparable, see Table 1
Protection against contamination	Low risk	No evidence of contamination
Incorrect analysis	Low risk	Generalised estimating equations to accommodate cluster-level covariates
Risk of bias overall	Low risk	Low risk across all important domains

Bertoni 2009

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family/internal medicine N health professionals: NR N patients: 5057

Bertoni 2009 *(Continued)*

Interventions	Number of arms: 2 Description of groups: Intervention 1: guidelines + lecture + A&F + academic detailing; Intervention 2: personal digital assistant + guidelines + lecture + A&F + academic detailing
Outcomes	Clinical behaviour category: prescribing, screening Format of Audit & Feedback: verbal (interactive) Source: computerised search of charts Frequency: annually Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomisation method not described
Allocation concealment (selection bias)	Low risk	According to the study flow diagram, it appears likely all clusters were allocated at the same time.
Blinding of outcome assessment (detection bias)	Low risk	Medical record abstractors were blind to the allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Practices lost to follow-up were balanced between groups.
Selective reporting (reporting bias)	Low risk	All outcomes appear to be reported but no protocol available to check against
Recruitment bias	Low risk	Recruitment occurred prior to randomisation.
Baseline imbalance in outcome measurements or characteristics	Low risk	No important differences in # of practices, patients. Differences in # of providers, but we don't think this is important.
Protection against contamination	Low risk	Not blinded, but randomised by practice. Unlikely practices communicating to contaminate
Incorrect analysis	Unclear risk	No clustering at the level of the practitioner, only the practice. Although reasons explained, it is unclear what effect this might have had.
Risk of bias overall	Low risk	Low risk across all important domains

Bhatia 2014

Study characteristics

Methods	Design: parallel cluster-RCT

Bhatia 2014 *(Continued)*

Participants	Country: USA Setting: quaternary care centre (outpatient cardiology and internal medicine clinics) Speciality: cardiology N health professionals: 112 N patients: 1213
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: lecture + written materials + A&F
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: written Source: computerised search of charts Frequency: monthly Instructions for improvement: not reported
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random number generator used to divide fellows and residents into control or intervention groups
Allocation concealment (selection bias)	Unclear risk	No information on allocation concealment
Blinding of outcome assessment (detection bias)	Low risk	Blinding of assessors and unlikely blinding could have been broken.
Incomplete outcome data (attrition bias) All outcomes	Low risk	It appears that all outcomes were reported for one group (cardiology fellows) and that additional data were reported online only for residents.
Selective reporting (reporting bias)	Low risk	No study register or protocol - however, it appears that all expected outcomes listed in methods were reported in the results.
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	High risk	Patients in intervention group had higher rates of previous myocardial infarction and hospitalisation in the last year.
Protection against contamination	Unclear risk	In a single-site study, there may have been cross communication between physicians in training.
Incorrect analysis	Low risk	Clustering accounted for in analysis
Risk of bias overall	High risk	Due to baseline imbalances

Bhatia 2017

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Canada/USA Setting: hospital outpatient clinic Speciality: cardiology N health professionals: 196 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: 20-min video lecture + decision-support phone app + A&F
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: written Source: manual chart review Frequency: monthly Instructions for improvement: not reported
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Randomisation sequence likely adequate as a data manager was in charge of random sequence
Allocation concealment (selection bias)	Low risk	As per flow chart and a central body determining randomisation and allocation; allocation was likely concealed.
Blinding of outcome assessment (detection bias)	Low risk	Outcome assessors were blinded to allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	ITT used in analysis. Losses and missing outcomes also balanced and explained.
Selective reporting (reporting bias)	Low risk	Consistent measures with registered protocol - https://www.clinicaltrials.gov/ct2/show/study/NCT02038101?cond=NCT02038101&draw=2&rank=1
Recruitment bias	Low risk	Unlikely physicians recruited to the trial after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unclear if there were important differences characteristics between groups. Stratification was used to reduce risk of imbalance.

Bhatia 2017 *(Continued)*

Protection against contamination	Unclear risk	Physicians allocated within each service and possible that communication between groups may have occurred.
Incorrect analysis	Low risk	Clustering taken into account in analysis
Risk of bias overall	Low risk	Low risk across all important domains

Blais 2008a

Study characteristics	
Methods	Design: Parallel Cluster-RCT
Participants	Country: Canada Setting: community pharmacy Speciality: NA N health professionals: 60 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F letters
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: computerised search of charts Frequency: Quarterly Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial, allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Administrative data used
Incomplete outcome data (attrition bias) All outcomes	Low risk	Dropouts comparable in both groups (p. 229)

Blais 2008a *(Continued)*

Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	Recruitment did not occur after randomisation.
Baseline imbalance in outcome measurements or characteristics	Low risk	Table 2 and Table 3
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	High risk	Did not account for clustering
Risk of bias overall	High risk	Due to incorrect analysis

Blais 2008b

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Canada Setting: primary care Speciality: GP/family N health professionals: 71 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F letters
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: computerised search of charts Frequency: quarterly Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial, allocation after recruitment completed

Blais 2008b *(Continued)*

Blinding of outcome assessment (detection bias)	Low risk	Administrative data used
Incomplete outcome data (attrition bias) All outcomes	Low risk	Dropouts comparable in both groups (p. 229)
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	Recruitment did not occur after randomisation.
Baseline imbalance in outcome measurements or characteristics	Low risk	Table 2 and Table 3
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	High risk	Did not account for clustering
Risk of bias overall	High risk	Due to incorrect analysis

Bloos 2017

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Germany Setting: hospital inpatient Speciality: emergency medicine N health professionals: NR N patients: 4183
Interventions	Number of arms: 2 Description of groups: Control: lecture on sepsis care + sepsis-related publications; Intervention: lecture on sepsis care + sepsis-related publications + interprofessional quality improvement teams + A&F + reminders
Outcomes	Clinical behaviour category: prescribing, testing/exams Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: quarterly Instructions for improvement: general
Notes	

Bloos 2017 *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Likely that clusters were randomised and allocated centrally at one time (based on flow chart)
Blinding of outcome assessment (detection bias)	Low risk	Outcome data recorded in and collected from electronic records, so unlikely subject to bias
Incomplete outcome data (attrition bias) All outcomes	Low risk	The loss of clusters was reasonably balanced with reasons that do not appear related to the intervention. Missing outcome data were also balanced. ITT analysis was used.
Selective reporting (reporting bias)	Unclear risk	Appears to have been missing data on reporting of length of stay; however, this was not necessarily a key focus on the study.
Recruitment bias	Low risk	There was no evidence that other hospitals were recruited following randomisation.
Baseline imbalance in outcome measurements or characteristics	Low risk	Some baseline imbalances but analyses adjusted for baseline imbalances
Protection against contamination	Low risk	Intervention performed within hospitals - unlikely cross over between hospitals
Incorrect analysis	Low risk	Analysis adjusted for clustering
Risk of bias overall	Low risk	Low risk across all important domains

Boekeloo 1990

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital inpatient Speciality: internal medicine N health professionals: 29 N patients: 459
Interventions	Number of arms: 4 Description of groups: Control 1: no intervention; Control 2: reminders; Intervention 1: A&F; Intervention 2: A&F + reminders
Outcomes	Clinical behaviour category: testing/exams, treatment decision/action, referrals

Boekeloo 1990 *(Continued)*

	Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: not reported/not applicable Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial, allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unable to assess
Selective reporting (reporting bias)	Unclear risk	Unable to assess
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence there was low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	High risk	Patients differed in key variables (lab values, medical history).
Protection against contamination	High risk	Contamination between physicians likely
Incorrect analysis	Unclear risk	Unable to assess
Risk of bias overall	High risk	Due to baseline imbalance and protection against contamination

Boet 2018

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Canada Setting: hospital inpatient Speciality: anaesthesiology N health professionals: 45

Boet 2018 *(Continued)*

	N patients: 7846
Interventions	Number of arms: 3 Description of groups: Control: no intervention; Intervention 1: benchmarked A&F; Intervention 2: ranked A&F
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: written Source: administrative/billing data Frequency: monthly Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	"An independent statistician not otherwise involved in the study performed the randomization to protect allocation concealment." It appears that allocation was done centrally and concealed.
Blinding of outcome assessment (detection bias)	Low risk	It is unclear if the outcome assessors were blinded. Data on patient outcomes and process of care (warmer use) were collected objectively, using electronic medical record systems, so it is unlikely that the lack of blinding would affect the outcome.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Figure 3 shows no loss to follow-up of providers in each arm of the trial.
Selective reporting (reporting bias)	Unclear risk	The trial registration says the primary outcome is "Patient peri-operative temperature" without describing this as a dichotomous outcome (percentage with hypothermia). The predefined secondary outcome, "Intraoperative warming device usage perioperative" also appears to be split into two outcomes in this trial publication. It is unclear whether these outcomes were just poorly defined in the trial registration or if these were deviations. Patient postoperative surgical site infection (SSI) rate is listed as one of the main study outcomes in the registered protocol. In the published full text, the study authors explain that it was too costly and potentially meaningless to obtain these data, so they did not include this outcome in the published paper. The original second secondary outcome in the trial registration was not measured and trialists described this due to the financial burden and seeing no difference in the primary outcome. This explanation does not suggest a risk of bias; however, it would have been better if this was pre-planned (i.e. only obtained data on postoperative surgical site infection if there was a difference in the primary outcome).
Recruitment bias	Low risk	There is no evidence of further recruitment after the enroled clusters were randomised.

Boet 2018 *(Continued)*

Baseline imbalance in outcome measurements or characteristics	Unclear risk	Trial did not discuss whether there were any significant baseline imbalances. The data presented suggests there could potentially be imbalances.
Protection against contamination	Low risk	Allocation by cluster (anaesthesiologist) and unlikely control clusters received the intervention.
Incorrect analysis	Low risk	Analysis was done at the patient level but "adjusted for clustering by provider and for all the pre-specified covariates".
Risk of bias overall	Low risk	Low risk across all important domains

Bond 2011

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: dialysis centres Speciality: nephrology N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + educational materials developed for past influenza vaccination campaigns; Intervention 2: A&F + educational materials + educational seminars + assistance with and review of centre-specific action plans + monthly calls with monitors
Outcomes	Clinical behaviour category: immunisation Format of Audit & Feedback: not reported/not applicable Source: administrative/billing data Frequency: once Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	The method of random sequence generation was not described.
Allocation concealment (selection bias)	Low risk	Central allocation

Bond 2011 *(Continued)*

Blinding of outcome assessment (detection bias)	High risk	Knowledge of intervention likely to influence outcome measurement
Incomplete outcome data (attrition bias) All outcomes	Low risk	Low attrition rates in both groups
Selective reporting (reporting bias)	Low risk	All prespecified outcomes in the methods section were reported in the results.
Recruitment bias	Low risk	No report that participant centres were recruited after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	No baseline differences found
Protection against contamination	High risk	Risk of cross talk between groups
Incorrect analysis	High risk	Clustering not taken into account in analysis
Risk of bias overall	High risk	Protection against contamination and incorrect analysis judged as high risk

Bonds 2009

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: NR N patients: 2111
Interventions	Number of arms: 2 Description of groups: Intervention 1: JNC 7 blood pressure guideline + academic detailing + automatic BP machines + A&F; Intervention 2: ATPIII cholesterol management guideline + CME session + academic detailing + educational materials + A&F
Outcomes	Clinical behaviour category: prescribing, diagnosis, counselling Format of Audit & Feedback: written Source: manual chart review Frequency: once Instructions for improvement: general
Notes	

Bonds 2009 *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	The method of random sequence generation was not described.
Allocation concealment (selection bias)	Low risk	Though not specified, based on the study flow diagram, it appears likely that clusters were allocated at the same time.
Blinding of outcome assessment (detection bias)	Low risk	The outcome assessors were blinded to group allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition rates were similar across groups - both groups lost 1 practice to follow-up.
Selective reporting (reporting bias)	Low risk	Although the trial protocol was not found and the trial was not registered, all prespecified outcomes were measured and adequately reported in the paper.
Recruitment bias	Low risk	There was no evidence to suggest any further recruitment took place after randomisation.
Baseline imbalance in outcome measurements or characteristics	Low risk	There was no baseline imbalance between the 2 groups.
Protection against contamination	Low risk	Although the control group was exposed to a similar programme as the intervention one, there was no record of contamination of the control group by the GLAD Heart intervention.
Incorrect analysis	Low risk	The analysis took the clustering design into account and data were analysed and presented at the practice level.
Risk of bias overall	Low risk	Low risk across all important domains

Bonevski 1999

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Australia Setting: primary care Speciality: GP/family N health professionals: 19 N patients: 3091
Interventions	Number of arms: 2

Bonevski 1999 *(Continued)*

	Description of groups: Control: CME + consensus on standards of care + individual goals for care + practitioner performance rates; Intervention: CME + consensus on standards of care + individual goals for care + practitioner performance rates + A&F
Outcomes	Clinical behaviour category: screening, testing/exams Format of Audit & Feedback: electronic dashboard Source: computerised search of charts Frequency: not reported/not applicable Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Unclear risk	Unable to assess
Blinding of outcome assessment (detection bias)	Low risk	Patients appear to be unaware of intervention arm
Incomplete outcome data (attrition bias) All outcomes	Low risk	High participation rates, no evidence of variation between groups
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	Recruitment did not occur after randomisation.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Low risk	Logistic regression with a generalised estimating equation
Risk of bias overall	Unclear risk	Unclear risk across important domains

Borgiel 1999

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Canada

Borgiel 1999 *(Continued)*

	Setting: primary care Speciality: GP/family N health professionals: 59 N patients: 2395
Interventions	Number of arms: 2 Description of groups: Control: Practice Assessment Report (PAR) - medical chart audits + global assessment of quality of care without specific recommendations for change; Intervention: Continuing Medical Education Plan (CMEP) - structured feedback + practice report + visit from a mentor
Outcomes	Clinical behaviour category: immunisation, testing/exams, counselling, referrals Format of Audit & Feedback: verbal (didactic) Source: manual chart review, patient-reported data Frequency: once Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Stratification and block randomisation
Allocation concealment (selection bias)	Unclear risk	Unable to assess
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	Equal numbers of subjects in both groups completed the study.
Selective reporting (reporting bias)	Unclear risk	Unable to assess
Recruitment bias	Unclear risk	Physicians assigned to groups in the order that they were recruited
Baseline imbalance in outcome measurements or characteristics	Low risk	Physicians and patients comparable (Table 1 and 2)
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	High risk	Unpaired t-test
Risk of bias overall	High risk	Due to incorrect analysis

Brady 1988a

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital outpatient clinic Speciality: internal medicine N health professionals: 45 N patients: 811
Interventions	Number of arms: 3 Description of groups: Intervention 1: group audit + education session; Intervention 2: group audit + education session + self-audit (flu vaccine); Intervention 3: group audit + education session + self-audit (mammography)
Outcomes	Clinical behaviour category: immunisation Format of Audit & Feedback: not reported/not applicable Source: manual chart review Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	A random number table was used to generate the random sequence.
Allocation concealment (selection bias)	Low risk	No description of allocation concealment. Seems likely from figure 1 that all residents were randomised at the same time point
Blinding of outcome assessment (detection bias)	Low risk	Blinding of outcome assessors not clearly stated but 'investigators were generally unaware of each residents group assignment' and they were using routinely collected measures.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data collection was performed on all records available (so no attrition).
Selective reporting (reporting bias)	Low risk	Trial was conducted in 1985-86 prior to trial registry commencement. The results for all prespecified outcomes were presented in the paper.
Recruitment bias	Low risk	From figure 1, it appears that all participants were recruited at the same time.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Separate baseline measures for the 3 groups have not been provided.

Brady 1988a *(Continued)*

Protection against contamination	Low risk	Unlikely that the control group participants performed their own audit and the trial was designed to avoid this.
Incorrect analysis	High risk	No mention of clustering has been made in the analysis.
Risk of bias overall	High risk	Incorrect analysis - high

Brady 1988b

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital outpatient clinic Speciality: internal medicine N health professionals: 45 N patients: 728
Interventions	Number of arms: 3 Description of groups: Intervention 1: group audit + education session; Intervention 2: group audit + education session + self-audit (flu vaccine); Intervention 3: group audit + education session + self-audit (mammography)
Outcomes	Clinical behaviour category: screening Format of Audit & Feedback: not reported/not applicable Source: manual chart review Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	A random number table was used to generate the random sequence.
Allocation concealment (selection bias)	Low risk	No description of allocation concealment. Seems likely from figure 1 that all residents were randomised at the same time point
Blinding of outcome assessment (detection bias)	Low risk	Blinding of outcome assessors not clearly stated but 'investigators were generally unaware of each residents group assignment' and they were using routinely collected measures.
Incomplete outcome data (attrition bias)	Low risk	Data collection was performed on all records available (so no attrition).

Brady 1988b *(Continued)*

All outcomes		
Selective reporting (reporting bias)	Low risk	Trial was conducted in 1985-86 prior to trial registry commencement. The results for all prespecified outcomes were presented in the paper.
Recruitment bias	Low risk	From figure 1, it appears that all participants were recruited at the same time.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Separate baseline measures for the 3 groups have not been provided.
Protection against contamination	Low risk	Unlikely that the control group participants performed their own audit and the trial was designed to avoid this.
Incorrect analysis	High risk	No mention of clustering has been made in the analysis.
Risk of bias overall	High risk	Incorrect analysis - high; baseline imbalance - unclear

Bregnhøj 2009

Study characteristics		
Methods	Design: parallel cluster-RCT	
Participants	Country: Denmark Setting: primary care Speciality: GP/family N health professionals: 41 N patients: 212	
Interventions	Number of arms: 3 Description of groups: Control: no intervention; Intervention 1: interactive educational meeting; Intervention 2: interactive educational meeting + A&F	
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: administrative/billing data Frequency: once Instructions for improvement: co-developed plans	
Notes		
Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	The GPs were randomised using a computer-generated randomisation list prepared by a statistician.

Bregnhøj 2009 *(Continued)*

Allocation concealment (selection bias)	Low risk	Though not specified, based on the study flow diagram, it appears likely that clusters were allocated on a single occasion. GPs were informed of group assignment after baseline data collected implies allocation concealment.
Blinding of outcome assessment (detection bias)	Low risk	Evaluator groups blinded to intervention group of GPs and identity of patients
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Patient loss to follow-up greater in combined intervention group (38%) than in single intervention group (13%) or control group (11%) (Fig 1, p. 3). Explanations given: death, information not provided by GP, changed GP during intervention period
Selective reporting (reporting bias)	Low risk	The protocol was not available but the main outcome (MAI score) was reported as prespecified in the Methods (p. 5 & Table 3, p. 6). Selected text is a Covidence image.
Recruitment bias	Low risk	Only single practice GPs were recruited, so no-one else in the cluster. Also GPs only told of randomisation after baseline and none recruited afterwards
Baseline imbalance in outcome measurements or characteristics	Low risk	Patient MAI and number of medications imbalanced at baseline, but accounted for in analysis
Protection against contamination	Low risk	All GPs working in single practice - unlikely to be contamination between groups. Only time they were together was for the education session, where the next stage of the combined intervention came after that.
Incorrect analysis	High risk	Analysis adjusted for MAI and number of medications but not GP
Risk of bias overall	High risk	Due to incorrect analysis

Brown 1994

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Australia Setting: dental clinic Speciality: NA N health professionals: 24 practices N patients: 2142
Interventions	Number of arms: 3 Description of Groups: Control: no intervention; Intervention 1: without hygienists: 1-day seminar + bi-monthly newsletter + individualised comparative feedback + technical assistance; Intervention 2: with hygienists: 1-day seminar + bi-monthly newsletter + individualised comparative feedback + technical assistance
Outcomes	Clinical behaviour category: diagnosis, counselling, treatment decision/action Format of Audit & Feedback: both written and verbal

Brown 1994 *(Continued)*

	Source: manual chart review Frequency: quarterly Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	High risk	Group 1 was all practices that agreed to participate that employed a dental hygienist - no randomisation. Groups 2 and 3 were randomised from those that did not employ a dental hygienist, but no details of sequence generation provided
Allocation concealment (selection bias)	Unclear risk	No description of allocation concealment. Randomisation performed when consent performed, so may not have been all be at one time point
Blinding of outcome assessment (detection bias)	Unclear risk	Blinding of outcome assessors was not described.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Loss to follow-up and incomplete outcome data appears balanced between groups.
Selective reporting (reporting bias)	Low risk	The study date precluded trial registration. Results for all prespecified outcomes presented in the paper
Recruitment bias	Unclear risk	It is unclear whether patients were enroled in each practice after randomisation.
Baseline imbalance in outcome measurements or characteristics	High risk	Table 1 has a number of differences between groups. Other variables not reported
Protection against contamination	Low risk	Allocation was by practice - as all individual private practices, unlikely control group got the intervention
Incorrect analysis	Unclear risk	Unclear risk of bias as, although the dental practice was the unit of analysis, it is not clear whether the effect of clustering was taken into account.
Risk of bias overall	High risk	Baseline variability

Brown 2018

Study characteristics

Methods	Design: step wedge
Participants	Country: Australia Setting: hospital inpatient Speciality: urology

Brown 2018 *(Continued)*

	N health professionals: 37 N patients: 1071
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: interactive education session + printed guidelines + peer support and reinforcement from clinic leaders + A&F
Outcomes	Clinical behaviour category: referrals Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: variable Instructions for improvement: not reported
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation was "determined by a computer-generated random number sequence".
Allocation concealment (selection bias)	Unclear risk	The method of allocation was not described.
Blinding of outcome assessment (detection bias)	High risk	"Clinical data were extracted by independent research assistants, blinded to the date of intervention commencement", and it appears unlikely data would be recorded incorrectly/subjectively. However, clinician knowledge and attitude was self-reported in a survey and clinicians were aware of their allocation. It is likely bias may be introduced into this outcome.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Incomplete data appears balanced (with similar reasons) across groups.
Selective reporting (reporting bias)	High risk	Some changes to pre-planned outcomes (from trial registration) suggests potential risk of bias. Two primary outcomes in the trial registration (referral to radiology and referral to RAVES trial) have been merged into one primary outcome. The third primary outcome (time between surgery and radiotherapy) does not appear in the trial publication. Enrolment into RAVES trial was part of a primary outcome in the trial registration, but is a secondary outcome in the trial publication. Another secondary outcome was added pre-analysis, but this was unlikely to have introduced bias.
Recruitment bias	Low risk	No recruitment of urologists after allocation. No recruitment process for patient data
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Baseline characteristics and outcome measures of patients appear mostly balanced (despite some significant differences noted in patient characteristics). Characteristics of urologists not reported so unclear if potential baseline imbalances for urologists. Baseline outcome measures per cluster not reported so unclear

Brown 2018 *(Continued)*

Protection against contamination	Unclear risk	Locations of the participating sites not described. Unclear if there was potential for contamination between sites
Incorrect analysis	Low risk	Appears appropriate analysis for clustering was used - "Generalised linear regression models with Poisson distribution, log link and generalised estimating equation (GEE) adjustment for the clustering of patients within urologists"
Risk of bias overall	High risk	Due to blinding and selective outcome reporting

Brunette 2015

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: mental health clinic Speciality: psychiatry N health professionals: 85 N patients: 14,615
Interventions	Number of arms: 2 Description of groups: Intervention 1: in-person educational outreach + A&F; Intervention 2: videoconference educational outreach + A&F
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: administrative/billing data Frequency: semi-annually Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Stated as computer-generated randomisation table
Allocation concealment (selection bias)	Unclear risk	No details and it is unclear if clusters were allocated at the same time.
Blinding of outcome assessment (detection bias)	Low risk	Unclear if outcome assessors are blinded, but collecting Medicaid data as primary outcome means that it is unlikely that any bias would affect results.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	< 30% difference in sample size of mental health centres, and both groups had large sample sizes of recipients. However, unclear what the attrition rate of either group was, which may be consequential for evaluating each intervention,

Brunette 2015 *(Continued)*

		especially as higher dropout is a known issue for telemedicine vs F2F interventions.
Selective reporting (reporting bias)	Low risk	Outcome data appears to be consistently reported in text, tables and chart.
Recruitment bias	Unclear risk	No mention of timing of recruitment to randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Although groups mainly similar, limited baseline characteristics were assessed and some significant differences in mental health disorders between groups may have impacted results.
Protection against contamination	Low risk	'At the five centres randomly assigned to in-person education.' Implies randomisation by centre not practitioner
Incorrect analysis	Unclear risk	Although regression models were mentioned, adjustment for clustering was not discussed.
Risk of bias overall	Low risk	Low risk across all important domains

Buffington 1991

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 45 N patients: 10,525
Interventions	Number of arms: 3 Description of groups: Control: no intervention; Intervention 1: vaccine administration graph (self-tracked and recorded by each physician); Intervention 2: vaccine administration graph (self-tracked and recorded by each physician) + postcard reminders sent to patients
Outcomes	Clinical behaviour category: immunisation Format of Audit & Feedback: written Source: other Frequency: weekly Instructions for improvement: not reported
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement

Buffington 1991 *(Continued)*

Random sequence generation (selection bias)	Unclear risk	Method of sequence generation not reported (unable to annotate)
Allocation concealment (selection bias)	Unclear risk	It is unclear if all clusters might have been allocated at the same time.
Blinding of outcome assessment (detection bias)	High risk	Outcome assessors (practice staff) were not blinded to group allocation for measurement of influenza vaccine rates and were asked to tabulate and calculate outcomes themselves.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Loss of outcome data or participants with reasons not clearly described
Selective reporting (reporting bias)	Low risk	Sufficient information provided in methods; outcomes all reported
Recruitment bias	Unclear risk	Not described
Baseline imbalance in outcome measurements or characteristics	Unclear risk	No reporting of characteristics of the 3 clusters
Protection against contamination	Low risk	Allocation by practice and unlikely the control group received the interventions
Incorrect analysis	Unclear risk	The method of analysis (including taking into account the clusters) was not reported.
Risk of bias overall	High risk	Due to blinding

Campbell 2006

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Australia Setting: obstetrics and gynaecology clinic or ward Speciality: obstetrics N health professionals: NR N patients: 5849
Interventions	Number of arms: 2 Description of groups: Control: educational information; Intervention: educational information + clinic data (A&F) + programme training and support from midwife facilitator + computerised clinic feedback (from patient surveys) + copy of smoking cessation policy
Outcomes	Clinical behaviour category: screening, counselling Format of Audit & Feedback: written

Campbell 2006 *(Continued)*

	Source: other Frequency: monthly Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Method of sequence generation was not reported.
Allocation concealment (selection bias)	Low risk	Cluster-randomised trial and strata-based allocation - cluster and stratified at the same time; therefore, this reduces risk of bias associated with allocation concealment.
Blinding of outcome assessment (detection bias)	Unclear risk	Outcomes were self-reported by patients and it is unclear if they were aware of the trial and intervention.
Incomplete outcome data (attrition bias) All outcomes	Low risk	1/12 cluster missing from analysis from SD group. This does not appear to be significant or due to the intervention/outcome.
Selective reporting (reporting bias)	Low risk	No protocol or trial register.
Recruitment bias	Low risk	No additional recruitment after cluster had been set. p. 98 under Design: "random allocation of clinics was undertaken within strata based on clinic size and patient smoking rates as determined at the baseline survey to achieve balance between groups".
Baseline imbalance in outcome measurements or characteristics	Low risk	No important differences in outcomes or characteristics across study groups were present.
Protection against contamination	Low risk	Allocation by cluster (e.g. clinic) and it is therefore unlikely that the groups received alternate intervention.
Incorrect analysis	Low risk	Appears that clustering effects have been addressed in the analyses.
Risk of bias overall	Unclear risk	Unclear risk across important domains

Cánovas 2009

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Spain Setting: primary care Speciality: GP/family

Cánovas 2009 *(Continued)*

	N health professionals: NR N patients: 500
Interventions	Number of arms: 4 Description of groups: Control 1: no intervention; Control 2: internal commitment to perform quality assurance cycle + quality improvement education course; Intervention 1: A&F + quality assurance training; Intervention 2: A&F
Outcomes	Clinical behaviour category: prescribing, testing/exams Format of Audit & Feedback: written Source: manual chart review Frequency: semi-annually Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Method of sequence generation was not reported.
Allocation concealment (selection bias)	Unclear risk	No details or study flow diagram to determine if clusters likely allocated at the same time
Blinding of outcome assessment (detection bias)	Low risk	The outcomes were not fully known to the clinicians who recorded in medical records from which the outcomes were derived.
Incomplete outcome data (attrition bias) All outcomes	High risk	2/3 of group 1 were excluded with explanations from the authors. This is a significant proportion (even if unintentional) and leaves only one centre contributing to the outcome and potentially biases the results.
Selective reporting (reporting bias)	Unclear risk	Unclear risk of selective outcome reporting bias as the trial was not registered, study protocol was not found, and the outcomes were decided by the intervention group participants after commencement of the trial.
Recruitment bias	Low risk	It does not appear that any centres were recruited after allocation.
Baseline imbalance in outcome measurements or characteristics	High risk	No reporting of baseline characteristics, but baseline outcome data do appear to differ significantly. While not clearly reported, it does not appear that trialists adjusted for imbalances.
Protection against contamination	Low risk	Allocation was by cluster (health centres) and thus unlikely that clinicians received intervention/s from another group
Incorrect analysis	Unclear risk	Not enough information on how the trialists performed their analyses.
Risk of bias overall	High risk	High due to baseline imbalances and incomplete outcome data

Carney 2012

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: radiology clinic Speciality: radiology N health professionals: 74 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: education; Intervention: A&F + education
Outcomes	Clinical behaviour category: screening Format of Audit & Feedback: electronic dashboard Source: computerised search of charts Frequency: once Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Method of sequence generation was not reported.
Allocation concealment (selection bias)	Unclear risk	No details or study flow diagram to determine if clusters likely allocated at the same time
Blinding of outcome assessment (detection bias)	High risk	Outcomes self-reported by participants (positive mammograms) - could be subjective and influenced by knowledge of allocation.
Incomplete outcome data (attrition bias) All outcomes	High risk	Per-protocol analysis was used. The flow of study participants is poorly reported (e.g. allocation of radiologists who were excluded due to no follow-up data) and a large proportion of participants did not complete the CME intervention, with analysis separated into those who completed the CME intervention and those who did not. Even if this was pre-planned, there is a possibility that the participants who completed the CME intervention may have differed from those who did not.
Selective reporting (reporting bias)	High risk	Unadjusted recall rates for each time point reported without variance.
Recruitment bias	Low risk	No evidence of recruitment bias
Baseline imbalance in outcome measurements or characteristics	Low risk	See table 1, baseline characteristics and outcomes balanced between groups

Carney 2012 *(Continued)*

Protection against contamination	Unclear risk	Some radiologists may have worked at the same facility; there may have been some cross-communication.
Incorrect analysis	Unclear risk	Not enough information
Risk of bias overall	High risk	Due to blinding, incomplete outcome data, selective outcome reporting

Chaillet 2015

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Canada Setting: hospital inpatient Speciality: obstetrics N health professionals: NR N patients: 184,592
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: onsite training + A&F + educational outreach + local opinion leader
Outcomes	Clinical behaviour category: treatment decision/action, prescribing Format of Audit & Feedback: verbal (interactive) Source: manual chart review Frequency: quarterly Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	To avoid imbalance in the size of the two groups, we used computer-generated, blocked randomisation within each stratum, with blocks consisting of four centres or, for strata with fewer than eight hospitals, two centres.
Allocation concealment (selection bias)	Low risk	Allocation was performed centrally and at one time.
Blinding of outcome assessment (detection bias)	Low risk	Data collectors were not blinded but the measures being collected were objectively abstracted from medical records. Outcome assessors (physicians) filled in medical records, and it is unlikely these would be subject to risk of bias.
Incomplete outcome data (attrition bias)	Low risk	"No hospital or woman was lost to follow-up".

Chaillet 2015 *(Continued)*

All outcomes		
Selective reporting (reporting bias)	Low risk	Protocol available, and all expected outcomes were reported.
Recruitment bias	Low risk	No evidence of hospitals recruited after randomisation. All eligible patient participants were included.
Baseline imbalance in outcome measurements or characteristics	Low risk	Hospitals were randomly assigned using strata/cluster sampling to minimise baseline imbalances, and participants at intervention and control hospitals were similar
Protection against contamination	Low risk	Intervention was delivered by cluster; physicians rarely travelled between hospitals; contamination bias was assessed annually.
Incorrect analysis	Low risk	Analyses appropriately accounted for clustering of data.
Risk of bias overall	Low risk	Low risk across all important domains

Charrier 2008

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Italy Setting: hospital inpatient Speciality: NA N health professionals: 160 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + facilitation
Outcomes	Clinical behaviour category: other Format of Audit & Feedback: not reported/not applicable Source: not reported/not applicable Frequency: quarterly Instructions for improvement: not reported
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Method of sequence generation was not reported.

Charrier 2008 *(Continued)*

Allocation concealment (selection bias)	Unclear risk	No details or study flow diagram to determine if clusters likely allocated at the same time
Blinding of outcome assessment (detection bias)	High risk	Auditors were not blind to group allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	It does not appear any clusters were lost to follow-up.
Selective reporting (reporting bias)	Low risk	Results reported as described in the methods. No protocol available
Recruitment bias	Low risk	Unlikely participants were recruited to the trial after clusters were randomised
Baseline imbalance in outcome measurements or characteristics	Low risk	Similar characteristics for participants and patients between intervention and control groups
Protection against contamination	Low risk	Cluster randomisation performed on selected departments to prevent risk of contamination
Incorrect analysis	Unclear risk	Unclear if clusters taken into account in analysis
Risk of bias overall	High risk	Due to blinding

Chassin 1986

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital inpatient Speciality: obstetrics N health professionals: 1483 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: Education + A&F
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: written Source: manual chart review Frequency: monthly Instructions for improvement: general
Notes	

Chassin 1986 *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Hospitals were randomly assigned however the method of generating the random sequence was not described
Allocation concealment (selection bias)	Unclear risk	No details or study flow diagram to determine if clusters likely allocated at the same time.
Blinding of outcome assessment (detection bias)	Unclear risk	It is unclear whether outcome assessors were blinded to the treatment groups or not.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Study flow not described so unclear about how many recruited to begin with
Selective reporting (reporting bias)	Low risk	The paper presents the results of all pre-specified outcomes. As the study was conducted prior to 2005, trial could not be registered.
Recruitment bias	Unclear risk	not described
Baseline imbalance in outcome measurements or characteristics	Low risk	Although there was a slight increase in the number of hospitals with obstetrical residences in the intervention group, this did not affect the results in a significant way as shown in table 2.
Protection against contamination	Low risk	"Control hospitals received neither the educational program nor the monitoring. These hospitals were thus not contacted at all by the study between the baseline data collection activity in June 1980 and the final data collection activity." Low risk of contamination bias.
Incorrect analysis	Low risk	Analysis at cluster level
Risk of bias overall	Unclear risk	Unclear risk across important domains

Cheater 2006

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: NA N health professionals: 176 N patients: 1078
Interventions	Number of arms: 4 Description of groups: Control 1: educational materials; Control 2: educational outreach; Intervention 1: A&F; Intervention 2: A&F + educational outreach

Cheater 2006 *(Continued)*

Outcomes	Clinical behaviour category: screening, prescribing, testing/exams, diagnosis, treatment decision/action, counselling, referrals Format of Audit & Feedback: written Source: manual chart review Frequency: monthly Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Stratified randomisation, blocks of 4
Allocation concealment (selection bias)	Low risk	Quote "concealed randomization"
Blinding of outcome assessment (detection bias)	Low risk	Quote "data collectors blind to allocation"
Incomplete outcome data (attrition bias) All outcomes	Low risk	All practices followed up
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Groups comparable (p. 545)
Protection against contamination	Low risk	No evidence of contamination
Incorrect analysis	Low risk	Clustering accounted for in analysis
Risk of bias overall	Low risk	Low risk across all important domains

Clarke 2020

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Germany/Ireland/Italy Setting: hospital inpatient Speciality: obstetrics

Clarke 2020 *(Continued)*

	N health professionals: NR N patients: 1956
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: patient education session + clinician education session + access to optional online educational resources and mobile/computer applications + appointment of midwife and obstetric opinion leaders + A&F + joint discussion by women and clinicians
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: verbal (interactive) Source: manual chart review Frequency: monthly Instructions for improvement: not reported
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	"The randomisations were done in the presence of a witness who was unaware of which site was associated with which code within the pairs and triplets until after the allocations were complete" (protocol). Allocation likely done centrally at one time
Blinding of outcome assessment (detection bias)	Low risk	Due to the nature of the outcomes, it is unlikely bias could be introduced in recording the outcomes.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Missing data appears balanced between arms with some of the reasons provided (do not suggest the reasons are related to the intervention).
Selective reporting (reporting bias)	Low risk	Trial was registered and trial protocol was published, no deviations from the protocol were found.
Recruitment bias	Low risk	No clusters recruited after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline imbalances were reported and adjusted for in the analyses.
Protection against contamination	Low risk	As investigators took adequate precautions against contamination (by avoiding an intervention and control cluster in the same city), it is presumed there is a low risk of contamination bias in this trial.
Incorrect analysis	Low risk	Clustering taken into account in analysis
Risk of bias overall	Low risk	Low risk across all important domains

Clyne 2015

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Ireland Setting: primary care Speciality: GP/family N health professionals: NR N patients: 196
Interventions	Number of arms: 2 Description of groups: Intervention 1: academic detailing with a pharmacist + A&F + tailored patient information leaflets; Intervention 2: usual care + A&F
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Practices were allocated to intervention and control groups by an independent researcher using minimisation, an allocation method commonly used in cluster-randomised controlled trials to ensure balanced allocation of important cluster-level attributes such as practice size when cluster numbers are small.
Allocation concealment (selection bias)	Low risk	Central allocation using minimisation - "Practices were allocated to intervention and control groups by an independent researcher using minimization".
Blinding of outcome assessment (detection bias)	Low risk	As outcome assessors were blinded to group allocation, there is a low risk of detection bias.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Results reflect the outcomes described.
Selective reporting (reporting bias)	Low risk	Trial registered (ISRCTN41694007). Authors published all outcomes as per protocol. https://trialsjournal.biomedcentral.com/articles/10.1186/1745-6215-14-72
Recruitment bias	Low risk	There was no recruitment of participants into the trial after randomisation.

Clyne 2015 *(Continued)*

Baseline imbalance in outcome measurements or characteristics	Low risk	Balanced allocation of important cluster-level attributes such as practice size when cluster numbers are small
Protection against contamination	Low risk	The control group was not exposed to the intervention.
Incorrect analysis	Low risk	Clustering taken into account and adjusted - "analyzed using random effects logistic regression with the individual as the unit of analysis and the practice included as the random effect, to control for the effects of clustering."
Risk of bias overall	Low risk	Low risk across all important domains

Colón-Emeric 2007

Study characteristics	
Methods	Design:parallel cluster-RCT
Participants	Country: USA Setting: nursing homes Speciality: NA N health professionals: NR N patients: 606
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + education (modules) + teleconferences + academic detailing + printed educational material
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: manual chart review Frequency: Variable Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random number generator used for sequence generation
Allocation concealment (selection bias)	Unclear risk	No details or study flow diagram to determine if clusters likely allocated at the same time
Blinding of outcome assessment (detection bias)	Low risk	"Trained data collectors, blinded to intervention status, abstracted data from the medical record" and they would not have introduced any bias. It seems un-

Colón-Emeric 2007 *(Continued)*

		likely that nursing home staff would incorrectly record prescriptions given to patients.
Incomplete outcome data (attrition bias) All outcomes	Low risk	All enroled and randomised nursing homes provided outcome data.
Selective reporting (reporting bias)	Unclear risk	All outcomes specified in the methods were reported; however, no variance was reported.
Recruitment bias	Low risk	Inclusion of patient participants was dependent on an objective set of criteria, including having been at the nursing home for at least four weeks; therefore, it is unlikely that patient recruitment could have been biased after the nursing home was allocated.
Baseline imbalance in outcome measurements or characteristics	Low risk	There are a few baseline imbalances picked up by trialists: "Intervention residents were more likely to be African American, younger, and use tobacco; and less likely to have previous fracture or dysphagia." The difference in previous fractures is especially notable as it relates to patient inclusion (and potentially the outcomes). Multivariate analysis was conducted adjusting for baseline imbalances (P < 0.2) and no difference was detected; therefore, these baseline imbalances do not affect the results.
Protection against contamination	Low risk	Allocation was at the level of nursing homes. No evidence that health professional participants in the control group would have received the intervention. It may be possible, but seems unlikely.
Incorrect analysis	Low risk	Analysis was performed at the cluster level using Generalised Estimating Equation and appears appropriate - "multivariable logistic regression models with generalized estimating equation adjustment for repeated binary outcomes were constructed with backwards selection, using all variables with a baseline univariate P < 0.20 and variables with high clinical significance (age, race, tate [?]."
Risk of bias overall	Low risk	Low risk across all important domains

Coma 2019

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Spain Setting: primary care Speciality: GP/family N health professionals: 6600 N patients: 645,845
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F; Intervention 2: A&F + chart alerts
Outcomes	Clinical behaviour category: screening, prescribing, testing/exams, counselling, immunisation

Coma 2019 *(Continued)*

	Format of Audit & Feedback: electronic dashboard Source: computerised search of charts Frequency: monthly/weekly Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	As per flow chart, it is likely that allocation was done centrally at one time.
Blinding of outcome assessment (detection bias)	Low risk	Outcome measurement is not likely to be influenced by lack of blinding.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No mention of loss to follow-up or missing outcome data so reporting is unclear.
Selective reporting (reporting bias)	Low risk	Reported consistently with protocol
Recruitment bias	Low risk	All enroled at the same time.
Baseline imbalance in outcome measurements or characteristics	Low risk	Comparable groups from randomisation
Protection against contamination	Low risk	Computer pop-ups for intervention group would not be shown to control group. Cluster design means contamination is unlikely.
Incorrect analysis	Low risk	Analysis accounted for clusters
Risk of bias overall	Low risk	Low risk across all important domains

Crotty 2004

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Australia Setting: primary care Speciality: GP/family N health professionals: 98

Crotty 2004 *(Continued)*

	N patients: 715
Interventions	Number of arms: 2 Description of groups: Control: audit of clinical practice but no feedback; Intervention: academic detailing + A&F + pharmacist visit
Outcomes	Clinical behaviour category: prescribing, treatment decision/action Format of Audit & Feedback: verbal (didactic) Source: manual chart review Frequency: not reported/not applicable Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation by computer-generated random allocation program
Allocation concealment (selection bias)	Low risk	Central allocation was used, so it is likely that allocation was concealed - "random allocation program by a person external to the project".
Blinding of outcome assessment (detection bias)	Low risk	Nurses who conducted case notes audits were blinded to allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No clusters lost to follow-up. Attrition of patient participants appeared to be balanced between groups and reasons for attrition do not appear related to the intervention.
Selective reporting (reporting bias)	Unclear risk	No previously published protocol and no information was on the retrospectively published trial registration relating to outcomes. It is unclear if all previously planned outcomes are reported here.
Recruitment bias	Low risk	Audit included data from all patients of the participating residential facilities.
Baseline imbalance in outcome measurements or characteristics	Low risk	Some baseline imbalances were reported, but the analysis was adjusted to take these into account.
Protection against contamination	Low risk	Allocation by residential facility selected at random and matched to a facility in another region to avoid overlap of physicians
Incorrect analysis	Low risk	Clustering was taken into account for analysis.
Risk of bias overall	Low risk	Low risk across all important domains

Cundill 2015

Study characteristics

Cundill 2015 *(Continued)*

Methods	Design: parallel cluster-RCT
Participants	Country: Tanzania Setting: primary care Speciality: GP/family N health professionals: 104 N patients: 44,121
Interventions	Number of arms: 3 Description of groups: Control: standard training; Intervention 1: A&F SMS + standard training + workshop + SMS reminders; Intervention 2: A&F SMS + standard training + workshop + patient education materials + SMS reminders
Outcomes	Clinical behaviour category: prescribing, testing/exams Format of Audit & Feedback: written Source: administrative/billing data, patient-reported data Frequency: monthly Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Central random allocation
Blinding of outcome assessment (detection bias)	High risk	Outcomes were self-reported to interviewers, so there is a high risk of bias.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No incomplete or excluded outcome data. ITT was used.
Selective reporting (reporting bias)	Unclear risk	It's unclear whether outcomes were reported selectively, because there is no protocol and the secondary outcome is vaguely described.
Recruitment bias	Low risk	Unlikely there was recruitment after allocation
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline mostly comparable and analyses adjusted for a number of covariates (though not specified) so likely okay
Protection against contamination	Low risk	Randomisation at practice level over a large geographical area; no reason to suspect contamination
Incorrect analysis	Low risk	Analysis adjusted for clustering

Cundill 2015 *(Continued)*

Risk of bias overall	High risk	Due to blinding

Curtis 2005

Study characteristics	
Methods	Design: Parallel Cluster-RCT
Participants	Country: USA Setting: managed care organisation - primary care clinics and specialised rheumatology clinics Speciality: GP/family/internal medicine N health professionals: 101 N patients: 680
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + educational materials (literature & webblink)
Outcomes	Clinical behaviour category: prescribing, testing/exams Format of Audit & Feedback: written Source: manual chart review Frequency: monthly Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial, allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Data abstractors blind to treatment
Incomplete outcome data (attrition bias) All outcomes	Low risk	No difference between groups
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	Recruitment did not occur after randomisation.

Curtis 2005 *(Continued)*

Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 1
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Low risk	Generalised estimating equations to account for cluster randomisation
Risk of bias overall	Low risk	Low risk across all important domains

Curtis 2007

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 153 N patients: 949
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + educational modules + QI toolbox
Outcomes	Clinical behaviour category: testing/exams, prescribing Format of Audit & Feedback: written Source: administrative/billing data Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Method for randomisation not described, but likely it was computer-generated from study website (the use of 'state-of-the-art technology in dynamic randomization' is mentioned in the Comment section).
Allocation concealment (selection bias)	Low risk	Randomisation, and therefore allocation, was done when a physician logged on to computer for the first time for that practice. Therefore, allocation was concealed.
Blinding of outcome assessment (detection bias)	Low risk	Outcome assessment is not clearly reported, but objective data was collected and analysed (counting medication use) so bias unlikely

Curtis 2007 *(Continued)*

Incomplete outcome data (attrition bias) All outcomes	Low risk	Differences in completion between arms, but both ITT (primary analyses) and PP were used.
Selective reporting (reporting bias)	Unclear risk	No protocol, not possible to assess
Recruitment bias	Unclear risk	Physicians recruited and randomised over a period of time as they logged onto website. Unclear if there is potential for recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Baseline performance balanced. Some imbalances between treatment and control groups, including proportion of male patients and several glucocorticoid-use associated diseases (rheumatoid arthritis, systemic lupus, COPD, asthma, IBD) and other comorbidities.
Protection against contamination	Low risk	Due to the cluster design, contamination is unlikely.
Incorrect analysis	Low risk	Authors did check effect of clustering and did not present this due to very small differences.
Risk of bias overall	Low risk	Low risk across all important domains

Curtis 2011

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital inpatient Speciality: ICU N health professionals: 309 N patients: 2318
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + education + local champion
Outcomes	Clinical behaviour category: treatment decision/action, counselling Format of Audit & Feedback: both written and verbal Source: manual chart review, patient-reported data Frequency: once Instructions for improvement: general
Notes	
Risk of bias	

Curtis 2011 *(Continued)*

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	It is unclear how the random numbers were generated, but the study appears well conducted with an experienced team and multiple publications and reports (e.g. grant application) to document the methods, all suggesting the trialists likely used an adequate method.
Allocation concealment (selection bias)	Low risk	It is likely that all sites were allocated centrally at one time.
Blinding of outcome assessment (detection bias)	High risk	It is unclear if families of patients were aware of the site's allocation and, if so, this could bias the self-reported questionnaire. Nurses also completed a self-reported questionnaire and were aware of the trial allocation; therefore, this outcome is subject to bias.
Incomplete outcome data (attrition bias) All outcomes	High risk	Although methods stated ITT, questionnaire outcomes were reported for participants who returned the questionnaire (with response rate below 50% for all groups). This was balanced between groups for family members, but differed for nurses between intervention and control groups (baseline 46% vs 29% available, follow-up 39% vs 27% available), suggesting that availability of outcome data was affected and may have been biased.
Selective reporting (reporting bias)	Unclear risk	All outcomes prespecified in the (retrospective) trial registration are reported. The original grant application specifies NASQ (nurse satisfaction), ICU-QOC-4 (measures of quality of care and self-efficacy), and nursing self-efficacy items, which are not described here. It is unclear when this change was made.
Recruitment bias	Low risk	There is no evidence of recruitment after randomisation and allocation.
Baseline imbalance in outcome measurements or characteristics	Low risk	There were some baseline imbalances, but the methods taken to adjust outcome data were likely adequate to address imbalances.
Protection against contamination	High risk	"Second, because this is a randomized intervention within a single community, it is possible that there will be contamination across sites. All of the hospitals use some agency ICU nurse and respiratory therapist staff. Three of the hospitals (Providence, Swedish-First Hill, Swedish-Ballard) share some of their critical care staff. However, we do not feel this is a significant threat to the study because the majority of nurses and physicians only care for patients at one of the hospitals. In addition, agency nurses and respiratory therapists make up only a small part of the ICU staff on any one shift at each of the hospitals". Although the authors do not think contamination was a significant threat, it appears that some control participants would have received the intervention and therefore there is high risk of bias.
Incorrect analysis	Low risk	Likely appropriate analyses
Risk of bias overall	High risk	Due to protection against contamination, blinding and incomplete outcome data

DeVore 2015

Study characteristics	
Methods	Design: parallel cluster-RCT

DeVore 2015 *(Continued)*

Participants	Country: USA Setting: hospital inpatient Speciality: cardiology N health professionals: NR N patients: 71,829
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + teleconferences/webinars
Outcomes	Clinical behaviour category: prescribing, testing/exams, treatment decision/action, counselling, immunisation Format of Audit & Feedback: both written and verbal Source: administrative/billing data Frequency: quarterly Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	It is unclear how the random sequence was generated.
Allocation concealment (selection bias)	Low risk	Not described but, based on study flow diagram, it would appear all sites allocated at one time.
Blinding of outcome assessment (detection bias)	Unclear risk	No information on blinding of outcome assessors was provided in the paper, hence there is an unclear risk of detection bias.
Incomplete outcome data (attrition bias) All outcomes	Low risk	The number of cluster dropouts and reasons for withdrawal were similar across the two groups.
Selective reporting (reporting bias)	Unclear risk	No protocol available, but outcomes are reported as prespecified in Methods.
Recruitment bias	Low risk	All hospitals were randomised into the study simultaneously and no patients appeared to have been recruited to the study after completion of randomisation.
Baseline imbalance in outcome measurements or characteristics	Low risk	No baseline imbalances between groups were apparent.
Protection against contamination	Low risk	It seems unlikely that control hospitals would receive the intervention, or that staff would overlap between intervention and control hospitals.
Incorrect analysis	Unclear risk	The paper does not describe how clustering was accounted for in the analysis.

DeVore 2015 *(Continued)*

Risk of bias overall	Unclear risk	Unclear risk across important domains

Diamantouros 2017

Study characteristics	
Methods	Design: step wedge
Participants	Country: Canada Setting: hospital inpatient Speciality: internal medicine N health professionals: NR N patients: 720
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + order sets + reminders + clinician education + guidelines
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: not reported/not applicable Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Unclear risk	Randomisation occurred in two phases.
Blinding of outcome assessment (detection bias)	Low risk	Data collected from chart review
Incomplete outcome data (attrition bias) All outcomes	Low risk	No loss of clusters
Selective reporting (reporting bias)	Unclear risk	Retrospectively registered protocol; however, all outcomes reported
Recruitment bias	Low risk	No recruitment after randomisation

Diamantouros 2017 *(Continued)*

Baseline imbalance in outcome measurements or characteristics	Unclear risk	Not reported
Protection against contamination	Low risk	Unlikely that the control group participants received intervention and the trial was designed to avoid this
Incorrect analysis	Low risk	GEE used to account for clustering
Risk of bias overall	Low risk	Low risk across most domains

Dijkstra 2005

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: hospital outpatient clinic Speciality: internal medicine N health professionals: 42 N patients: 769
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F+ clinician education + patient education
Outcomes	Clinical behaviour category: testing/exams, counselling, referrals Format of Audit & Feedback: both written and verbal Source: manual chart review, patient-reported data Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Not enough information provided
Allocation concealment (selection bias)	Low risk	Random allocation was done by a person outside the research group and concealed from the investigators until the start of the intervention.
Blinding of outcome assessment (detection bias)	Low risk	Laboratory results and medical records used for data collection

Dijkstra 2005 *(Continued)*

Incomplete outcome data (attrition bias) All outcomes	High risk	> 10% lost in each arm
Selective reporting (reporting bias)	Unclear risk	No protocol available, but outcomes are reported as prespecified in Methods.
Recruitment bias	Low risk	No recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	No significant differences between groups
Protection against contamination	Low risk	Unlikely that the control group participants received intervention and the trial was designed to avoid this
Incorrect analysis	Low risk	The Glimmix procedure (SAS 6.12), which controls for clustering effects, was used for the multilevel (hospital–physician–patient) logistic regression analysis.
Risk of bias overall	High risk	Due to incomplete outcome data

Dormuth 2012

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Canada Setting: primary care Speciality: GP/family N health professionals: 2725 N patients: 2,117,959
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + evidence-based messages; Intervention 2: delayed audit & feedback + evidence-based messages
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: administrative/billing data Frequency: once Instructions for improvement: general
Notes	
Risk of bias	

Dormuth 2012 *(Continued)*

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	"Pairs of locations were randomised to heads or tails in spreadsheet". Unit of randomisation was the practice community. Matched pairs (size, location, mix of practitioners)
Allocation concealment (selection bias)	Low risk	Method of allocation is not described; however, based on the steps taken by the trialists to blind physician details, it is likely the study concealed allocation.
Blinding of outcome assessment (detection bias)	Low risk	The outcome was extracted from prescription records and objective, therefore, not subject to bias if allocation was known by the assessor/s.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	There is no information about flow of study participants. It is not reported how many physicians were randomised.
Selective reporting (reporting bias)	Unclear risk	There is no previously published protocol or trial registration. It is unclear if all preplanned outcomes were measured and reported as preplanned. The pilot study and other similar RCTs by the trialists differ in outcome/s reported (and exact nature of the intervention) so it is not possible, based on the available information, to judge that all expected outcomes were measured and reported in the expected manner here.
Recruitment bias	Low risk	Physicians were recruited and then the communities they worked in were randomised.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	All but one baseline characteristics are balanced between groups. It is unclear if this is an important characteristic that would affect the outcomes.
Protection against contamination	Unclear risk	It seems likely that the risk of contamination bias was reduced by randomising communities. It is not clear how much contact there was between the various community physicians.
Incorrect analysis	High risk	Data were adjusted to take physician variables into account. However, each practice was a cluster and the analysis was performed with the physician as the unit of analysis.
Risk of bias overall	High risk	Due to incorrect analysis

Dreischulte 2016

Study characteristics	
Methods	Design: step wedge
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: NR N patients: 66,394

Dreischulte 2016 *(Continued)*

Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: clinician education + financial incentives + A&F
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: variable Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Performed by a randomisation service and highly likely the method was adequate. See appendix - "Allocation of practices was independently carried out by the randomisation service provided by the UK Clinical Research Network registered Tayside Clinical Trials Unit. Recruited practices were stratified by list size tertile and randomly allocated within strata to one of 10 starting dates".
Allocation concealment (selection bias)	Unclear risk	Despite wording, it is unclear if clusters were actually aware of their allocation (intervention start date) beforehand. See appendix - "Due to the nature of the study, allocation concealment was not possible for practices (who received the intervention), the core research team (who delivered the intervention) or the trial analyst (since analysis requires knowledge of the date that each practice started the intervention)."
Blinding of outcome assessment (detection bias)	Low risk	Outcomes measures were recorded in medical records and unlikely to be recorded with bias.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No practices withdrew/lost and all eligible patients included in outcome data
Selective reporting (reporting bias)	Low risk	Trial registered and protocol described in detail in the supplementary appendix, and no obvious deviations from the study protocol found, hence, there is low risk of reporting bias.
Recruitment bias	Low risk	All clusters were recruited into the study as per the stepped-wedge design; there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline data not presented, but data adjusted for baseline/pre-intervention period, so seems that imbalances would have been adjusted for
Protection against contamination	Low risk	Cluster design was chosen in order to avoid contamination.
Incorrect analysis	Low risk	Clustering taken into account during analysis
Risk of bias overall	Low risk	Low risk across all important domains

Dudzinski 2016

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: cardiology clinic Speciality: cardiology N health professionals: 66 N patients: 16,025
Interventions	Number of arms: 2 Description of groups: Control: education (lecture); Intervention: A&F + education
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: written Source: manual chart review, computerised search of charts Frequency: monthly Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	A random number generator was used to generate the random sequence.
Allocation concealment (selection bias)	Low risk	Not described, but based on study flow diagram, it is likely all clusters were allocated at the same time.
Blinding of outcome assessment (detection bias)	Low risk	Outcome assessors for classifying TTE were blinded to allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	There was only 1 withdrawal form the intervention group due to retirement from practice, hence, low risk of attrition bias
Selective reporting (reporting bias)	Low risk	Trial was registered and study protocol was found. Results for all prespecified outcomes were reported.
Recruitment bias	Low risk	Not applicable
Baseline imbalance in outcome measurements or characteristics	Low risk	No baseline imbalance between groups

Dudzinski 2016 *(Continued)*

Protection against contamination	Low risk	Although there was a possibility of contamination, this was not considered to be significant enough to influence the results in a tangible way.
Incorrect analysis	Low risk	Not applicable
Risk of bias overall	Low risk	Low risk across all important domains

Eccles 2001

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: NR N patients: 1693
Interventions	Number of arms: 4 Description of groups: Control 1: no intervention; Control 2: reminders; Intervention 1: A&F; Intervention 2: A&F + reminders
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: written Source: manual chart review Frequency: semi-annually Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random number table
Allocation concealment (selection bias)	Low risk	Cluster trial, allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Radiology departments not aware of randomisation
Incomplete outcome data (attrition bias) All outcomes	Low risk	Minimal dropout

Eccles 2001 *(Continued)*

Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No new participants recruited after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Low risk	Randomised by practice
Incorrect analysis	Unclear risk	Although the study was a cluster-randomised trial, the main analysis was undertaken on summary statistics corresponding to the unit of randomisation, and clustering was not a problem.
Risk of bias overall	Low risk	Low risk across all important domains

Elouafkaoui 2016

Study characteristics	
Methods	Design: factorial
Participants	Country: Scotland Setting: dental clinic Speciality: NA N health professionals: NR N patients: 2142
Interventions	Number of arms: 9 Description of groups: Control: No feedback; Intervention: AF with various components to dentists about antibiotics
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: administrative/billing data Frequency: semi-annually Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation

Elouafkaoui 2016 *(Continued)*

Allocation concealment (selection bias)	Low risk	Allocation by trial statistician at start of study
Blinding of outcome assessment (detection bias)	Low risk	The statistician was blinded to the identity of the practices.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No missing data
Selective reporting (reporting bias)	Low risk	All outcomes reported
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline characteristics and outcomes balanced between arms
Protection against contamination	Low risk	Cluster design to reduce contamination
Incorrect analysis	Low risk	Clustering accounted for in analysis
Risk of bias overall	Low risk	Low across all domains

Eltayeb 2005

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Sudan Setting: primary care Speciality: GP/family N health professionals: 37 N patients: 60
Interventions	Number of arms: 4 Description of groups: Control: no intervention; Intervention 1: A&F; Intervention 2: A&F + guidelines + education (seminars); Intervention 3: A&F + guidelines + academic detailing
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: manual chart review Frequency: monthly Instructions for improvement: general

Eltayeb 2005 *(Continued)*

Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Unclear risk	Unable to assess
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unable to assess
Selective reporting (reporting bias)	Unclear risk	Unable to assess
Recruitment bias	Unclear risk	Not enough information
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 2
Protection against contamination	Low risk	Health centres separate
Incorrect analysis	High risk	Did not account for clustering
Risk of bias overall	High risk	Due to incorrect analysis

Estrada 2011

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 205 N patients: 1182
Interventions	Number of arms: 2 Description of groups: Intervention: A&F + education + guidelines + reminders; Control: education + guidelines
Outcomes	Clinical behaviour category: screening

Estrada 2011 *(Continued)*

	Format of Audit & Feedback: electronic dashboard Source: manual chart review Frequency: not reported/not applicable Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Method for sequence generation is not described.
Allocation concealment (selection bias)	Low risk	Physicians were randomised and allocated upon enrolment (visiting the website) and this was also concealed from investigators and statisticians.
Blinding of outcome assessment (detection bias)	Low risk	Physicians were masked to the arms of the trial and data abstraction of patient records was done blinded to allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Missing outcome data at follow-up appear balanced with reasons in the two arms and do not seem related to the intervention and outcomes.
Selective reporting (reporting bias)	High risk	The secondary outcomes from the trial registration are different to the secondary outcomes described in this publication. The primary outcomes in the trial registration were also not defined, so it is unclear when the specific definition of each primary outcome was made.
Recruitment bias	High risk	The non-random sampling of physicians may have introduced selection bias. Second, physicians were asked to provide records of consecutively seen patients, and while we were unable to monitor compliance with this request, the wide range of A1c, BP and LDL values suggests that not only well-controlled patients were selected, but poorly-controlled patients as well.
Baseline imbalance in outcome measurements or characteristics	Low risk	A few baseline imbalances were noted, but analysis was conducted adjusting for baseline covariates that differed.
Protection against contamination	Low risk	Due to the cluster design of the study, only one physician from each practice participated in the trial, so there is little risk of control physicians receiving the intervention.
Incorrect analysis	Low risk	Clustering was taken into account during analysis.
Risk of bias overall	High risk	Due to selective outcome reporting and recruitment bias

Everett 1983

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA

Everett 1983 *(Continued)*

	Setting: hospital inpatient Speciality: internal medicine N health professionals: 30 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: educational materials + A&F
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: verbal (interactive) Source: manual chart review Frequency: bi-weekly Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Unclear risk	Unable to assess
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unable to assess
Selective reporting (reporting bias)	Unclear risk	Unable to assess
Recruitment bias	Low risk	No recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	High risk	See Table 1
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	High risk	Student's t-test
Risk of bias overall	High risk	Due to incorrect analysis and baseline imbalance

Fabbri 2019

Study characteristics		
Methods	Design: parallel cluster-RCT	
Participants	Country: India Setting: community care Speciality: obstetrics N health professionals: NR N patients: 3519	
Interventions	Number of arms: 4 Description of groups: Control: no intervention; Intervention 1: A&F report cards + meeting with providers; Intervention 2: A&F report cards + meeting with community members; Intervention 3: A&F report cards + meeting with community members + meeting with providers	
Outcomes	Clinical behaviour category: counselling, immunisation, referrals, treatment decision/action, other Format of Audit & Feedback: both written and verbal Source: patient-reported data Frequency: variable Instructions for improvement: co-developed plans	
Notes		
Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	As per flow chart, it is likely that allocation was done centrally at one time.
Blinding of outcome assessment (detection bias)	Low risk	Fieldworkers who collected data were masked to the group allocations of the clusters, and they operated independently of the team who were involved in the delivery of the report card intervention.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No loss of clusters; missing outcome data are balanced with reasons reported. ITT analysis used
Selective reporting (reporting bias)	Unclear risk	No protocol or trial registration; unclear reporting
Recruitment bias	Low risk	No evidence of recruitment after randomisation and allocation
Baseline imbalance in outcome measurements or characteristics	Low risk	The characteristics of women in the four treatment groups and in pairs of intervention groups at baseline are shown in table 1. Participants in the four groups were similar in age, caste, religion, and wealth. There were small differences in educational attainment and residence between groups. Across the

Fabbri 2019 *(Continued)*

		groups, women also had similar maternal and neonatal health coverage indicators. No important baseline differences between groups
Protection against contamination	Low risk	Due to cluster design, contamination is unlikely.
Incorrect analysis	Low risk	Clustering was taken into account in analysis.
Risk of bias overall	Low risk	Low risk across all important domains

Fairbrother 1999

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: community care Speciality: paediatrics N health professionals: 60 N patients: 3000
Interventions	Number of arms: 4 Description of groups: Control: no intervention; Intervention 1: A&F + financial bonus; Intervention 2: A&F + enhanced fee for service; Intervention 3: A&F only
Outcomes	Clinical behaviour category: immunisation Format of Audit & Feedback: written Source: manual chart review Frequency: monthly Instructions for improvement: not reported
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Unclear risk	Unable to assess
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess

Fairbrother 1999 *(Continued)*

Incomplete outcome data (attrition bias) All outcomes	Low risk	Only one dropout
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence, there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	High risk	Rates of immunisation differ.
Protection against contamination	Low risk	Separate practices
Incorrect analysis	Low risk	Logistic and linear regression and standard errors to correct for clustering
Risk of bias overall	High risk	Due to baseline imbalances

Feder 1995a

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: 39 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + education
Outcomes	Clinical behaviour category: screening, counselling, prescribing, testing/exams Format of Audit & Feedback: verbal (interactive) Source: manual chart review, computerised search of charts Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**

Feder 1995a *(Continued)*

Random sequence generation (selection bias)	Unclear risk	Study authors say that practices were stratified by a number of factors and then randomised, but no information is provided on how the random sequence was generated or how the process was conducted.
Allocation concealment (selection bias)	Unclear risk	No information is provided on how practices were allocated to their respective treatment groups.
Blinding of outcome assessment (detection bias)	Low risk	No information is provided regarding blinding of the outcome assessors. Outcome data were obtained from the clinical records of a random sample of patients from each of the practices.
Incomplete outcome data (attrition bias) All outcomes	Low risk	None of the 24 practices were lost to follow-up after randomisation.
Selective reporting (reporting bias)	Unclear risk	Outcomes reported as per methods, but no protocol means it is unclear if all preplanned outcomes have been reported.
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	As per table 1 p. 1474, the groups were fairly evenly balanced across a number of factors.
Protection against contamination	Low risk	While two practices were registered as single-handed practices, this was due to the other partner not participating in the trial, so those partners would not have an effect on the trial and outcomes. It appears the trial kept track of all practice clinicians, so unlikely they would have worked across multiple sites.
Incorrect analysis	Unclear risk	Not enough information
Risk of bias overall	Unclear risk	Unclear risk across important domains

Feder 1995b

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: 39 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + education
Outcomes	Clinical behaviour category: screening, testing/exams Format of Audit & Feedback: written

Feder 1995b *(Continued)*

	Source: manual chart review, computerised search of charts Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Study authors say that practices were stratified by a number of factors and then randomised, but no information is provided on how the random sequence was generated or how the process was conducted.
Allocation concealment (selection bias)	Unclear risk	No information is provided on how practices were allocated to their respective treatment groups.
Blinding of outcome assessment (detection bias)	Low risk	No information is provided regarding blinding of the outcome assessors. Outcome data were obtained from the clinical records of a random sample of patients from each of the practices.
Incomplete outcome data (attrition bias) All outcomes	Low risk	None of the 24 practices were lost to follow-up after randomisation.
Selective reporting (reporting bias)	Unclear risk	Outcomes reported as per methods, but no protocol means it is unclear if all preplanned outcomes have been reported.
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	As per table 1 p. 1474, the groups were fairly evenly balanced across a number of factors.
Protection against contamination	Low risk	While two practices were registered as single-handed practices, this was due to the other partner not participating in the trial, so those partners would not have an effect on the trial and outcomes. It appears the trial kept track of all practice clinicians, so unlikely they would have worked across multiple sites.
Incorrect analysis	Unclear risk	Not enough information
Risk of bias overall	Unclear risk	Unclear risk across important domains

Ferguson 2003a

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital inpatient Speciality: cardiology

Ferguson 2003a *(Continued)*

	N health professionals: NR N patients: 179,625
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: quality champion + call to action letter + education materials + A&F
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: electronic dashboard Source: not reported/not applicable Frequency: variable Instructions for improvement: not reported
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Participants unaware
Incomplete outcome data (attrition bias) All outcomes	Low risk	Roughly equal dropouts in 3 groups
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters; hence, there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Low risk	Demographics similar in 3 groups
Protection against contamination	Low risk	Separate hospitals
Incorrect analysis	Low risk	Hierarchical logistic regression with random effects to account for clustering
Risk of bias overall	Low risk	Low risk across all important domains

Ferguson 2003b

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital inpatient Speciality: cardiology N health professionals: NR N patients: 179,083
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: quality champion + call to action letter + education materials + A&F
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: electronic dashboard Source: not reported/not applicable Frequency: variable Instructions for improvement: not reported
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Participants unaware
Incomplete outcome data (attrition bias) All outcomes	Low risk	Roughly equal dropouts in 3 groups
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters; hence, there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Low risk	Demographics similar in 3 groups

Ferguson 2003b *(Continued)*

Protection against contamination	Low risk	Separate hospitals
Incorrect analysis	Low risk	Hierarchical logistic regression with random effects to account for clustering
Risk of bias overall	Low risk	Low risk across all important domains

Fiks 2013

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: paediatrics N health professionals: NR N patients: 22,486
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention 1: EHR-based alerts + clinician education + A&F + patient reminders + patient education Intervention 2: EHR-based alerts + clinician education + A&F; Intervention 3: patient reminders + patient education
Outcomes	Clinical behaviour category: immunisation Format of Audit & Feedback: written Source: computerised search of charts Frequency: quarterly Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	High risk	A statistician (Dr Localio) generated the allocation sequence and implemented the randomisation. Actual randomisation process not described
Allocation concealment (selection bias)	Low risk	Allocation done centrally as it was done by a statistician.
Blinding of outcome assessment (detection bias)	Low risk	While participants were not blind to their allocation, the recording of vaccination is objective and unlikely to be biased from knowledge of intervention.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No practices were lost to follow-up. Balanced losses to follow-up in patient participants.

Fiks 2013 *(Continued)*

Selective reporting (reporting bias)	Unclear risk	All specified outcomes are reported; however, vaccination rates were not specifically reported for each arm and only reported as hazard ratios (with two arms compared).
Recruitment bias	Low risk	All eligible patients included in study after clusters randomised to allocated intervention
Baseline imbalance in outcome measurements or characteristics	Low risk	No baseline imbalances detected
Protection against contamination	Low risk	Practices randomised, and unlikely contamination between groups
Incorrect analysis	Low risk	The study used "Cox proportional hazard regression models accounting for the clustered design and including covariates". No issues detected with cluster analysis
Risk of bias overall	High risk	Due to random sequence generation

Fiks 2017

Study characteristics		
Methods	Design: parallel cluster-RCT	
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 105 N patients: 790	
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: education + expert consultations + A&F	
Outcomes	Clinical behaviour category: screening Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: bi-monthly Instructions for improvement: general	
Notes		
Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**

Fiks 2017 *(Continued)*

Random sequence generation (selection bias)	Low risk	Random sequence was generated by a statistician.
Allocation concealment (selection bias)	Low risk	Statistician performed randomisation, so allocation was likely central. Based on study flow diagram, it is likely clusters were allocated at the same time.
Blinding of outcome assessment (detection bias)	High risk	Assessors were not blinded and performed a manual chart review - bias could have affected outcomes reported. Consensus of reporting was necessary, implying some subjective judgements.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No loss of practitioners reported. Numbers reported in results consistent with those randomised
Selective reporting (reporting bias)	Unclear risk	No protocol available to see if all preplanned outcomes were reported
Recruitment bias	Low risk	Recruitment occurred after randomisation.
Baseline imbalance in outcome measurements or characteristics	Low risk	Groups mostly balanced, and imbalances controlled for in analysis
Protection against contamination	Low risk	Allocation by cluster (e.g. practice) and it is unlikely that the control group received the intervention.
Incorrect analysis	Low risk	Results were also described by practice. To control for differences in patient and clinician covariates between study arms, we used separate logistic regression models for each outcome, with independent variables including study arm, time period (baseline versus intervention), and an interaction between study arm and time period. Models were adjusted for covariates as provider type. Statistical Analysis: The study population was described using means and standard deviations (SD) for continuous variables and frequencies with proportions for categorical variables. To evaluate the effect of the intervention on EBPs for ADHD, we calculated the proportion of patients in both study arms with parent and teacher rating scales sent and received during the baseline and intervention periods. We then calculated the percentage point change between periods within each arm, and the difference-in-difference between arms.
Risk of bias overall	High risk	Due to blinding

Filardo 2009

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital inpatient Speciality: internal medicine N health professionals: NR

Filardo 2009 *(Continued)*

	N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: education + A&F
Outcomes	Clinical behaviour category: prescribing, testing/exams, immunisation Format of Audit & Feedback: verbal (didactic) Source: computerised search of charts Frequency: not reported/not applicable Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial, allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	2 of 47 hospitals dropped out.
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Low risk	Separate hospitals
Incorrect analysis	Low risk	Clustering accounted for in analysis
Risk of bias overall	Unclear risk	Unclear risk across important domains

Finkelstein 2001

Study characteristics

Methods	Design: parallel cluster-RCT

Finkelstein 2001 *(Continued)*

Participants	Country: USA Setting: multi-speciality clinics within two managed care organisations Speciality: GP/family N health professionals: 157 N patients: 8815
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + education + academic detailing
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: administrative/billing data Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	No information about how the randomisation was done
Allocation concealment (selection bias)	Unclear risk	No details or study flow diagram to determine if clusters were likely allocated at the same time
Blinding of outcome assessment (detection bias)	Low risk	Knowledge of allocation would not influence the outcome data which were collected from claims data.
Incomplete outcome data (attrition bias) All outcomes	Low risk	All 12 practices that were randomised were also included in analysis.
Selective reporting (reporting bias)	Unclear risk	Outcomes reported as prespecified in methods but no protocol available
Recruitment bias	Low risk	Recruitment occurred before randomisation.
Baseline imbalance in outcome measurements or characteristics	Low risk	Used pair-matched randomisation. Baseline differences accounted for in analysis
Protection against contamination	Low risk	Hard to guess. Parents got educational materials, which might have been shared around. Physicians who got intervention might communicate with those who didn't. But risk of both seems low, so my guess would be that any significant contamination was unlikely.
Incorrect analysis	Low risk	Clustering accounted for
Risk of bias overall	Unclear risk	Unclear risk across important domains

Foster 2007

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: educational workshop + practice development plan + educational materials; Intervention: educational workshop + practice development plan + A&F + educational materials
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: both written and verbal Source: manual chart review, computerised search of charts Frequency: semi-annually Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random number allocation
Allocation concealment (selection bias)	Low risk	Central randomisation
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	High risk	Lost half of intervention group
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence, there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 1

Foster 2007 *(Continued)*

Protection against contamination	Low risk	Practices separate
Incorrect analysis	Unclear risk	Says clustering taken into account with analysis of covariance but few details were given
Risk of bias overall	High risk	Due to incomplete outcome data

Foy 2004

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: obstetrics and gynaecology clinic or ward Speciality: gynaecology N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: guidelines; Intervention: A&F + guidelines + education meetings + education materials
Outcomes	Clinical behaviour category: screening, prescribing, counselling, referrals Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: once Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Randomised pairs by independent statistician
Allocation concealment (selection bias)	Low risk	Cluster trial, allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	1 unit with no cases, otherwise complete data

Foy 2004 *(Continued)*

Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 1
Protection against contamination	Low risk	Separate hospitals
Incorrect analysis	Low risk	Generalised estimating equations
Risk of bias overall	Low risk	Low risk across all important domains

Fretheim 2006

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Norway Setting: primary care Speciality: GP/family N health professionals: 501 N patients: 3052
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: educational outreach visit + computerised reminders + guidelines + A&F
Outcomes	Clinical behaviour category: screening, prescribing Format of Audit & Feedback: verbal (didactic) Source: computerised search of charts Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random sequence was generated from computer software.
Allocation concealment (selection bias)	Low risk	Allocation was concealed and provided by one central person. Authors "gave her identification numbers representing each recruited practice, and she in-

Fretheim 2006 *(Continued)*

		formed us whether the practice was allocated to the intervention or control group."
Blinding of outcome assessment (detection bias)	Low risk	Investigators assessing outcomes and conducting analyses were blinded to the allocation of practices (p. 4).
Incomplete outcome data (attrition bias) All outcomes	Low risk	3/73 and 4/73 practices in the intervention and control arm, respectively, were lost to follow-up for similar reasons. Response rate from patient questionnaires is similar between groups.
Selective reporting (reporting bias)	Low risk	All prespecified outcomes reported
Recruitment bias	Low risk	No clusters recruited after allocation. Patients were included on the basis of their eligibility for each outcome (data abstracted from medical records by blind investigators) and not actively recruited.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	The trialists reported that "the practices and patients in the intervention and control groups were similar with regards to baseline characteristics", however, only one baseline characteristic was measured for health professionals and two baseline characteristics measured for patients. These do not appear to be enough to determine if there were baseline imbalances or not.
Protection against contamination	Low risk	"We decided that randomizing practices rather than physicians or patients would be the most feasible design for practical reasons, and also the most appropriate to avoid contamination from the intervention to the control group." (p. 2). Assuming that physicians did not work in more than one practice, the risk of bias is low.
Incorrect analysis	Low risk	Data were adjusted for baseline differences and clustering effects.
Risk of bias overall	Low risk	Unclear based on baseline imbalances and protection against contamination

Frijling 2002

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: primary care Speciality: GP/family physician N health professionals: 185 N patients: NR
Interventions	Number of arms: 2 Description of Groups: Control: no intervention; Intervention: A&F + education materials + facilitator support
Outcomes	Clinical behaviour category: testing/exams, prescribing, counselling, referrals Format of Audit & Feedback: both written and verbal

Frijling 2002 *(Continued)*

	Source: manual chart review Frequency: 1.4 months Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random number generator, blocks of 4
Allocation concealment (selection bias)	Low risk	Quote "person responsible for the randomization process was blind to identities".
Blinding of outcome assessment (detection bias)	High risk	Physician report data
Incomplete outcome data (attrition bias) All outcomes	Low risk	3 practices lost to follow-up
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence, there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Low risk	Table 1
Protection against contamination	Low risk	Separate practices
Incorrect analysis	Low risk	Multilevel logistic regression
Risk of bias overall	High risk	Due to blinding

Frijling 2003

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: primary care Speciality: GP/family physician N health professionals: 185 N patients: NR

Frijling 2003 *(Continued)*

Interventions	Number of arms: 2 Description of Groups: Control: no intervention; Intervention: A&F + facilitator support
Outcomes	Clinical behaviour category: screening, prescribing, counselling, referrals Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: 7 months Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote "randomly allocated...random number generator"
Allocation concealment (selection bias)	Low risk	Quote "person responsible for the randomization process was blind to identities".
Blinding of outcome assessment (detection bias)	High risk	Physician report data
Incomplete outcome data (attrition bias) All outcomes	Low risk	3 of 124 practices dropped out
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence, there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 1
Protection against contamination	Low risk	Separate practices
Incorrect analysis	Low risk	Multilevel logistic regression
Risk of bias overall	High risk	Due to blinding

Fuller 2012

Study characteristics

Methods	Design: step wedge
Participants	Country: UK

Fuller 2012 *(Continued)*

	Setting: hospital inpatient Speciality: GP/family/internal medicine N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F
Outcomes	Clinical behaviour category: other Format of Audit & Feedback: verbal (interactive) Source: other Frequency: bi-monthly Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random sequence was generated from computer software.
Allocation concealment (selection bias)	Low risk	Clusters were allocated at the same time.
Blinding of outcome assessment (detection bias)	Low risk	Hand hygiene was measured by covert direct observation by an observer blinded to ward allocation or randomisation of the intervention.
Incomplete outcome data (attrition bias) All outcomes	Low risk	60 study wards in the 16 randomised hospitals, of which 33 (22 ACE and 11 ITU) in 13 hospitals went on to implement the intervention. Eight wards began implementation very late, and for these the end of the trial was extended to December 31st 2009 to ensure that they had a year of data collection post-implementation. Intention-to-treat analysis was conducted for the 60 wards randomised into the intervention, and per-protocol analysis was performed for the 33 implementing wards.
Selective reporting (reporting bias)	Low risk	"There was a predefined protocol (www.idrn.org/nosec.php) which was followed except for three violations. Firstly, it was not possible to perform the MRSA prevalence screening. Secondly, a questionnaire measuring ward culture was filled out by so few nurses that this was dropped from the protocol. Thirdly, delayed Research and Development registration shortened the baseline pre-randomisation phase from twelve months to nine in the first hospitals randomised to the intervention. Both the protocol and supporting CONSORT checklist are available as supporting information: see Protocol S1 and Checklist S1)."
Recruitment bias	Low risk	Hospitals were recruited and then randomised to enter the intervention, so low risk of recruitment bias

Fuller 2012 *(Continued)*

Baseline imbalance in outcome measurements or characteristics	Low risk	In the stepped-wedge design, participating wards act as their own control, so risk of bias due to baseline imbalance is low.
Protection against contamination	Low risk	Risk of bias due to contamination was minimised due to clustering and hospitals were only aware of their own time of randomisation into the intervention.
Incorrect analysis	Low risk	Generalised linear mixed (multilevel) models were used to analyse the data. The model takes into account the hierarchical nature of the design (measurements nested within hospitals, and, if appropriate, hospitals within trusts).
Risk of bias overall	Low risk	Average of all 'other' biases

Ganz 2005

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: provider organisations (PO) within a managed healthcare plan Speciality: GP/family/other N health professionals: NR N patients: 1850
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: workshop + educational materials + QI program + A&F
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: not reported/not applicable Source: manual chart review Frequency: monthly Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Method of random sequence generation is not reported.
Allocation concealment (selection bias)	Low risk	Clusters were allocated at the same time.
Blinding of outcome assessment (detection bias)	Low risk	Unclear if abstractors were blind, but outcomes were abstracted from audit of medical charts (and discrete options), and abstractors were trained and "Over-

Ganz 2005 *(Continued)*

		reads were done for each reviewer until there was a 100% match between reviewer and research staff chart review forms". Unlikely that bias was introduced in outcome assessment
Incomplete outcome data (attrition bias) All outcomes	Low risk	2/19 POs in the intervention group and 1/17 POs in the control group dissolved (balanced and unlikely to be due to the intervention and outcome). 1/19 PO in the intervention group refused chart review. There is no further explanation, but it does not appear to be substantial enough to introduce bias (5% of POs) in the outcome data.
Selective reporting (reporting bias)	Low risk	All expected outcomes included in the methods were reported: patient and PO characteristics; rate of CRC with any test; rate of ever had test within guidelines; comparison of rates between intervention and control; logistic regression model
Recruitment bias	Low risk	There is no evidence of recruitment after randomisation and allocation.
Baseline imbalance in outcome measurements or characteristics	High risk	There were some significant baseline imbalances between groups in sampled patients and potentially some differences in mean number of PCPs in POs.
Protection against contamination	Unclear risk	Unclear if there was potential for staff in POs to be involved with more than one PO
Incorrect analysis	High risk	The analyses for most outcomes, including comparisons of the rate of testing in intervention and control POs do not account for clustering of the data except for in the logistic regression - in which data that account for clustering are not shown.
Risk of bias overall	High risk	Baseline imbalance and incorrect analysis judged as high risk; protection against contamination judged as unclear risk

Geary 2019

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Sweden Setting: primary care Speciality: GP/family N health professionals: NR N patients: 12,766
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F report + guidelines
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: administrative/billing data

Geary 2019 *(Continued)*

	Frequency: monthly Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Lottery method was used for randomisation.
Allocation concealment (selection bias)	Unclear risk	No details or study flow diagram for clusters to determine if they were likely all allocated at the same time
Blinding of outcome assessment (detection bias)	Unclear risk	Outcomes assessment not clearly reported but likely unblinded. Unclear on the process of recording medication dispensation and whether this could be biased
Incomplete outcome data (attrition bias) All outcomes	Low risk	6/104 intervention groups and 1/103 control groups excluded with reasons provided. These are not related to the intervention, and attrition is still reasonably balanced.
Selective reporting (reporting bias)	Low risk	Outcomes consistent with registered protocol NCT01942031
Recruitment bias	Low risk	No evidence of recruitment after randomisation and allocation
Baseline imbalance in outcome measurements or characteristics	Low risk	Some baseline imbalances reported, but data were adjusted for baseline characteristics, so likely low risk
Protection against contamination	Low risk	Due to cluster design, contamination is unlikely.
Incorrect analysis	Unclear risk	No information about adjusting for clustering
Risk of bias overall	Unclear risk	Unable to assess bias in important domains

Gerber 2014

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 162 N patients: 185,212
Interventions	Number of arms: 2

Gerber 2014 *(Continued)*

	Description of groups: Control: no intervention; Intervention: clinician education + A&F
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: manual chart review, computerised search of charts Frequency: tri-annually Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Not specifically reported, but a statistician performed randomisation, so it is likely an adequate method was used
Allocation concealment (selection bias)	Low risk	Statistician performed randomisation centrally.
Blinding of outcome assessment (detection bias)	Low risk	Data collected from routine electronic medical records; unlikely for bias to be introduced
Incomplete outcome data (attrition bias) All outcomes	Low risk	No practices lost to follow-up; 81/84 clinicians in each arm analysed, which is balanced and not significant
Selective reporting (reporting bias)	Low risk	Results reported as in methods and trial registration
Recruitment bias	Low risk	Unlikely participants/clusters recruited to the trial after clusters randomised
Baseline imbalance in outcome measurements or characteristics	Low risk	Similar at baseline or adjusted for in analysis
Protection against contamination	Low risk	Clinics randomised to avoid contamination
Incorrect analysis	Low risk	Clustering taken into account
Risk of bias overall	Low risk	Low risk across all domains

Gilkey 2014

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care

Gilkey 2014 *(Continued)*

	Speciality: NA N health professionals: 91 N patients: NR
Interventions	Number of arms: 3 Description of groups: Control: no intervention; Intervention 1: A&F + in-person training + goal setting; Intervention 2: A&F + virtual training + goal setting
Outcomes	Clinical behaviour category: immunisation Format of Audit & Feedback: verbal (interactive) Source: computerised search of charts Frequency: monthly Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomisation process not well described
Allocation concealment (selection bias)	High risk	Allocation performed by author
Blinding of outcome assessment (detection bias)	Low risk	Data were extracted from the North Carolina Immunization Registry
Incomplete outcome data (attrition bias) All outcomes	Low risk	No losses reported
Selective reporting (reporting bias)	Low risk	Prospectively registered protocol; all outcomes reported
Recruitment bias	Low risk	No evidence of recruitment after randomisation and allocation
Baseline imbalance in outcome measurements or characteristics	Low risk	See table 1, characteristics and outcomes balanced between groups
Protection against contamination	Low risk	Cluster design means contamination is unlikely.
Incorrect analysis	Low risk	Clustering taken into account
Risk of bias overall	High risk	Due to allocation concealment

Gilkey 2019

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: paediatrics N health professionals: 77 N patients: 22,983
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: training session + video vignettes + A&F + educational materials
Outcomes	Clinical behaviour category: immunisation Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: quarterly Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not enough information to determine sequence generation
Allocation concealment (selection bias)	Low risk	As per flow chart, allocation was likely done at one time, centrally
Blinding of outcome assessment (detection bias)	Low risk	Outcomes assessment (electronic medical records) unlikely to be biased even with knowledge of allocation
Incomplete outcome data (attrition bias) All outcomes	Low risk	Primary analysis used ITT, and sensitivity analysis was also performed with "questionable" data collection excluded.
Selective reporting (reporting bias)	Unclear risk	No protocol or trial registration to check against. Appears to be complete reporting from what is provided
Recruitment bias	Low risk	No evidence of recruitment after randomisation and allocation
Baseline imbalance in outcome measurements or characteristics	High risk	Vaccination coverage higher in the intervention group at baseline - this could affect results of the QI program.

Gilkey 2019 *(Continued)*

Protection against contamination	Low risk	Allocation to intervention or control was by cluster and involved attendance at training sessions as well as assessment - so unlikely that physicians in the control (waiting-list) group received the intervention
Incorrect analysis	Low risk	Analysis accounted for clusters
Risk of bias overall	High risk	Baseline imbalance judged to be high risk

Gjelstad 2013

Study characteristics		
Methods	Design: parallel cluster-RCT	
Participants	Country: Norway Setting: primary care Speciality: GP/family N health professionals: 489 N patients: NR	
Interventions	Number of arms: 2 Description of groups: Intervention 1 (antibiotic prescribing): academic detailing + education + A&F; Intervention 2 (appropriate prescribing): academic detailing + education + A&F	
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: once/monthly Instructions for improvement: general	
Notes		
Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Method of sequence generation is unclear.
Allocation concealment (selection bias)	Low risk	Though not explicitly described, it appears all practices were allocated at one time.
Blinding of outcome assessment (detection bias)	Low risk	Doctors in both arms received an intervention, so it is unlikely they knew which behaviour the trial was targetting. Outcome measures are collected from patient records and not likely subject to bias by abstractors (whether they knew allocation or not).
Incomplete outcome data (attrition bias)	Low risk	Incomplete data from dropouts appear balanced between groups and reasons do not appear to suggest risk of bias from intervention and to outcomes.

Gjelstad 2013 *(Continued)*

All outcomes		
Selective reporting (reporting bias)	Low risk	Outcomes align with reported outcomes in the study protocol.
Recruitment bias	Low risk	The randomisation was stratified by five geographical regions to reconcile the number of available peer academic detailers needed in each arm of the study within each region. Within each stratum, the continuing medical education group number and the type of intervention were drawn separately by staff not engaged in the project. After the randomisation, each member of the continuing medical education groups gave written consent to participate in the study.
Baseline imbalance in outcome measurements or characteristics	Low risk	No important differences in outcomes or characteristics across study groups were present.
Protection against contamination	Unclear risk	Professionals were allocated within a geographical region and it is possible that communication between intervention and control professionals could have occurred (unclear if it was possible or not).
Incorrect analysis	Low risk	Data were adjusted for covariates.
Risk of bias overall	Low risk	Low risk across most domains

Goderis 2010

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Belgium Setting: primary care Speciality: GP/family N health professionals: 142 N patients: 2495
Interventions	Number of arms: 2 Description of groups: Intervention 1: treatment protocol + A&F + education + case-coaching; Intervention 2: treatment protocol + A&F + education + case-coaching + intensified follow-up with GPs + shared-care protocol
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: manual chart review Frequency: twice/quarterly Instructions for improvement: specific
Notes	

Goderis 2010 *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation
Allocation concealment (selection bias)	Low risk	Clusters were likely allocated at one time.
Blinding of outcome assessment (detection bias)	High risk	Main outcomes were at patient level using blood analyses and blood pressures, with the latter outcome subject to risk of bias due to knowledge of the intervention.
Incomplete outcome data (attrition bias) All outcomes	High risk	18 physicians in UQIP and 4 physicians in AQIP dropped out after allocation and before intervention (significant difference); a further 2 UQIP physicians dropped out after the intervention (non-significant difference). Analysis was described as ITT, but trialists only analysed practices that remained.
Selective reporting (reporting bias)	Low risk	Results are reported consistent with the published protocol.
Recruitment bias	Low risk	Central recruitment done before trial commenced
Baseline imbalance in outcome measurements or characteristics	Low risk	Baselines appeared balanced, and they used pairwise randomisation.
Protection against contamination	Low risk	Both arms received same basic programme. The intervention arm got extra stuff, like educational intervention about SDM and patient-centred comms. Even if people in the arm did communicate with control, it seems unlikely this material could easily be shared to contaminate trial.
Incorrect analysis	Low risk	Taken into account in analysis
Risk of bias overall	High risk	Based on blinding and incomplete outcome data

Goff 2003

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 605 N patients: 3224
Interventions	Number of arms: 2

Goff 2003 *(Continued)*

	Description of groups: Control: no intervention; Intervention: A&F + educational materials (recommendations) + patient chart reminders
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: administrative/billing data Frequency: annually Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Administrative databases used for data collection
Incomplete outcome data (attrition bias) All outcomes	Low risk	Similar numbers of dropouts in both groups due to closings
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No recruitment after allocation
Baseline imbalance in outcome measurements or characteristics	Low risk	Patient groups comparable at baseline
Protection against contamination	Low risk	Practices separate
Incorrect analysis	Low risk	Generalised estimating equations
Risk of bias overall	Low risk	Low risk of bias across all important domains

Goldberg 1998

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care

Goldberg 1998 *(Continued)*

	Speciality: GP/family N health professionals: 95 N patients: 4995
Interventions	Number of arms: 3 Description of groups: Control: no intervention; Intervention 1: A&F + academic detailing; Intervention 2: audit & feedback + academic detailing + continuous quality improvement (education and resources)
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: computerised search of charts, patient-reported data Frequency: not reported/not applicable Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	15 small group practices of providers and patients were randomised in blocks. Unclear how randomisation was determined
Allocation concealment (selection bias)	Unclear risk	No details or study flow diagram to determine allocation concealment of clusters
Blinding of outcome assessment (detection bias)	High risk	Not addressed and used a mixture of chart audit and patient survey. Likely that bias could be introduced in outcome assessment, since it is likely allocations were known
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Flow of participants not clearly reported and difficult to determine missing outcomes per arm
Selective reporting (reporting bias)	Unclear risk	No protocol or trial registration
Recruitment bias	Unclear risk	Unclear the number of providers per group and when the randomisation occurred and if anyone came later
Baseline imbalance in outcome measurements or characteristics	Low risk	No differences in practitioners between groups at baseline. Only difference was in the percentage of males in patients
Protection against contamination	Unclear risk	Practices randomised, but multiple practices at one clinical site may have been randomised to different groups and unclear if there is any interaction between these
Incorrect analysis	Low risk	The primary study analysis was carried out across clinics at the level of the group practice, the unit of randomisation to study arms
Risk of bias overall	High risk	Due to blinding

Gonzales 2013

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 155 N patients: 12,776
Interventions	Number of arms: 3 Description of groups: Control: no intervention; Intervention 1: A&F + education + printed decision support; Intervention 2: A&F + education + computerised decision support
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: verbal (interactive) Source: administrative/billing data Frequency: monthly Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Method of random sequence generation is not reported.
Allocation concealment (selection bias)	Low risk	No details but, based on study flow diagram, it is likely clusters were allocated at the same time.
Blinding of outcome assessment (detection bias)	Low risk	All office visits and prescription data were extracted from the electronic health record system by individuals who were unaware of the group allocation of the study site.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Although it is not clearly described, there does not appear to be any loss of clusters in any of the groups at the end of the intervention period.
Selective reporting (reporting bias)	Low risk	Study protocol available. The proposed outcomes match the outcomes reported in the published full text.
Recruitment bias	Low risk	Practices were selected prior to randomisation.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Baseline characteristics of practices and physicians are not reported.

Gonzales 2013 *(Continued)*

Protection against contamination	Low risk	The cluster design reduces the risk of contamination between groups. The study area was quite large, so it is unlikely that there was contamination between clusters.
Incorrect analysis	Low risk	Analyses were adjusted for clustering.
Risk of bias overall	Unclear risk	Unclear due to baseline imbalances and random sequence generation

Grady 1997

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 95 N patients: 11,426
Interventions	Number of arms: 3 Description of groups: Control 1: clinician education + educational materials; Control 2: educational materials + posters + chart stickers; Intervention: educational materials + posters + chart stickers + A&F + financial incentive
Outcomes	Clinical behaviour category: referrals Format of Audit & Feedback: written Source: manual chart review Frequency: variable Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation
Allocation concealment (selection bias)	Unclear risk	Unable to assess
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias)	Low risk	87% follow-up of physicians

Grady 1997 *(Continued)*

All outcomes		
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence, there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Low risk	Separate practices
Incorrect analysis	Unclear risk	Repeated measures ANOVA
Risk of bias overall	Unclear risk	Unable to assess bias

Guadagnoli 2000

Study characteristics		
Methods	Design: parallel cluster-RCT	
Participants	Country: USA Setting: hospital inpatient Speciality: oncology N health professionals: NR N patients: 2314	
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + local opinion leaders: Intervention 2: A&F only	
Outcomes	Clinical behaviour category: counselling Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: once Instructions for improvement: general	
Notes		
Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess

Guadagnoli 2000 *(Continued)*

Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	All hospitals with at least 7 cases participated in follow-up.
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No evidence of recruitment after randomisation and allocation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Low risk	Separate hospitals
Incorrect analysis	High risk	Wilcoxon Rank Sum test
Risk of bias overall	High risk	High due to incorrect analysis

Gude 2016a

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: cardiology clinic Speciality: cardiology N health professionals: NR N patients: 14,847
Interventions	Number of arms: 2 Description of groups: Intervention 1 (psychological rehab): A&F + goal setting/action planning + educational outreach; Intervention 2 (physical rehab): A&F + goal setting/action planning + educational outreach
Outcomes	Clinical behaviour category: counselling Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: quarterly Instructions for improvement: co-developed plans

Gude 2016a *(Continued)*

Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Concealment occurred in the initial enrolment and allocation
Blinding of outcome assessment (detection bias)	Low risk	Outcome assessment unlikely to be biased as both arms receiving intervention and data collected from electronic records
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Incomplete outcome data are not clearly reported.
Selective reporting (reporting bias)	Low risk	Reported in line with trial registration and published protocol
Recruitment bias	Low risk	Allocation concealed to those enroling centres; no centres enroled later
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline characteristics similar between groups, except for type of centre which was controlled for in analysis
Protection against contamination	Low risk	Due to cluster design, contamination is unlikely.
Incorrect analysis	High risk	No reporting of taking clustering into account for analysis
Risk of bias overall	High risk	Incorrect analysis judged to be high risk

Gude 2016b

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: cardiology clinic Speciality: cardiology N health professionals: NR N patients: 14,847
Interventions	Number of arms: 2 Description of groups: Intervention 1 (psychological rehab): A&F + goal setting/action planning + educational outreach; Intervention 2 (physical rehab): A&F + goal setting/action planning + educational outreach

Gude 2016b *(Continued)*

Outcomes	Clinical behaviour category: testing/exams, counselling Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: quarterly Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Concealment occurred in the initial enrolment and allocation
Blinding of outcome assessment (detection bias)	Low risk	Outcome assessment unlikely to be biased as both arms receiving intervention and data collected from electronic records
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Incomplete outcome data are not clearly reported.
Selective reporting (reporting bias)	Low risk	Reported in line with trial registration and published protocol
Recruitment bias	Low risk	Allocation concealed to those enroling centres; no centres enroled later
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline characteristics similar between groups, except for type of centre which was controlled for in analysis
Protection against contamination	Low risk	Due to cluster design, contamination is unlikely.
Incorrect analysis	High risk	No reporting of taking clustering into account for analysis
Risk of bias overall	High risk	Incorrect analysis judged to be high risk

Guldberg 2011

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Denmark Setting: primary care Speciality: GP/family

Guldberg 2011 *(Continued)*

	N health professionals: 158 N patients: 2716
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F
Outcomes	Clinical behaviour category: prescribing, testing/exams Format of Audit & Feedback: electronic dashboard Source: computerised search of charts Frequency: variable Instructions for improvement: not reported
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	High risk	No allocation concealment
Blinding of outcome assessment (detection bias)	Low risk	Unlikely to influence the biologic measures used
Incomplete outcome data (attrition bias) All outcomes	Low risk	Low risk of attrition bias as intention-to-treat analysis was performed. Further analysis after removing the 25 practices showed no significant change in results.
Selective reporting (reporting bias)	Unclear risk	The trial was not registered and no study protocol was found.
Recruitment bias	Low risk	No recruitment after allocation
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline appears balanced.
Protection against contamination	Low risk	Cluster design means contamination is unlikely.
Incorrect analysis	Low risk	Clustering was accounted for in the analysis which was done at the practice level.
Risk of bias overall	High risk	Due to allocation concealment

Gulliford 2019

Study characteristics

Gulliford 2019 *(Continued)*

Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + educational materials + decision support + webinar
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: monthly Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Allocation performed centrally at one time
Blinding of outcome assessment (detection bias)	Unclear risk	Potentially outcome assessment may be differentially recorded at different sites.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Similar number of withdrawals from both groups for similar reasons, hence, there is a low risk of attrition bias.
Selective reporting (reporting bias)	Low risk	The trial was registered and protocol published; all deviations from the protocol were clearly outlined in the paper.
Recruitment bias	Low risk	No information on participants being recruited to the trial after randomisation had taken place
Baseline imbalance in outcome measurements or characteristics	Low risk	No obvious baseline differences between groups
Protection against contamination	Low risk	Due to cluster design, contamination is unlikely. Authors note both trial arms may have been contaminated by a system-wide initiative, but this would not be differential between arms. "The NHS introduced an incentive known as the ‘Quality Premium’ to encourage reduced antibiotic utilisation. Antimicrobial

Gulliford 2019 *(Continued)*

		stewardship interventions became available from a number of national and local sources. Perhaps as a result of some of these interventions, the English Surveillance Programme for Antimicrobial Utilisation and Resistance (ESPAUR) found that there was a 13% decrease in the number of antibiotic prescriptions dispensed from GP settings between 2012 and 2016. These system-wide interventions will have 'contaminated' both trial arms of the study. It is possible that a greater intervention effect might have been observed in their absence."
Incorrect analysis	Low risk	Cluster-level analysis was originally planned, but deviations made with explanations. Cluster-level analysis was still performed as secondary analysis
Risk of bias overall	Low risk	Low risk across all important domains

Gullion 1988

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family physicians N health professionals: 84 N patients: 2583
Interventions	Number of arms: 4 Description of groups: Control: no intervention; Intervention: A&F + educational material + conference discussion
Outcomes	Clinical behaviour category: testing/exams, treatment decision/action Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote: "Stratified random assignment"
Allocation concealment (selection bias)	Unclear risk	Unable to assess
Blinding of outcome assessment (detection bias)	Low risk	Quote: "medical abstractors, blinded to conditions"

Gullion 1988 *(Continued)*

Incomplete outcome data (attrition bias) All outcomes	Low risk	Minimal number (5/111) lost to follow-up and at least one from all 4 groups
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence, there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Low risk	Patients comparable at baseline, see Table 3
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Unclear risk	ANCOVA was used but few details given
Risk of bias overall	Low risk	Low risk across most domains

Guthrie 2016

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: NR N patients: 170,659
Interventions	Number of arms: 3 Description of groups: Intervention 1: A&F; Intervention 2: A&F + behavioural change materials
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: computerised search of charts Frequency: quarterly Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**

Guthrie 2016 *(Continued)*

Random sequence generation (selection bias)	Low risk	A random number generator was used to generate the random sequence.
Allocation concealment (selection bias)	Low risk	Allocation of practices to the three arms was conducted by an independent unit and concealed from project teams, hence, there is a low risk of selection bias.
Blinding of outcome assessment (detection bias)	Low risk	As the outcomes were measured using routinely recorded data which could not be influenced by the research team, hence, there is a low risk of detection bias.
Incomplete outcome data (attrition bias) All outcomes	Low risk	The Consort flow diagram showed that number of withdrawals (2 in one arm and 1 each in the other 2 arms) and reasons for withdrawal (practice split/merge) were similar across all three arms.
Selective reporting (reporting bias)	Low risk	All prespecified outcomes are reported.
Recruitment bias	Low risk	All practices were recruited at the same time and no patients were enroled in the study after randomisation; there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Low risk	Though not explicitly reported, baseline appears balanced. Analyses adjusted for health board and high risk prescribing at baseline
Protection against contamination	Low risk	Practices were the unit, so contamination unlikely
Incorrect analysis	Unclear risk	Unclear about adjusting for clustering
Risk of bias overall	Low risk	Low risk across most important domains

Hallsworth 2016

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: 3227 N patients: NR
Interventions	Number of arms: 4 Description of groups: Control 1: no intervention; Control 2: patient-education materials; Intervention 1: A&F + patient-education materials; Intervention 2: A&F + patient-education materials (leaflet, poster, letter)
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written

Hallsworth 2016 *(Continued)*

	Source: administrative/billing data Frequency: once Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Eligible practices were randomly assigned (1:1) into two groups by a computer-generated allocation sequence.
Allocation concealment (selection bias)	Low risk	As all GP practices were randomised at the same time, there is low risk of bias due to allocation concealment.
Blinding of outcome assessment (detection bias)	Low risk	The outcomes are from audit data and unlikely to be subject to bias when extracted by study personnel. Physicians were also unaware of the trial despite being aware of their intervention, thus it is unlikely there would be bias introduced into recording antibiotic prescriptions.
Incomplete outcome data (attrition bias) All outcomes	Low risk	See figure 1, less than 10% lost in each arm
Selective reporting (reporting bias)	Low risk	Trial was registered and a change from the protocol to the secondary outcome was justified in the results paper.
Recruitment bias	Low risk	No evidence of recruitment bias
Baseline imbalance in outcome measurements or characteristics	Low risk	Outcomes and characteristics appear similar between arms.
Protection against contamination	Low risk	Due to clustering, unlikely contamination
Incorrect analysis	Low risk	Clustering taken into account
Risk of bias overall	Low risk	Low risk across all domains

Harris 2013

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Canada Setting: primary care Speciality: GP/family N health professionals: 154

Harris 2013 *(Continued)*

	N patients: 5646
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + report cards (individual) + diabetes specialist/educator consultation support + pharmacist support + education; Intervention 2: A&F + report cards (individual) + education
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: not reported/not applicable Source: manual chart review Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	It appears that there has been a random sequence allocation generator.
Allocation concealment (selection bias)	Low risk	Physicians were randomly allocated to treatment group by the Coordinating Centre.
Blinding of outcome assessment (detection bias)	Unclear risk	It is unclear how the primary outcome (physician's insulin prescribing rate (IPR), the number of insulin starts per the 12-month intervention of insulin-eligible patients) was assessed (i.e. chart review or questionnaire) so the risk of detection bias is unclear. Secondary clinical outcome data were extracted from data from the audit of consenting patients' charts so are less likely to be affected by risk of detection bias.
Incomplete outcome data (attrition bias) All outcomes	Low risk	154 randomised; 151 included in analysis (2 lost from intervention group and 1 lost from control group) - very small loss to follow-up
Selective reporting (reporting bias)	Low risk	Other than modifications to the recruitment, it appears the study aligns with the trial registration: https://clinicaltrials.gov/ct2/show/NCT00593489.
Recruitment bias	Low risk	No evidence of recruitment bias
Baseline imbalance in outcome measurements or characteristics	High risk	Intervention physicians were older, saw more patients per day, had been in practice longer and had lower level of diabetes-related CME attendance.
Protection against contamination	Unclear risk	Not clear if physicians worked in the same facilities
Incorrect analysis	Low risk	Analysis at cluster level
Risk of bias overall	High risk	Due to baseline imbalance

Harris 2015

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Australia Setting: primary care Speciality: GP/family N health professionals: 122 N patients: 21,848
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + practice facilitation + group training
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: once Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation sequence was used.
Allocation concealment (selection bias)	Low risk	From protocol: "Randomisation will be conducted centrally, after completion of the baseline general practice and practice nurse (PN) surveys, by one of the investigators, a statistician not involved in the data collection or intervention (UJ)" Randomisation and subsequent allocation were performed centrally and allocation was concealed.
Blinding of outcome assessment (detection bias)	High risk	Practices were aware of their allocation. As there is an element of self-reporting in some of the outcomes, it is possible that these are subject to bias.
Incomplete outcome data (attrition bias) All outcomes	Low risk	The protocol described that the analysis would use ITT; however, in this publication, the analysis excluded practices before baseline data collection (1/16 intervention practices and 1/16 control practices, due to reasons that do not appear to be related to outcomes) and due to data collection issues (3/16 control practices only). The reasons for more control clusters lost do not appear to be related to the intervention and outcomes.
Selective reporting (reporting bias)	Low risk	Though the outcomes are arranged differently to the previously published protocol, it appears all the important outcomes are reported.
Recruitment bias	Low risk	Unlikely that physicians were recruited to the trial after randomisation.

Harris 2015 *(Continued)*

Baseline imbalance in outcome measurements or characteristics	Low risk	There were no significant differences between intervention and control groups with respect to gender or in the number of practices that had staff aged 55 years or older, or those who had been working in general practice for 10 or more years. There were significant differences in the years in general practice and years working at the enroled practice (which was longer for the control group; table 1).
Protection against contamination	Low risk	Allocation was by practice and unlikely that the control group received the intervention
Incorrect analysis	Low risk	Analysis adjusted for clustering at patients within practice level
Risk of bias overall	High risk	Due to blinding

Hayashino 2016

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Japan Setting: primary care Speciality: GP/family N health professionals: 192 N patients: 2199
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + reminders
Outcomes	Clinical behaviour category: testing/exams, counselling Format of Audit & Feedback: written Source: manual chart review Frequency: monthly Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	The method of sequence generation was adequate to ensure randomisation.
Allocation concealment (selection bias)	Low risk	Allocation to groups seems to have been concealed from study personnel, and all clusters were randomised to the two groups at the same time, hence, there is low risk of selection bias.

Hayashino 2016 *(Continued)*

Blinding of outcome assessment (detection bias)	Low risk	Clinicians were aware of allocation, but recording in medical records was unlikely to be biased. Patient outcomes were collected by research personnel and also unlikely to be biased.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Incomplete data appear balanced between groups and do not appear to be affected by the intervention or influencing the outcomes.
Selective reporting (reporting bias)	Low risk	Trial was registered and protocol was published; there were no deviations from the protocol and results for all prespecified outcomes were published.
Recruitment bias	Low risk	Unlikely that recruitment happened after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Since multinomial logistic regression was used to adjust for categorical confounders, it is likely that the baseline difference in diabetic medication use between the two groups was adjusted for in the analysis.
Protection against contamination	Low risk	It appears there is low risk of bias due to contamination between groups, as the authors have stated it as one of the strengths of this study.
Incorrect analysis	Low risk	The effects of clustering were accounted for in the data analysis.
Risk of bias overall	Low risk	Low risk across all domains

Hayes 2001

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital inpatient Speciality: internal medicine N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F (from physicians) + guidelines + QI plan; Intervention 2: A&F (written) + guidelines + QI plan
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: once Instructions for improvement: co-developed plans
Notes	

Hayes 2001 *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random number table
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Quote: "Abstractors were not informed of the hospital's intervention status."
Incomplete outcome data (attrition bias) All outcomes	Low risk	Only one of 29 hospitals dropped out.
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No participants were enrolled post-randomisation of clusters, hence, there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 1
Protection against contamination	Low risk	Separate hospitals
Incorrect analysis	Low risk	Chi-square adjusted for clustering
Risk of bias overall	Low risk	Low risk of bias across all important domains

Hayes 2002

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital inpatient Speciality: cardiology N health professionals: NR N patients: 2365
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F (from physicians) + guidelines + QI plan; Intervention 2: A&F (written) + guidelines + QI plan
Outcomes	Clinical behaviour category: prescribing, diagnosis

Hayes 2002 *(Continued)*

	Format of Audit & Feedback: electronic dashboard Source: manual chart review Frequency: once Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial, allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	Minimal number lost to follow-up
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Low risk	Separate hospitals
Incorrect analysis	Low risk	Generalised estimating equations
Risk of bias overall	Unclear risk	Unclear risk in all important domains

Hemkens 2017

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Switzerland Setting: primary care Speciality: GP/family N health professionals: 2900

Hemkens 2017 *(Continued)*

	N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + guidelines
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: administrative/billing data Frequency: quarterly Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Allocation to groups was performed by an independent researcher and seemed to be done at one time.
Blinding of outcome assessment (detection bias)	Low risk	There is a low risk of detection bias as outcome assessors were blinded to the groups' status.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Low risk of attrition bias as withdrawals and reasons for withdrawals were similar across groups. Both ITT and PP analyses were performed.
Selective reporting (reporting bias)	Low risk	The trial was registered and study protocol was published; there were no changes from the protocol.
Recruitment bias	Low risk	No active recruitment as all eligible physicians were included in the study without knowledge
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline is balanced.
Protection against contamination	Low risk	Due to the cluster design of the trial, it is unlikely that control sites received the intervention.
Incorrect analysis	Low risk	Clustering was taken into account for analysis.
Risk of bias overall	Low risk	Low risk across all domains

Hemminiki 1992

Study characteristics

Methods	Design: parallel cluster-RCT

Hemminiki 1992 *(Continued)*

Participants	Country: Finland Setting: hospital inpatient Speciality: obstetrics N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of Groups: Control: no intervention; Intervention: A&F
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: written Source: manual chart review Frequency: once Instructions for improvement: NR
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Data collected from registers
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data on 48 of 52 hospitals
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence, there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Low risk	Separate hospitals
Incorrect analysis	Unclear risk	Not clear what the analysis was, if any
Risk of bias overall	Low risk	Low risk across most important domains

Hendryx 1998

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital inpatient Speciality: internal medicine N health professionals: NR N patients: 224
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + education
Outcomes	Clinical behaviour category: prescribing, testing/exams, treatment decision/action Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial, allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	All hospitals followed up
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence, there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess

Hendryx 1998 *(Continued)*

Protection against contamination	Low risk	Separate hospitals
Incorrect analysis	High risk	No adjustments done
Risk of bias overall	High risk	Due to incorrect analysis

Herbert 2004

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Canada Setting: primary care Speciality: GP/family N health professionals: 200 N patients: 12,598
Interventions	Number of arms: 4 Description of groups: Intervention 1: education on thiazide diuretics + A&F; Intervention 2: education on topics unrelated to main goal of study + A&F; Control 1: education on thiazide diuretics; Control 2: education on topics unrelated to main goal of study
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: administrative/billing data Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Appears random with matching
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Arbitrary codes used for labelling
Incomplete outcome data (attrition bias) All outcomes	Low risk	See Figure 1

Herbert 2004 *(Continued)*

Selective reporting (reporting bias)	Unclear risk	Unable to assess
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence, there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Low risk	Separate hospitals
Incorrect analysis	Low risk	Clustering accounted for in analysis
Risk of bias overall	Low risk	Low risk across most important domains

Hermans 2013

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Belgium, Greece, Luxembourg, Portugal, Spain, UK Setting: primary care + hospital outpatient clinics Speciality: GP/family N health professionals: 477 N patients: 4027
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F
Outcomes	Clinical behaviour category: screening, counselling Format of Audit & Feedback: written Source: computerised search of charts Frequency: tri-annually Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	A computer program was used to generate randomisation sequence.

Hermans 2013 *(Continued)*

Allocation concealment (selection bias)	Low risk	Randomisation sequence was generated centrally and appears to be concealed until assignment.
Blinding of outcome assessment (detection bias)	Low risk	The majority of outcomes are laboratory tests which would not be subject to bias even if allocation was known. For other outcomes, e.g. provision of dietary advice (which would be self-recorded by physicians), it is unlikely that knowledge of allocation would bias these outcomes as physicians in both arms were receiving feedback and may not be aware if they were in the intervention or control arm.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Missing outcome data are not described with reasons to assess whether they were balanced between groups (only total missing data reported).
Selective reporting (reporting bias)	Low risk	All outcomes specified in the trial registration were reported.
Recruitment bias	Low risk	There is no evidence of sites being recruited after randomisation and allocation.
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline characteristics and outcomes data appear similar between groups. Confounding variables were also taken into account in the models for analysis.
Protection against contamination	Low risk	Due to the cluster design of the trial, it is unlikely that control sites received the intervention.
Incorrect analysis	Low risk	Clustering was taken into account for analysis.
Risk of bias overall	Low risk	Low risk across most domains

Herrin 2006

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 92 N patients: 1891
Interventions	Number of arms: 3 Description of groups: Intervention 1: A&F on diabetes processes of care, based on Medicare claims data; Intervention 2: A&F on Medicare claims + feedback on clinical measures from medical record; Intervention 3: A&F on Medicare claims + feedback on clinical measures from medical record + access to a diabetes resource nurse
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: not reported/not applicable

Herrin 2006 *(Continued)*

	Source: manual chart review, patient-reported data Frequency: not reported/not applicable Instructions for improvement: not reported
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Performed on all units at start
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	See Figure
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence, there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 1
Protection against contamination	Low risk	Separate practices
Incorrect analysis	Low risk	Clustering accounted for in analysis
Risk of bias overall	Unclear risk	Unable to assess risk in important domains

Hillman 1998

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: NR N patients: NR

Hillman 1998 *(Continued)*

Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + financial incentive
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: written Source: manual chart review Frequency: semi-annually Instructions for improvement: not reported
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Stratified randomisation
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data collected from all sites
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Low risk	Separate sites
Incorrect analysis	High risk	Clustering was not taken into account in the analysis.
Risk of bias overall	High risk	Due to incorrect analysis

Hillman 1999

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA

Hillman 1999 *(Continued)*

	Setting: primary care Speciality: GP/family/paediatrics N health professionals: NR N patients: NR
Interventions	Number of arms: 3 Description of groups: Control: no feedback; Intervention 1: feedback alone; Intervention 2: feedback and financial incentives
Outcomes	Clinical behaviour category: immunisation, screening Format of Audit & Feedback: written Source: manual chart review Frequency: semi-annually Instructions for improvement: not reported
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Unclear risk	Unable to assess
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	No missing data
Selective reporting (reporting bias)	Low risk	All outcomes reported
Recruitment bias	Unclear risk	Unable to assess
Baseline imbalance in outcome measurements or characteristics	High risk	Significant differences in baseline characteristics and outcomes between arms
Protection against contamination	Low risk	Randomised by site; contamination unlikely
Incorrect analysis	High risk	Clustering not accounted for in analysis
Risk of bias overall	Unclear risk	Due to randomisation sequence generation and allocation concealment

Hocking 2018

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Australia Setting: primary care Speciality: GP/family N health professionals: 586 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: education only; Intervention: A&F + education + guidelines + reminders
Outcomes	Clinical behaviour category: screening Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: quarterly Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Central allocation
Blinding of outcome assessment (detection bias)	Low risk	Labs assessing outcome measure not aware of allocation
Incomplete outcome data (attrition bias) All outcomes	Low risk	ITT analysis used
Selective reporting (reporting bias)	Low risk	Reported according to methods and protocol
Recruitment bias	High risk	New clinics/GPs in towns were recruited after randomisation.
Baseline imbalance in outcome measurements or characteristics	Low risk	Similar baseline between the groups and analyses also adjusted for baseline imbalances
Protection against contamination	Low risk	Unlikely contamination between towns

Hocking 2018 *(Continued)*

Incorrect analysis	Low risk	Analysis adjusted for clustering
Risk of bias overall	High risk	Recruitment bias judged to be high

Hogg 2008

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Canada Setting: primary care Speciality: GP/family N health professionals: NR N patients: 1620
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + outreach facilitation
Outcomes	Clinical behaviour category: testing/exams, counselling, immunisation Format of Audit & Feedback: verbal (interactive) Source: manual chart review, patient-reported data Frequency: once Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Matched randomisation using statistical package
Allocation concealment (selection bias)	Low risk	Allocation sequence unavailable until assignment
Blinding of outcome assessment (detection bias)	Low risk	Physicians unaware of exact outcomes, and data collection/analysis performed by personnel unaware of allocation
Incomplete outcome data (attrition bias) All outcomes	Low risk	Results reported for all included clinics
Selective reporting (reporting bias)	Unclear risk	Analysis plan inadequately described, and no protocol, so unclear whether selective outcome reporting occurred

Hogg 2008 *(Continued)*

Recruitment bias	Unclear risk	There was no explicit description of randomisation occurring only at one point or before the trial commenced. Reading between the lines, it is likely that both were the case, but I can't quite be sure.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Differences in 2 domains; these don't appear to have been accounted for in analysis
Protection against contamination	Low risk	Contamination unlikely as randomisation occurred at practice level.
Incorrect analysis	Unclear risk	No mention of adjusting analysis for clusters was made.
Risk of bias overall	Low risk	Due to baseline imbalance and incorrect analysis

Holm 1990

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Denmark Setting: primary care Speciality: GP/family N health professionals: 356 N patients: 460,076
Interventions	Number of arms: 3 Description of groups: Control 1: no intervention; Control 2: educational materials; Intervention: A&F + educational materials
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: administrative/billing data Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	"Drawing lots" used as method of randomisation
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed

Holm 1990 *(Continued)*

Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	Minimal number of dropouts
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 1
Protection against contamination	Low risk	Separate practices
Incorrect analysis	High risk	Clustering not taken into account
Risk of bias overall	High risk	Due to incorrect analysis

Horbar 2004

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital inpatient Speciality: obstetrics N health professionals: NR N patients: 6182
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + education (workshop) + evidence reviews + faculty support; Intervention 2: audit & feedback (centre-specific)
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: administrative/billing data Frequency: once Instructions for improvement: general
Notes	
Risk of bias	

Horbar 2004 *(Continued)*

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation was achieved using computer program
Allocation concealment (selection bias)	Low risk	Enroled hospitals were allocated to treatment arm at one time point and after enroling in the study.
Blinding of outcome assessment (detection bias)	Unclear risk	Unclear if some outcome measures could be subject to bias in recording (e.g. timing) as doctors were not blind to their allocation
Incomplete outcome data (attrition bias) All outcomes	Low risk	There does not appear to be any missing data at the patient level, and there was no attrition of clusters.
Selective reporting (reporting bias)	Low risk	All prespecified outcomes appear to have been reported.
Recruitment bias	Low risk	It appears all recruited hospitals were randomised and allocated at one time. There is no evidence of enroling more sites after randomisation.
Baseline imbalance in outcome measurements or characteristics	Low risk	As per annotation
Protection against contamination	Unclear risk	While treatment was allocated at the hospital level, it is unclear whether clinicians travelled between hospitals. It is unclear whether the researchers attempted to control for potential contamination.
Incorrect analysis	Low risk	Analyses appear to adequately control for clustering of data.
Risk of bias overall	Low risk	Low across most domains

Houston 2015

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: NR N patients: 4789
Interventions	Number of arms: 2 Description of groups: Control: paper referral; Intervention: paper referral + patient e-referral + audit & feedback + educational materials + implementation coordinators + incentives (financial & CME)
Outcomes	Clinical behaviour category: referrals Format of Audit & Feedback: electronic dashboard

Houston 2015 *(Continued)*

	Source: computerised search of charts, patient-reported data Frequency: not reported/not applicable Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	(See Houston 2010). Online randomisation table developed by statisticians was used.
Allocation concealment (selection bias)	Low risk	(See Houston 2010) Allocation was managed centrally by two statisticians, and allocation of clusters was automatically done when the first provider in a cluster registered; therefore, performed centrally and concealed.
Blinding of outcome assessment (detection bias)	Low risk	While some measures were objective, others were self-reported by patients (e.g. smoking cessation). However, practices were unaware of the three trial arms and patients were unaware of the intervention allocated, so risk of bias for outcome assessment is low.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition is significantly higher in the control arm; however, trialists recognised this and adjusted smoking cessation analysis for completed cases, assuming missing data were smokers, and on the selection model. Incomplete outcome data appear adequately dealt with.
Selective reporting (reporting bias)	Low risk	All outcomes reported on
Recruitment bias	Low risk	No evidence of recruitment bias
Baseline imbalance in outcome measurements or characteristics	Low risk	Got baseline imbalance due to incomplete/dropout, but accounted for this in analysis.
Protection against contamination	Low risk	Due to cluster design, contamination is unlikely.
Incorrect analysis	Low risk	Clustering taken into account for analysis
Risk of bias overall	Low risk	Low risk across all domains

Huffman 2018

Study characteristics

Methods	Design: step wedge
Participants	Country: India Setting: hospital inpatient Speciality: cardiology

Huffman 2018 *(Continued)*

	N health professionals: NR N patients: 22,557
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + education + guidelines
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: monthly Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Concealed and central allocation
Blinding of outcome assessment (detection bias)	Low risk	Most outcomes were objective or unlikely subject to bias even with knowledge of allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Clear information provided on the 5 steps with numbers enroled, loss to follow-up etc. Primary outcome follow-up 99%
Selective reporting (reporting bias)	Low risk	Outcome measures consistent with registered protocol - https://www.clinicaltrials.gov/ct2/show/NCT02256657?cond=NCT02256657&draw=2&rank=1
Recruitment bias	Low risk	Unlikely participants/clusters recruited to the trial after clusters had been randomised
Baseline imbalance in outcome measurements or characteristics	Low risk	Some observed differences in baseline characteristics; however, analysis adjusted for baseline characteristics
Protection against contamination	Low risk	Allocated by hospital and unlikely the control group received the intervention
Incorrect analysis	Low risk	Analysis includes cluster-adjusted and cluster and temporal-adjusted primary and secondary trial outcome event rates including within-hospital clustering adjustments.
Risk of bias overall	Low risk	Overall low risk of bias across all domains

Huis 2013

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: hospital inpatient Speciality: NA N health professionals: 2733 N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: education + reminders + A&F+ targeting adequate products and facilities; Intervention 2: education + reminders + A&F + targeting adequate products and facilities + commitment from management + modelling from leaders + norms + target setting
Outcomes	Clinical behaviour category: other Format of Audit & Feedback: written Source: other Frequency: semi-annually Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Allocation occurred centrally and at one time prior to commencement.
Blinding of outcome assessment (detection bias)	Low risk	The outcome assessment was made by observers who were blinded to allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	There is some confusion over the flow of participants (it says initially 37 wards were in the SAS arm but then it says 47) but the analyses used both ITT and per protocol.
Selective reporting (reporting bias)	Low risk	Data reported as per clinical trial registration
Recruitment bias	Low risk	Recruitment occurred at once prior to trial commencement.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Baseline balance between groups not clearly reported; baseline outcome measures appear similar between groups

Huis 2013 *(Continued)*

Protection against contamination	Unclear risk	Although nursing wards were allocated, it is unclear if there would be contamination between wards.
Incorrect analysis	Low risk	Authors adjusted for clustering
Risk of bias overall	Low risk	Low risk across most important domains

Hurlimann 2015

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Switzerland Setting: primary care Speciality: GP/family N health professionals: NR N patients: 28,952
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + guidelines
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: computerised search of charts Frequency: semi-annually Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Randomisation method not described
Allocation concealment (selection bias)	Unclear risk	No details or study flow diagram to determine if clusters were likely allocated at the same time
Blinding of outcome assessment (detection bias)	Low risk	Unclear but outcome data were collected in medical records, so unlikely to be subject to bias
Incomplete outcome data (attrition bias) All outcomes	Low risk	Loss of practices were reasonably balanced though reasons were not reported.

Hurlimann 2015 *(Continued)*

Selective reporting (reporting bias)	Low risk	Outcomes consistent with registered protocol - https://www.clinicaltrials.gov/ct2/show/NCT01358916?cond=NCT01358916&draw=2&rank=1
Recruitment bias	Unclear risk	There was no study flow diagram and there was insufficient information provided to make a clear judgement regarding this domain.
Baseline imbalance in outcome measurements or characteristics	Low risk	There were very few baseline differences between groups that were adequately adjusted for in the analysis.
Protection against contamination	Low risk	Allocated by practice and deemed unlikely the control group received the intervention
Incorrect analysis	Low risk	Appropriate method of analysis was used; hence, there is a low risk of bias due to incorrect analysis.
Risk of bias overall	Unclear risk	Due to sequence generation, allocation concealment and recruitment bias

Hux 1999

Study characteristics		
Methods	Design: parallel cluster-RCT	
Participants	Country: Canada Setting: primary care Speciality: GP/family N health professionals: 251 N patients: NR	
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + educational bulletins	
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: administrative/billing data Frequency: bi-monthly Instructions for improvement: general	
Notes		
Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess

Hux 1999 *(Continued)*

Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Unique identifiers used
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data available for 250/251 physicians
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No evidence of recruitment after randomisation and allocation
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 1
Protection against contamination	Low risk	Physicans recruited from separate addresses
Incorrect analysis	High risk	Clustering not taken into account
Risk of bias overall	High risk	Due to incorrect analysis

Ivers 2013

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Canada Setting: primary care Speciality: GP/family N health professionals: 53 N patients: 4617
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + worksheet; Intervention 2: A&F
Outcomes	Clinical behaviour category: prescribing, testing/exams Format of Audit & Feedback: written Source: computerised search of charts, administrative/billing data Frequency: semi-annually Instructions for improvement: general
Notes	

Ivers 2013 *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer program used for randomisation by minimisation
Allocation concealment (selection bias)	Low risk	Allocation was done at one time.
Blinding of outcome assessment (detection bias)	Low risk	In the study protocol, the following description appears: "A research coordinator who is otherwise unaffiliated with the study will use a master list with study IDs and participant names to prepare the intervention materials."
Incomplete outcome data (attrition bias) All outcomes	Low risk	Analysis by ITT
Selective reporting (reporting bias)	Low risk	Primary and secondary outcomes reported
Recruitment bias	Low risk	This was information from the study protocol: "Although minimization has been criticised for increasing the risk for selection bias. . ., the benefits of achieving balance at baseline through the use of minimization outweighs the minor risks involved (especially for small trials) and therefore has been widely recommended. . .. We believe that risk of selection bias is low in this case because the recruitment of the first fourteen practices (clusters) was completed prior to allocation of the practices using the minimization software".
Baseline imbalance in outcome measurements or characteristics	Low risk	Practices were allocated using minimisation (conducted by the study analyst using the free software, MINIM) to achieve balance on baseline values of the primary outcomes and on the number of eligible patients in each cluster.
Protection against contamination	Low risk	To reduce the risk of contamination, randomisation was at the level of the primary care clinic.
Incorrect analysis	Low risk	The primary analysis used intention-to-treat.
Risk of bias overall	Low risk	Low risk across all domains

Kaboré 2019

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Burkina Faso Setting: hospital inpatient Speciality: obstetrics N health professionals: NR N patients: 4174

Kaboré 2019 *(Continued)*

Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + SMS reminders + computer algorithms
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: variable Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation was achieved using computer-generated blocked randomisation.
Allocation concealment (selection bias)	Low risk	"All participating hospitals were randomly allocated simultaneously to minimize the risk of allocation bias". Allocation was concealed and done centrally at one time.
Blinding of outcome assessment (detection bias)	Low risk	"Data were collected daily using a standardized questionnaire, in the same way at both the intervention and control hospitals. Data completeness and quality were assessed during daily maternity staff meetings, through quarterly on-site visits and queries sent to on-site data collectors to resolve discrepancies identified by the data manager. Data collectors were aware of the randomization assignments but were not involved in outcome assessments." Data collectors used standardised forms and were monitored. It is unlikely outcome assessment was biased due to knowledge of allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Analysis used ITT. 1/11 clusters in the control arm was lost to follow-up because the operating room was not operational post-intervention. There is no reason to suspect there is risk of bias.
Selective reporting (reporting bias)	Low risk	Results are presented at baseline and for the intervention and control groups for all indicators as indicated.
Recruitment bias	Low risk	Figure 1 demonstrates that assignment occurred following randomisation.
Baseline imbalance in outcome measurements or characteristics	Low risk	Similar characteristics between the two groups are reported.
Protection against contamination	Low risk	Due to the cluster nature of the trial, it is unlikely that control sites received the intervention. Trialists monitored for quality improvement programmes being carried out at the same time that could impact caesarean rate and staff transfers into and out of hospitals. It is acknowledged that staff could potentially have forwarded SMS reminders to staff in the intervention arm, though this would have resulted in a bias towards the null anyway.
Incorrect analysis	Low risk	Authors used intention-to-treat analyses.

Kaboré 2019 *(Continued)*

Risk of bias overall	Low risk	Low risk across all domains

Kahan 2009

Study characteristics		
Methods	Design: parallel cluster-RCT	
Participants	Country: Israel Setting: primary care Speciality: GP/family N health professionals: 298 N patients: NR	
Interventions	Number of arms: 4 Description of groups: Control 1: no intervention; Control 2: clinician education (lecture); Intervention 1: A&F; Intervention 2: A&F + clinician education (lecture)	
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: computerised search of charts Frequency: once Instructions for improvement: general	
Notes		
Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Performed on all units at start
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unable to assess
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No recruitment after randomisation

Kahan 2009 *(Continued)*

Baseline imbalance in outcome measurements or characteristics	Low risk	Table 4
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Unclear risk	Method of analysis not reported
Risk of bias overall	Unclear risk	Unable to assess bias for important domains

Kaminski 2016

Study characteristics		
Methods	Design: parallel cluster-RCT	
Participants	Country: Poland Setting: screening service site Speciality: gastroenterology N health professionals: 176 N patients: NR	
Interventions	Number of arms: 2 Description of groups: Intervention 1: education + A&F; Intervention 2: A&F	
Outcomes	Clinical behaviour category: screening Format of Audit & Feedback: written Source: administrative/billing data Frequency: variable Instructions for improvement: general	
Notes		
Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random method was used for sequence generation.
Allocation concealment (selection bias)	Low risk	It appears all centres were allocated at the same time, so there would not have been a risk of foreseeing allocation.
Blinding of outcome assessment (detection bias)	Low risk	All colonoscopists were aware of being routinely monitored. Screening centre leaders were not aware of the full extent, aims and outcomes of the trial. Data were recorded in databases and unlikely to be biased between groups.

Kaminski 2016 *(Continued)*

Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition minimal and accounted for
Selective reporting (reporting bias)	Unclear risk	One minor change from the trial registration is: "Screening centre leader's proximal and distal adenoma detection rate" was listed in the trial registration but only "proximal ADR" was reported in the publication. This change is not explained, and it is unclear if bias was introduced or not.
Recruitment bias	Low risk	No recruitment appeared to occur after randomisation (Fig. 1).
Baseline imbalance in outcome measurements or characteristics	Low risk	Minimal baseline differences between groups
Protection against contamination	Low risk	Clinicians in intervention group had to undergo a training session that control group clinicians were not able to attend. As the clinicians were all leaders of their own centres, it is unlikely one of the intervention group clinicians would have subsequently trained one of the control group clinicians.
Incorrect analysis	Low risk	Clustering was accounted for in the analysis.
Risk of bias overall	Low risk	Low risk across most domains

Katz 2004

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: family practice and internal medicine clinic Speciality: GP/family N health professionals: 75 N patients: 2163
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + counselling + education + reminders
Outcomes	Clinical behaviour category: counselling Format of Audit & Feedback: written Source: manual chart review, patient-reported data Frequency: twice Instructions for improvement: general
Notes	

Katz 2004 *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random number generator used to assign pairs of clinics to intervention/control
Allocation concealment (selection bias)	Low risk	It appears a statistician allocated practices at one time.
Blinding of outcome assessment (detection bias)	Low risk	Outcomes were self-reported by patients (likely unaware of their physician's allocation) and collected by research personnel - unclear if personnel were blind at exit interviews, but personnel collecting 2 and 6-month data were blinded. Unlikely that outcome assessment was biased
Incomplete outcome data (attrition bias) All outcomes	Low risk	No practices were lost to follow-up; loss of patients between groups appear balanced (with reasons provided), but analysis by ITT (counting patients lost to follow-up as not quitting smoking)
Selective reporting (reporting bias)	Low risk	Comprehensive reporting of results in alignment with methods. No protocol to check against
Recruitment bias	Low risk	All recruitment performed before randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline variables largely similar
Protection against contamination	Low risk	Single clinics all randomised to one group. Unlikely that the control groups received the intervention
Incorrect analysis	Low risk	Clustering at clinic and clinician level accounted for in analysis
Risk of bias overall	Low risk	Low risk across all domains

Kaufmann-Kolle 2011

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Germany Setting: primary care Speciality: GP/family N health professionals: 96 N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: Benchmarking Quality Circles+ A&F; Intervention 2: Traditional Quality Circles + A&F
Outcomes	Clinical behaviour category: prescribing

Kaufmann-Kolle 2011 *(Continued)*

	Format of Audit & Feedback: both written and verbal Source: administrative/billing data Frequency: bi-monthly Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Not reported
Allocation concealment (selection bias)	Unclear risk	Not reported
Blinding of outcome assessment (detection bias)	Low risk	Prescription data
Incomplete outcome data (attrition bias) All outcomes	High risk	Imbalanced losses between groups (5 vs 32)
Selective reporting (reporting bias)	High risk	Retrospectively registered protocol; not all outcomes reported
Recruitment bias	Low risk	Recruitment did not occur after randomisation.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Not reported
Protection against contamination	Low risk	Cluster design; unlikely to be contaminated
Incorrect analysis	Unclear risk	Analysis not described
Risk of bias overall	High risk	Due to incomplete outcome data and selective reporting

Kennedy 2015

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Canada Setting: nursing homes Speciality: long-term care N health professionals: NR

Kennedy 2015 *(Continued)*

	N patients: 5591
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + education + action planning
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: computerised search of charts, administrative/billing data Frequency: semi-annually Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence generation
Allocation concealment (selection bias)	Low risk	Appears clusters were allocated at one time, as offsite investigator performed randomisation and allocation, so unlikely that allocation would have been known
Blinding of outcome assessment (detection bias)	Low risk	Database manager and analysts blinded
Incomplete outcome data (attrition bias) All outcomes	Low risk	Used ITT analysis
Selective reporting (reporting bias)	Low risk	Protocol stated they will analyse prescribed vitamin D, calcium and other osteoporosis medication, and number of fractures or falls. All except falls analysed; lack of falls analysis justified.
Recruitment bias	Low risk	In protocol, clear that recruitment occurred before randomisation
Baseline imbalance in outcome measurements or characteristics	High risk	Baseline imbalance for key variables including vitamin D use at baseline, osteoporosis diagnosis, prevalence of hip fractures etc.
Protection against contamination	Low risk	Clustering performed by centre; risk of contamination low
Incorrect analysis	Low risk	GEE analysis used to account for clustering
Risk of bias overall	High risk	Due to baseline imbalance

Kerry 2000

Study characteristics

Kerry 2000 *(Continued)*

Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: 175 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + guidelines
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: written Source: administrative/billing data Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Stratified randomisation used
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Data collected from central computer registry
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data collected on all practices
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence, there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Low risk	Table 1
Protection against contamination	Low risk	Practices separate
Incorrect analysis	High risk	t-test

Kerry 2000 *(Continued)*

Risk of bias overall	High risk	Due to incorrect analysis

Kiefe 2001

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: family medicine, internal medicine, endocrinology Speciality: GP/family/internal medicine N health professionals: 97 N patients: 3291
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + achievable benchmark; Intervention 2: A&F
Outcomes	Clinical behaviour category: immunisation, testing/exams Format of Audit & Feedback: written Source: manual chart review, administrative/billing data Frequency: variable Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	Equal numbers in both groups lost to follow-up (Figure 1)
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence, there is low risk of recruitment bias.

Kiefe 2001 *(Continued)*

Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Low risk	Generalised linear models
Risk of bias overall	Unclear risk	Unable to assess bias in important domains

Kim 1999

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 41 N patients: 1810
Interventions	Number of arms: 2 Description of groups: Control: clinician education; Intervention: A&F + academic detailing + clinician education
Outcomes	Clinical behaviour category: immunisation, testing/exams, counselling Format of Audit & Feedback: both written and verbal Source: manual chart review, patient-reported data Frequency: semi-annually Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess

Kim 1999 *(Continued)*

Incomplete outcome data (attrition bias) All outcomes	High risk	Only reviewed charts of patients who responded
Selective reporting (reporting bias)	Unclear risk	Unable to assess
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence, there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Low risk	Mixed model; nested ANOVA
Risk of bias overall	High risk	Due to incomplete outcome data

Kinsinger 1998

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 134 N patients: 2887
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + facilitation
Outcomes	Clinical behaviour category: counselling, testing/exams Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: variable Instructions for improvement: co-developed plans
Notes	
Risk of bias	

Bias	Authors' judgement	Support for judgement

Kinsinger 1998 *(Continued)*

Random sequence generation (selection bias)	Low risk	Randomisation conducted by statistician
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Research assistants who collected data were blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	58/62 practices provided data.
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Table 1 and Table 2
Protection against contamination	Low risk	Separate practices
Incorrect analysis	Low risk	Generalised estimating equations
Risk of bias overall	Low risk	Low risk across all domains

Kritchevsky 2008

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital inpatient Speciality: internal medicine N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + intervention meetings/conference calls; Intervention 2: A&F only
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: variable

Kritchevsky 2008 *(Continued)*

	Instructions for improvement: co-developed plans	
Notes		

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Paired, blinded randomisation
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Authors reported high reliability of main outcomes, so judged to be low risk
Incomplete outcome data (attrition bias) All outcomes	Low risk	All hospitals provided data.
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Table 2
Protection against contamination	Low risk	Separate hospitals
Incorrect analysis	Unclear risk	Method of analysis not described
Risk of bias overall	Low risk	Low risk across most important domains

Lafata 2007

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 294 N patients: 19,892
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + academic detailing

Lafata 2007 *(Continued)*

Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: both written and verbal Source: computerised search of charts, administrative/billing data Frequency: once Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Method for sequence generation was not reported.
Allocation concealment (selection bias)	Low risk	Allocation method was not described but likely done at the same time
Blinding of outcome assessment (detection bias)	Low risk	Administrative data collected
Incomplete outcome data (attrition bias) All outcomes	Low risk	No detail but administrative data used, so unlikely to be an issue
Selective reporting (reporting bias)	Low risk	While there was no previously published protocol or trial registration, the method of choosing outcomes was not up to the trialists (based on interviews and other published research) and therefore unlikely subject to selective reporting.
Recruitment bias	Low risk	No evidence of active recruitment after allocation, as all physicians from three organisations were recruited
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Minor differences, especially for digoxin group - unclear if variables that were different would introduce much bias
Protection against contamination	Low risk	Sessions performed within randomised practices, so unlikely contamination
Incorrect analysis	Low risk	GEE equations used to account for clustering
Risk of bias overall	Low risk	Low risk across most domains

Lakshminarayan 2010

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: mixed hospital departments

Lakshminarayan 2010 *(Continued)*

	Speciality: neurology N health professionals: NR N patients: 2305
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + customised feedback + barrier identification; Intervention 2: A&F only
Outcomes	Clinical behaviour category: prescribing, treatment decision/action, counselling Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: variable Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Abstractors blinded
Incomplete outcome data (attrition bias) All outcomes	Low risk	See Figure 1
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence, there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Low risk	Separate hospitals
Incorrect analysis	Low risk	Multi-level random-effects modeling
Risk of bias overall	Low risk	Low risk across most domains

Lemelin 2001

Study characteristics		
Methods	Design: parallel cluster-RCT	
Participants	Country: Canada Setting: primary care Speciality: GP/family N health professionals: NR N patients: 4000	
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + consensus building + opinion leaders and networking + academic detailing + educational materials + reminders	
Outcomes	Clinical behaviour category: screening, prescribing, testing/exams, counselling, immunisation Format of Audit & Feedback: verbal (interactive) Source: manual chart review Frequency: not reported/not applicable Instructions for improvement: general	
Notes		
Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	The method for sequence generation was not reported.
Allocation concealment (selection bias)	Low risk	Though not explicitly described, it appears all sites were randomised at the same time.
Blinding of outcome assessment (detection bias)	Low risk	Clinicians were aware of their allocation, but it is unlikely that there would be a sustained biased recording in patient medical charts over two years. Auditors collecting the data from medical charts (for some of the outcomes) were blind to allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	1/23 intervention clusters was lost to follow-up (due to moving practice) so this does not appear to be as a result of the intervention and affecting the outcome.
Selective reporting (reporting bias)	Unclear risk	There was no trial registration and the protocol paper published in the same year does not document the preplanned patient outcomes. Therefore, it is unclear if all preplanned outcomes and analyses were reported.
Recruitment bias	Low risk	All participants enroled before randomisation.
Baseline imbalance in outcome measurements or characteristics	Low risk	Similar number of practices per group and characteristics; intervention groups tended to be from larger communities.

Lemelin 2001 *(Continued)*

Protection against contamination	Low risk	Facilitator only interacted with the intervention group, not the control group.
Incorrect analysis	Unclear risk	It is unclear if clustering was taken into account during analysis.
Risk of bias overall	Low risk	Low risk across most domains

Lesuis 2018

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: rheumatology clinic Speciality: rheumatology N health professionals: 20 N patients: 990
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + clinician education; Intervention 2: A&F + clinician education + computer decision-support
Outcomes	Clinical behaviour category: screening, prescribing, testing/exams, treatment decision/action, referrals Format of Audit & Feedback: verbal (didactic) Source: manual chart review, computerised search of charts Frequency: monthly Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	A computer-generated random sequence was used to randomise participants into the two groups.
Allocation concealment (selection bias)	Low risk	Sealed opaque envelopes were used to conceal allocation into the two groups.
Blinding of outcome assessment (detection bias)	Unclear risk	Data collection was not blinded and potentially subject to bias from knowledge of allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No loss to follow-up

Lesuis 2018 *(Continued)*

Selective reporting (reporting bias)	Low risk	The trial was registered (NTR4449) and results for primary outcomes were given; the trial registry also mentions few secondary outcomes (possible predictive factors are derived from earlier studies done in our centre and all focus on individual clinician factors: cognitive bias (Inventory of Cognitive Bias in Medicine; ICBM) personality (Big Five Inventory; BFI) and thinking styles), however the secondary outcomes mentioned in the paper are adherence percentages of separate guideline indicators and mean difference change scores in SSS before and after the intervention. Results for these secondary outcomes are given in the paper.
Recruitment bias	Low risk	All eligible attendees were allotted interventions. All analysed participants selected from the overall RA who attended the study clinic.
Baseline imbalance in outcome measurements or characteristics	Low risk	No significant differences at baseline for clinicians or patients
Protection against contamination	Low risk	CDSS was designed so control clinicians could not see the intervention.
Incorrect analysis	Low risk	Group allocation variable
Risk of bias overall	Low risk	Low risk across most domains

Levi 2020

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Australia Setting: hospital inpatient Speciality: neurology N health professionals: NR N patients: 22,384
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + group meetings + workshops + education
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: verbal (interactive) Source: computerised search of charts Frequency: variable Instructions for improvement: specific
Notes	

Levi 2020 *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation by statistician implies adequate random sequence generation.
Allocation concealment (selection bias)	Low risk	It would not be possible to foresee allocation, as this was done centrally and randomly in one event via Stata.
Blinding of outcome assessment (detection bias)	High risk	Clinicians not blinded and collected the outcome data in the database
Incomplete outcome data (attrition bias) All outcomes	Low risk	No clusters were lost to follow-up.
Selective reporting (reporting bias)	Low risk	Outcome reporting was consistent with trial registration including thrombolysis rates in intervention and control hospitals at follow-up as well as intracranial haemorrhage rates and functional outcomes at 3 months.
Recruitment bias	Low risk	Unlikely individuals were recruited to the trial after clusters were randomised
Baseline imbalance in outcome measurements or characteristics	Low risk	Baselines were even; accounted for criteria like strokes, cluster size (see figure 2)
Protection against contamination	Low risk	Not likely a site received the wrong intervention; randomisation at site level.
Incorrect analysis	Low risk	Analysis done at cluster levels Protocol: "Analysis will involve cluster-level summary data. The difference in post-intervention thrombolysis rates between intervention and control groups will be compared via a two-stage approach to adjust for baseline thrombolysis rates, using a stratified t-test".
Risk of bias overall	High risk	High risk due to blinding

Leviton 1999

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital inpatient Speciality: obstetrics N health professionals: NR N patients: 6798
Interventions	Number of arms: 2

Leviton 1999 *(Continued)*

	Description of groups: Control: no intervention; Intervention: A&F + group discussion + lecture + chart reminders + influential colleague
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: monthly Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	A random number table was used to generate the random sequence.
Allocation concealment (selection bias)	Low risk	As this is a cluster RCT where all institutions were randomly assigned to intervention or control at the same time, there is low risk of biased allocation to treatment groups.
Blinding of outcome assessment (detection bias)	Unclear risk	It is not clear whether the trained medical data collectors were blind to treatment allocation or not.
Incomplete outcome data (attrition bias) All outcomes	Low risk	There was no attrition bias in both groups; data were analysed in all randomised institutions.
Selective reporting (reporting bias)	Low risk	Although there was no trial protocol and the trial was conducted prior to the establishment of trial registries, results were provided for the prespecified study outcome.
Recruitment bias	Low risk	No patients were recruited for the study once randomisation was completed.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Authors reported no baseline differences except for significant differences in frequency of abnormal foetal conditions or foetal distress. No data were provided though.
Protection against contamination	Low risk	Authors stated that there was no overlapping of physicians across control and intervention hospitals. Further, the 2 residents who shared hospitals were both in the intervention group. Hence, there is low risk of contamination between the control and intervention groups.
Incorrect analysis	Unclear risk	Although authors have reported hospital-level results, it is unclear whether clustering was accounted for in the analysis as it was not reported clearly.
Risk of bias overall	Low risk	Baseline imbalance and incorrect analysis judged as unclear, but all others low risk

Lim 2018

Study characteristics

Lim 2018 *(Continued)*

Methods	Design: parallel cluster-RCT
Participants	Country: Malaysia Setting: primary care Speciality: GP/family N health professionals: NR N patients: NR
Interventions	Number of arms: 3 Description of groups: Control: no intervention; Intervention 1: structured prescription review + full A&F; Intervention 2: structured prescription review + partial A&F
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: monthly Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	"Simple randomisation" was used but no information on the methods used to generate the sequence
Allocation concealment (selection bias)	Low risk	Not explicitly stated but, based on the study flow diagram, it appears likely that all clusters were allocated at the same time.
Blinding of outcome assessment (detection bias)	High risk	Pharmacists were the outcome reporters and they were aware of group allocations. Data analysts checked the pharmacists' reporting to ensure accuracy (which was fed back to pharmacists) but they were also aware of group allocations.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	It appears no clusters were lost to follow-up. Percentage of prescribers evaluated at each time point is not clearly reported with reasons. There does not appear to be a significant dropoff of prescribers evaluated over time, and it is unclear why levels were much higher in the full intervention group at all time points.
Selective reporting (reporting bias)	Low risk	All outcomes specified in the methods were reported, though the primary outcome is the only one presented entirely graphically (with no numerical data).
Recruitment bias	Low risk	Recruitment did not occur after randomisation.
Baseline imbalance in outcome measurements or characteristics	High risk	There appears to be significant baseline differences in outcomes (e.g. primary outcome) and characteristics (e.g. prescribers evaluated, prescriber turnover rate) that suggests there are baseline imbalances between trial groups.

Lim 2018 *(Continued)*

Protection against contamination	Low risk	Allocation is at the health clinic (cluster) level and training was provided to participants separated by allocation to avoid contamination. Contamination (though not impossible if a prescriber worked at more than one clinic) appears unlikely.
Incorrect analysis	High risk	Trial does not appear to take clustering into account for analysis.
Risk of bias overall	High risk	Baseline imbalance and incorrect analysis

Linder 2010

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 573 N patients: 136,633
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + electronic dashboard
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: electronic dashboard Source: computerised search of charts, administrative/billing data Frequency: monthly Instructions for improvement: not reported
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Only says "Matched pairs were randomized" but no information on how random sequence was generated
Allocation concealment (selection bias)	Low risk	No details but, based on the study flow diagram, it appears likely that all clusters were allocated at the same time.
Blinding of outcome assessment (detection bias)	Low risk	Outcome was prescriptions recorded on electronic records and unlikely to be at risk of bias even if physicians were not blinded
Incomplete outcome data (attrition bias) All outcomes	Low risk	Appears to be complete; accounted for and not of concern. No missing data reported

Linder 2010 *(Continued)*

Selective reporting (reporting bias)	Low risk	The study protocol is not available for comparison; however, all of the outcomes reported on in the methods were reported on in the results.
Recruitment bias	Low risk	Practices were recruited and then randomised, so low risk of recruitment bias
Baseline imbalance in outcome measurements or characteristics	Low risk	There were no significant differences between intervention and control practices in the number of years using the EHR, mean visits per year, the baseline antibiotic prescribing rate, or the baseline antibiotic prescribing rate for ARIs (data not shown). There were no significant differences in clinician or patient characteristics between intervention and control groups.
Protection against contamination	Low risk	It is not clear if any of the physicians worked in multiple practices. If not, then risk of contamination was low.
Incorrect analysis	Low risk	The primary outcome was the antibiotic prescribing rate for ARIs, based on electronic prescribing using the EHR, in an intent-to-intervene analysis, adjusted for clustering by practice.
Risk of bias overall	Low risk	Low risk across most domains

Lobach 1996

Study characteristics		
Methods	Design: parallel cluster-RCT	
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 45 N patients: NR	
Interventions	Number of arms: 2 Description of groups: Control: no feedback; Intervention: feedback	
Outcomes	Clinical behaviour category: other Format of Audit & Feedback: written Source: computerised search of charts Frequency: monthly Instructions for improvement: not reported	
Notes		
Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**

Lobach 1996 *(Continued)*

Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Unclear risk	Unable to assess
Blinding of outcome assessment (detection bias)	Low risk	Data taken from electronic patient record
Incomplete outcome data (attrition bias) All outcomes	Low risk	No missing data
Selective reporting (reporting bias)	Low risk	All outcomes reported
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	High risk	Audit performed by the same clinician
Incorrect analysis	High risk	Clustering not accounted for in analysis
Risk of bias overall	Unclear risk	Due to randomisation sequence generation and allocation concealment

Lomas 1991

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Canada Setting: hospital inpatient Speciality: obstetrics N health professionals: 76 N patients: 2496
Interventions	Number of arms: 3 Description of groups: Control: no intervention; Intervention 1: A&F; Intervention 2: education
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: both written and verbal Source: manual chart review, computerised search of charts Frequency: quarterly

Lomas 1991 *(Continued)*

	Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Chart audits done by trained staff
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data on all hospitals provided
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters, hence, there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Low risk	Table 1
Protection against contamination	Low risk	Separate hospitals
Incorrect analysis	Low risk	Nested ANOVA with weighted outcome rates
Risk of bias overall	Low risk	Low risk across most domains

Lopez-Picazo 2011

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Spain Setting: primary care Speciality: GP/family N health professionals: 265 N patients: 430,525
Interventions	Number of arms: 4

Lopez-Picazo 2011 *(Continued)*

	Description of groups: Control: no intervention; Intervention 1: A&F; Intervention 2: educational session + A&F; Intervention 3: face-to-face educational session + A&F
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: monthly Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Method for sequence generation was not reported.
Allocation concealment (selection bias)	Unclear risk	No information regarding allocation concealment and whether this occurred at the same time or centrally
Blinding of outcome assessment (detection bias)	Low risk	Physicians were not blinded to their allocation but, as they did not know the study's outcomes, it is unlikely that intervention groups would record clinical records differently to control physicians.
Incomplete outcome data (attrition bias) All outcomes	Low risk	The trial stated that 245 physicians were randomised and data were available for all 245 physicians.
Selective reporting (reporting bias)	Low risk	All prespecified outcomes (mean number of drug interactions and cost-effectiveness) appear to have been reported.
Recruitment bias	Low risk	No evidence that any physicians were actively recruited after randomisation. All eligible physicians were eligible to participate in trial.
Baseline imbalance in outcome measurements or characteristics	Low risk	No significant differences amongst the study groups
Protection against contamination	High risk	Likely the control group contaminated
Incorrect analysis	Low risk	Analyses appear to have adequately adjusted for clustering.
Risk of bias overall	High risk	Protection against contamination judged as high risk

Luitjes 2018

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands

Luitjes 2018 *(Continued)*

	Setting: hospital inpatient Speciality: obstetrics N health professionals: NR N patients: 2294
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + computerised decision-support; Intervention 2: A&F only
Outcomes	Clinical behaviour category: prescribing, counselling, testing/exams Format of Audit & Feedback: written Source: manual chart review Frequency: monthly Instructions for improvement: not reported
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation procedure by drawing sealed, opaque envelopes
Allocation concealment (selection bias)	Low risk	Concealed opaque envelopes were used to conceal allocation.
Blinding of outcome assessment (detection bias)	Low risk	Outcomes collected from medical records, and it is unlikely these would be subject to bias when recorded by physicians
Incomplete outcome data (attrition bias) All outcomes	Low risk	ITT analysis
Selective reporting (reporting bias)	Low risk	All outcomes reported
Recruitment bias	Low risk	No evidence of recruitment bias as there was no recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Trialist report groups were balanced.
Protection against contamination	Low risk	Due to cluster design, contamination is unlikely.
Incorrect analysis	Low risk	Clustering taken into account in analysis
Risk of bias overall	Low risk	Low risk across all domains

Lundborg 1999a

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Sweden Setting: primary care Speciality: GP/family N health professionals: 100 N patients: 4721
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + clinician education
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: verbal (interactive) Source: manual chart review, administrative/billing data Frequency: once Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	The random sequence was generated using the SPSS 6.0 random generator (information extracted from the protocol paper).
Allocation concealment (selection bias)	Low risk	Clusters were allocated at the same time.
Blinding of outcome assessment (detection bias)	Low risk	Doctors likely aware of their allocation; however, it is unlikely the prescription data would be subject to risk of bias.
Incomplete outcome data (attrition bias) All outcomes	Low risk	All randomised groups completed the study and contributed to the final data analysis.
Selective reporting (reporting bias)	Low risk	All prespecified outcomes were analysed and results were presented adequately.
Recruitment bias	Low risk	There is no evidence of any additional recruitment into the trial after randomisation was conducted.
Baseline imbalance in outcome measurements or characteristics	Low risk	No baseline differences were found between the two groups.
Protection against contamination	Unclear risk	It is unclear whether there was any contamination between groups or not.

Lundborg 1999a *(Continued)*

Incorrect analysis	Low risk	The multi-level regression analysis takes into account data at the cluster level as well as nested data (individual patient level) which seems appropriate for the type of intervention which is aiming to achieve change at both practice and patient level.
Risk of bias overall	Low risk	Protection against contamination judged as unclear, but low risk across all other domains

Lundborg 1999b

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Sweden Setting: primary care Speciality: GP/family N health professionals: 104 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + clinician education
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: verbal (interactive) Source: manual chart review, administrative/billing data Frequency: once Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	The random sequence was generated using the SPSS 6.0 random generator (information extracted form the protocol paper).
Allocation concealment (selection bias)	Low risk	Clusters were allocated at the same time.
Blinding of outcome assessment (detection bias)	Low risk	Doctors likely aware of their allocation; however, it is unlikely the prescription data would be subject to risk of bias.
Incomplete outcome data (attrition bias) All outcomes	Low risk	All randomised groups completed the study and contributed to the final data analysis.

Lundborg 1999b *(Continued)*

Selective reporting (reporting bias)	Low risk	All prespecified outcomes were analysed and results were presented adequately.
Recruitment bias	Low risk	There is no evidence of any additional recruitment into the trial after randomisation was conducted.
Baseline imbalance in outcome measurements or characteristics	Low risk	No baseline differences were found between the two groups.
Protection against contamination	Unclear risk	It is unclear whether there was any contamination between groups or not.
Incorrect analysis	Low risk	The multi-level regression analysis takes into account data at the cluster level as well as nested data (individual patient level) which seems appropriate for the type of intervention which is aiming to achieve change at both practice and patient level.
Risk of bias overall	Low risk	Protection against contamination judged as unclear, but low risk across all other domains

Lynch 2016

Study characteristics		
Methods	Design: parallel cluster-RCT	
Participants	Country: Australia Setting: hospital inpatient Speciality: neurology N health professionals: NR N patients: 586	
Interventions	Number of arms: 2 Description of groups: Control: education; Intervention: education + A&F + reminders + opinion leaders + plan and strategy development	
Outcomes	Clinical behaviour category: screening Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: monthly Instructions for improvement: co-developed plans	
Notes		
Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**

Lynch 2016 *(Continued)*

Random sequence generation (selection bias)	Low risk	Generated by computer program
Allocation concealment (selection bias)	Low risk	Allocation was performed by a blinded third party using a coded list.
Blinding of outcome assessment (detection bias)	Low risk	Blinded outcome assessment
Incomplete outcome data (attrition bias) All outcomes	Low risk	No clusters were lost to follow-up or excluded. Patients who had to be excluded from analysis were balanced between groups with reasons explained (that do not appear related to the intervention/outcome).
Selective reporting (reporting bias)	High risk	Both secondary outcomes listed in the trial registration are not reported in this publication: proportions of patients who accessed stroke rehabilitation following discharge from acute hospital and self-reported rehabilitation assessment and referral practices at each stroke unit. This is not discussed and no reason given.
Recruitment bias	Low risk	It is possible health professionals at the hospital were recruited and involved after clusters were randomised, but unlikely to increase bias in this study as both intervention and control hospitals were likely affected equally.
Baseline imbalance in outcome measurements or characteristics	Low risk	Over 300 patient participants, and baseline patient characteristics and outcome measures similar to judge as low risk
Protection against contamination	Low risk	Unlikely the control hospitals received the intervention
Incorrect analysis	Low risk	Clustering is taken into account for analyses.
Risk of bias overall	High risk	Due to selective reporting

Machline-Carrion 2019a

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Brazil Setting: primary care and hospital outpatient clinics Speciality: GP/family/internal medicine N health professionals: NR N patients: 1619
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: case management + A&F + decision-support + educational materials
Outcomes	Clinical behaviour category: prescribing, counselling

Machline-Carrion 2019a *(Continued)*

	Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: monthly Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Web-based computerised randomisation applied by a statistician
Allocation concealment (selection bias)	Low risk	Allocation was done centrally at one time (see protocol).
Blinding of outcome assessment (detection bias)	Low risk	It is unlikely that there would be bias in the recording of patient medical records. Data collectors were blinded to allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Trialists analysed outcomes using ITT.
Selective reporting (reporting bias)	Low risk	All prespecified outcomes are reported.
Recruitment bias	Low risk	Randomisation occurred at the start of the study all at once and there were separate hospitals randomised to intervention or routine practice.
Baseline imbalance in outcome measurements or characteristics	Low risk	Characteristics of clusters and patients were observed to be similar at baseline.
Protection against contamination	Low risk	Due to the cluster nature of the trial, it is unlikely that the control arm received the intervention.
Incorrect analysis	Low risk	Analyses was performed with clustering taken into account.
Risk of bias overall	Low risk	Overall low risk for all domains

Machline-Carrion 2019b

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Brazil, Argentina, Peru Setting: hospital inpatient Speciality: neurology N health professionals: NR

Machline-Carrion 2019b *(Continued)*

	N patients: 1658
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: individual case prompts + educational materials + A&F + on-site training + web and phone-based training
Outcomes	Clinical behaviour category: screening, prescribing, treatment decision/action, counselling Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: variable Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Web-based computerised randomisation applied by a statistician
Allocation concealment (selection bias)	Low risk	Allocation was done centrally at one time (see protocol).
Blinding of outcome assessment (detection bias)	Low risk	It is unlikely that there would be bias in the recording of patient medical records. Data collectors were blinded to allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Trialists analysed outcomes using ITT.
Selective reporting (reporting bias)	Low risk	All prespecified outcomes are reported.
Recruitment bias	Low risk	Randomisation occurred at the start of the study all at once and there were separate hospitals randomised to intervention or routine practice.
Baseline imbalance in outcome measurements or characteristics	Low risk	Characteristics of clusters and patients were observed to be similar at baseline.
Protection against contamination	Low risk	Due to the cluster nature of the trial, it is unlikely that the control arm received the intervention.
Incorrect analysis	Low risk	Analyses was performed with clustering taken into account.
Risk of bias overall	Low risk	Overall low risk for all domains

Maddocks 2011

Study characteristics

Maddocks 2011 *(Continued)*

Methods	Design: parallel cluster-RCT
Participants	Country: Canada Setting: primary care Speciality: GP/family N health professionals: 22 N patients: 23,688
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + clinician education + training
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: both written and verbal Source: computerised search of charts, administrative/billing data Frequency: annually Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	A researcher "randomly allocated", but there is no detail on the method of sequence generation.
Allocation concealment (selection bias)	Unclear risk	No details or study flow diagram to determine if all clusters were allocated at the same time
Blinding of outcome assessment (detection bias)	Low risk	It is unclear whether outcome assessors were aware of allocations, but as the outcomes were extracted from medical records and the extraction method appears objective, it is unlikely bias could have been introduced in outcome assessment.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	There is no information on the proportion (if all) of the physician participants remained for the duration of the study. Two control arm practices were removed due to low availability of one outcome (mammography screening) and it is unknown why this was the case.
Selective reporting (reporting bias)	Low risk	The outcomes specified in the methods are reported in full.
Recruitment bias	Low risk	It appears that randomisation occurred after the practices and physicians were recruited and agreed to participate in the study.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Baseline outcome measures are reported but baseline characteristics of the practices and physicians are not described.
Protection against contamination	Low risk	Allocations are by practice and it is unlikely that the control group participants received the intervention.

Maddocks 2011 *(Continued)*

Incorrect analysis	High risk	Trial does not appear to take clustering into account for analysis.
Risk of bias overall	High risk	Incorrect analysis

Mainous 2000

Study characteristics		
Methods	Design: parallel cluster-RCT	
Participants	Country: USA Setting: Primary care Speciality: GP/family/paediatrics/internal medicine N health professionals: 216 N patients: NR	
Interventions	Number of arms: 4 Description of groups: Control: No feedback; Intervention 1: patient education; Intervention 2: feedback; Intervention 3: feedback plus patient education	
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: administrative/billing data Frequency: once Instructions for improvement: not reported	
Notes		
Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Unclear risk	Unable to assess
Blinding of outcome assessment (detection bias)	Low risk	Assessors were not blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No missing data
Selective reporting (reporting bias)	Low risk	All outcomes reported

Mainous 2000 *(Continued)*

Recruitment bias	Low risk	No evidence of cluster recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline characteristics and outcomes balanced between arms
Protection against contamination	High risk	Randomisation by physician; some may have been in the same practice
Incorrect analysis	High risk	Clustering not accounted for in analysis
Risk of bias overall	Unclear risk	Due to randomisation sequence generation and allocation concealment

Manfredi 1998

Study characteristics		
Methods	Design: parallel cluster-RCT	
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: NR N patients: NR	
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + clinician education + reminders	
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: not reported/not applicable Instructions for improvement: general	
Notes		
Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Study authors matched and randomly assigned medical practices to the intervention groups, but it is not clear how the random sequence was generated.
Allocation concealment (selection bias)	Unclear risk	No details or study flow diagram to determine if all clusters were allocated at the same time

Manfredi 1998 *(Continued)*

Blinding of outcome assessment (detection bias)	Low risk	Although it is unclear if the outcome assessors were blinded to the practice treatment allocation, the outcome data were extracted through reviews of randomly selected patients' medical charts.
Incomplete outcome data (attrition bias) All outcomes	Low risk	51 practices agreed to participate in the study; between pre-intervention and post-intervention, 4 practices were lost to follow-up (details in the text). This is a 7.8% loss to follow-up. It is unlikely that this loss to follow-up would bias the outcome.
Selective reporting (reporting bias)	Unclear risk	No prior trial reg/protocol, so unclear if all predefined outcomes are measured and in the same way
Recruitment bias	Unclear risk	Not enough information as the flow of cluster recruitment and randomisation is not clear
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline and post-intervention patient samples for the intervention and control group were similar in terms of patient sex, age, type of insurance and continuity of care at the same practice.
Protection against contamination	Unclear risk	Randomisation by cluster, but not enough information on staff in practices, so unclear if study ensured staff were not working across multiple sites
Incorrect analysis	Unclear risk	It does not appear that the cluster component was accounted for in the analyses.
Risk of bias overall	Unclear risk	Recruitment bias; protection against contamination and incorrect analysis judged as unclear

Martin 1980

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital inpatient Speciality: internal medicine N health professionals: 24 N patients: NR
Interventions	Number of arms: 3 Description of groups: Control: no intervention; Intervention 1: financial incentive; Intervention 2: A&F
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: verbal (interactive) Source: manual chart review Frequency: weekly Instructions for improvement: specific

Martin 1980 *(Continued)*

Notes		

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	No attrition on process-of-care outcomes
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters; hence, there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Low risk	Table 1
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Unclear risk	Multiple regression and 2-way ANOVA were used, but not clear if clustering was taken into account
Risk of bias overall	Unclear risk	Unable to assess biases across important domains

Marton 1985

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family/internal medicine N health professionals: 57 N patients: NR
Interventions	Number of arms: 4

Marton 1985 *(Continued)*

	Description of groups: Control 1: no intervention; Control 2: clinician educational materials; Intervention 1: A&F; Intervention 2: A&F + clinician educational materials
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: written Source: manual chart review Frequency: bi-weekly Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No description of whether subjects lost to follow-up
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters; hence, there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 1
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	High risk	2-way ANOVA and Kruskal Wallis/t-test. Seems like clustering was not taken into account
Risk of bias overall	High risk	Due to incorrect analysis

McCartney 1997

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: UK

McCartney 1997 *(Continued)*

	Setting: primary care Speciality: GP/family N health professionals: NR N patients: 2813
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: verbal (didactic) Source: vomputerised search of charts Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed. Sealed envelopes used
Blinding of outcome assessment (detection bias)	Low risk	Data collected by computer searches
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data from all 28 practices
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Figure 1
Protection against contamination	Low risk	Separate practices
Incorrect analysis	High risk	Clustering not taken into account
Risk of bias overall	High risk	Due to incorrect analysis

McClellan 2003

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 477 N patients: 22,971
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + guidelines + educational materials + guideline implementation aids
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: written Source: administrative/billing data Frequency: semi-monthly Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random number table
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Data collected from database
Incomplete outcome data (attrition bias) All outcomes	Low risk	Similar dropouts in both groups
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Table 1
Protection against contamination	Low risk	Separate practices

McClellan 2003 *(Continued)*

Incorrect analysis	High risk	Clustering was not taken into account in the analysis.
Risk of bias overall	High risk	Due to incorrect analysis

McClellan 2004

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: outpatient end-stage renal disease treatment centre Speciality: nephrology N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + clinician & patient educational materials + quarterly monitoring + assistance by network staff; Intervention 2: A&F
Outcomes	Clinical behaviour category: prescribing, treatment decision/action Format of Audit & Feedback: both written and verbal Source: other Frequency: not reported/not applicable Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random number allocation
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Lab values blinded
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data from all centres in study
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured

McClellan 2004 *(Continued)*

Recruitment bias	Low risk	No recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 1
Protection against contamination	Low risk	Separate centres
Incorrect analysis	Low risk	Analysed at cluster level; unit of analysis same as unit of allocation
Risk of bias overall	Low risk	Low risk across all domains

McCluskey 2016

Study characteristics		
Methods	Design: parallel cluster-RCT	
Participants	Country: Australia Setting: stroke rehabilitation centre Speciality: NA N health professionals: NR N patients: 263	
Interventions	Number of arms: 2 Description of groups: Control: mailed guidelines; Intervention: training workshop + educational materials + A&F	
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: once Instructions for improvement: specific	
Notes		
Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	An 'independent randomisation service' was used to allocate clusters to experimental or control groups. Minimisation was used, a form of covariate adaptive randomisation to ensure a balance of variables; location (centre or home-based teams), funding (public or private), volume of caseload and level of outings at baseline

McCluskey 2016 *(Continued)*

Allocation concealment (selection bias)	Low risk	Authors stated 'concealed allocation.' As minimisation was used to randomise teams, the sites would have been recruited before the random sequence was generated, reducing selection bias.
Blinding of outcome assessment (detection bias)	Low risk	Patient files were audited by an individual blinded to team allocation. Therapists documenting in files may have been unblinded, but this is unlikely to affect reporting of outcome (whether a community outing occurred or not).
Incomplete outcome data (attrition bias) All outcomes	Low risk	No loss to follow-up in outcome data of primary outcome due to method (audit of patient files). Loss to follow-up of observed stroke survivors of 13% in both intervention and control groups; this population related to secondary outcomes only. Staff turnover high ("up to 50%") during study but did not impact data due to patient measures used as outcomes 1 cluster lost to follow-up due to service cessation
Selective reporting (reporting bias)	Low risk	Reports primary and secondary outcome data as outlined in published protocol
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Adequate similarity; stroke survivors similar in terms of age, sex and stroke severity between groups. Characteristics of teams similar at baseline in terms of funding; minor differences between groups in location; and control group had more outpatient settings than intervention group. Similar baseline characteristics: low risk, adequate similarity; stroke survivors similar in terms of age, sex and stroke severity between groups. Characteristics of teams similar at baseline in terms of funding; minor differences between groups in location; control group had more outpatient settings than intervention group. Control group providing fewer outings at baseline but direction of potential bias unlikely to impact results
Protection against contamination	Low risk	Allocation was by community rehabilitation teams. Unlikely that the control group received the intervention.
Incorrect analysis	Low risk	Analysis was performed taking cluster effect into account.
Risk of bias overall	Low risk	Low risk across all domains

McConnell 1982

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: ambulatory care clinic Speciality: GP/family N health professionals: 35 N patients: 801
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + educational materials

McConnell 1982 *(Continued)*

Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: verbal (didactic) Source: administrative/billing data Frequency: once Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Medicaid data used
Incomplete outcome data (attrition bias) All outcomes	Low risk	33/35 physicians followed up
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	High risk	Clustering was not taken into account in the analysis.
Risk of bias overall	High risk	Due to incorrect analysis

Meeker 2016

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: Primary care Speciality: GP/Internal medicine/family

Meeker 2016 *(Continued)*

	N health professionals: 248 N patients: NR
Interventions	Number of arms: 8 Description of groups: Control: no intervention; Intervention 1: suggested alternatives (CDS alert to physician with alternatives to antibiotics); Intervention 2: accountable justification (CDS alert prompting physician to enter justification for prescribing abx); Intervention 3: peer comparison A&F; Intervention 4: suggested alternatives + accountable justification; Intervention 5: suggested alternatives + peer comparison A&F; Intervention 6: accountable justification + peer comparison A&F: Intervention 7: suggested alternatives + accountable justification + peer comparison A&F
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: computerised search of charts Frequency: monthly Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomization conducted in R
Allocation concealment (selection bias)	Low risk	Clusters randomised before start of trial
Blinding of outcome assessment (detection bias)	Low risk	Objective data source (EHR or billing records)
Incomplete outcome data (attrition bias) All outcomes	Low risk	No loss to follow-up at cluster level
Selective reporting (reporting bias)	Low risk	Prospectively registered protocol
Recruitment bias	Low risk	No evidence that recruitment occurred after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline characteristics appear mostly similar.
Protection against contamination	Low risk	Factorial design reduces contamination across clusters.
Incorrect analysis	Unclear risk	Adjustments for potential clustering effect not mentioned
Risk of bias overall	Low risk	Low risk across most domains

Mertens 2015

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 556 N patients: 639,613
Interventions	Number of arms: 3 Description of groups: Control: no intervention; Intervention 1: education + alcohol screening training session for primary care physicians + A&F; Intervention 2: education + alcohol screening training session for non-physician providers and medical assistants + A&F
Outcomes	Clinical behaviour category: screening, counselling Format of Audit & Feedback: both written and verbal Source: computerised search of charts, administrative/billing data Frequency: quarterly Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	It is unclear how the random sequence was generated.
Allocation concealment (selection bias)	Low risk	Though not explicitly described, it would appear all clusters were allocated at one time.
Blinding of outcome assessment (detection bias)	Unclear risk	Some outcomes (not primary) were based on self-report and participants were aware of their allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	3/201 in PCP group lost to follow-up (none in other two arms). Unlikely to bias outcomes
Selective reporting (reporting bias)	Unclear risk	Outcomes listed in clinical trials record were all reported; however, the proportion of brief interventions and proportion of referrals to treatment were combined - uncertain how this may have impacted results
Recruitment bias	Low risk	Recruitment was done at one stage and no clinics or PCPs were recruited after the commencement of the study.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Baseline characteristics for each group not reported

Mertens 2015 *(Continued)*

Protection against contamination	Unclear risk	Post hoc analysis for medical centre 9 found that a few physicians in the NPP and MA arm delivered BI/RT themselves to their few patients screening positive; these physicians were responsible for the vast majority of BI/ RTs delivered in this clinic. Some movement of MAs from the NPP and MA arm to other arms occurred in these medical centres; in some sites, MAs covered duties for absent MAs in other clinics.
Incorrect analysis	Low risk	Analysis accounted appropriately for clustering
Risk of bias overall	Unclear risk	Due to baseline imbalances and risk of contamination

Metlay 2007

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: emergency departments Speciality: emergency medicine N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + clinician education sessions + patient education
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: womputerised search of charts Frequency: annually Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Coin flip randomisation used
Allocation concealment (selection bias)	Unclear risk	No details or study flow diagram to determine if clusters were likely allocated at the same time

Metlay 2007 *(Continued)*

Blinding of outcome assessment (detection bias)	Low risk	Outcomes (collected from medical records) unlikely to be influenced by knowledge of allocation. Self-reporting by patients also unlikely to be affected because it is unlikely they were aware of their allocation
Incomplete outcome data (attrition bias) All outcomes	Low risk	1/8 control sites lost to follow-up (unable to identify appropriate staff to participate in the study). This is unlikely to be a significant imbalance between groups.
Selective reporting (reporting bias)	Unclear risk	No trial registration or protocol found
Recruitment bias	Low risk	All practices recruited before randomisation
Baseline imbalance in outcome measurements or characteristics	High risk	Table 2 - differences observed between control and intervention groups in year 1 for type of hospital, age, sex, previous ARI, radiography ordered, and provider type
Protection against contamination	Low risk	Allocated by cluster (hospitals) and it is unlikely that the control hospitals received the intervention
Incorrect analysis	Low risk	Clustering taken into account in primary analysis
Risk of bias overall	High risk	Baseline imbalance at high risk

Mitchell 2005

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: NR N patients: 30,345
Interventions	Number of arms: 3 Description of groups: Control: no intervention; Intervention 1: A&F; Intervention 2: A&F + risk information
Outcomes	Clinical behaviour category: treatment decision/action, testing/exams Format of Audit & Feedback: written Source: computerised search of charts Frequency: variable Instructions for improvement: not reported
Notes	

Mitchell 2005 *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Administrative data
Incomplete outcome data (attrition bias) All outcomes	Low risk	52/54 GPs provided data.
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No participants were enroled post-randomisation of clusters; hence, there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 1
Protection against contamination	Low risk	Separate GPs
Incorrect analysis	Low risk	Clustering accounted for in analysis
Risk of bias overall	Low risk	Low risk across all domains

Moher 2001

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: NR N patients: 1906
Interventions	Number of arms: 3 Description of groups: Intervention 1: A&F; Intervention 2: A&F + assistance setting up disease register and systematic recall of patients to general practitioner; Intervention 3: A&F + assistance setting up disease register and systematic recall of patients to nurse-led clinic
Outcomes	Clinical behaviour category: screening, prescribing, testing/exams

Moher 2001 *(Continued)*

	Format of Audit & Feedback: verbal (interactive) Source: manual chart review, computerised search of charts Frequency: not reported/not applicable Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data on all 21 practices
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Unclear risk	Limited information on recruitment process (79 practices invited but "eventually recruited 21 practices")
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 1
Protection against contamination	Low risk	Separate practices
Incorrect analysis	Unclear risk	No information on statistical methodology; unclear whether clustering was considered in analysis
Risk of bias overall	Unclear risk	Unclear risk across important domains

Mold 2008

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: NR

Mold 2008 *(Continued)*

	N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F; Intervention 2: A&F + academic detailing + practice facilitation + IT support
Outcomes	Clinical behaviour category: screening, testing/exams, immunisation Format of Audit & Feedback: both written and verbal Source: manual chart review, computerised search of charts Frequency: not reported/not applicable Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Coin tosses
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Interviewers blinded
Incomplete outcome data (attrition bias) All outcomes	Low risk	All clinicians followed up
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No evidence of recruiting clusters after randomisation
Baseline imbalance in outcome measurements or characteristics	High risk	Table 1
Protection against contamination	Low risk	Separate practices
Incorrect analysis	Low risk	Logistic regression
Risk of bias overall	High risk	Due to baseline imbalance

Mold 2014

Study characteristics

Methods	Design: parallel cluster-RCT

Mold 2014 *(Continued)*

Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: NR N patients: 1016
Interventions	Number of arms: 4 Description of groups: Intervention 1: A&F + academic detailing + asthma guidelines + asthma toolkit + action plan templates; Intervention 2: A&F + academic detailing + asthma guidelines + asthma toolkit + action plan templates + assistance from practice facilitator; Intervention 3: A&F + academic detailing + asthma guidelines + asthma toolkit + action plan templates + local learning collaborative; Intervention 4: A&F + academic detailing + asthma guidelines + asthma toolkit + action plan templates + assistance from practice facilitator + local learning collaborative
Outcomes	Clinical behaviour category: screening, prescribing, counselling Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: monthly Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Random assignment, but method of sequence generation not reported
Allocation concealment (selection bias)	Unclear risk	No information regarding allocation concealment and whether this occurred at the same time or centrally
Blinding of outcome assessment (detection bias)	Unclear risk	Outcome assessors at two levels - practitioners recording asthma management (unlikely to be affected by knowledge of allocation - low risk) and data extraction for outcomes (unclear if assessors were blinded and though standardised forms were used, the data collection process was not entirely objective and could be at risk of bias)
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No information about flow of participation. 2 practices did not provide outcome data and unclear which groups they were from as # randomised to each group at the beginning not reported
Selective reporting (reporting bias)	Unclear risk	Protocol not available; however, all outcomes reported
Recruitment bias	Unclear risk	No information about recruitment (and subsequent randomisation) methods, so timing of recruitment is unclear
Baseline imbalance in outcome measurements or characteristics	Unclear risk	There appeared to be baseline imbalances in some of the outcomes between groups (table 3) though this is not described by the trial, so it is unclear if this was significant.

Mold 2014 *(Continued)*

Protection against contamination	Unclear risk	Not enough information to assess
Incorrect analysis	High risk	Trial did not adjust for clustering.
Risk of bias overall	High risk	Incorrect analysis judged as high risk; recruitment bias, baseline imbalance, selective reporting and protection against contamination judged as unclear

Morrison 2015

Study characteristics	
Methods	Design: step wedge
Participants	Country: Canada Setting: hospital inpatient Speciality: emergency medicine N health professionals: NR N patients: 934
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + local detailing (education)
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: variable Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	A computerised randomisation scheme was used.
Allocation concealment (selection bias)	Low risk	Allocation centralised; appears to have been finished before study started and after 4 hospitals couldn't partake.
Blinding of outcome assessment (detection bias)	Low risk	Medical records unlikely to be subjectively recorded. Data abstractors were blind to allocations.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Modified ITT used due to 5 hospitals dropping out (unclear on allocation) due to reasons that were not related to the intervention

Morrison 2015 *(Continued)*

Selective reporting (reporting bias)	High risk	No odds ratio for active vs baseline data reported protocol: https://implementationscience.biomedcentral.com/articles/10.1186/1748-5908-6-4
Recruitment bias	Low risk	Recruitment occurred before trial started.
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline balanced
Protection against contamination	Low risk	This does not appear to have been contamination with different hospitals being separated into clusters.
Incorrect analysis	Low risk	All cluster trials should allow for correlations between individuals in the same cluster. A consequence is that a parallel cluster trial will require a larger sample size than a corresponding individually randomised trial. The standard approach here assumes that any two observations from the same cluster will have a constant correlation between them, a quantity known as the intra-cluster correlation. Then the increase in sample size for the simple parallel cluster trial is determined by a simple multiplicative factor (the design effect), which depends both on the magnitude of the intra-cluster correlation and on the number of subjects in each cluster.
Risk of bias overall	High risk	Due to selective reporting

Myers 2004

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: NR N patients: 2992
Interventions	Number of arms: 2 Description of groups: Control: reminders; Intervention: A&F + CME reminders + educational outreach
Outcomes	Clinical behaviour category: diagnosis Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: variable Instructions for improvement: co-developed plans
Notes	
Risk of bias	

Myers 2004 *(Continued)*

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random number generation used
Allocation concealment (selection bias)	Low risk	Randomisation was done at the practice level, and appears to have been conducted at one time and prior to the study commencing. https://www.ncbi.nlm.nih.gov/pubmed/11570828
Blinding of outcome assessment (detection bias)	Low risk	Outcome assessors at two levels - physicians (filling in internal chart audit form) and researchers extracting data from ICA for the outcome of interest (CDE recommendation). Despite participants knowing their allocation, it is unlikely there would be a difference in how they fill in the ICA). Researchers assessing the ICA data developed a set of decision rules, previously validated this method in a pilot study, and the data were dichotomous (likely low risk)..
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No information about whether practices and PCPs remained for the duration of the study or if some left
Selective reporting (reporting bias)	High risk	Myers 2001 (study design, implementation and baseline data publication) says their secondary outcomes are the intention to recommend and perform CDE, which they report baseline data for. This is missing in the 2004 publication.
Recruitment bias	Low risk	Randomisation performed after practices were recruited. No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Similar baselines as per table 2 and 3, showing no significant differences
Protection against contamination	Unclear risk	Allocation at the cluster level but not enough information to determine if trial ensured practitioners did not work in other practices etc.
Incorrect analysis	Low risk	Generalised estimating equation (GEE) methods were used to adjust for the clustering of patients within practices.
Risk of bias overall	High risk	Due to selective reporting

Nace 2020

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: nursing homes Speciality: long-term care N health professionals: NR N patients: NR
Interventions	Number of arms: 2

Nace 2020 *(Continued)*

	Description of groups: Control: no intervention; Intervention: A&F + educational materials + web-based coaching
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: not reported/not applicable Frequency: not reported/not applicable Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Randomisation method not described
Allocation concealment (selection bias)	Low risk	No details but based on the study flow diagram, all clusters were likely allocated at the same time.
Blinding of outcome assessment (detection bias)	Unclear risk	Collection of outcome data are not clearly described, so it is difficult to determine whether the process could be subject to bias.
Incomplete outcome data (attrition bias) All outcomes	High risk	Three out of 13 (23%) control sites were lost to follow-up (2 were withdrawals but no further elaboration) compared to 0 in the intervention arm. Incomplete outcome data are imbalanced and potentially related to the intervention and outcome.
Selective reporting (reporting bias)	Unclear risk	The trial was not registered and no study protocol was found.
Recruitment bias	Low risk	No evidence of recruiting clusters after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	No statistically significant differences between groups at baseline, and stratified in randomisation to minimise differences
Protection against contamination	Low risk	Randomised by practice and unlikely contamination
Incorrect analysis	Low risk	Appropriate method of analysis which accounted for clustering effects was used.
Risk of bias overall	High risk	High risk due to incomplete outcome data

Naughton 2007

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Ireland

Naughton 2007 *(Continued)*

	Setting: primary care Speciality: GP/family N health professionals: 110 N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F only; Intervention 2: A&F + academic detailing
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: administrative/billing data Frequency: once Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Administrative data used
Incomplete outcome data (attrition bias) All outcomes	Low risk	1 dropout of 98 GPs
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No evidence of recruiting clusters after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Table 1
Protection against contamination	Low risk	Separate GPs
Incorrect analysis	Low risk	"The prescribing differences between the randomised groups were examined using multivariate regression with a random effects model for each therapy, e.g. statin therapy in patients with CVD. This allowed for adjustment of GP clustering within practices, baseline prescribing and confounding factors such as practice participation in the Heartwatch programme and patient population structure."

Naughton 2007 *(Continued)*

Risk of bias overall	Low risk	Low risk across most domains

Naughton 2009

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Ireland Setting: primary care Speciality: GP/family N health professionals: 110 N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F only; Intervention 2: A&F + academic detailing
Outcomes	Clinical behaviour category: Format of Audit & Feedback: Source: administrative/billing data Frequency: once Instructions for improvement:
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Randomisation process not described
Allocation concealment (selection bias)	Unclear risk	Allocation concealment not described
Blinding of outcome assessment (detection bias)	Low risk	Data collected from database of pharmacy claims
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Losses not reported
Selective reporting (reporting bias)	Unclear risk	Protocol not available; however, all outcomes reported
Recruitment bias	Low risk	No evidence that recruitment occurred after randomisation

Naughton 2009 *(Continued)*

Baseline imbalance in outcome measurements or characteristics	Low risk	No significant differences in characteristics or overall prescribing patterns
Protection against contamination	Low risk	Cluster design; unlikely to be contamination
Incorrect analysis	Unclear risk	No evidence of adjustment for clustering effect
Risk of bias overall	Unclear risk	Unclear risk across most domains

Navarro 2012

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Australia Setting: primary care Speciality: GP N health professionals: 275 N patients: 5529
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + educational materials
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: administrative/billing data Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	A blocked randomisation sequence was generated using Stata Statistical Software (information sourced from http://anzctr.org.au); hence there is a low risk of selection bias.
Allocation concealment (selection bias)	Low risk	"Allocation to intervention/control is performed later by the holder of the allocation schedule who is at the central administration site i.e. allocation is concealed" (information from http://anzctr.org.au); hence there is a low risk of selection bias.

Navarro 2012 *(Continued)*

Blinding of outcome assessment (detection bias)	Low risk	Unclear if assessor/s were blinded, but pharmacy cost data and hospitalisation rates were collected, so unlikely to matter if not
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	It is unclear if GPs were recruited to participate or they were all included; therefore, excluded/missing GP data are unclear.
Selective reporting (reporting bias)	Low risk	The trial was registered and no deviations from protocol were found; hence, there is a low risk of reporting bias.
Recruitment bias	Unclear risk	As there was no study flow diagram or written text reporting on whether recruitment continued post-randomisation or not, there is an unclear risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Low risk	There is a low risk of bias due to baseline imbalance as baseline characteristics between groups were comparable.
Protection against contamination	Low risk	There is a low risk of contamination bias, as adequate steps were taken to minimise contamination between groups.
Incorrect analysis	Low risk	Communities were units of analysis.
Risk of bias overall	Low risk	Low risk across most domains

Navathe 2020

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/internal medicine/family/paediatrics N health professionals: 88 N patients: 74,778
Interventions	Number of arms: 3 Description of groups: Intervention 1: individual A&F; Intervention 2: peer comparison A&F; Intervention 3: peer comparison A&F + financial incentive
Outcomes	Clinical behaviour category: screening, testing/exams, treatment decision/action, counselling, immunisation Format of Audit & Feedback: electronic dashboard Source: administrative/billing data Frequency: semi-monthly/variable Instructions for improvement: general

Navathe 2020 *(Continued)*

Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Not enough information to determine method of randomisation
Allocation concealment (selection bias)	Low risk	Per supplementary document, it appears clusters were allocated at the same time.
Blinding of outcome assessment (detection bias)	Low risk	Outcome data extracted from medical records and unlikely to be biased
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No clear description of flow of participants throughout study. Participation in the third arm was low.
Selective reporting (reporting bias)	High risk	Primary outcome was changed on clinical trials registry after data analysis (though authors specify in the appendix this decision was made before data analysis). Two arms were also combined due to low engagement.
Recruitment bias	Low risk	As per study timeline in appendix, no recruitment after randomisation and allocation
Baseline imbalance in outcome measurements or characteristics	Low risk	Analyses adjusted for baseline imbalances.
Protection against contamination	High risk	Authors acknowledge likelihood of contamination (and bias towards the null) due to providers within the same organisation being randomised to different arms.
Incorrect analysis	Low risk	Clustering taken into account in analyses
Risk of bias overall	High risk	Protection against contamination - high risk

Nejad 2016

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Iran Setting: primary care Speciality: GP N health professionals: 906 N patients: NR
Interventions	Number of arms: 3

Nejad 2016 *(Continued)*

	Description of groups: Intervention 1: paper letter A&F on ratio of antibiotics, parenteral steroids, and systemic steroids; Intervention 2: paper letter A&F on ratio of antibiotics, parenteral steroids; Intervention 3: text message A&F on parenteral steroids
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: computerised search of charts Frequency: quarterly Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	High risk	Sequence generation not reported. Control arm was not randomly allocated and appears to be GPs that the intervention could not be sent to; however, this is poorly described
Allocation concealment (selection bias)	Low risk	Based on participant flow diagram, it seems likely all participants were allocated centrally at one time.
Blinding of outcome assessment (detection bias)	Low risk	Outcome data were extracted from medical records and unlikely to be recorded with bias.
Incomplete outcome data (attrition bias) All outcomes	Low risk	An ITT analysis was also performed for excluded cases.
Selective reporting (reporting bias)	Unclear risk	Outcomes seem to be reported but no trial registration or protocol to check
Recruitment bias	Low risk	Fig 1: All recruitment done before randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Characteristics similar at baseline
Protection against contamination	Unclear risk	GP appears to be the cluster - no information provided about whether multiple GPs from the same clinics were randomised differently
Incorrect analysis	Low risk	Analysis adjusted for clusters
Risk of bias overall	High risk	Due to random sequence generation

Nilsson 2001a

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Sweden

Nilsson 2001a *(Continued)*

	Setting: primary care Speciality: GP/family N health professionals: 40 N patients: 95,140
Interventions	Number of arms: 3 Description of groups: Control: no intervention; Intervention: A&F + educational session and materials
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Prescription data used
Incomplete outcome data (attrition bias) All outcomes	Low risk	40 GPs provided data.
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	Randomisation occurred after recruitment was complete.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Low risk	Separate GPs
Incorrect analysis	Unclear risk	No adjustment for clustering mentioned (t-tests used to compare changes in mean differences, but units of analysis and randomization are both with individual GPs)
Risk of bias overall	Low risk	Low risk across all important domains

Nilsson 2001b

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Sweden Setting: primary care Speciality: GP/family N health professionals: 40 N patients: 95,140
Interventions	Number of arms: 3 Description of groups: Control: no intervention; Intervention: A&F + educational session and materials
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Prescription data used
Incomplete outcome data (attrition bias) All outcomes	Low risk	40 GPs provided data.
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	Randomisation occurred after recruitment was complete.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Low risk	Separate GPs

Nilsson 2001b *(Continued)*

Incorrect analysis	Unclear risk	No adjustment for clustering mentioned (t-tests used to compare changes in mean differences, but units of analysis and randomisation are both with individual GPs)
Risk of bias overall	Low risk	Low risk across all important domains

O'Connell 1999

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Australia Setting: primary care Speciality: GP/family N health professionals: 2440 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + educational newsletter
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: administrative/billing data Frequency: semi-annually Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Stratification with block size of 4
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Prescribing rates objective data
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data from all subjects
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured

O'Connell 1999 *(Continued)*

Recruitment bias	Low risk	All eligible general practices were randomised (based on postcode).
Baseline imbalance in outcome measurements or characteristics	Low risk	Table
Protection against contamination	Low risk	Avoided by postal codes
Incorrect analysis	Low risk	Adjustments for clustering described
Risk of bias overall	Low risk	Low risk across all domains

O'Connor 2009

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family physicians N health professionals: 123 N patients: 3703
Interventions	Number of arms: 4 Description of Groups: Control: no intervention; Intervention 1: patient-focused intervention; Intervention 2: A&F + physician reminders + educational materials (physician only); Intervention 3: A&F + educational materials (physician and patients)
Outcomes	Clinical behaviour category: prescribing, testing/exams Format of Audit & Feedback: written Source: computerised search of charts Frequency: tri-annually Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed

O'Connor 2009 *(Continued)*

Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unable to assess
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Unclear risk	Recruitment process not discussed in detail
Baseline imbalance in outcome measurements or characteristics	Low risk	Table 1
Protection against contamination	Low risk	Separate GPs
Incorrect analysis	Low risk	Adjustments for clustering described
Risk of bias overall	Unclear risk	Unable to assess bias across important domains

Ornstein 2004

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 61 N patients: 87,291
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + academic detailing + guidelines/best practices
Outcomes	Clinical behaviour category: screening, prescribing Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: quarterly Instructions for improvement: general
Notes	
Risk of bias	

Ornstein 2004 *(Continued)*

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Baseline adaptive randomisation scheme
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Data from computerised source
Incomplete outcome data (attrition bias) All outcomes	Low risk	3/20 subjects dropped out
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Low risk	Separate GPs
Incorrect analysis	Low risk	Random effects for practices used in mixed models to adjust for clustering
Risk of bias overall	Low risk	Low risk across most domains

Ornstein 2010

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: NR N patients: 68,150
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + academic detailing + guidelines/best practices
Outcomes	Clinical behaviour category: screening Format of Audit & Feedback: written Source: computerised search of charts

Ornstein 2010 *(Continued)*

	Frequency: quarterly Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	As per study flow diagram, allocation was likely done centrally at one time.
Blinding of outcome assessment (detection bias)	Low risk	Outcome measurement is not likely to be influenced by lack of blinding.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No missing data noted
Selective reporting (reporting bias)	Unclear risk	Protocol not available. Outcomes reported in Table 4 as prespecified in Methods
Recruitment bias	Low risk	No evidence of recruitment after randomisation and allocation
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline characteristics and outcomes balanced between groups, see tables 3 and 4.
Protection against contamination	Low risk	Control practices unlikely to receive intervention
Incorrect analysis	Low risk	Clustering accounted for in analysis
Risk of bias overall	Low risk	Low risk across most domains

Overbeek 2010

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: laboratories Speciality: pathology N health professionals: NR N patients: 266
Interventions	Number of arms: 2

Overbeek 2010 *(Continued)*

	Description of groups: Control: no intervention; Intervention: A&F + guidelines + education + reminders
Outcomes	Clinical behaviour category: testing/exams, referrals Format of Audit & Feedback: written Source: computerised search of charts Frequency: quarterly Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomisation method not described
Allocation concealment (selection bias)	Unclear risk	No details provided
Blinding of outcome assessment (detection bias)	Unclear risk	It was unclear who the outcome assessors were and then even how they were blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No clusters lost/excluded. Some patients were excluded for reasons provided and not in relation to the intervention.
Selective reporting (reporting bias)	Unclear risk	There is no protocol available, but all outcomes are reported as prespecified in the Methods.
Recruitment bias	Low risk	No recruitment of hospitals after allocation. All patients meeting criteria were eligible - no proactive recruitment
Baseline imbalance in outcome measurements or characteristics	Unclear risk	It is unclear whether adjustments were made at baseline. Baseline characteristics may have some imbalances, but this is not discussed.
Protection against contamination	Low risk	Due to cluster design with hospitals as the unit, contamination is unlikely.
Incorrect analysis	Low risk	ORs adjusted for clustering for identification of patients at risk for Lynch syndrome and for applying MSI testing
Risk of bias overall	Unclear risk	Unclear risk across important domains

Palmer 1985a

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA

Palmer 1985a *(Continued)*

	Setting: primary care Speciality: GP/family/paediatrics N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: clinician educational meeting + clinician educational materials + A&F
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: both written and verbal Source: manual chart review, administrative/billing data Frequency: once Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Blocked randomisation
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	High turnover of professionals; impact unclear
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No evidence of additional recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Low risk	Separate clusters
Incorrect analysis	Low risk	GLM with mixed effects
Risk of bias overall	Low risk	Low risk across most important domains

Palmer 1985b

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family/paediatrics N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: clinician educational meeting + clinician educational materials + A&F
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: both written and verbal Source: manual chart review, administrative/billing data Frequency: once Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Blocked randomisation
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	High turnover of professionals; impact unclear
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No evidence of additional recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Low risk	Separate clusters

Palmer 1985b *(Continued)*

Incorrect analysis	Low risk	GLM with mixed effects
Risk of bias overall	Low risk	Low risk across most important domains

Papadakis 2018

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Canada Setting: primary care Speciality: GP/family N health professionals: 166 N patients: 1990
Interventions	Number of arms: 2 Description of groups: Control: education + reminders + patient materials; Intervention: A&F + provider education + reminders + patient materials
Outcomes	Clinical behaviour category: screening, prescribing, counselling Format of Audit & Feedback: both written and verbal Source: patient-reported data Frequency: once Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	A methods group performed randomisation; thus, it is likely that sequence generation was adequate.
Allocation concealment (selection bias)	Low risk	Allocation was done centrally by a methods group and likely concealed.
Blinding of outcome assessment (detection bias)	Low risk	Outcomes were self-reported by patients, who were blinded to their practice's allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	73% of OMSC group and 81% of OMSC+ group contributed follow-up data and there are no reasons described for missing outcome data. However trial was by ITT.
Selective reporting (reporting bias)	Low risk	All outcomes specified in the protocol are reported.

Papadakis 2018 *(Continued)*

Recruitment bias	Low risk	There is no evidence of recruitment after sites were randomised and allocated.
Baseline imbalance in outcome measurements or characteristics	Low risk	No major differences between groups
Protection against contamination	Low risk	Due to the cluster design, it is unlikely that the OMSC sites received the enhanced intervention.
Incorrect analysis	Low risk	Clustering taken into account in analysis
Risk of bias overall	Low risk	Low risk across all domains

Pape 2011

Study characteristics		
Methods	Design: parallel cluster-RCT	
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 68 N patients: 6963	
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + decision-support + team-based care; Intervention 2: A&F + decision-support	
Outcomes	Clinical behaviour category: prescribing, testing/exams Format of Audit & Feedback: electronic dashboard Source: computerised search of charts Frequency: not reported/not applicable Instructions for improvement: general	
Notes		
Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Method of sequence generation is not reported.
Allocation concealment (selection bias)	Low risk	Secure cluster allocation used

Pape 2011 *(Continued)*

Blinding of outcome assessment (detection bias)	Low risk	Outcomes were collected by medical records and unlikely to be biased by knowledge of group allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No clusters lost to follow-up, but proportions of patients changed: Intervention group total remained approx same: 28% (lost) + 26% (new) Control group increased by approx 18%: 30% (lost) + 48% (new) (Fig ?, p. 2).
Selective reporting (reporting bias)	Unclear risk	There is no trial registration or previously published protocol, so it is unclear if all preplanned outcomes were reported in this publication.
Recruitment bias	Low risk	No evidence that any recruitment occurred after allocation as all eligible clinics were invited to participate
Baseline imbalance in outcome measurements or characteristics	Low risk	"Clinics were randomised in a 1:2 intervention-to-control schedule using secure cluster allocation. Unequal allocation was used to accommodate pharmacist workload capacity... Clinics were matched based on the panel size of patients with DM, mean age of the DM panel, and proportion of patients with commercial insurance." (p. 2)
Protection against contamination	Low risk	"Cluster randomisation was used to minimise confusion on the part of physicians and staff and to limit contamination bias". (p. 6)
Incorrect analysis	Low risk	Analyses was performed adjusted for clustering.
Risk of bias overall	Low risk	Low risk across most domains

Patel M 2018

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 96 N patients: 4774
Interventions	Number of arms: 3 Description of groups: Control: no intervention; Intervention 1: A&F (no comparison); Intervention 2: A&F (peer comparison)
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: electronic dashboard Source: computerised search of charts Frequency: once Instructions for improvement: general

Patel M 2018 *(Continued)*

Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Electronic randomisation is likely to have been set up by statisticians and hence adequate.
Allocation concealment (selection bias)	Low risk	While not explicitly stated, given that randomisation was performed centrally and trialists report that "All investigators, statisticians, and data analysts were blinded to arm assignments until the study and analysis were completed", it seems highly likely that the study ensured allocation was performed centrally and concealed.
Blinding of outcome assessment (detection bias)	Low risk	All personnel involved in outcome assessment were unaware of allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No missing outcomes and ITT analyses performed
Selective reporting (reporting bias)	Low risk	Prespecified outcome reported
Recruitment bias	Low risk	No recruitment of PCPs/patients after allocation. Eligible PCPs and patients were included in the study without needing to inform them.
Baseline imbalance in outcome measurements or characteristics	Low risk	Patients differed slightly across arms in a few sociodemographic and baseline characteristics but were mostly well balanced (Table 2) in the active as the clustering unit. To test the robustness of our findings, we conducted a sensitivity analysis in which the main model was estimated by adjusting for patient characteristics, including age, sex, race/ethnicity, insurance type, median annual household income by zip code, Charlson comorbidity index score, body mass index, most recent LDL-C level measurement, 10-year ASCVD risk score, clinical ASCVD, congestive heart failure, diabetes, hypertension, and smoking history, as well as PCP panel size. In post hoc exploratory analyses, the sensitivity model was further adjusted for PCP degree and using interaction terms of the study arm variables with race/ethnicity and separately with insurance type. To obtain the adjusted difference, 4774 patients were randomised (Figure). The PCPs differed slightly across arms for type of medical degree, but all other characteristics were well balanced (Table 1).
Protection against contamination	Low risk	PCPs were unaware they were involved in the trial, and it is unlikely that there would be contamination between arms.
Incorrect analysis	Low risk	Cluster as the unit was used in the analysis.
Risk of bias overall	Low risk	Low risk across all domains

Patel S 2018

Study characteristics

Methods	Design: parallel cluster-RCT

Patel S 2018 *(Continued)*

Participants	Country: USA Setting: hospital inpatient Speciality: internal medicine N health professionals: 118 N patients: 1069
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F only; Intervention 2: A&F + online dashboard + in-person data review
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: electronic dashboard Source: computerised search of charts Frequency: bi-weekly Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Coin tossing to generate random sequence
Allocation concealment (selection bias)	Unclear risk	Unclear if clusters were recruited before randomisation or if they occurred on a single occasion
Blinding of outcome assessment (detection bias)	Low risk	Primary and secondary outcome data were automatically recorded from electronic records and unlikely subject to any bias. Study investigators recorded attendance at STAT rounds (unlikely subject to bias) and surveyed participants on perception of the intervention (knowledge of allocation would not influence reporting of experience of intervention). Analyses were performed and conducted by personnel blinded to group allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Small loss to follow-up and reasons given
Selective reporting (reporting bias)	High risk	The trial registration has four primary outcomes (which is split into one primary outcome and three secondary outcomes in the trial publication), and four secondary outcomes (only one of which is described in the methods and reported in the results; three preplanned outcomes not reported - total phlebotomy sticks, percent of patients on telemetry until discharge, and proportion of patients discharged by noon). One outcome not in the trial registration added - 30-day patient readmission
Recruitment bias	Low risk	No evidence of recruitment to the trial after the start of the trial

Patel S 2018 *(Continued)*

Baseline imbalance in outcome measurements or characteristics	Low risk	The clinical characteristics of patients discharged from the intervention and control groups did not differ significantly between groups.
Protection against contamination	High risk	It appears that within medical teams, different physicians may have been allocated to different trial arms so there may have been risk of contamination.
Incorrect analysis	Low risk	Analysis performed at the cluster level
Risk of bias overall	High risk	High due to potential risk of contamination

Pavese 2014

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: France Setting: hospital inpatient Speciality: internal medicine/paediatrics/emergency medicine N health professionals: NR N patients: 249
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F only; Intervention 2: A&F + training session
Outcomes	Clinical behaviour category: prescribing, testing/exams Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: once Instructions for improvement: not reported
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Not enough information to determine if a truly random method was used to allocate stratified groups
Allocation concealment (selection bias)	Unclear risk	No details or study flow diagram to determine if allocation concealment was likely done, even though trialists stated that cluster-RCT consort criteria were met
Blinding of outcome assessment (detection bias)	Low risk	A specialist with no knowledge of group allocation collected data from electronic records onto standardised data collection forms.

Pavese 2014 *(Continued)*

Incomplete outcome data (attrition bias) All outcomes	Low risk	Different cohorts and numbers of patient participants were included for each audit (therefore no missing outcome data, as available patient data at each time point obtained and analysed). No evidence that medical departments were lost to follow-up
Selective reporting (reporting bias)	Unclear risk	There is no previously published protocol or trial registration, so it is unclear if all preplanned outcomes and analysis were reported.
Recruitment bias	Low risk	No participants recruited after group recruitment
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Baseline imbalances in characteristics and outcome data are not described, but there appear to be differences in characteristics (e.g. type of department) and outcomes (e.g. sufficient volume of blood). It is unclear if these differences are significant.
Protection against contamination	High risk	Although this was a cluster randomised trial, all clusters (medical departments) were located in one hospital. It is possible that medical professionals may have discussed the intervention across different clusters.
Incorrect analysis	High risk	Trialists did not perform analysis at the cluster level, and it is unclear if their analyses were sufficient.
Risk of bias overall	High risk	High risk due to contamination and incorrect analysis

Peiris 2015

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Australia Setting: primary care Speciality: GP/family N health professionals: NR N patients: 38,877
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + training + computerised decision-support
Outcomes	Clinical behaviour category: screening, prescribing, testing/exams Format of Audit & Feedback: electronic dashboard Source: computerised search of charts Frequency: daily Instructions for improvement: general
Notes	

Peiris 2015 *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Sequence generation was random as it was generated by a web-based platform (via trial registration).
Allocation concealment (selection bias)	Low risk	Allocation was performed centrally.
Blinding of outcome assessment (detection bias)	Low risk	Outcomes were collected from de-identified patient data by personnel blinded to allocation, using an extraction tool. Unlikely that outcome assessment was subject to bias
Incomplete outcome data (attrition bias) All outcomes	Low risk	1/61 randomised but not analysed
Selective reporting (reporting bias)	Low risk	All outcomes prespecified in the trial registration are reported.
Recruitment bias	Low risk	No evidence of recruitment occurring after sites were allocated
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline patient characteristics - no significant differences between intervention and usual care groups
Protection against contamination	Low risk	Allocated by service
Incorrect analysis	Low risk	Clustering was taken into account in analysis.
Risk of bias overall	Low risk	Low risk across all domains

Persell 2020

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: clinical decision-support + A&F + workflow management + patient education; Intervention 2: clinical decision-support + A&F + workflow management + patient education + population management
Outcomes	Clinical behaviour category: screening, prescribing

Persell 2020 *(Continued)*

	Format of Audit & Feedback: verbal (interactive) Source: manual chart review, computerised search of charts, other Frequency: annually Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation algorithm applied by a statistician likely adequate
Allocation concealment (selection bias)	Low risk	Central allocation and likely at the same time (per flow chart)
Blinding of outcome assessment (detection bias)	Unclear risk	Not enough information about outcome assessment, particularly as some forms of data could potentially be subject to bias
Incomplete outcome data (attrition bias) All outcomes	Low risk	Similar numbers from Intervention and Control groups lost to follow-up for similar reasons
Selective reporting (reporting bias)	Low risk	Outcomes reported as specified in trial protocol
Recruitment bias	Low risk	No recruitment after randomisation and allocation
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline characteristics and outcome data appear balanced.
Protection against contamination	Low risk	Due to cluster design, contamination is unlikely.
Incorrect analysis	High risk	No mention of adjustment for clustering in paper or protocol
Risk of bias overall	High risk	Incorrect analysis - high

Petersen 2013

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital outpatient clinic Speciality: GP/family N health professionals: 125

Petersen 2013 *(Continued)*

	N patients: NR
Interventions	Number of arms: 4 Description of groups: Intervention 1: A&F + education + individual-level financial incentives; Intervention 2: A&F + education + practice-level financial incentives; Intervention 3: A&F + education + individual and practice-level financial incentives; Intervention 4: A&F + education
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: manual chart review Frequency: tri-annually Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	An adequate method was used for sequence generation.
Allocation concealment (selection bias)	Low risk	Based on the flowchart of enrolment, and as a data analyst performed randomisation, it is likely that allocation was done centrally and concealed.
Blinding of outcome assessment (detection bias)	Low risk	Outcomes were recorded in patient charts (unlikely that physicians would be biased in their recording) and a random sample was abstracted by data collectors who were blinded to allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	1/20 physicians left the study due to leaving the practice. No other physicians left. There does not appear to be any issues with missing outcome data.
Selective reporting (reporting bias)	High risk	None of the secondary outcomes specified in the trial registration are reported here (unclear if these outcomes are reported elsewhere but no other results papers are included in the additional files).
Recruitment bias	Low risk	There is no evidence of recruitment after randomisation and allocation.
Baseline imbalance in outcome measurements or characteristics	Low risk	There are some significant baseline imbalances reported, but analyses were adjusted for predetermined baseline covariates and likely accounted for most of these differences.
Protection against contamination	Low risk	Due to the cluster design by hospital, it is unlikely that there was contamination between groups.
Incorrect analysis	Low risk	The trial analysed with the physician as the unit of analysis
Risk of bias overall	High risk	Due to selective reporting

Peters-Klimm 2008

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Germany Setting: primary care Speciality: GP/family N health professionals: 37 N patients: 168
Interventions	Number of arms: 2 Description of groups: Control: educational session; Intervention: A&F + educational sessions
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: manual chart review, computerised search of charts Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	"After recruitment of practices and patients, stratified randomisation of practices was carried out by an external third party using the "single-coin method" based on a computer-generated list." (Peters-Klimm 2009) A random method was used for sequence generation.
Allocation concealment (selection bias)	Low risk	"Intervention allocations were concealed from participating GPs and the intervention team by the third party until shortly before the intervention started in October 2005." Allocation was adequately concealed from participants.
Blinding of outcome assessment (detection bias)	Low risk	GPs in both groups received an educational intervention and would not necessarily know what the trial's aim/hypothesis was; therefore, it is unlikely that documentation and extraction of the outcome was biased.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Similar losses to follow-up and exclusion from analysis in both groups (explained by deaths and contraindications): TTT intervention group: GPs: 0 lost to follow-up & 1 excluded from analysis Patients: 6 lost to follow-up & 11 not included in analysis (80/91 included in analysis) Standard control group: GPs: 0 lost to follow-up; all included in analysis Patients: 9 lost to follow-up & 2 excluded from analysis (66/77 included in analysis)

Peters-Klimm 2008 *(Continued)*

Selective reporting (reporting bias)	Low risk	Peters-Klimm 2009 publication reports all outcomes retrospectively specified in the trial registration. Outcomes in Peters-Klimm 2008 were not specified in the (retrospective) trial registration but appear to be logical outcomes for inclusion and not suggestive of selective outcome reporting.
Recruitment bias	Low risk	No evidence of recruitment occurring after practices were randomised and allocated
Baseline imbalance in outcome measurements or characteristics	Low risk	All baseline characteristics and outcome data (except for hypertension) appear balanced between groups.
Protection against contamination	Low risk	A cluster design was used with GPs as the cluster, so it is unlikely that "standard" group participants received the TTT intervention.
Incorrect analysis	Low risk	Analyses were conducted with clustering taken into account.
Risk of bias overall	Low risk	Low risk across all domains

Pettersson 2011

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Sweden Setting: nursing homes Speciality: NA N health professionals: NR N patients: 2537
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + educational sessions and materials
Outcomes	Clinical behaviour category: prescribing, treatment decision/action Format of Audit & Feedback: written Source: manual chart review Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation sequence

Pettersson 2011 *(Continued)*

Allocation concealment (selection bias)	Low risk	No details, but based on the study flow diagram, it appears likely that all clusters were allocated at the same time
Blinding of outcome assessment (detection bias)	Low risk	Unlikely that knowledge of allocation influenced the recording of patient data and extraction of outcome data for the trial
Incomplete outcome data (attrition bias) All outcomes	High risk	More control nursing homes were lost to follow-up (9/29 vs 3/29).
Selective reporting (reporting bias)	Unclear risk	No outcomes were specified in the trial registration (which was not prospective), so it is unclear if all preplanned outcomes were measured and reported here.
Recruitment bias	Low risk	There is no evidence of recruitment after randomisation and allocation.
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline characteristics and outcome data appear balanced between groups.
Protection against contamination	Low risk	Cluster design means that it is unlikely that the control arm received the intervention.
Incorrect analysis	Low risk	Analysis was performed at the patient level with clustering taken into account.
Risk of bias overall	High risk	Due to incomplete outcome data

Pimlott 2003

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Canada Setting: primary care Speciality: GP/family N health professionals: 374 N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + educational materials (for benzodiazepines); Intervention 2: A&F + educational materials (for antihypertensive drugs)
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: administrative/billing data Frequency: bi-monthly Instructions for improvement: general

Pimlott 2003 *(Continued)*

Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Data collected from central database
Incomplete outcome data (attrition bias) All outcomes	Low risk	All physicians followed up
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Table 1
Protection against contamination	Low risk	Separate GPs
Incorrect analysis	Unclear risk	Unclear if clustering was considered
Risk of bias overall	Low risk	Low risk across most domains

Price-Haywood 2014

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 18 N patients: 168
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + communication-skills training; Intervention 2: A&F only
Outcomes	Clinical behaviour category: screening, testing/exams

Price-Haywood 2014 *(Continued)*

	Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: semi-annually Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated
Allocation concealment (selection bias)	Low risk	Not explicitly described, but seems like all clusters allocated at one time
Blinding of outcome assessment (detection bias)	Unclear risk	Standardised patients who recorded the outcomes and were not blinded to allocation. Potential for knowledge to influence outcome reporting. Unclear if chart auditors were blinded
Incomplete outcome data (attrition bias) All outcomes	Low risk	ITT used
Selective reporting (reporting bias)	Unclear risk	Trial registry was retrospectively posted; 12 and 24-month data for cancer screening rates would be measured but only one time point (unclear on time point) was reported. Patient cancer screening knowledge appears to be a new outcome.
Recruitment bias	Low risk	Appear to be enroled at the same time with no added participants after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Similar characteristics between patients and physicians
Protection against contamination	Low risk	Randomised at practice level, and only 2 patients crossed over groups; therefore, unlikely to impact outcomes
Incorrect analysis	Unclear risk	Unclear if clustering taken into account in analyses
Risk of bias overall	Low risk	Low risk across most domains

Quanbeck 2018

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care

Quanbeck 2018 *(Continued)*

	Speciality: GP/family N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + academic detailing + organisational coaching
Outcomes	Clinical behaviour category: screening, prescribing, testing/exams, counselling Format of Audit & Feedback: verbal (interactive) Source: computerised search of charts Frequency: monthly Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Randomisation by computerised random number generator (p. 5)
Allocation concealment (selection bias)	High risk	Allocation was not concealed given that a clinic within a pair was invited to be the intervention arm and, if they declined, the other in the pair was then assigned the intervention arm.
Blinding of outcome assessment (detection bias)	Low risk	Personnel and participants were not blinded to their allocation and the control group received no intervention. It is unlikely that this would have affected the patient's outcomes versus if the intervention was given outside of a trial.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No clinics reported as lost to follow-up
Selective reporting (reporting bias)	Unclear risk	While the trial registration lists most patient outcomes up to 3 years, this publication reports up to 12 months. Provider dropout (secondary outcome) was not explicitly reported.
Recruitment bias	High risk	Sites were recruited after initial recruitment and randomisation when some sites declined to participate. Of the control group, 3/4 were recruited late.
Baseline imbalance in outcome measurements or characteristics	Low risk	Matched-pair design (p. 4)
Protection against contamination	Low risk	Clinics were randomised - it is unlikely that control clinics received the intervention.
Incorrect analysis	Unclear risk	No adjustment for clustering mentioned
Risk of bias overall	High risk	Due to allocation concealment and recruitment bias

Quinley 2004

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + patient educational materials; Intervention 2: A&F + patient educational materials + expert outreach/facilitation
Outcomes	Clinical behaviour category: immunisation Format of Audit & Feedback: both written and verbal Source: administrative/billing data Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Data collected from central claims
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data collected from Medicare; very small number had insufficient data
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	See page 108

Quinley 2004 *(Continued)*

Protection against contamination	Low risk	Randomisation by practice
Incorrect analysis	Unclear risk	No adjustment for clustering mentioned (t-tests used to compare changes in mean differences, but units of analysis and randomisation are both individual GPs)
Risk of bias overall	Low risk	Low risk across most domains

Raasch 2000

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Australia Setting: primary care Speciality: GP/family N health professionals: 46 N patients: 1366
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F
Outcomes	Clinical behaviour category: diagnosis Format of Audit & Feedback: both written and verbal Source: other Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random number table
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	4/46 lost to follow-up

Raasch 2000 *(Continued)*

Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	High risk	"Analysis at the level of the doctors alone could not account for variation introduced by the office practice, nonstandardised setting of the study."
Risk of bias overall	High risk	Due to incorrect analysis

Raja 2015

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital inpatient Speciality: emergency medicine N health professionals: 43 N patients: 109,793
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: written Source: computerised search of charts Frequency: quarterly Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	A random method was used for sequence generation.

Raja 2015 *(Continued)*

Allocation concealment (selection bias)	Unclear risk	No details or study flow diagram to determine if it was likely all clusters were allocated at the same time
Blinding of outcome assessment (detection bias)	Low risk	Researchers checked that physicians were not recording their behaviour differently and biasing the outcome data.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Physicians who left were excluded from analysis. Trialists do not report the numbers recruited and randomised, so it is unclear how many physicians (clusters) were subsequently excluded from analysis.
Selective reporting (reporting bias)	Unclear risk	There is no trial registration or previously published protocol, so it is unclear if all preplanned outcomes and analyses were reported.
Recruitment bias	Low risk	We included all attending emergency physicians.
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline characteristics and outcome data were balanced between groups.
Protection against contamination	High risk	As all participants were located on one site, it is likely (as trialists also discuss) that physicians in the control arm were aware of or influenced by the intervention received by physicians in the intervention arm.
Incorrect analysis	Unclear risk	Methods of analyses are insufficiently reported, so unclear.
Risk of bias overall	High risk	Due to contamination

Ralph 2018

Study characteristics	
Methods	Design: step wedge
Participants	Country: Australia Setting: community care Speciality: GP/family N health professionals: NR N patients: 402
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + monitoring + continuous quality improvement activities
Outcomes	Clinical behaviour category: Format of Audit & Feedback: Source: Frequency: Instructions for improvement:

Ralph 2018 *(Continued)*

Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	A random method was used for sequence generation.
Allocation concealment (selection bias)	Low risk	Allocation was done centrally at the NHMRC CTC as noted above; hence, low risk of selection bias
Blinding of outcome assessment (detection bias)	Low risk	It is unlikely that knowledge of allocation influenced outcome data recording and extraction. Data were also checked for accuracy.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No clusters lost to follow-up and all eligible patient-level outcome data were included.
Selective reporting (reporting bias)	Unclear risk	Primary outcome is expanded upon in the data reported, and multiple secondary outcomes were added in this publication (compared to the one primary and three secondary outcomes listed in the trial registration). It is unclear if these decisions were made to make the intervention look better.
Recruitment bias	Low risk	All eligible health centres were invited to participate and there is no evidence of recruitment after allocation.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Table 1 shows few baseline differences between clusters; however, there is no mention of these differences being accounted for in the analyses.
Protection against contamination	Low risk	Some patients (number not specified) moved between sites. However, the outcome is measured for the healthcare received by a patient within a health centre, so it is unlikely that patients have an influence on the trial outcomes.
Incorrect analysis	High risk	Analysis was performed at the patient-level and no evidence that data were adjusted for cluster effect
Risk of bias overall	High risk	Due to incorrect analysis

Rantz 2001

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: nursing homes Speciality: long-term care N health professionals: NR N patients: NR
Interventions	Number of arms: 3

Rantz 2001 *(Continued)*

	Description of groups: Control: no intervention; Intervention 1: educational workshops + educational materials + A&F; Intervention 2: educational workshops + educational materials + A&F + access to telephone and/or on-site clinical nurse specialist
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: both written and verbal Source: other Frequency: quarterly Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data analysed as intention-to-treat
Selective reporting (reporting bias)	Unclear risk	Unable to assess
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 1
Protection against contamination	Low risk	Separate nursing homes
Incorrect analysis	Low risk	Adjustments for clustering described (adjustment for resident case mix)
Risk of bias overall	Unclear risk	Unclear risk across important domains

Rask 2001

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA

Rask 2001 *(Continued)*

	Setting: primary care Speciality: GP/family/internal medicine N health professionals: 28 N patients: 491
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F; Intervention 2: A&F + nurse facilitator + clinical practice guidelines + patient reminders
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: both written and verbal Source: administrative/billing data Frequency: quarterly Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data on all 4 clinics
Selective reporting (reporting bias)	Unclear risk	Unable to assess
Recruitment bias	Low risk	No evidence of additional recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Low risk	Logistic regression
Risk of bias overall	Unclear risk	Unable to assess risk in important domains

Robling 2002

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: NR N patients: NR
Interventions	Number of arms: 4 Description of groups: Control 1: education seminar; Control 2: guidelines; Intervention 1: A&F; Intervention 2: A&F + education seminar
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: once Instructions for improvement: not reported
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	High risk	Random numbers table used and likely adequate for first trial, then stratified and randomised for second trial (likely also adequate). However in trial 1, 7 practices chose their own assignment and formed a third arm.
Allocation concealment (selection bias)	Low risk	No details but based on the study flow diagram, it appears likely that all clusters were allocated at the same time.
Blinding of outcome assessment (detection bias)	High risk	GPs and researchers were aware of allocation and GPs self-reported their requesting behaviour and other relevant clinical data during interviews in order for a panel (who were blinded to allocation) to categorise the requests as guideline concordant or not. There is potential for bias in self-reporting.
Incomplete outcome data (attrition bias) All outcomes	High risk	2 practices declined to participate in trial two and it is unclear which arms they were in in trial one. Outcome data analysed varied between arms in trial one (79-85% in trial 1) and trial two (47-75%) and most were due to "no follow-up", suggesting there was bias in incomplete outcome data.
Selective reporting (reporting bias)	Unclear risk	There is no prepublished protocol or trial registration, so it is unclear if all preplanned outcomes were reported.
Recruitment bias	Low risk	There is no evidence recruitment was after randomisation and allocation.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	No baseline characteristics of practices were reported, so it is unclear.

Robling 2002 *(Continued)*

Protection against contamination	Low risk	Due to the cluster design, it is unlikely that there was contamination between sites.
Incorrect analysis	Low risk	Clustering was taken into account in the analyses.
Risk of bias overall	High risk	Due to sequence generation, incomplete outcome data and blinding

Rodriguez 2015

Study characteristics	
Methods	Design: step wedge
Participants	Country: Argentina Setting: hospital inpatient Speciality: emergency medicine N health professionals: 705 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + leadership commitment + surveillance of materials needed to comply with hand hygiene + reminders + storyboard of the project
Outcomes	Clinical behaviour category: other Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: monthly Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Table of random numbers
Allocation concealment (selection bias)	Low risk	Concealed table of allocation
Blinding of outcome assessment (detection bias)	High risk	Outcome assessors/site coordinators not able to be blinded
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	2 sites dropped out and reasons do not seem to have relation to the intervention. Outcome data were reported for them even though trial stated they were excluded from analysis.

Rodriguez 2015 *(Continued)*

Selective reporting (reporting bias)	Unclear risk	No prior trial registration was available that defined the outcomes to make accurate assessment, and also main outcome (table 3) raw data not reported
Recruitment bias	Low risk	It appears that sites were allocated centrally; recruitment bias unlikely
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess; not described
Protection against contamination	High risk	Contamination could not be ruled out, because every site coordinator knew already that hand hygiene had to be improved.
Incorrect analysis	Low risk	Analysis at cluster level
Risk of bias overall	High risk	Due to contamination and blinding

Roos-Blom 2019

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: hospital inpatient Speciality: emergency medicine N health professionals: NR N patients: 25,141
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + action planning template; Intervention 2: A&F + action planning template + action implementation toolbox
Outcomes	Clinical behaviour category: screening Format of Audit & Feedback: electronic dashboard Source: administrative/billing data Frequency: monthly Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence

Roos-Blom 2019 *(Continued)*

Allocation concealment (selection bias)	Low risk	Allocation concealed by researcher unaffiliated with the study
Blinding of outcome assessment (detection bias)	Unclear risk	Unclear if trial personnel were involved in outcome assessment (as they were unblinded)
Incomplete outcome data (attrition bias) All outcomes	Low risk	All ICUs randomised were followed up and analysed.
Selective reporting (reporting bias)	Low risk	Primary and secondary outcomes consistent with registered protocol
Recruitment bias	Low risk	No evidence of recruitment after randomisation and allocation
Baseline imbalance in outcome measurements or characteristics	Low risk	No significant differences in table 1
Protection against contamination	Low risk	Allocated by ICUs (clusters) across the Netherlands and unlikely that the control group received the intervention
Incorrect analysis	Low risk	Analysed with clustering taken into account
Risk of bias overall	Low risk	Overall low risk of bias

Ruangkanchanasetr 1993a

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Thailand Setting: primary care Speciality: paediatrics N health professionals: 9 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + clinician education
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: verbal (interactive) Source: manual chart review Frequency: semi-weekly Instructions for improvement: general
Notes	

Ruangkanchanasetr 1993a *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unable to assess; no information about follow-up provided
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Unclear risk	No adjustment for clustering mentioned (non-parametric tests used, but unit of analysis and randomisation are both at the level of individual GPs)
Risk of bias overall	Unclear risk	Unclear across important domains

Ruangkanchanasetr 1993b

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Thailand Setting: primary care Speciality: paediatrics N health professionals: 9 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + clinician education
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: verbal (interactive)

Ruangkanchanasetr 1993b *(Continued)*

	Source: manual chart review Frequency: semi-weekly Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unable to assess; no information about follow-up provided
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Unclear risk	No adjustment for clustering mentioned (non-parametric tests used, but unit of analysis and randomisation are both at the level of individual GPs)
Risk of bias overall	Unclear risk	Unclear across important domains

Ruangkanchanasetr 1993c

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Thailand Setting: primary care Speciality: paediatrics N health professionals: 18 N patients: NR

Ruangkanchanasetr 1993c *(Continued)*

Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + clinician education
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: verbal (interactive) Source: manual chart review Frequency: semi-weekly Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unable to assess; no information about follow-up provided
Selective reporting (reporting bias)	Low risk	Appropriate outcomes measured
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Unclear risk	No adjustment for clustering mentioned (non-parametric tests used, but unit of analysis and randomisation are both at the level of individual GPs)
Risk of bias overall	Unclear risk	Unclear across important domains

Rubin 2001

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Australia

Rubin 2001 *(Continued)*

	Setting: hospital inpatient Speciality: internal medicine N health professionals: NR N patients: 1117
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F; Intervention 2: A&F + staff meeting
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: once Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random number table
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Outcome assessors blinded to intervention
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unable to assess
Selective reporting (reporting bias)	Unclear risk	Unable to assess
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Low risk	Separate hospitals
Incorrect analysis	Unclear risk	No adjustment for clustering mentioned
Risk of bias overall	Low risk	Low risk across most domains

Rust 1999

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: resident continuity clinic Speciality: paediatrics N health professionals: 32 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F
Outcomes	Clinical behaviour category: immunisation Format of Audit & Feedback: written Source: manual chart review, other Frequency: monthly Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Unclear risk	Unable to assess
Blinding of outcome assessment (detection bias)	High risk	Not done
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unable to assess
Selective reporting (reporting bias)	Low risk	Cluster trial; allocation after recruitment completed
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Unclear risk	Unable to assess

Rust 1999 *(Continued)*

Incorrect analysis	Unclear risk	No adjustment for clustering mentioned
Risk of bias overall	High risk	Due to blinding

Ryan 2014

Study characteristics	
Methods	Design: step wedge
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: computerised decision-support + A&F + technical assistance visits + financial incentives; Intervention 2: computerised decision-support + A&F + technical assistance visits
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: quarterly Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	No details on method of random sequence generation
Allocation concealment (selection bias)	Unclear risk	No details or study flow diagram to determine if clusters were likely allocated at the same time
Blinding of outcome assessment (detection bias)	Low risk	Outcome measures were obtained from electronic health records. Unlikely subject to bias
Incomplete outcome data (attrition bias) All outcomes	Low risk	Does not appear any clusters were lost
Selective reporting (reporting bias)	Unclear risk	No protocol to compare outcomes against

Ryan 2014 *(Continued)*

Recruitment bias	Low risk	Practices randomised but more added for year 2; same process, so probably not an issue
Baseline imbalance in outcome measurements or characteristics	Low risk	No difference in practice characteristics, only financial incentives which were the study design
Protection against contamination	Low risk	Allocation by cluster; nothing to suggest contamination risk
Incorrect analysis	Low risk	Analysis at cluster level
Risk of bias overall	Unclear risk	Unclear across important domains

Ryskina 2018

Study characteristics		
Methods	Design: parallel cluster-RCT	
Participants	Country: USA Setting: hospital inpatient Speciality: internal medicine N health professionals: 114 N patients: 2042	
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F	
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: electronic dashboard Source: computerised search of charts Frequency: bi-weekly Instructions for improvement: general	
Notes		
Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random number generator was used.
Allocation concealment (selection bias)	Low risk	Not described but appears likely all clusters allocated at one time

Ryskina 2018 *(Continued)*

Blinding of outcome assessment (detection bias)	Low risk	The investigators were blinded to randomisation and during data collection and analysis.
Incomplete outcome data (attrition bias) All outcomes	Low risk	No missing outcome data and ITT analysis used
Selective reporting (reporting bias)	High risk	Outcomes were reported for the primary outcome and consistent with registered trial. Secondary outcomes from registered trial (e.g. rate of imaging orders) not presented though
Recruitment bias	Low risk	During each 2-week block, we randomised three of the clusters to the experimental group and three to the control group using a random number generator.
Baseline imbalance in outcome measurements or characteristics	Low risk	There were no statistically significant differences in physician characteristics, patient census per physician per day, or non-targeted laboratory tests between intervention and control groups.
Protection against contamination	High risk	Single-centre study and difficult to protect against this risk of bias
Incorrect analysis	Low risk	Analysis took into account clustering.
Risk of bias overall	High risk	Due to potential risk in protection against contamination

Sauaia 2000

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital inpatient Speciality: cardiology N health professionals: NR N patients: 929
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F (in-person + written); Intervention 2: A&F (written)
Outcomes	Clinical behaviour category: prescribing, treatment decision/action, counselling Format of Audit & Feedback: both written and verbal Source: administrative/billing data Frequency: once Instructions for improvement: specific
Notes	

Sauaia 2000 *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer randomisation
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Quality indicators collected centrally
Incomplete outcome data (attrition bias) All outcomes	Low risk	18 hospitals included in data collection
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Table 2
Protection against contamination	Low risk	Separate hospitals
Incorrect analysis	Low risk	"Multiple logistic regression models (patient as unit of analysis) and the Laird-Ware linear model (hospital as the unit of analysis) were applied to the data".
Risk of bias overall	Low risk	Low across all domains

Scales 2011

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Canada Setting: hospital inpatient Speciality: emergency medicine N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + educational outreach/materials + reminders
Outcomes	Clinical behaviour category: prescribing, treatment decision/action

Scales 2011 *(Continued)*

	Format of Audit & Feedback: written Source: computerised search of charts, other Frequency: monthly Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer randomisation, performed by statistician, performed after all ICUs recruited (protocol)
Allocation concealment (selection bias)	Low risk	Allocation was concealed.
Blinding of outcome assessment (detection bias)	High risk	Data collectors were a nurse in each ICU (likely aware of their allocation).
Incomplete outcome data (attrition bias) All outcomes	Low risk	No loss of clusters
Selective reporting (reporting bias)	Low risk	Analysed in accordance with that planned in published protocol (ref 13)
Recruitment bias	Low risk	The 15 ICUs were randomised into two groups and no further recruitment of ICUs was conducted.
Baseline imbalance in outcome measurements or characteristics	Low risk	Table 3 showed that the characteristics of participating ICUs were similar across both groups except for minor differences in pair 2 which was accounted for in the analysis.
Protection against contamination	Low risk	Randomisation at the ICU so unlikely risk of contamination
Incorrect analysis	Low risk	Clustering accounted for in analysis
Risk of bias overall	High risk	Due to blinding

Scales 2016

Study characteristics

Methods	Design: step wedge
Participants	Country: Canada Setting: hospital inpatient Speciality: emergency medicine N health professionals: NR

Scales 2016 *(Continued)*

	N patients: 905
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + educational outreach/materials + reminders
Outcomes	Clinical behaviour category: diagnosis Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: monthly Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer randomisation
Allocation concealment (selection bias)	Low risk	Sounds like randomisation to blocks had occurred at one time point
Blinding of outcome assessment (detection bias)	Low risk	Data collectors blinded; electronic medical records also used
Incomplete outcome data (attrition bias) All outcomes	Low risk	Low data in follow-up from 1/18 hospitals. Unlikely this significantly affected outcomes
Selective reporting (reporting bias)	Low risk	Outcomes reported in line with clinical trial registration
Recruitment bias	Low risk	No clinics recruited after randomisation.
Baseline imbalance in outcome measurements or characteristics	Low risk	Characteristics similar across groups (table 1)
Protection against contamination	Low risk	Randomised by hospital to prevent contamination
Incorrect analysis	Low risk	Adjustment for clustering in analysis
Risk of bias overall	Low risk	Low risk across all domains

Schectman 1995

Study characteristics

Methods	Design: parallel cluster-RCT

Schectman 1995 *(Continued)*

Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 63 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: educational materials; Intervention: A&F + educational materials
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: administrative/billing data Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	All physicians included in analysis
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	See Figure 1
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Low risk	Unit of analysis was the physician.
Risk of bias overall	Unclear risk	Unable to assess risk across important domains

Schectman 2003

Study characteristics		
Methods	Design: parallel cluster-RCT	
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 120 N patients: NR	
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + clinician education	
Outcomes	Clinical behaviour category: testing/exams, referrals Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: variable Instructions for improvement: specific	
Notes		
Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Sealed envelopes used
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Abstractors blinded
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unable to assess
Selective reporting (reporting bias)	Unclear risk	Unable to assess
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess

Schectman 2003 *(Continued)*

Protection against contamination	Low risk	Separate sites for GPs
Incorrect analysis	Low risk	Nested model (physician within site) was utilised.
Risk of bias overall	Low risk	Low risk across important outcomes

Schmidt-Mende 2017

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Sweden Setting: primary care Speciality: GP/family N health professionals: NR N patients: 119,910
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + educational outreach/materials + reminders
Outcomes	Clinical behaviour category: prescribing, treatment decision/action, other Format of Audit & Feedback: both written and verbal Source: administrative/billing data Frequency: not reported/not applicable Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer random number generator
Allocation concealment (selection bias)	Low risk	Though not explicitly described, it appears all practices were allocated at one time.
Blinding of outcome assessment (detection bias)	Low risk	Data retrieved from electronic records and unlikely to be subject to bias
Incomplete outcome data (attrition bias) All outcomes	Low risk	1/69 practices dropped out; this would not suggest significant imbalance in incomplete outcome data between groups.

Schmidt-Mende 2017 *(Continued)*

Selective reporting (reporting bias)	Low risk	Outcomes consistent with clinical trial registration
Recruitment bias	Low risk	No recruitment after baseline
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline characteristics had high similarity (table 1).
Protection against contamination	Low risk	Whole practices randomised; unlikely contamination
Incorrect analysis	Low risk	Accounted for clustering in analysis
Risk of bias overall	Low risk	Low risk across all domains

Schneider 2008

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Germany Setting: primary care Speciality: GP/family N health professionals: 96 N patients: 314
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F; Intervention 2: A&F with benchmark
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: Manual chart review, patient-reported data Frequency: not reported/not applicable Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed

Schneider 2008 *(Continued)*

Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	High risk	Many lost to follow-up
Selective reporting (reporting bias)	High risk	Only secondary outcomes
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Unclear risk	No adjustment for clustering mentioned
Risk of bias overall	High risk	Due to incomplete outcome data and selective reporting

Scholes 2006

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 204 N patients: 12,691
Interventions	Number of arms: 2 Description of groups: Control: guidelines; Intervention: A&F + education + reminders
Outcomes	Clinical behaviour category: screening Format of Audit & Feedback: both written and verbal Source: computerised search of charts, administrative/billing data Frequency: quarterly Instructions for improvement: co-developed plans
Notes	
Risk of bias	

Scholes 2006 *(Continued)*

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Database used for data
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data from all clinics presented
Selective reporting (reporting bias)	Unclear risk	Unable to assess
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Table 2
Protection against contamination	Low risk	Separate clinics
Incorrect analysis	Low risk	GEEs used to adjust for within-clinic correlations
Risk of bias overall	Low risk	Low risk across most domains

Shen 2018

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: China Setting: primary care Speciality: GP/family N health professionals: 65 N patients: 1048
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F
Outcomes	Clinical behaviour category: prescribing, counselling Format of Audit & Feedback: written Source: patient-reported data, other Frequency: quarterly

Shen 2018 *(Continued)*

	Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Not stated whether randomisation occurred at one or several temporal points. However, it sounds like it was done centrally; likely low RoB.
Blinding of outcome assessment (detection bias)	Unclear risk	Unclear if data collectors were aware of allocation
Incomplete outcome data (attrition bias) All outcomes	Low risk	No loss of clusters
Selective reporting (reporting bias)	Unclear risk	No trial registration or protocol found
Recruitment bias	Low risk	The study sites were determined via a 3-step clustered randomisation. Step 1 divided all the 55 rural counties in Anhui province into north (17 counties), middle (16), and south (22). Step 2 randomly selected: (1) 4 counties from each of the regions, (2) 1 township from each of the selected counties, and (3) 2 administrative villages from each of the selected townships. Step 3 randomly assigned the 2 villages within each township into intervention and control arms. All the 24 village clinics in the selected villages agreed to participate.
Baseline imbalance in outcome measurements or characteristics	Low risk	Table 1 describes balanced baselines.
Protection against contamination	Low risk	From appendix 5: "Only the village doctors with the 12 village clinics on the intervention arm were allowed to use JITIF-WBA. They needed not pay for accessing the application. But each of the participating village doctors was asked to register a personal account and was only able to record and view service encounters with his/her patients".
Incorrect analysis	Unclear risk	Controls for clustering not described. Table 2 & 3 look like analysis was at participant level.
Risk of bias overall	Low risk	Low risk across most domains

Sinclair 1982

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Canada Setting: mental health clinic

Sinclair 1982 *(Continued)*

	Speciality: paediatrics/mental health N health professionals: 11 N patients: 65
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + educational training
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: not reported/not applicable Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Data collected by blinded assessor
Incomplete outcome data (attrition bias) All outcomes	Low risk	Follow-up for all professionals
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Unclear risk	Limited details on analysis
Risk of bias overall	Low risk	Low risk across most important domains

Siriwardena 2002

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + educational outreach; Intervention 2: A&F only
Outcomes	Clinical behaviour category: immunisation Format of Audit & Feedback: both written and verbal Source: manual chart review, computerised search of charts Frequency: once Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Data collected as part of national campaign
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data from all practices provided
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Groups similar at baseline
Protection against contamination	Low risk	Separate practices

Siriwardena 2002 *(Continued)*

Incorrect analysis	Unclear risk	No adjustment for clustering mentioned
Risk of bias overall	Low risk	Low risk across most domains

Smith 1998

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: pharmacy and outpatient clinics Speciality: GP/family N health professionals: NR N patients: 222
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + guidelines
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: administrative/billing data Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer randomisation
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Data collected through Medicaid
Incomplete outcome data (attrition bias) All outcomes	Low risk	Fewer than 10% stopped taking drugs.
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of cluster recruitment after randomisation

Smith 1998 *(Continued)*

Baseline imbalance in outcome measurements or characteristics	Low risk	Table 1
Protection against contamination	Low risk	Randomised by cluster
Incorrect analysis	Unclear risk	No adjustment for clustering mentioned
Risk of bias overall	Low risk	Low risk across most domains

Smith-Bindman 2020

Study characteristics		
Methods	Design: step wedge	
Participants	Country: USA/Japan/Germany/Netherlands/Switzerland/England Setting: imaging facilities Speciality: NA N health professionals: NR N patients: 745,802	
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F; Intervention 2: A&F + multicomponent QI initiative	
Outcomes	Clinical behaviour category: other Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: once Instructions for improvement: specific	
Notes		
Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Sequence generation is not clearly reported.
Allocation concealment (selection bias)	Low risk	Central allocation by an independent person
Blinding of outcome assessment (detection bias)	Unclear risk	Outcome data assessment not described in enough detail but it appears that de-identified CT scans were assessed so detection bias is not likely.
Incomplete outcome data (attrition bias)	Unclear risk	Loss of facilities ranged from 15%, 20% and 30% in the three arms, with reasons reported and these do not appear related to the intervention. There is po-

Smith-Bindman 2020 *(Continued)*

All outcomes		tential concern for the arm that lost 30% of facilities (compared to the other two - 15% and 20%).
Selective reporting (reporting bias)	Low risk	Outcomes reported as per trial registration
Recruitment bias	Low risk	No evidence of recruitment after randomisation and allocation
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline appears balanced.
Protection against contamination	Unclear risk	Authors acknowledge facilities are interconnected and efforts were made to reduce contamination, but it is unclear if this was successful between facilities within an organisation.
Incorrect analysis	Low risk	Clustering was accounted for in the analysis.
Risk of bias overall	Unclear risk	Protection against contamination - unclear

Socolar 1998

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: providers prepared to perform examinations for social service clients (likely in the community) Speciality: unclear N health professionals: 147 N patients: 811
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + educational materials
Outcomes	Clinical behaviour category: screening Format of Audit & Feedback: written Source: manual chart review Frequency: once Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random number assignment

Socolar 1998 *(Continued)*

Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Documentation assessed by blinded reviewers
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unable to assess
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of cluster recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Table 1
Protection against contamination	Low risk	Separate professionals
Incorrect analysis	Low risk	Appropriate adjustment for clustering described (linear regression with ICC)
Risk of bias overall	Low risk	Low risk across most domains

Soleymani 2019

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Iran Setting: general medicine, paediatrics, infectious disease outpatient clinics Speciality: GP/family/paediatrics/infectious disease N health professionals: 820 N patients: NR
Interventions	Number of arms: 4 Description of groups: Control 1: educational materials; Control 2: no intervention; Intervention 1: regular A&F; Intervention 2: new A&F
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: administrative/billing data Frequency: once Instructions for improvement: specific

Soleymani 2019 *(Continued)*

Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random number generator
Allocation concealment (selection bias)	Low risk	Based on the study flow diagram, it appears likely that all clusters were allocated at the same time.
Blinding of outcome assessment (detection bias)	Unclear risk	Not enough information on outcome assessment and data collection
Incomplete outcome data (attrition bias) All outcomes	Low risk	Few withdrawals and, though not described which arm these withdrawals were from, the low number would be unlikely to bias the outcomes.
Selective reporting (reporting bias)	Low risk	Outcomes and time frames were consistent with registered protocol.
Recruitment bias	Low risk	Unlikely that physicians were recruited to the trial after clusters were randomised
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline balanced as described in table 1
Protection against contamination	Unclear risk	It is possible that the physicians who received the printed educational material could have communicated between physicians allocated to other groups.
Incorrect analysis	Low risk	We conducted the analyses at the strata level (physicians), as the intended unit of inference was the physician.
Risk of bias overall	Low risk	Low risk across most domains

Solomon 2004

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: arthritis clinic Speciality: rheumatology N health professionals: 32 N patients: 373
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + education + reminder

Solomon 2004 *(Continued)*

Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: manual chart review, administrative/billing data Frequency: variable Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unable to assess
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of cluster recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	One physician lost to follow-up; not related to intervention. See Table 1
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Low risk	"Because the intervention was directed at rheumatologists and patients were clustered within rheumatologists' practices, we adjusted the analyses for treating rheumatologist by including a term for the doctor in the multivariate model using the generalised estimating equation (GEE)."
Risk of bias overall	Unclear risk	Unclear risk in important domains

Soumerai 1998

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital inpatient

Soumerai 1998 *(Continued)*

	Speciality: cardiology N health professionals: NR N patients: 5347
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + opinion leader education + educational materials + administrative support; Intervention 2: A&F only
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: once Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	No description of sequence generation
Allocation concealment (selection bias)	Low risk	No description of allocation methodology, but as randomisation was performed on one day, and based on a flow chart, it appears likely that clusters were allocated at one time
Blinding of outcome assessment (detection bias)	Low risk	Unclear if data abstractors were blinded. But there were multiple research nurses doing this, with ongoing inter-rater agreement maintained and a random sample audited, so it is unlikely that biases were introduced into data abstraction. Data for medical records were put in by healthcare professionals unaware of the study arms and hypothesis.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	1/17 control sites lost due to the hospital closing down, which was outside of study's control and not influenced by the intervention. Some hospitals excluded from drug-specific analyses with explanation, but unclear if this was planned prior.
Selective reporting (reporting bias)	Unclear risk	No protocol or trial registry to determine if all preplanned outcomes were reported in the preplanned manner
Recruitment bias	Low risk	No evidence of recruitment after randomisation and allocation
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline characteristics and outcome data appear balanced.
Protection against contamination	Low risk	Due to the cluster design (including randomising by geography), contamination is unlikely.
Incorrect analysis	Unclear risk	How trialists analysed hospitals as a unit is unclear in the methods and in the data presented.

Soumerai 1998 *(Continued)*

Risk of bias overall	Low risk	Low risk across most important domains

Spoorenberg 2015

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: hospital inpatient Speciality: internal medicine/urology N health professionals: NR N patients: 3991
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + education + reminders; Intervention 2: A&F only
Outcomes	Clinical behaviour category: prescribing, testing/exams, treatment decision/action Format of Audit & Feedback: both written and verbal Source: manual chart review, computerised search of charts Frequency: variable/once Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Randomisation by minimisation by statistician; likely okay
Allocation concealment (selection bias)	Low risk	Allocation likely central and done at one time
Blinding of outcome assessment (detection bias)	Low risk	Low risk of detection bias as the assessors were blinded to treatment groups.
Incomplete outcome data (attrition bias) All outcomes	Low risk	All recruited clusters completed the study and were analysed.
Selective reporting (reporting bias)	Unclear risk	The trial registration stated that cost-effectiveness is a secondary outcome; however, the paper does not report the reason for not collecting or reporting this outcome.
Recruitment bias	Low risk	No interventions at the patient level were done and patient data were analysed in a retrospective design anonymously. CFS: patients enroled from October 15, 2010: enrolment in post-intervention measurement at least 6

Spoorenberg 2015 *(Continued)*

		months after providing the feedback reports. MFS: patient enrolment in post-intervention measurement depended on the individual implementation schedule: at least 3 months after the hospital's kick-off meeting, and at least 6 months after providing the first feedback report.
Baseline imbalance in outcome measurements or characteristics	Low risk	No baseline differences between groups
Protection against contamination	Low risk	Allocation at practice level
Incorrect analysis	Low risk	Appropriate analysis method used which accounted for clustering effects
Risk of bias overall	Low risk	Low risk across most domains

Stewardson 2016

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Switzerland Setting: hospital inpatient Speciality: internal medicine N health professionals: NR N patients: NR
Interventions	Number of arms: 3 Description of groups: Control: no intervention; Intervention 1: A&F; Intervention 2: A&F + patient participation
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: both written and verbal Source: other Frequency: variable Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated sequence
Allocation concealment (selection bias)	Low risk	Central allocation and likely done at one time

Stewardson 2016 *(Continued)*

Blinding of outcome assessment (detection bias)	Unclear risk	Unclear if nurses assessing the outcomes were aware of allocation
Incomplete outcome data (attrition bias) All outcomes	Low risk	1/21 control arm lost to follow-up with reason provided. Unlikely missing outcome data are biased
Selective reporting (reporting bias)	Low risk	The outcomes reported in the trial were consistent with the retrospective registered protocol.
Recruitment bias	Low risk	These strata grouped together wards with similar patient charac teristics.
Baseline imbalance in outcome measurements or characteristics	Low risk	No important baseline differences were noted in primary or secondary outcome or characteristics of wards.
Protection against contamination	High risk	First, although we attempted to minimise cross-contamination between study groups by using a cluster randomised design, such cross-contamination was difficult to avoid because of the movement of healthcare workers between wards (particularly physicians, allied health-care professionals, and shared institutional leadership). Second, exclusion from intervention groups motivated some control wards to develop their own hand hygiene initiatives, including patient participation. Although this so-called study effect in the control group was a success from a patient safety perspective, it undermined the study design.
Incorrect analysis	Low risk	Analyses between groups took into account clustering design in the RCT.
Risk of bias overall	High risk	Due to potential risk in protection against contamination

Søndergaard 2002

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Denmark Setting: primary care Speciality: GP/family N health professionals: 292 N patients: 1650
Interventions	Number of arms: 3 Description of groups: Control 1: A&F on unrelated topic; Intervention 1: A&F (patient count); Intervention 2: A&F (aggregated)
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: manual chart review, administrative/billing data Frequency: quarterly

Søndergaard 2002 *(Continued)*

	Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Data collected from billing database
Incomplete outcome data (attrition bias) All outcomes	Low risk	All prescriptions assessed
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of cluster recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Quote: "There was no statistically significant differences between the intervention and the control groups....at the onset of the trial."
Protection against contamination	Low risk	Practices randomised
Incorrect analysis	Low risk	Appropriate adjustment for clustering described (cox regression with robust variance estimates)
Risk of bias overall	Low risk	Low risk across most domains

Søndergaard 2003

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Denmark Setting: primary care Speciality: GP/family N health professionals: 299 N patients: NR
Interventions	Number of arms: 2

Søndergaard 2003 *(Continued)*

	Description of groups: Control: no intervention; Intervention: A&F + educational material (clinical guidelines)
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: administrative/billing data Frequency: once Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Data collected from billing database
Incomplete outcome data (attrition bias) All outcomes	Low risk	All prescriptions assessed
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of cluster recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Table 1
Protection against contamination	Low risk	Practices randomised
Incorrect analysis	Unclear risk	No adjustment for clustering mentioned
Risk of bias overall	Low risk	Low risk across most domains

Søndergaard 2006

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Denmark Setting: primary care

Søndergaard 2006 *(Continued)*

	Speciality: GP/family N health professionals: 30 N patients: 633
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + educational outreach/materials
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: not reported/not applicable Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random number table
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	High risk	Audit done by participants
Incomplete outcome data (attrition bias) All outcomes	Low risk	Equal dropouts in both groups; see Figure 1
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of cluster recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 1
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Low risk	"The association between intervention and changes in outcome variables was assessed using logistic regression analyses. We adjusted for clustering at the level of the individual GP, thus accounting for non-independence of observations within each practice."
Risk of bias overall	High risk	Due to blinding

Tadrous 2020

Study characteristics		
Methods	Design: parallel cluster-RCT	
Participants	Country: Canada Setting: nursing homes Speciality: long-term care N health professionals: NR N patients: 5363	
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + academic detailing; Intervention 2: A&F only	
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: administrative/billing data Frequency: quarterly Instructions for improvement: general	
Notes		
Risk of bias		
Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Randomisation occurred at one time, likely centrally.
Blinding of outcome assessment (detection bias)	Unclear risk	The article states that nursing home staff (aware of allocation) complete RAI (Resident Assessment Instrument) assessments within 14 days of admission, but it is not clear who makes the 3, 6 and 12-month assessments (using data from the administrative claims databases).
Incomplete outcome data (attrition bias) All outcomes	Low risk	No nursing homes were lost to follow-up in either intervention or control group (Figure 1). ITT analyses used
Selective reporting (reporting bias)	Low risk	Table 2 - outcomes reported as planned in protocol (available in supplementary material)
Recruitment bias	Unclear risk	Care homes could not be blinded to their allocation and staff could choose whether to participate - may have led to recruitment bias?
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Some baseline imbalances reported and it does not appear adjustments were made

Tadrous 2020 *(Continued)*

Protection against contamination	Unclear risk	Nursing homes were not blinded to their allocation and it is unclear if staff may be able to move between sites.
Incorrect analysis	Low risk	Clustering taken into account in analysis
Risk of bias overall	Low risk	Recruitment bias, baseline imbalance and protection against contamination judged as unclear; all else low risk

Thomas 2006

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: 370 N patients: NR
Interventions	Number of arms: 4 Description of groups: Control 1: no intervention; Intervention 1: A&F; Control 2: reminders; Intervention 2: A&F + reminders
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: written Source: administrative/billing data Frequency: quarterly Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Quote: "cluster randomization...with a minimization procedure"
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Quote: "The laboratory personnel who processed the requests were unaware of the intervention-group status."
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data from all sites randomised included in analysis

Thomas 2006 *(Continued)*

Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of cluster recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Low risk	Separate practices
Incorrect analysis	Unclear risk	No adjustment for clustering mentioned
Risk of bias overall	Low risk	Low risk across most domains

Thomas 2007

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: community-based continuity clinic Speciality: internal medicine N health professionals: 78 N patients: 483
Interventions	Number of arms: 2 Description of groups: Control: education; Intervention: A&F + education + patient reminders + instruction on patient registry
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: written Source: computerised search of charts Frequency: quarterly Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess

Thomas 2007 *(Continued)*

Allocation concealment (selection bias)	Unclear risk	Unable to assess
Blinding of outcome assessment (detection bias)	Unclear risk	Unclear if blood pressure assessors were blinded
Incomplete outcome data (attrition bias) All outcomes	Low risk	All subjects included in analysis
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of cluster recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Quote "Resident demographic data (age, sex, and year in training) were similar between groups."
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Low risk	Appropriate adjustment for clustering described (GEE with repeated measurements per subject)
Risk of bias overall	Unclear risk	Unable to assess risk in most domains

Tianviwat 2016

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Thailand Setting: dental clinic Speciality: NA N health professionals: 45 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + clinical education
Outcomes	Clinical behaviour category: other Format of Audit & Feedback: both written and verbal Source: other Frequency: semi-annually Instructions for improvement: co-developed plans

Tianviwat 2016 *(Continued)*

Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Based on this and flowchart, allocation was likely done centrally at one time..
Blinding of outcome assessment (detection bias)	Unclear risk	Data analyst blinded, external examiner not blinded, and dental nurses blinded but aware of the outcome being assessed. It is unclear who provided the outcome data, but it may be dental nurses.
Incomplete outcome data (attrition bias) All outcomes	Low risk	There was no loss to follow-up from both groups, hence, there is low risk of attrition bias.
Selective reporting (reporting bias)	Unclear risk	Reported as per methods and trial registration but registration performed retrospectively
Recruitment bias	Low risk	No evidence of recruitment after randomisation and allocation
Baseline imbalance in outcome measurements or characteristics	High risk	Baseline imbalances noted, and it is unclear what adjustments were made for these as not clearly described
Protection against contamination	Unclear risk	The authors noted that participants came together once a month and there may have been discussion about the intervention, even if the Control arm could not receive the tailored intervention.
Incorrect analysis	Low risk	Adjustment for clustering performed
Risk of bias overall	High risk	Baseline imbalance - high; protection against contamination - unclear

Tierney 1986a

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: internal medicine N health professionals: 64 N patients: 6045
Interventions	Number of arms: 2 Description of groups: Control 1: A&F + chart reminders for compliance with Protocol B preventative measures; Control 2: chart reminders for compliance with Protocol A preventative measures; Inter-

Tierney 1986a *(Continued)*

	vention 1: A&F for compliance with Protocol A preventative measures; Intervention 2: A&F + chart reminders for compliance with Protocol A preventative measures
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: written Source: computerised search of charts Frequency: monthly Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Unclear risk	Unable to assess
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unable to assess
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of cluster recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	No baseline data presented
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Low risk	The unit of analysis was the physician.
Risk of bias overall	Unclear risk	Unclear across important domains

Tierney 1986b

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care

Tierney 1986b *(Continued)*

	Speciality: internal medicine N health professionals: 71 N patients: 6045
Interventions	Number of arms: 2 Description of groups: Control 1: A&F + chart reminders for compliance with Protocol A preventative measures; Control 2: chart reminders for compliance with Protocol B preventative measures; Intervention 1: A&F for compliance with Protocol B preventative measures; Intervention 2: A&F + chart reminders for compliance with Protocol B preventative measures
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: written Source: computerised search of charts Frequency: monthly Instructions for improvement: specific
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Unclear risk	Unable to assess
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unable to assess
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of cluster recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	No baseline data presented
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Low risk	The unit of analysis was the physician.
Risk of bias overall	Unclear risk	Unclear across important domains

Tija 2015

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: nursing homes Speciality: GP/family/internal medicine N health professionals: NR N patients: 5502
Interventions	Number of arms: 3 Description of groups: Intervention 1: toolkit only; Intervention 2: toolkit plus feedback; Intervention 3: toolkit plus feedback plus detailing
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: administrative/billing data Frequency: once Instructions for improvement: NR
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Unclear risk	Unable to assess
Blinding of outcome assessment (detection bias)	Low risk	Outcome unlikely to be affected by knowledge of assessor
Incomplete outcome data (attrition bias) All outcomes	Low risk	No missing data
Selective reporting (reporting bias)	Low risk	All outcomes reported
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline characteristics and outcomes balanced between arms

Tija 2015 *(Continued)*

Protection against contamination	Low risk	Separate practices unlikely to have communication between practices
Incorrect analysis	Unclear risk	Unable to assess
Risk of bias overall	Unclear risk	Due to incorrect analysis, randomisation generation and concealment

Trent 2019

Study characteristics	
Methods	Design: step wedge
Participants	Country: USA Setting: hospital inpatient Speciality: emergency medicine N health professionals: 23 N patients: 469
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: written Source: manual chart review Frequency: variable Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated randomisation
Allocation concealment (selection bias)	Low risk	Allocation was concealed until just before the intervention.
Blinding of outcome assessment (detection bias)	Unclear risk	Unclear if data collector was aware of allocation
Incomplete outcome data (attrition bias) All outcomes	Low risk	No physicians left the study and no data were missing.

Trent 2019 *(Continued)*

Selective reporting (reporting bias)	Unclear risk	Reported in line with methods but no protocol to check against
Recruitment bias	Low risk	In stepped-wedge studies with recruitment extended over time or without blinding to the intervention, this may not be possible, creating a risk of selection bias, even though clusters are allocated randomly over time.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Table 1 - some differences between groups of patient variables (for all patients, not just baseline period). Baseline not specifically analysed for patients or physicians
Protection against contamination	High risk	Given that this study was performed at a single institution, contamination could have occurred if physicians who had received the intervention discussed the intervention with physicians who had not previously received the intervention. Beyond requesting that physicians who received the intervention not discuss their data with other physicians, the study team could not reasonably control for contamination that occurred through regular discourse.
Incorrect analysis	Low risk	Researchers took into account stepped-wedge clustering effect and also adjusted for cluster-time intervention.
Risk of bias overall	High risk	Due to protection against contamination

Trietsch 2017a

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: primary care Speciality: GP/family N health professionals: 197 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + educational meetings + educational materials + opinion leaders
Outcomes	Clinical behaviour category: prescribing, testing/exams Format of Audit & Feedback: both written and verbal Source: administrative/billing data Frequency: not reported/not applicable Instructions for improvement: co-developed plans
Notes	
Risk of bias	

Trietsch 2017a *(Continued)*

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Allocation sequence was computer-generated by an independent research assistant.
Allocation concealment (selection bias)	Low risk	Allocation was done centrally by an independent researcher and concealed.
Blinding of outcome assessment (detection bias)	Low risk	"The researcher was blinded until all data analyses had been completed." (p. 4)
Incomplete outcome data (attrition bias) All outcomes	Low risk	Clusters that dropped out appear balanced in both arms for similar reasons. Analyses were performed by ITT as well as per-protocol. Some data were excluded from analyses due to unforeseen issues and unlikely due to bias and unlikely introduced bias (e.g. new guidelines launched and significant public awareness contaminating data, insufficient data, etc.).
Selective reporting (reporting bias)	High risk	Multiple outcomes listed in the trial registration and published protocol are not published here or in the process evaluation paper - e.g. inter-doctor variation in prescribing, costs, etc. Other time points specified in the trial registration (12 and 18 months) are also missing from this publication.
Recruitment bias	Low risk	Patients not recruited to trial. Individual GPs already members of LQIC groups (p. 2)
Baseline imbalance in outcome measurements or characteristics	Low risk	There were no significant baseline characteristic imbalances. Between-group effect estimates were adjusted for baseline.
Protection against contamination	Low risk	The set of five topics presented to groups A & B differed between the two arms.
Incorrect analysis	Low risk	Linear mixed model with LQIC as random effect used to account for clustering of practices
Risk of bias overall	High risk	Due to selective reporting

Trietsch 2017b

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: primary care Speciality: GP/family N health professionals: 197 N patients: NR
Interventions	Number of arms: 2

Trietsch 2017b *(Continued)*

	Description of groups: Control: no intervention; Intervention: A&F + educational meetings + educational materials + opinion leaders
Outcomes	Clinical behaviour category: prescribing, testing/exams Format of Audit & Feedback: both written and verbal Source: administrative/billing data Frequency: not reported/not applicable Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Allocation sequence was computer-generated by an independent research assistant.
Allocation concealment (selection bias)	Low risk	Allocation was done centrally by an independent researcher and concealed.
Blinding of outcome assessment (detection bias)	Low risk	"The researcher was blinded until all data analyses had been completed." (p. 4)
Incomplete outcome data (attrition bias) All outcomes	Low risk	Clusters that dropped out appear balanced in both arms for similar reasons. Analyses were performed by ITT as well as per-protocol. Some data were excluded from analyses due to unforeseen issues and unlikely due to bias and unlikely introduced bias (e.g. new guidelines launched and significant public awareness contaminating data, insufficient data, etc.).
Selective reporting (reporting bias)	High risk	Multiple outcomes listed in the trial registration and published protocol are not published here or in the process evaluation paper - e.g. inter-doctor variation in prescribing, costs, etc. Other time points specified in the trial registration (12 and 18 months) are also missing from this publication.
Recruitment bias	Low risk	Patients not recruited to trial. Individual GPs already members of LQIC groups (p. 2)
Baseline imbalance in outcome measurements or characteristics	Low risk	There were no significant baseline characteristic imbalances. Between-group effect estimates were adjusted for baseline.
Protection against contamination	Low risk	The set of five topics presented to groups A & B differed between the two arms.
Incorrect analysis	Low risk	Linear mixed model with LQIC as random effect used to account for clustering of practices
Risk of bias overall	High risk	Due to selective reporting

Tu 2009

Study characteristics

Tu 2009 *(Continued)*

Methods	Design: parallel cluster-RCT
Participants	Country: Canada Setting: hospital inpatient Speciality: cardiology N health professionals: NR N patients: 20,039
Interventions	Number of arms: 2 Description of groups: Intervention 1: early A&F; Intervention 2: delayed A&F
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: written Source: manual chart review Frequency: once Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Stratified and performed by a statistician
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Blinded based on communication with author
Incomplete outcome data (attrition bias) All outcomes	Low risk	See Figure
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of cluster recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 1
Protection against contamination	Low risk	Hospitals separate
Incorrect analysis	Low risk	"The primary analyses of the study data took into account the clustered nature of the data and examined the difference in the mean hospital-specific perfor-

Tu 2009 *(Continued)*

		mance between the 2 study groups, using methods appropriate for the analysis of repeated cross-sectional cluster randomized trials."
Risk of bias overall	Low risk	Low risk across all domains

Van Bruggen 2008

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: primary care Speciality: GP/family N health professionals: NR N patients: 1640
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + facilitated support + education
Outcomes	Clinical behaviour category: prescribing, testing/exams Format of Audit & Feedback: not reported/not applicable Source: computerised search of charts Frequency: once Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random number table by an independent researcher
Allocation concealment (selection bias)	Low risk	Because an independent researcher carried out randomisation, and the study flow diagram suggests it is likely that all clusters were allocated at the same time, allocation was likely concealed.
Blinding of outcome assessment (detection bias)	Unclear risk	Some outcomes could be subject to bias from knowledge of allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Balanced attrition; unlikely to be related to intervention
Selective reporting (reporting bias)	Unclear risk	No protocol to check reported outcomes against

Van Bruggen 2008 *(Continued)*

Recruitment bias	Low risk	Recruitment occurred prior to randomisation.
Baseline imbalance in outcome measurements or characteristics	Low risk	Participating general practices were randomised into an intervention and control group. Prior to randomisation, practices were divided into groups according to the following criteria: practice type (single-handed, duo or group practice) and presence of a specialised nurse.
Protection against contamination	Low risk	Allocation by practice
Incorrect analysis	Low risk	Controlled for cluster design
Risk of bias overall	Low risk	Low risk across most domains

Van der Velden 2016

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: primary care Speciality: GP/family N health professionals: 169 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + patient educational materials + educational meeting + clinical practice guidelines
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: verbal (interactive) Source: administrative/billing data, other Frequency: annually Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Random allocation by web-based randomisation module
Allocation concealment (selection bias)	Low risk	Allocation was undertaken centrally via a web-based randomisation module - assumed low risk of bias

Van der Velden 2016 *(Continued)*

Blinding of outcome assessment (detection bias)	Low risk	Outcomes collected from a database and unlikely subject to bias if abstractor was aware of allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Missing GPs appear balanced between groups and reasons provided to not suggest it was due to the intervention
Selective reporting (reporting bias)	Unclear risk	No protocol to check reported outcomes against
Recruitment bias	Low risk	GPs in the practices were informed about their allocation after they had promised to participate.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Table shows that the socioeconomic status of the patient population in the Intervention arm was lower than in the control arm; there were more group practices in the Intervention arm; fewer practices in the south in the Intervention arm; male/female GPs were split 50/50 in the Intervention arm, but there were more female GPs in the Control arm. These may have been important differences.
Protection against contamination	Unclear risk	Allocation to Intervention or Control was by cluster (practice). It is possible that there was an overlap between the arms (shared staff maybe) or that the Control group GPs' prescribing behaviour was affected by their knowledge of the Intervention.
Incorrect analysis	High risk	It appeared that the authors did not use recommended analyses for clusters, between cluster and even intra-cluster effects.
Risk of bias overall	High risk	Due to incorrect analysis

Van der Weijden 1999

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: primary care Speciality: GP/family physicians N health professionals: 32 N patients: 3950
Interventions	Number of arms: 2 Description of Groups: Control: guidelines; Intervention: A&F + clinician & patient educational materials + educational meetings + clinical practice guidelines + local opinion leader
Outcomes	Clinical behaviour category: screening Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: NR

Van der Weijden 1999 *(Continued)*

	Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Blocked randomisation
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Blinded data collectors
Incomplete outcome data (attrition bias) All outcomes	Low risk	20/20 practices
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of cluster recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 2
Protection against contamination	Low risk	Practices separate
Incorrect analysis	Low risk	Appropriate adjustment for clustering described (multilevel analysis + random-effects log regression)
Risk of bias overall	Low risk	Low risk across all domains

Vellinga 2016

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Ireland Setting: primary care Speciality: GP/family N health professionals: 71 N patients: NR
Interventions	Number of arms: 3

Vellinga 2016 *(Continued)*

	Description of groups: Control: coding workshop; Intervention 1: workshop + A&F + reminder; Intervention 2: workshop + A&F + reminder + educational materials
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: monthly Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	A computer-generated random sequence was used.
Allocation concealment (selection bias)	Low risk	Each practice was sequentially assigned to a study arm according to a computer-generated schedule.
Blinding of outcome assessment (detection bias)	Low risk	Outcomes based on data extracted from patient management software
Incomplete outcome data (attrition bias) All outcomes	Low risk	No practices lost to follow-up
Selective reporting (reporting bias)	Low risk	The trial was registered and protocol published. Although the protocol states qualitative data will be collected from GPs as well as costs data, these were not done, and authors did not give a reason for not collecting this data. However, given the study aim of antimicrobial prescribing behaviour, it is unlikely this would pose a threat to selective reporting of outcomes. Deviation from protocol explained with reasoning
Recruitment bias	Low risk	Patients not recruited to trial (clusters = GP practices)
Baseline imbalance in outcome measurements or characteristics	Low risk	There did not seem to be any significant differences in baseline characteristics between the 3 groups; further, the study analysis adjusted for patient and practice level characteristics, which would have accounted for differences, if any.
Protection against contamination	Low risk	Since the GPs were confined to their respective practices within each of the 3 study arms, there is low risk of contamination between practices.
Incorrect analysis	Low risk	ICC performed and analysis done at practice (unit of randomisation) level
Risk of bias overall	Low risk	Low risk across all domains

Veninga 1999

Study characteristics

Veninga 1999 *(Continued)*

Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands, Norway, Sweden, Slovakia Setting: Specialist and GP clinics Speciality: GP/family N health professionals: 665 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + education meetings
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: variable Frequency: variable Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Unclear risk	Unable to assess
Blinding of outcome assessment (detection bias)	Low risk	Outcome data unlikely to be subject to assessor bias
Incomplete outcome data (attrition bias) All outcomes	Low risk	No missing data
Selective reporting (reporting bias)	Low risk	All outcomes reported
Recruitment bias	Low risk	No evidence of cluster recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Characteristics and outcomes balanced between arms
Protection against contamination	Low risk	Cluster design to reduce contamination
Incorrect analysis	Low risk	"A multilevel model was used taking into account the hierarchical structure of the data (prescriptions were nested within patients, patients were nested

Veninga 1999 *(Continued)*

		within doctors, and doctors were nested within groups of doctors) by including random effects."
Risk of bias overall	Unclear risk	Unclear due to randomisation generation and concealment

Verstappen 2003a

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: primary care Speciality: GP/family N health professionals: 174 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + clinical practice guidelines + educational meetings
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: both written and verbal Source: administrative/billing data Frequency: not reported/not applicable Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Blocked randomisation
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Tests ordered objective data.
Incomplete outcome data (attrition bias) All outcomes	Low risk	1/26 practices lost to follow-up
Selective reporting (reporting bias)	Unclear risk	Unable to assess

Verstappen 2003a *(Continued)*

Recruitment bias	Low risk	No evidence of cluster recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 2
Protection against contamination	Low risk	Independent clinics
Incorrect analysis	Low risk	Appropriate adjustment for clustering described (3-level model)
Risk of bias overall	Low risk	Low risk across most domains

Verstappen 2003b

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: primary care Speciality: GP/family N health professionals: 174 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + clinical practice guidelines + educational meetings
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: both written and verbal Source: administrative/billing data Frequency: not reported/not applicable Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Blocked randomisation
Allocation concealment (selection bias)	Low risk	Cluster trial, allocation after recruitment completed

Verstappen 2003b *(Continued)*

Blinding of outcome assessment (detection bias)	Low risk	Tests ordered objective data.
Incomplete outcome data (attrition bias) All outcomes	Low risk	1/26 practices lost to follow-up
Selective reporting (reporting bias)	Unclear risk	Unable to assess
Recruitment bias	Low risk	No evidence of cluster recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 2
Protection against contamination	Low risk	Independent clinics
Incorrect analysis	Low risk	Appropriate adjustment for clustering described (3-level model)
Risk of bias overall	Low risk	Low risk across all important domains

Verstappen 2004

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: primary care Speciality: GP/family N health professionals: 194 N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + clinical practice guidelines + educational meetings; Intervention 2: A&F only
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: both written and verbal Source: administrative/billing data Frequency: not reported/not applicable Instructions for improvement: co-developed plans
Notes	
Risk of bias	

Verstappen 2004 *(Continued)*

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Blocked randomisation
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Tests ordered objective data.
Incomplete outcome data (attrition bias) All outcomes	Low risk	26/27 sites competed trial.
Selective reporting (reporting bias)	Unclear risk	Unable to assess
Recruitment bias	Low risk	No evidence of cluster recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 2; non-significant differences
Protection against contamination	Low risk	Independent sites
Incorrect analysis	Low risk	Appropriate adjustment for clustering described (3-level model)
Risk of bias overall	Low risk	Low risk across most domains

Vervloet 2016

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: primary care Speciality: GP/family N health professionals: 77 N patients: 154,250
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + electronic prescribing system (ICT) + educational meeting
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: verbal (interactive) Source: computerised search of charts

Vervloet 2016 *(Continued)*

	Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Method of sequence generation is not described.
Allocation concealment (selection bias)	Low risk	Clusters were paired and subsequently allocated their group. It appears likely allocation was done centrally and concealed.
Blinding of outcome assessment (detection bias)	Low risk	Patient records of prescriptions (outcome) unlikely to be biased due to knowledge of allocation
Incomplete outcome data (attrition bias) All outcomes	Low risk	No PTAM group or family physicians lost to follow-up (p. 2)
Selective reporting (reporting bias)	High risk	Trial registration has two outcomes that are not reported in this publication - occurrence of complications and patients' judgement of quality of care. Omitting the occurrence of complications in particular may be biasing the results.
Recruitment bias	Low risk	No evidence of recruitment after allocation. All eligible groups were invited to participate.
Baseline imbalance in outcome measurements or characteristics	Low risk	Despite the matching procedure used in assigning PTAMs to the intervention or control group, the total FP practice size varied largely between PTAM groups. Overall, the intervention PTAM groups had smaller practice sizes than their controls. The variability in practice size conceals the intervention effect as it complicates estimation of the variability that can be attributed to the effect of the intervention. By standardising the number of prescriptions (per 1000 patients), this variability was reduced, providing a more precise estimation of the intervention effect.
Protection against contamination	Low risk	Participants were in pre-existing PTAM groups. The control arm did not receive the intervention until after the trial.
Incorrect analysis	Low risk	Analyses performed with clustering taken into account
Risk of bias overall	High risk	High due to selective reporting

Vingerhoets 2001

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: primary care Speciality: GP/family

Vingerhoets 2001 *(Continued)*

	N health professionals: 60 N patients: 7286
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + educational materials
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: written Source: patient-reported data Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated list of random numbers
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Quote "Patients were blinded for the intervention.."
Incomplete outcome data (attrition bias) All outcomes	Low risk	Less than 10% dropouts; 4 physicians dropped out of control, one out of intervention
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of cluster recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Patient groups did not differ between arms.
Protection against contamination	Low risk	43 separate practices
Incorrect analysis	Low risk	Appropriate adjustment for clustering described (multilevel regression model)
Risk of bias overall	Low risk	Low risk across all domains

Von Lengerke 2019

Study characteristics

Von Lengerke 2019 *(Continued)*

Methods	Design: parallel cluster-RCT
Participants	Country: Germany Setting: hospital inpatient Speciality: emergency medicine N health professionals: 1087 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: education session; Intervention: A&F + tailored education sessions
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: verbal (interactive) Source: computerised search of charts Frequency: not reported/not applicable Instructions for improvement: not reported
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	No information on how sequence was generated
Allocation concealment (selection bias)	Low risk	See protocol. Based on the study flow diagram, it appears likely that all clusters were allocated at the same time.
Blinding of outcome assessment (detection bias)	High risk	Outcome assessors for NI blinded, but not hand hygiene compliance
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Not enough information to judge completeness of outcome data for MDRO infection, NI incidence and compliance rates
Selective reporting (reporting bias)	Low risk	Outcomes consistent with registered protocol (DRKS00010960) - compliance rates, and infections
Recruitment bias	Low risk	No evidence of recruitment after randomisation and allocation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Some baseline imbalances in characteristics and performances
Protection against contamination	Unclear risk	Nurses and physicians were allocated within a single-centre hospital and possible to communicate across wards/ICUs
Incorrect analysis	Low risk	Analysed by cluster-level

Von Lengerke 2019 *(Continued)*

Risk of bias overall	High risk	Due to blinding

Wadland 2007

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family/internal medicine/obstetrics and gynaecology N health professionals: 308 N patients: 1992
Interventions	Number of arms: 2 Description of groups: Control: reminders; Intervention: A&F
Outcomes	Clinical behaviour category: referrals Format of Audit & Feedback: written Source: patient-reported data, other Frequency: quarterly Instructions for improvement: not reported
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	Over 300/308 clinicians provided data.
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of cluster recruitment after randomisation

Wadland 2007 *(Continued)*

Baseline imbalance in outcome measurements or characteristics	Low risk	See Table 1 and 2
Protection against contamination	Low risk	Separate clinics
Incorrect analysis	Unclear risk	No adjustment for clustering mentioned (but repeated measures study with time being the unit of analysis)
Risk of bias overall	Unclear risk	Unable to assess risk in important domains

Wahlström 2003

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Laos Setting: internal medicine, paediatrics, outpatient Speciality: internal medicine/paediatrics N health professionals: 122 N patients: 23,156
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: verbal (interactive) Source: manual chart review Frequency: monthly Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess

Wahlström 2003 *(Continued)*

Incomplete outcome data (attrition bias) All outcomes	Low risk	Data from all hospital departments provided
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of cluster recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Unclear risk	No adjustment for clustering mentioned
Risk of bias overall	Unclear risk	Unable to assess risk in most domains

Wald 2014

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital inpatient Speciality: inpatient services N health professionals: NR N patients: 13,499
Interventions	Number of arms: 2 Description of groups: Intervention 1: education + 1 round of A&F; Intervention 2: education + 2 rounds of A&F
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: both written and verbal Source: manual chart review, computerised search of charts Frequency: semi-annually/once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**

Wald 2014 *(Continued)*

Random sequence generation (selection bias)	Unclear risk	Randomisation method not described
Allocation concealment (selection bias)	Unclear risk	No details or study flow diagram to determine whether all clusters were likely allocated at the same time
Blinding of outcome assessment (detection bias)	Unclear risk	Although outcome collection would not appear likely subject to bias (unlikely that data was recorded with bias), there is a lack of information.
Incomplete outcome data (attrition bias) All outcomes	Low risk	It appears all allocated units were included in analysis.
Selective reporting (reporting bias)	Unclear risk	No trial registration. Results report on objectives.
Recruitment bias	Low risk	Ongoing enrolment but no evidence of recruiting after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Some baseline differences were noted but the analyses appear to have adjusted for patient, hospital and unit characteristics.
Protection against contamination	Low risk	Randomisation and hospital level; not likely to be contamination
Incorrect analysis	Low risk	Clustering was taken into account in analyses.
Risk of bias overall	Unclear risk	Unable to assess risk in important domains

Walsh 2007

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital inpatient Speciality: obstetrics N health professionals: NR N patients: 2871
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + multimodal quality improvement training + benchmarking
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: variable

Walsh 2007 *(Continued)*

	Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated
Allocation concealment (selection bias)	Low risk	Allocation was concealed adequately using appropriate methods; hence, there is a low risk of selection bias.
Blinding of outcome assessment (detection bias)	Low risk	Recording of neonatal data unlikely subject to bias as intervention and control groups were unaware of the exact nature of the two trial arms. It is unclear if the research nurse abstracting data was masked to allocation, but data were quality checked and unlikely biased (even if allocation was known).
Incomplete outcome data (attrition bias) All outcomes	Low risk	The consort flow diagram shows there was no loss to follow-up or withdrawal in either arm; low risk of attrition bias
Selective reporting (reporting bias)	Unclear risk	Trial registration (written at the inception of the trial) stated one primary and one secondary outcome. Multiple secondary outcomes were added in this publication and it is possible these were added retrospectively to show improvement in the intervention arm (some of these are highlighted in the abstract).
Recruitment bias	Low risk	As per figure 1 in the paper, no infants were recruited into the study post-randomisation; there is low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Low risk	There is low risk of other bias due to baseline differences between groups as this did not affect outcomes in any way.
Protection against contamination	Low risk	Control centres were also masked to work at the intervention centres; low risk of contamination bias
Incorrect analysis	Low risk	Appropriate analytical methods used
Risk of bias overall	Low risk	Low risk across most domains

Wang 2018

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: China Setting: hospital inpatient Speciality: internal medicine N health professionals: NR

Wang 2018 *(Continued)*

	N patients: 4800
Interventions	Number of arms: 2 Description of groups: Control: usual practice + stroke registry participation; Intervention: evidence-based clinical pathway + written care protocols + quality coordinator + A&F
Outcomes	Clinical behaviour category: prescribing, screening Format of Audit & Feedback: verbal (interactive) Source: computerised search of charts Frequency: variable Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computerised random number generator was used.
Allocation concealment (selection bias)	Low risk	Allocation was concealed.
Blinding of outcome assessment (detection bias)	Low risk	Outcome evaluators described as independent and blinded
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No clusters lost. The authors do not explain the loss of patients to follow-up (397 in the Intervention arm, 423 in the Control arm).
Selective reporting (reporting bias)	Low risk	Outcomes reported in Table 2 as preplanned in protocol (2014)
Recruitment bias	Unclear risk	Consecutive eligible patients were recruited - although this happened after the hospitals had been randomised, it is unlikely that patients would be aware of the allocation.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Baseline data appear for the whole 40 hospitals and not for the Intervention baseline and Control group baseline. It was not clear whether there were any significant differences at baseline between the two groups.
Protection against contamination	Low risk	Allocation to Intervention or Control was by cluster (hospital) and it is unlikely that Control group hospitals received the intervention.
Incorrect analysis	Low risk	Table 2: adherence to evidence-based performance measures include an intra-cluster correlation coefficient calculation for each measure.
Risk of bias overall	Low risk	Low risk across most domains

Ward 1996

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Australia Setting: primary care Speciality: GP/family N health professionals: 160 N patients: 386
Interventions	Number of arms: 3 Description of groups: Intervention 1: A&F + clinical practice guidelines; Intervention 2: A&F + clinical practice guidelines + interview with GP; Intervention 3: A&F + clinical practice guidelines + interview with nurse
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: both written and verbal Source: manual chart review Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Quote "randomly allocated...stratified by number of patients recruited"
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unable to assess
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of cluster recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess

Ward 1996 *(Continued)*

Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Unclear risk	No adjustment for clustering mentioned
Risk of bias overall	Low risk	Low risk across important domains

Wathne 2018

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Norway Setting: hospital inpatient Speciality: infectious disease/pulmonary medicine/gastroenterology N health professionals: NR N patients: 1802
Interventions	Number of arms: 3 Description of groups: Control 1: no intervention; Control 2: academic detailing; Intervention: A&F
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: verbal (interactive) Source: manual chart review Frequency: once Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Random draw
Allocation concealment (selection bias)	Unclear risk	No details or study flow diagram to determine whether it is likely all clusters were allocated at the same time
Blinding of outcome assessment (detection bias)	Low risk	Assessment was performed blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	8 clusters/wards allocated to interventions; results relate to 8 clusters (no loss to follow-up). Cluster level analysis not performed, as there were fewer patients than anticipated

Wathne 2018 *(Continued)*

Selective reporting (reporting bias)	Unclear risk	No protocol to check report outcomes against. Reported results in line with methods - did change planned analysis due to low number of patients in post-intervention period
Recruitment bias	Low risk	Patients included consecutively so physicians were not able to be selective about which patients they recruited
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Baseline characteristics between groups not specifically commented on - may not make sense that they would be similar as coming from different areas of pathology
Protection against contamination	Unclear risk	Wards allocated, but different wards within the same hospital allocated to different groups - may be possible cross-contamination
Incorrect analysis	High risk	Clusters were not accounted for in the analysis.
Risk of bias overall	High risk	Due to incorrect analysis

Wattal 2015

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: India Setting: hospital inpatient Speciality: internal medicine N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: computerised search of charts Frequency: monthly Instructions for improvement: not reported
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	No information about the method of random sequence generation

Wattal 2015 *(Continued)*

Allocation concealment (selection bias)	Unclear risk	No details or study flow diagram to determine whether it is likely all clusters were allocated at the same time
Blinding of outcome assessment (detection bias)	Unclear risk	Outcomes assessment and data collection not described and unclear if bias was minimised
Incomplete outcome data (attrition bias) All outcomes	Low risk	It appears there was no loss to follow-up.
Selective reporting (reporting bias)	Unclear risk	No protocol is available to judge whether data were selectively reported.
Recruitment bias	Low risk	No evidence of recruitment bias
Baseline imbalance in outcome measurements or characteristics	High risk	The intervention/control groups stratified by unit type have big variance between n counts, i.e. cardiology has treatment A (n = 124), treatment C has n = 157 & control B has n = 132.
Protection against contamination	Low risk	Tried to account for contamination
Incorrect analysis	Low risk	Analysis at cluster level
Risk of bias overall	High risk	Due to baseline imbalance

Wei 2017

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: China Setting: primary care Speciality: paediatrics N health professionals: 42 N patients: 769
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + guidelines + education + patient materials
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: manual chart review, computerised search of charts Frequency: monthly Instructions for improvement: general

Wei 2017 *(Continued)*

Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer program in R used for randomisation
Allocation concealment (selection bias)	Unclear risk	Pilot trial with 3 interventions and 3 control clusters
Blinding of outcome assessment (detection bias)	Low risk	Objective data source (EHR)
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No clusters lost, some data excluded, but data for analysis were randomly selected subsets.
Selective reporting (reporting bias)	Low risk	Prospectively registered protocol; minimal change to outcomes reported
Recruitment bias	Unclear risk	More doctors at endpoint than baseline for intervention group
Baseline imbalance in outcome measurements or characteristics	Low risk	Baseline characteristics appear mostly similar.
Protection against contamination	High risk	Some contamination between intervention and control clusters might have occurred because the townships in different groups were sometimes next to each other.
Incorrect analysis	Low risk	Clustering effect considered in analysis
Risk of bias overall	High risk	Due to protection against contamination

Wells 2000

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 181 N patients: 1356
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + clinician/patient education
Outcomes	Clinical behaviour category: prescribing, counselling

Wells 2000 *(Continued)*

	Format of Audit & Feedback: verbal (didactic) Source: manual chart review Frequency: monthly Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	A random number table was used to allocate clusters to Intervention and Control.
Allocation concealment (selection bias)	Low risk	No details, but based on the study flow diagram, it appears likely that all clusters were allocated at the same time.
Blinding of outcome assessment (detection bias)	High risk	Some outcomes self-reported by patients (not blinded to allocation) so potential for bias
Incomplete outcome data (attrition bias) All outcomes	Low risk	Losses to follow-up were mostly similar (as proportions) in both groups and explained by death, non-response, or dropouts.
Selective reporting (reporting bias)	Unclear risk	The protocol for this study is not available. Process of care and health outcomes are reported as planned in the Methods (Results and Tables 4 & 5). Employment outcomes are reported only in the Comment section (not Results).
Recruitment bias	Unclear risk	Although physicians were recruited to the trial prior to knowing about their intervention status, there was no information to confirm that subsequent recruitment of the patients was affected by physicians' knowledge of the intervention.
Baseline imbalance in outcome measurements or characteristics	Low risk	Annotated text (not visible in Covidence) reports that clinic clusters were grouped into matched blocks based on patient demographics, clinician speciality and distance to mental health providers. Table 2 shows baseline characteristics of patients, which are similar between intervention and control groups.
Protection against contamination	Low risk	Allocation to Intervention or Control was by cluster (clinic) and it is unlikely that Control group clinics received the intervention.
Incorrect analysis	Low risk	Annotated text (not visible in Covidence) shows that the authors considered clustering effects but decided to report unadjusted results as intraclass correlations are close to zero.
Risk of bias overall	High risk	Due to blinding

Welschen 2004

Study characteristics

Methods	Design: parallel cluster-RCT

Welschen 2004 *(Continued)*

Participants	Country: Netherlands Setting: primary care Speciality: GP/family N health professionals: 100 N patients: 3620
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F clinician education + patient education
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: written Source: administrative/billing data Frequency: tri-annually Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Tossed coin used to ensure random allocation
Allocation concealment (selection bias)	Low risk	Allocation likely done at one time
Blinding of outcome assessment (detection bias)	Low risk	Physicians (aware of allocation) recorded in patient files and blinded researchers abstracted data from files. Unlikely to be problematic
Incomplete outcome data (attrition bias) All outcomes	Low risk	No groups lost to follow-up. Similar numbers of GPs lost to follow-up in either arm & authors provided explanation for this.
Selective reporting (reporting bias)	Unclear risk	Outcome (antibiotic prescribing rate for RTIs) reported as specified in Methods. Protocol not available
Recruitment bias	Unclear risk	It was unclear exactly the timing of the recruitment, and it may have occurred following the selection of the clusters.
Baseline imbalance in outcome measurements or characteristics	Low risk	At baseline, mean antibiotic prescription rates for registered encounters for respiratory tract symptoms did not differ significantly between the two groups (27% v 29%).
Protection against contamination	Low risk	Allocation to Intervention or Control was by cluster (peer review group) and it is unlikely that there was contamination between the two arms.
Incorrect analysis	Low risk	The authors did take into account the intra-class coefficients.
Risk of bias overall	Low risk	Low risk across most domains

Whidden 2018

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Mali Setting: community care Speciality: NA N health professionals: 148 N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: performance dashboard + supervision + A&F; Intervention 2: supervision + A&F
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: electronic dashboard Source: manual chart review, patient-reported data Frequency: monthly Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Although stratified randomisation is mentioned, there was no description of the method of random sequence generation.
Allocation concealment (selection bias)	Low risk	No details, but based on the study flow diagram and study timeline, it appears likely that all clusters were allocated at the same time
Blinding of outcome assessment (detection bias)	Low risk	Although supervisors responsible for collecting data forms were not blinded, the data clerks who entered the data into databases were blinded. Bias appears unlikely.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Only one CHW lost due to retirement. Data are otherwise complete.
Selective reporting (reporting bias)	Low risk	Outcomes match trial registration outcomes.
Recruitment bias	Low risk	No evidence of recruitment after allocation
Baseline imbalance in outcome measurements or characteristics	Low risk	No apparent baseline imbalances

Whidden 2018 *(Continued)*

Protection against contamination	High risk	Potential source of bias
Incorrect analysis	Low risk	Clustering taken into account in analyses
Risk of bias overall	High risk	Potential risk of contamination - high

Williams 2016

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital inpatient Speciality: neurology N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: quality improvement program + A&F; Intervention 2: A&F only
Outcomes	Clinical behaviour category: screening, prescribing, treatment decision/action, counselling Format of Audit & Feedback: verbal (didactic) Source: computerised search of charts, administrative/billing data Frequency: variable Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Randomisation method not described
Allocation concealment (selection bias)	Low risk	No details, but based on the study flow diagram, it appears likely that all clusters were allocated at the same time
Blinding of outcome assessment (detection bias)	Low risk	It is unclear whether assessors were blinded, but they were remotely collecting chart data, so even if not, it appears to present low risk.
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	One intervention site missing from post-I data collection; unclear why

Williams 2016 *(Continued)*

Selective reporting (reporting bias)	Unclear risk	There is no protocol available; not possible to assess
Recruitment bias	Low risk	Appears to be all enroled at the same time; no post-enrolment of sites
Baseline imbalance in outcome measurements or characteristics	Low risk	Some baseline imbalances but analyses adjusted for baseline parameters
Protection against contamination	Low risk	Due to cluster design, contamination is unlikely.
Incorrect analysis	Low risk	Clustering taken into account in analyses
Risk of bias overall	Low risk	Low risk across most domains

Willis 2020a

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: NR N patients: 99,829
Interventions	Number of arms: 3 Description of groups: Control: A&F on blood pressure control + reminders + education; Intervention: A&F on anticoagulation + reminders + education
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: quarterly Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence

Willis 2020a *(Continued)*

Allocation concealment (selection bias)	Low risk	Central allocation and likely at the same time (per flow chart)
Blinding of outcome assessment (detection bias)	Low risk	Blinded outcome assessment
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition of practices balanced, and all practices analysed per ITT
Selective reporting (reporting bias)	Low risk	Outcomes consistent with protocol (Willis 2016)
Recruitment bias	Low risk	All clinics recruited before randomisation.
Baseline imbalance in outcome measurements or characteristics	Low risk	Well-balanced for characteristics of practices and patients
Protection against contamination	Low risk	Allocation was by practice; unlikely contamination between practitioners at different practices
Incorrect analysis	Low risk	Analysis took clustering into account.
Risk of bias overall	Low risk	Overall low risk of bias

Willis 2020b

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: NR N patients: 99,829
Interventions	Number of arms: 4 Description of groups: Control: A&F on anticoagulation + reminders + education; Intervention: A&F on blood pressure control + reminders + education
Outcomes	Clinical behaviour category: screening Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: quarterly Instructions for improvement: co-developed plans

Willis 2020b *(Continued)*

Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Central allocation and likely at the same time (per flow chart)
Blinding of outcome assessment (detection bias)	Low risk	Blinded outcome assessment
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition of practices balanced, and all practices analysed per ITT
Selective reporting (reporting bias)	Low risk	Outcomes consistent with protocol (Willis 2016)
Recruitment bias	Low risk	All clinics recruited before randomisation.
Baseline imbalance in outcome measurements or characteristics	Low risk	Well-balanced for characteristics of practices and patients
Protection against contamination	Low risk	Allocation was by practice; unlikely contamination between practitioners at different practices
Incorrect analysis	Low risk	Analysis took clustering into account.
Risk of bias overall	Low risk	Overall low risk of bias

Willis 2020c

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: NR N patients: 67,475
Interventions	Number of arms: 5 Description of groups: Control: A&F on risky prescribing + reminders + education; Intervention: A&F on diabetes + reminders + education
Outcomes	Clinical behaviour category: testing/exams, screening, prescribing, immunisation, referrals

Willis 2020c *(Continued)*

	Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: quarterly Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Central allocation and likely at the same time (per flow chart)
Blinding of outcome assessment (detection bias)	Low risk	Blinded outcome assessment
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition of practices balanced, and all practices analysed per ITT
Selective reporting (reporting bias)	Low risk	Outcomes consistent with protocol (Willis 2016)
Recruitment bias	Low risk	All clinics recruited before randomisation.
Baseline imbalance in outcome measurements or characteristics	Low risk	Well-balanced for characteristics of practices and patients
Protection against contamination	Low risk	Allocation was by practice; unlikely contamination between practitioners at different practices
Incorrect analysis	Low risk	Analysis took clustering into account.
Risk of bias overall	Low risk	Overall low risk of bias

Willis 2020d

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: UK Setting: primary care Speciality: GP/family N health professionals: NR

Willis 2020d *(Continued)*

	N patients: 67,475
Interventions	Number of arms: 6 Description of groups: Control: A&F on diabetes + reminders + education; Intervention: A&F on risky prescribing + reminders + education
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: computerised search of charts Frequency: quarterly Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Computer-generated random sequence
Allocation concealment (selection bias)	Low risk	Central allocation and likely at the same time (per flow chart)
Blinding of outcome assessment (detection bias)	Low risk	Blinded outcome assessment
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition of practices balanced, and all practices analysed per ITT
Selective reporting (reporting bias)	Low risk	Outcomes consistent with protocol (Willis 2016)
Recruitment bias	Low risk	All clinics recruited before randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Well-balanced for characteristics of practices and patients
Protection against contamination	Low risk	Allocation was by practice; unlikely contamination between practitioners at different practices
Incorrect analysis	Low risk	Analysis took clustering into account.
Risk of bias overall	Low risk	Overall low risk of bias

Winickoff 1984

Study characteristics

Methods	Design: other

Winickoff 1984 *(Continued)*

Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 16 N patients: 1847
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: written Source: computerised search of charts Frequency: monthly Instructions for improvement: not reported
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomised after stratification
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Low risk	No losses reported
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Performance similar pre-intervention
Protection against contamination	High risk	Authors acknowledged the possibility.
Incorrect analysis	Unclear risk	Limited info on statistical methodology
Risk of bias overall	High risk	Due to protential contamination

Winkens 1995a

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: primary care Speciality: GP/family N health professionals: 79 N patients: 187,000
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: written Source: manual chart review Frequency: semi-annually Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	High risk	No blinding done
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data collected from central registry routinely
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess

Winkens 1995a *(Continued)*

Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Unclear risk	No adjustment for clustering mentioned
Risk of bias overall	High risk	Due to blinding

Winkens 1995b

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Netherlands Setting: primary care Speciality: GP/family N health professionals: 79 N patients: 187,000
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: written Source: manual chart review Frequency: semi-annually Instructions for improvement: specific
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	High risk	No blinding done
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data collected from central registry routinely

Winkens 1995b *(Continued)*

Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Unclear risk	No adjustment for clustering mentioned
Risk of bias overall	High risk	Due to blinding

Winslade 2016a

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Canada Setting: pharmacy Speciality: NA N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F
Outcomes	Clinical behaviour category: counselling Format of Audit & Feedback: written Source: manual chart review Frequency: once Instructions for improvement: general
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Randomisation conducted by statistician using a random numbers table.
Allocation concealment (selection bias)	Low risk	Allocation likely performed centrally using sequential randomisation

Winslade 2016a *(Continued)*

Blinding of outcome assessment (detection bias)	Low risk	Only the pharmacists' regulatory authority had access to look up codes for real pharmacy names.
Incomplete outcome data (attrition bias) All outcomes	Low risk	See online supp - incomplete data from pharmacies appears balanced and due to problems unrelated to the intervention/outcomes.
Selective reporting (reporting bias)	Unclear risk	The protocol is not available, but all outcomes are reported as prespecified in the Methods.
Recruitment bias	Low risk	There is no evidence of recruitment after randomisation and allocation.
Baseline imbalance in outcome measurements or characteristics	Low risk	Characteristics of pharmacies, pharmacists and patients seem to be well-balanced (Table 1).
Protection against contamination	Low risk	Due to the cluster design, it is unlikely there was contamination.
Incorrect analysis	High risk	Clustering was not taken into account during analysis.
Risk of bias overall	High risk	Incorrect analysis - high

Winslade 2016b

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: Canada Setting: pharmacy Speciality: NA N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F
Outcomes	Clinical behaviour category: counselling Format of Audit & Feedback: written Source: manual chart review Frequency: once Instructions for improvement: specific
Notes	
Risk of bias	

Winslade 2016b *(Continued)*

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomisation conducted by statistician using a random numbers table
Allocation concealment (selection bias)	Low risk	Allocation likely performed centrally using sequential randomisation
Blinding of outcome assessment (detection bias)	Low risk	Only the pharmacists' regulatory authority had access to look up codes for real pharmacy names.
Incomplete outcome data (attrition bias) All outcomes	Low risk	See online supp - incomplete data from pharmacies appears balanced and due to problems unrelated to the intervention/outcomes.
Selective reporting (reporting bias)	Unclear risk	The protocol is not available, but all outcomes are reported as prespecified in the Methods.
Recruitment bias	Low risk	There is no evidence of recruitment after randomisation and allocation.
Baseline imbalance in outcome measurements or characteristics	Low risk	Characteristics of pharmacies, pharmacists and patients seem to be well-balanced (Table 1).
Protection against contamination	Low risk	Due to the cluster design, it is unlikely there was contamination.
Incorrect analysis	High risk	Clustering was not taken into account during analysis.
Risk of bias overall	High risk	Incorrect analysis - high

Wones 1987

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: hospital inpatient Speciality: internal medicine N health professionals: 21 N patients: NR
Interventions	Number of arms: 3 Description of groups: Control: no intervention; Intervention 1 : A&F (individual feedback); Intervention 2: A&F (with group comparator)
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: written Source: computerised search of charts

Wones 1987 *(Continued)*

	Frequency: bi-weekly Instructions for improvement: general
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	Low risk	Utilisation data from central computer
Incomplete outcome data (attrition bias) All outcomes	Low risk	Data provided on all 21 residents (Table 2)
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Unable to assess
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Unclear risk	No adjustment for clustering mentioned
Risk of bias overall	Low risk	Low risk across all important domains

Wu 2019

Study characteristics

Methods	Design: step wedge
Participants	Country: China Setting: hospital inpatient Speciality: internal medicine N health professionals: NR N patients: 29,346
Interventions	Number of arms: 2

Wu 2019 *(Continued)*

	Description of groups: Control: no intervention; Intervention: Quality of Care Initiative team + staff training + clinical pathways + A&F + online technical support + patient education
Outcomes	Clinical behaviour category: treatment decision/action Format of Audit & Feedback: electronic dashboard Source: computerised search of charts Frequency: bi-annually Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Low risk	Central randomisation was done using a computerised random schedule (information given in the protocol), hence, there is a low risk of selection bias.
Allocation concealment (selection bias)	Low risk	Allocation concealed and done centrally
Blinding of outcome assessment (detection bias)	Low risk	No information whether data collector was blinded; however, outcomes are objective so detection bias is unlikely even if allocation was known. Most outcome data were objective and primary outcomes were also adjudicated by independent committee masked to allocation.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Two hospitals withdrew before the intervention. This attrition is likely not of concern.
Selective reporting (reporting bias)	Low risk	Trial was registered and study protocol was published. Three secondary outcomes were added after the protocol was published with reasons explained.
Recruitment bias	Low risk	Recruitment occurred before randomisation via stratification by province.
Baseline imbalance in outcome measurements or characteristics	Low risk	In analysis plan supp file, stated that used GEE to check for baseline imbalance; also described analytic steps to account for potential imbalance.
Protection against contamination	Low risk	Due to cluster design, contamination is unlikely.
Incorrect analysis	Low risk	Clustering was accounted for in the analysis.
Risk of bias overall	Low risk	Low risk across all domains

Yano 2008

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: USA

Yano 2008 *(Continued)*

	Setting: primary care Speciality: GP/family N health professionals: NR N patients: 1941
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + education + priority setting + QI plan + facilitation; Intervention 2: A&F + mailed guidelines
Outcomes	Clinical behaviour category: counselling, referrals Format of Audit & Feedback: written Source: administrative/billing data Frequency: quarterly Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Method of sequence generation is not described.
Allocation concealment (selection bias)	Low risk	No details but based on the study flow diagram, it appears likely that all clusters were allocated at the same time.
Blinding of outcome assessment (detection bias)	Unclear risk	Knowledge of allocation could have influenced outcome assessment.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Similar proportions of patients lost to follow-up in intervention and control groups and reasons given. ITT used.
Selective reporting (reporting bias)	Unclear risk	There was no prior publication or trial registration that defined the outcomes of interest.
Recruitment bias	Low risk	From Figure 1, it appears that the practices were randomised first into intervention and control practices.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Practices similar at baseline, apart from higher use of provider incentives and tracking of screening performance (intervention practices); more provider counselling feedback and pharmacotherapy prescription authority (control practices) Some baseline differences in patients: Control patients more likely to smoke every day, wake up to smoke, to have tried nicotine patches, attended a smoking cessation programme, and tried other ways to quit pre-intervention. Also reported lower general health status, greater receipt of disability/Social Security Income, less likely to live alone. More likely to use primary care providers rather than specialists. (annotated text not visible in Covidence)

Yano 2008 *(Continued)*

		Authors adjusted for patient baseline differences, but reported that this did not change the effect of intervention.
Protection against contamination	Low risk	Allocation to Intervention or Control was by cluster (practice) and it is unlikely that Control group practices received the intervention.
Incorrect analysis	Low risk	The authors considered using an intraclass correlation coefficient to address problem of clustering effect, but as it was calculated to be not statistically significant from zero an unadjusted analytic approach was adopted. (annotated text not visible in Covidence).
Risk of bias overall	Unclear risk	Unable to assess risk in important domains

Zafar 2019

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: community care Speciality: GP/family N health professionals: 108 N patients: 5398
Interventions	Number of arms: 2 Description of groups: Control: CDS alerts; Intervention: A&F
Outcomes	Clinical behaviour category: testing/exams Format of Audit & Feedback: written Source: computerised search of charts Frequency: quarterly Instructions for improvement: not reported
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	No clear description of how practices were randomised
Allocation concealment (selection bias)	Unclear risk	No description of how practices were allocated and no flow chart
Blinding of outcome assessment (detection bias)	Low risk	As both arms were receiving interventions, it is unlikely that knowledge of allocation biased outcome data collection/assessment.

Zafar 2019 *(Continued)*

Incomplete outcome data (attrition bias) All outcomes	High risk	50% (54/108) of PCPs remained (28 in one group and 26 in the other group) from 8 practices, but it is unclear how many PCPs started in each arm and reasons for PCPs dropping out. Due to the high rate of loss and lack of explanation, risk of bias is suspected.
Selective reporting (reporting bias)	Unclear risk	No protocol or trial registration, so unclear if all preplanned outcomes were reported.
Recruitment bias	Low risk	There was no study flow diagram given however table S1 stated that those PCPs who joined the study after randomisation were excluded; hence, presumably there is a low risk of recruitment bias.
Baseline imbalance in outcome measurements or characteristics	Unclear risk	Baseline data are not presented for separate arms. Some differences in characteristics of the practices (see table S1 e.g. 7 community PCPs in arm 1 vs 19 in arm 2; 0 PCPs in family medicine in arm 1 vs 9 in arm 2) are not discussed. No adjustments made
Protection against contamination	High risk	2/54 (4%) of PCPs worked in multiple locations. This could be significant as some intervention periods had as few as 4 PCPs participating at the time in one arm.
Incorrect analysis	High risk	Analyses do not adjust for clustering.
Risk of bias overall	High risk	Due to incomplete outcome data, protection against contamination and incorrect analysis

Ziemer 2006

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family N health professionals: 345 N patients: 4038
Interventions	Number of arms: 4 Description of groups: Control 1: no intervention; Control 2: reminders; Intervention 1: A&F; Intervention 2: A&F + reminders
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: verbal (interactive) Source: manual chart review, computerised search of charts Frequency: bi-weekly Instructions for improvement: co-developed plans
Notes	

Ziemer 2006 *(Continued)*

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Unclear risk	Unable to assess
Blinding of outcome assessment (detection bias)	Unclear risk	Unable to assess
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unable to assess
Selective reporting (reporting bias)	Unclear risk	Unable to assess
Recruitment bias	Unclear risk	Recruitment and randomisation were ongoing from Jan 2000 to Dec 2002, rather than sequential.
Baseline imbalance in outcome measurements or characteristics	Low risk	Patient groups similar, see p. 509
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Low risk	Appropriate adjustment for clustering described (random-effects variable; "patients nested within providers")
Risk of bias overall	Unclear risk	Unable to assess bias across most domains

Zimmerman 2014

Study characteristics	
Methods	Design: parallel cluster-RCT
Participants	Country: USA Setting: primary care Speciality: GP/family/paediatrics N health professionals: NR N patients: 81,599
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + provider education + reminders + patient materials
Outcomes	Clinical behaviour category: immunisation

Zimmerman 2014 *(Continued)*

	Format of Audit & Feedback: written Source: computerised search of charts Frequency: weekly Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomisation method not described
Allocation concealment (selection bias)	Low risk	Based on the study flow diagram, clusters were allocated at the same time. Trialists also reported that cluster-RCT CONSORT criteria were met.
Blinding of outcome assessment (detection bias)	Low risk	It was not very clear who the outcome assessors were, but it does appear to be an independent group who housed the data. It doesn't seem likely the outcome data would be subject to bias.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Table 1 and 2 reported numbers across clusters analysed. No clusters lost
Selective reporting (reporting bias)	Low risk	Outcomes prespecified in trial registration appear to be reported.
Recruitment bias	Low risk	It does not appear that individuals recruited for the trial after clusters randomised
Baseline imbalance in outcome measurements or characteristics	Low risk	Similar characteristics across intervention clusters
Protection against contamination	Low risk	Due to cluster design with practices as the cluster, contamination is unlikely.
Incorrect analysis	Low risk	Analysis took into account clustering.
Risk of bias overall	Low risk	Low risk across all important domains

Zwar 1999

Study characteristics

Methods	Design: parallel cluster-RCT
Participants	Country: Australia Setting: primary care Speciality: GP/family N health professionals: 157

Zwar 1999 *(Continued)*

	N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + guidelines + patient education + educational visit; Intervention 2: A&F + guidelines + patient education + educational visit
Outcomes	Clinical behaviour category: prescribing Format of Audit & Feedback: both written and verbal Source: other Frequency: semi-annually Instructions for improvement: co-developed plans
Notes	

Risk of bias

Bias	**Authors' judgement**	**Support for judgement**
Random sequence generation (selection bias)	Unclear risk	Unable to assess
Allocation concealment (selection bias)	Low risk	Cluster trial; allocation after recruitment completed
Blinding of outcome assessment (detection bias)	High risk	Self-report data
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Unable to assess impact of dropout
Selective reporting (reporting bias)	Low risk	Appropriate outcomes reported
Recruitment bias	Low risk	No evidence of recruitment after randomisation
Baseline imbalance in outcome measurements or characteristics	Low risk	Trainees similar at baseline
Protection against contamination	Unclear risk	Unable to assess
Incorrect analysis	Low risk	"GP trainees were the unit of randomization and the unit of analysis".
Risk of bias overall	High risk	High risk of bias due to blinding

A1c: glycohaemoglobin
abx: antibiotics
ACE: angiotensin-converting-enzyme (inhibitor)
ADHD: attention deficit hyperactivity disorder
A&F: audit and feedback
AMI: acute myocardial infarction
ANCOVA: analysis of covariance

ANOVA: analysis of variance
ARI: acute respiratory infection
ASCVD: atherosclerotic cardiovascular disease
ATPIII: National Cholesterol Education Program's Adult Treatment Panel III report
BB: beta blocker
BI: brain injury
BP: blood pressure
CABG: coronary artery bypass graft
CDE: certified diabetes educator
CDS: clinical decision support
CDSS: clinical decision support system
CHD: coronary heart disease
CHF: congestive heart failure
CME: continuing medical education
COPD:chronic obstructive pulmonary disease
CQI: continuous quality improvement
CT: computed topography
CV: cardiovascular
DVT: deep vein thrombosis
EBP: evidence-based practice
EHR: electronic health record
EMR: electronic medical record
ER: emergency room
FOBT: faecal occult blood test
GEE: generalised estimating equation
GLM: generalised linear model
GP: general practitioner
IBD: inflammatory bowel disease
ICA: internal carotid artery
ICU: intensive care unit
IT: information technology
ITT: intention-to-treat
JITIF-WBA: just-in-time information and feedback - web-based aid
JNC: Joint National Committee
LDL: low-density lipoprotein
MA: medical assistant
MI: myocardial infarction
MRI: magnetic resonance imaging
NA: not applicable
NPP: non-physician practitioner
NR: not reported
NSAID: non-steroidal inflammatory drug
OMSC: Ottawa Model for Smoking Cessation
OR: odds ratio
PCP: primary care provider
RCT: randomised controlled trial
RT: respiratory therapist
Rx: treatment
QA: quality assurance
QI: quality improvement
SAS: statistical analysis system
SMS: short message service
SSS: sick sinus syndrome
STAT: statistic
TTE: transthoracic echocardiogram
TTT: tilt table test
UTI: urinary tract infection
vs: versus
VTE: venous thromboembolism

Characteristics of excluded studies *[ordered by study ID]*

Study	Reason for exclusion
Alexander 2011	No usable data
Anderson 1996	Not clearly A&F
Berman 1998	Cost feedback only
Berwanger 2018	Conference abstract
Buntinx 1993	Not A&F
Carroll 2018	No usable data
Cheater 2004	Protocol
Claes 2005	Focused on patient outcomes, not professional practices
Cline 2007	Patient-specific
Cloutier 2012	No usable outcomes
Cohen 1982	Cost feedback only
Curran 2008	Patient outcomes only
De Almeida Neto 2000	Not A&F
Dudzinski 2015	Conference abstract
Finkelstein 2008	No usable outcomes
Fischer 2011	No relevant outcomes
Forrest 2013	No usable data
Gama 1992	Not enough clusters
Gehlbach 1984	Not A&F
Halterman 2014	Not A&F
Handy 2017	Conference abstract
Heller 2001	No usable data
Hershey 1986	Not A&F
Hershey 1988	Not A&F
Hwang 2019	Cannot find full text
Kirwin 2010	No feedback

Study	Reason for exclusion
Kogan 2003	No usable data
Lagerløv 2000	No usable data
Linn 1980	Not A&F
Lytle 2015	No usable data
Magrini 2014	No usable data; not clear which direction feedback was trying to shift prescribing
Mayer 1998	No usable data; not actual patients
Mayne 2014	Secondary analysis of Fiks
McAlister 1986	Not A&F
Millard 2008	Not usable data
Norton 1985	Not A&F
Ogedegbe 2014	No usable data
Palmay 2014	Not audit and feedback
Palmer 1996	Secondary analysis of Palmer 1985
Papaioannou 2013	Co-publication of Kennedy 2015
Parchman 2019	Not A&F
Peden 2019a	No usable data
Peden 2019b	No usable data
Phillips 2004	Not A&F
Phillips 2005	Not A&F
Reiter 2015	No process measures
Sandbaek 1999	No usable data
Sinnema 2015	Not A&F
Sommers 1984	No usable data
Stenehjem 2018	Not A&F
Svetkey 2009	No process measures
Tjia 2015	No usable data
Trent 2018	Incorrect reference; duplicate of Trent 2019

Study	Reason for exclusion
Tuot 2018	No usable data
Vaisson 2019	No usable data
Van den Hombergh 1999	No quality of care metrics in audit
Weitzman 2009	Patient-level outcomes only
Wiggers 2017	Not A&F
Williams 2014	Conference abstract
Yadav 2019	No usable data
Young 2003	Not A&F

A&F: audit and feedback
RCT: randomised controlled trial

Characteristics of studies awaiting classification *[ordered by study ID]*

Adusumalli 2023

Methods	Design: Parallel Cluster-RCT
Participants	Country: USA Setting: Primary care N health professionals: 3426 N patients: 835,277
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention 1: A&F + peer benchmarking
Outcomes	Clinical behaviour category: Prescribing
Notes	

Aghdassi 2020

Methods	Design: Parallel Cluster-RCT
Participants	Country: Germany Setting: Hospital inpatient N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F quarterly, plus education and dispensers
Outcomes	Clinical behaviour category: Other
Notes	

Aghlmandi 2023

Methods	Design: Parallel Cluster-RCT
Participants	Country: Switzerland Setting: Primary care N health professionals: 158 N patients: 4131
Interventions	Number of arms: 4 Description of groups: Control : no intervention; Intervention 1: clinical nudge (prescribing choice prompt + monthly peer comparison feedback report); Intervention 2: patient nudge (text messages to remind patients to discuss the statin during their GP consultation); Intervention 3: combined conditions in arm 2 and arm 3
Outcomes	Clinical behaviour category: Prescribing
Notes	

Andrade 2022

Methods	Design: Parallel Cluster-RCT
Participants	Country: Australia Setting: Primary care N health professionals: 2552 N patients: 3271
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F + educational materials in DIGITAL format; Intervention 2: A&F + educational materials in POSTAL letter format
Outcomes	Clinical behaviour category: Prescribing
Notes	

Ashworth 2021

Methods	Design: Parallel Cluster-RCT
Participants	Country: Canada Setting: Primary care N health professionals: 296 N patients: NR
Interventions	Number of arms: 4 Description of groups: Intervention 1: A&F one time with patient-level information; Intervention 2: A&F + letter from regulatory college; Intervention 3: A&F + letter and phone call from pharmacist at regulatory college; Intervention 4: A&F + letter and phone call from physician at regulatory college
Outcomes	Clinical behaviour category: Prescribing
Notes	

Carney 2023

Methods	Design: Parallel Cluster-RCT
Participants	Country: Canada Setting: Primary care N health professionals: 4833 N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F including a description of a common clinical encounter, E. coli resistance rate, prescribing benchmarking bar chart, and development process of the A&F; Intervention 2: same intervention but with one-year delay
Outcomes	Clinical behaviour category: Prescribing
Notes	

Catho 2022

Methods	Design: Parallel Cluster-RCT
Participants	Country: Switzerland Setting: Hospital inpatient N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: receiving: A&F + teaching sessions + benchmarking; Intervention: A&F + decision-support (for type and duration of treatment) + self-guided re-evaluation of initial prescription
Outcomes	Clinical behaviour category: Prescribing
Notes	

Chang 2020

Methods	Design: Cross-over
Participants	Country: China Setting: Primary care N health professionals: 164 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F every 10 days on antibiotic prescribing
Outcomes	Clinical behaviour category: Prescribing
Notes	

Chappell 2021

Methods	Design: Parallel Cluster-RCT
Participants	Country: New Zealand Setting: Primary care N health professionals: 1260 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F letter with social norm info
Outcomes	Clinical behaviour category: Prescribing
Notes	

Chidwick 2022

Methods	Design: Parallel Cluster-RCT
Participants	Country: Australia Setting: Primary care N health professionals: NR N patients: 4935
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F and one-hour facilitated discussion
Outcomes	Clinical behaviour category: Prescribing
Notes	

Coombs 2021

Methods	Design: Step-Wedge
Participants	Country: Australia Setting: Emergency rooms N health professionals: 269 N patients: 4491
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F, education, fast-track outpatient care
Outcomes	Clinical behaviour category: Testing/exams
Notes	

Crowley 2023

Methods	Design: Parallel Cluster-RCT
Participants	Country: USA Setting: Hospital inpatient N health professionals: 438 N patients: 21,653
Interventions	Number of arms: 4 Description of groups: Control: usual care; Intervention 1: A&F; Intervention 2: peer benchmarking; Intervention 3: A&F + peer benchmarking
Outcomes	Clinical behaviour category: Prescribing
Notes	

Curtis 2021

Methods	Design: Parallel Cluster-RCT
Participants	Country: England Setting: Primary care N health professionals: NR N patients: NR
Interventions	Number of arms: 3 Description of groups: Control: no intervention; Intervention 1: A&F with same message sent three times; Intervention 2: A&F with same message sent three times
Outcomes	Clinical behaviour category: Prescribing
Notes	

Daneman 2021

Methods	Design: Parallel Cluster-RCT
Participants	Country: Canada Setting: Nursing homes N health professionals: 343 N patients: 36,194
Interventions	Number of arms: 2 Description of groups: Intervention 1: Dynamic A&F; Intervention 2: Static A&F
Outcomes	Clinical behaviour category: Prescribing
Notes	

Daneman 2022

Methods	Design: Parallel Cluster-RCT
Participants	Country: Canada Setting: Other N health professionals: NR N patients: > 28,000
Interventions	Number of arms: 3 Description of groups: Intervention 1: A&F email with social peer comparison nudge; Intervention 2: A&F email with "a maintenance of professional certification incentive nudge"; Intervention 3: A&F email with a prior participation nudge
Outcomes	Clinical behaviour category: Prescribing
Notes	

Devore 2021

Methods	Design: Parallel Cluster-RCT
Participants	Country: USA Setting: Hospital inpatient N health professionals: NR N patients: 5647
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F, education, Continuous quality improvement
Outcomes	Clinical behaviour category: Other
Notes	

Dutcher 2022

Methods	Design: Step-Wedge
Participants	Country: USA Setting: Primary care N health professionals: 183 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F - peer benchmarking + educational sessions
Outcomes	Clinical behaviour category: Prescribing
Notes	

Du Yan 2021

Methods	Design: Parallel Cluster-RCT
Participants	Country: USA Setting: Primary care N health professionals: 45 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: education only; Intervention: A&F monthly dashboard and education
Outcomes	Clinical behaviour category: Prescribing
Notes	

Emond 2022

Methods	Design: Step-Wedge
Participants	Country: Netherlands Setting: Hospital inpatient N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: educational activities, audit and feedback, reminders, organisational, team-directed, and patient-mediated activities
Outcomes	Clinical behaviour category: Other
Notes	

Gold 2022a

Methods	Design: Other
Participants	Country: England Setting: Primary care N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F (social norm benchmarking)
Outcomes	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F (social norm benchmarking)
Notes	

Gold 2022b

Methods	Design: Other
Participants	Country: England Setting: Primary care N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: overall prescribing letter; Intervention 1: broad-spectrum letter A&F; Intervention 2: broad-spectrum letter A&F
Outcomes	Clinical behaviour category: Prescribing
Notes	

Haskell 2021

Methods	Design: Parallel Cluster RCT
Participants	Country: Australia and New Zealand Setting: Hospital inpatient N health professionals: NR N patients: 3727
Interventions	Number of arms: 2 Description of groups: Control: guideline dissemination; Intervention: site-based clinical leads, stakeholder meetings, a train-the-trainer workshop, targeted educational delivery, other educational and promotional materials, and audit and feedback
Outcomes	Clinical behaviour category: Other
Notes	

Hocking 2022

Methods	Design: Factorial
Participants	Country: Australia Setting: Primary care N health professionals: NR N patients: 49,525
Interventions	Number of arms: 4 Description of groups: Control: no intervention; Intervention 1: incentive; Intervention 2: A&F + no incentive; Intervention 3: A&F + incentive
Outcomes	Clinical behaviour category: Testing/exams
Notes	

James 2022

Methods	Design: Step-Wedge
Participants	Country: Canada Setting: Other N health professionals: 34 N patients: 7106
Interventions	Number of arms: 2 Description of groups: Control: usual care; Intervention: A&F + educational session + decision aid
Outcomes	Clinical behaviour category: Treatment decision/action
Notes	

Jones 2023

Methods	Design: Step-Wedge
Participants	Country: Australia Setting: Primary care N health professionals: NR N patients: 37,946
Interventions	Number of arms: 4 Description of groups: Intervention 1: 2 intervention clusters in 4 periods (each) and 2 control clusters in 1 period ; Intervention 2: 2 intervention clusters in 3 periods (each) and 2 control clusters in 2 periods (each); Intervention 3: 2 intervention clusters in 2 periods (each) and 2 control clusters in 3 periods (each); Intervention 4: 2 intervention clusters in 1 period and 2 control clusters in 4 periods (each)
Outcomes	Clinical behaviour category: Diagnosis
Notes	

Kallen 2021

Methods	Design: Parallel Cluster-RCT
Participants	Country: Netherlands Setting: Hospital inpatient N health professionals: NR N patients: 8840
Interventions	Number of arms: 3 Description of groups: Intervention 1: A&F on antibiotic volume and continuous quality improvement; Intervention 2: A&F on appropriateness of antibiotics and continuous quality improvement; Intervention 3: A&F brief on appropriateness and continuous quality improvement
Outcomes	Clinical behaviour category: Prescribing
Notes	

Kronman 2020

Methods	Design: Step-Wedge
Participants	Country: USA Setting: Primary care N health professionals: NR N patients: 29,762
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: online tutorials and webinars on evidence-based communication strategies and antibiotic prescribing, booster video vignettes, and individualised antibiotic prescribing feedback reports
Outcomes	Clinical behaviour category: Prescribing
Notes	

Larkin 2021

Methods	Design: Parallel Cluster-RCT
Participants	Country: USA Setting: Emergency rooms N health professionals: 51 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: de-identified graph of the group's baseline CT utilisation, asked to estimate wherein the distribution of their own CT order practices fell, and then shown their actual performance. All participants also received a brief educational intervention.
Outcomes	Clinical behaviour category: Testing/exams
Notes	

Leis 2020

Methods	Design: Step-Wedge
Participants	Country: Canada Setting: Hospital inpatient N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F weekly, plus education and dispensers, and continuous quality improvement
Outcomes	Clinical behaviour category: Other
Notes	

Lillo 2022

Methods	Design: Other
Participants	Country: Denmark Setting: Primary care N health professionals: NR N patients: 253,762
Interventions	Number of arms: 7 Description of groups: Control: no intervention; Intervention 1: guidelines; Intervention 2: A&F + guidelines; Intervention 3: guidelines + non-interruptive alert; Intervention 4: A&F + guidelines + non-interruptive alert; Intervention 5: guidelines + non-interruptive alert + interruptive alert; Intervention 6: A&F + guidelines + non-interruptive alert + interruptive alert
Outcomes	Clinical behaviour category: Testing/exams
Notes	

Llewelyn 2023

Methods	Design: Step-Wedge
Participants	Country: UK Setting: Hospital inpatient N health professionals: NR N patients: 7,160,421
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F + decision aid + online training + educational materials
Outcomes	Clinical behaviour category: Prescribing
Notes	

Loiacono 2021

Methods	Design: Parallel Cluster-RCT
Participants	Country: USA Setting: pharmacies N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F with peer comparison
Outcomes	Clinical behaviour category: Immunisation
Notes	

Manz 2020

Methods	Design: Step-Wedge
Participants	Country: USA Setting: other outpatient N health professionals: 78 N patients: 14,607
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F weekly with patient-specific data and point-of-care reminders
Outcomes	Clinical behaviour category: Counselling
Notes	

McCleary 2023

Methods	Design: Factorial
Participants	Country: Canada Setting: Other N health professionals: 267 N patients: 11,017
Interventions	Number of arms: 4 Description of groups: Intervention 1: 67 clusters (median benchmark, benefit framing); Intervention 2: 65 clusters (top quartile benchmark, benefit framing); Intervention 3: 75 clusters (median benchmark, risk framing); Intervention 4: 60 clusters (top quartile benchmark, risk framing)
Outcomes	Clinical behaviour category: Prescribing
Notes	

McCracken 2023

Methods	Design: Other
Participants	Country: Canada Setting: Primary care N health professionals: 5157 N patients: 55,080
Interventions	Number of arms: 2 Description of groups: Intervention 1: Early feedback via mail; Intervention 2: Delayed feedback via mail
Outcomes	Clinical behaviour category: Prescribing
Notes	

Meier 2022

Methods	Design: Parallel Cluster-RCT
Participants	Country: Switzerland Setting: Primary care N health professionals: 71 N patients: 3838
Interventions	Number of arms: 2 Description of groups: Intervention 1: Bimonthly feedback reports with financial incentives; Intervention 2: Bimonthly feedback reports without financial incentives
Outcomes	Clinical behaviour category: Testing/exams
Notes	

Mitchell 2021

Methods	Design: Parallel Cluster-RCT
Participants	Country: USA Setting: Nursing homes N health professionals: NR N patients: 426
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: in-person seminar, an online course, management algorithms (posters, pocket cards), communication tips (pocket cards), and feedback reports on prescribing of antimicrobials
Outcomes	Clinical behaviour category: Prescribing
Notes	

Mortrude 2021

Methods	Design: Step-Wedge
Participants	Country: USA Setting: Primary care N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: report cards and education and decision-support
Outcomes	Clinical behaviour category: Prescribing
Notes	

Navathe 2022

Methods	Design: Factorial
Participants	Country: USA Setting: Hospital inpatient N health professionals: 438 N patients: 294,962
Interventions	Number of arms: 4 Description of groups: Control: Usual care; Intervention 1: Peer comparison feedback; Intervention 2: Individual audit feedback
Outcomes	Clinical behaviour category: Prescribing
Notes	

O'Connor 2022

Methods	Design: Factorial
Participants	Country: Australia Setting: Primary care N health professionals: 3660 N patients: NR
Interventions	Number of arms: 5 Description of groups: Intervention: Standard visual display and 1 feedback occasion; Intervention 2: Standard visual display and 2 feedback occasions; Intervention 3: Enhanced visual display and 1 feedback occasion
Outcomes	Clinical behaviour category: Referral
Notes	

Perkins 2020

Methods	Design: Step-Wedge
Participants	Country: USA Setting: Primary care N health professionals: NR N patients: 16,136
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: interprofessional education and communication training for healthcare providers, data assessment and feedback, and creation of an action plan that included both provider- and system-level changes
Outcomes	Clinical behaviour category: Immunisation
Notes	

Poss-Doering 2021

Methods	Design: Parallel Cluster-RCT
Participants	Country: Germany Setting: Primary care N health professionals: NR N patients: NR
Interventions	Number of arms: 3 Description of groups: Intervention 1: A&F with quality circles, plus education and incentives; Intervention 2: A&F with quality circles, plus education and incentives, plus education for medical assistants and patient education; Intervention 3: A&F with quality circles, plus education and incentives plus decision-support
Outcomes	Clinical behaviour category: Prescribing
Notes	

Reynolds 2021

Methods	Design: Step-Wedge
Participants	Country: USA Setting: Hospital inpatient N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F and educational outreach
Outcomes	Clinical behaviour category: Other
Notes	

Riordan 2021

Methods	Design: Parallel Cluster-RCT
Participants	Country: Ireland Setting: Primary care N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: waiting list; Intervention: practice reimbursement, an audit of attendance, electronic prompts targeting professionals, General Practice-endorsed patient reminders and a patient information leaflet
Outcomes	Clinical behaviour category: Testing/exams
Notes	

Roquilly 2020

Methods	Design: Parallel Cluster-RCT
Participants	Country: France Setting: Hospital inpatient N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F
Outcomes	Clinical behaviour category: Prescribing
Notes	

Ruud 2021

Methods	Design: Parallel Cluster-RCT
Participants	Country: Norway Setting: other outpatient N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: guideline dissemination; Intervention: toolkits, clinical training, implementation facilitation, data-based feedback
Outcomes	Clinical behaviour category: Other
Notes	

Schmiemann 2023

Methods	Design: Parallel Cluster-RCT
Participants	Country: Germany Setting: Primary care N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: Practices in the control group were informed about their allocation in a study investigating the treatment of urinary tract infections in women and the upcoming data extraction after 12 months (Q4) but were unaware of the multimodal intervention; Intervention: guideline recommendations for GPs and patients, provision of regional resistance data, and quarterly feedback of individual antibiotic prescribing proportions, benchmarking, and telephone counselling
Outcomes	Clinical behaviour category: Prescribing
Notes	

Schwartz 2021

Methods	Design: Parallel Cluster-RCT
Participants	Country: Canada Setting: Primary care N health professionals: 3500 N patients: NR
Interventions	Number of arms: 3 Description of groups: Control: no intervention; Intervention 1: A&F; Intervention 2: A&F
Outcomes	Clinical behaviour category: Prescribing
Notes	

Schweitzer 2022

Methods	Design: Step-Wedge
Participants	Country: Netherlands Setting: Hospital inpatient N health professionals: NR N patients: 4084
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: Multifaceted antimicrobial stewardship intervention based on education on local guidelines and audit and feedback
Outcomes	Clinical behaviour category: Prescribing
Notes	

Singer 2022

Methods	Design: Parallel Cluster-RCT
Participants	Country: Canada Setting: Primary care N health professionals: 178 N patients: NR
Interventions	Number of arms: 3 Description of groups: Control: did not receive any data specific to their prescribing; Intervention 1: Standard feedback report with a one-page summary of the Choosing Wisely Canada (CWC) recommendations of interest; Intervention 2: Standard feedback report, CWC recommendation summary and practice-specific data related to their prescribing rates for the CWC recommendations of interest, compared to rates for other providers at their clinic, in their health region and in the province
Outcomes	Clinical behaviour category: Prescribing
Notes	

Skinner 2024

Methods	Design: Parallel Cluster-RCT
Participants	Country: France Setting: Hospital inpatient N health professionals: NR N patients: 155,362
Interventions	Number of arms: 2 Description of groups: Control: no specific actions implemented; Intervention: Quarterly with control charts for monitoring their surgical outcomes (inpatient death, intensive care stay, reoperation and severe complications)
Outcomes	Clinical behaviour category: Other
Notes	

Stanworth 2022

Methods	Design: Factorial
Participants	Country: UK Setting: Hospital N health professionals: NR N patients: 2714
Interventions	Number of arms: 4 Description of groups: Intervention 1: Standard A&F content (detailed report (approximately 50 pages long), a regional presentation slide set, and (sometimes) action plan templates) and standard support; Intervention 2: Standard A&F content and enhanced support (toolkit that disseminates findings to all relevant clinical staff involved in transfusion decision-making; localised, team-level goal-setting, problem-solving, and action planning to change practice; and continued local monitoring of practice); Intervention 3: Enhanced A&F content (a brief report highlighting comparative performance and recommendations for selected key audit standards (maximum of 6 pages), a longer report covering all audit standards (maximum of 34 pages), and the longer, detailed standard report) and standard support
Outcomes	Clinical behaviour category: Treatment decision/action
Notes	

Suffoletto 2020

Methods	Design: Parallel Cluster-RCT
Participants	Country: USA Setting: Emergency rooms N health professionals: 37 N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F; Intervention 2: A&F with peer norm info

Suffoletto 2020 *(Continued)*

Outcomes	Clinical behaviour category: Prescribing
Notes	

Tangri 2021

Methods	Design: Parallel Cluster-RCT
Participants	Country: Canada Setting: other outpatient N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F, education, detailing
Outcomes	Clinical behaviour category: Referral
Notes	

Tinmouth 2021

Methods	Design: Parallel Cluster-RCT
Participants	Country: Canada Setting: other outpatient N health professionals: 33 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F
Outcomes	Clinical behaviour category: Other
Notes	

Torrente 2020

Methods	Design: Parallel Cluster-RCT
Participants	Country: Argentina Setting: other outpatient N health professionals: 1811 N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F emails x2; Intervention 2: educational emails
Outcomes	Clinical behaviour category: Prescribing

Torrente 2020 *(Continued)*

Notes	

Van Schie 2022

Methods	Design: Parallel Cluster-RCT
Participants	Country: Netherlands Setting: Hospital N health professionals: NR N patients: 32,923
Interventions	Number of arms: 2 Description of groups: Control: usual care; Intervention: Monthly updated feedback, education to interpret the feedback combined with clear targets for improvement of specific indicators, action implementation toolbox including evidence-based quality improvement initiative for each indicator reported in the feedback; short survey was emailed every 2 months together with the feedback to evaluate adherence to the intervention
Outcomes	Clinical behaviour category: Other
Notes	

Van Vugt 2021

Methods	Design: Parallel Cluster-RCT
Participants	Country: Netherlands Setting: Primary care N health professionals: NR N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: A&F, education; Intervention 2: A&F, education, patient education
Outcomes	Clinical behaviour category: Testing/exams
Notes	

Vaughan 2023

Methods	Design: Parallel Cluster-RCT
Participants	Country: USA Setting: Other N health professionals: 1186 N patients: NR
Interventions	Number of arms: 2 Description of groups: Intervention 1: In-person A&F (one-to-one (1:1) academic detailing from a professional colleague that included in-person audit, feedback, and peer benchmarking and on-

Vaughan 2023 *(Continued)*

	site engagement); Intervention 2: Dashboard A&F (monthly provider feedback via an electronic dashboard with an emailed summary and link to view detailed prescribing audit, feedback, and peer benchmarking)
Outcomes	Clinical behaviour category: Prescribing
Notes	

Wallis 2022

Methods	Design: Parallel Cluster-RCT
Participants	Country: New Zealand Setting: Primary care N health professionals: NR N patients: 21,867
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: A&F, detailing, patient info
Outcomes	Clinical behaviour category: Prescribing
Notes	

Yang 2023

Methods	Design: Other
Participants	Country: China Setting: Primary care N health professionals: 335 N patients: NR
Interventions	Number of arms: 2 Description of groups: Control: no intervention; Intervention: real-time pop-up warning message, a 10-day antibiotic prescription feedback summary, and distribution of educational manuals
Outcomes	Clinical behaviour category: Prescribing
Notes	

A&F: audit and feedback
CWC: community care worker
GP: general practitioner
RCT: randomised control trial

ADDITIONAL TABLES

Table 1. Characteristics of Included Studies

Study Characteristics	# of Trials	Per cent (N = 292)
Publication Year		
2016-2020	76	26.0%
2011-2015	52	17.8%
2006-2010	46	15.8%
2001-2005	53	18.2%
1996-2000	33	11.3%
Before 1996	32	11.0%
Country		
USA	120	41.1%
UK or Ireland	37	12.7%
Canada	29	9.9%
Australia	18	6.2%
Other	88	30.1%
Unit of Allocation		
Many Providers/groups	221	75.7%
Provider	65	22.3%
Unclear	5	1.7%
Patients	1	0.3%
Unit of Analysis		
Patient (encounter/number of proportion)	260	89.0%
Provider	18	6.2%
Many Providers/groups	15	5.1%

Intervention Characteristics	# of Trials	Per cent (N = 292)
Targeted Behaviours (can be more than one)		
Prescribing	146	50.0%
Testing/exams	108	37.0%
Treatment decision/action	64	21.9%
Screening	3	1.0%
Counselling	50	17.1%
Immunisation	26	8.9%
Referrals	17	5.8%
Diagnosis	43	14.7%
Other	2	0.7%
Direction of Change		
Increase/improve behaviour	225	77.1%
Decrease/reduce behaviour	65	22.3%

Table 1. Characteristics of Included Studies *(Continued)*

Other	13	4.5%
Number of Arms in Trial		
Two	224	76.7%
Three	39	13.3%
Four	25	8.6%
More than four	4	1.4%
Clinical Setting		
Primary care	163	55.8%
Hospital inpatient	65	22.3%
Other outpatient clinic	25	8.6%
Other/mixed	21	7.2%
Community care	18	6.2%
Medical Speciality		
GP/Family physician	170	58.2%
Internists	32	11.0%
Other	90	30.8%
Targeted Health Professional (could include more than one)		
Physician	237	81.2%
Nurses	36	12.3%
Pharmacists	8	2.7%
Other	42	14.4%
Direction of Change		
Increase behaviour		
Decrease behaviour		
Comparisons		
A. A&F vs No A&F	177	60.6%
B. A&F alone vs. No intervention	14	4.8%
C. A&F with co-interventions vs. No intervention	95	32.5%

Table 1. Characteristics of Included Studies *(Continued)*

D. A&F vs. A&F	17	5.8%
E. A&F with co-interventions vs. A&F alone	65	22.3%
F. A&F with co-interventions vs. A&F with co-interventions	19	6.5%

A&F: Audit and Feedback

Table 2. EPOC Coding of Interventions

EPOC Code	# of Arms	Per cent (N = 678)	Co-intervention with A&F	
			# of Arms	Per cent (N = 435)
Implementation Strategies				
Audit and Feedback	435	64.2%	86	19.8%
Clinician education	377	55.6%	303	69.7%
Clinician reminders	100	14.7%	87	20.0%
Patient-mediated interventions	60	8.8%	52	12.0%
Managerial supervision	14	2.1%	12	2.8%
Organisational culture	3	0.4%	3	0.7%
Delivery Arrangements				
Information and Communication Technology	62	9.1%	52	12.0%
Teamwork	20	2.9%	20	4.6%
Coordination of Patients	9	1.3%	9	2.1%
Where care is provided	9	1.3%	6	1.4%
Who provides care	6	0.9%	6	1.4%
Enabling Access	3	0.4%	3	0.7%
How/When care is delivered	3	0.4%	3	0.7%
Financial Arrangements				
Pathways	49	7.2%	43	9.9%
Financial Interventions	20	2.9%	18	4.1%
Public Reporting	2	0.3%	2	0.5%

Table 2. EPOC Coding of Interventions *(Continued)*

Governance Arrangements				
Authority and Governance	13	1.9%	11	2.5%

EPOC: Effective Practice of Care

Table 3. Characteristics of Audit and Feedback Interventions

	# of Arms	% (N = 435)
Audit		
Number of indicators*		
1	296	71.0%
2	99	23.7%
3	48	11.5%
4	52	12.5%
5 to 9	105	25.2%
10+	55	13.2%
Aggregation of data		
Organisation or team level	269	61.8%
Individual health professional level	195	44.8%
Source of data		
Manual chart review	177	40.7%
Computerised search of charts	164	37.7%
Administrative/billing data	112	25.7%
Patient-reported data	21	4.8%
Other	23	5.3%
Feedback		
Frequency		
More than weekly	4	0.9%
Weekly	6	1.4%
Monthly	88	3.0%
Less than monthly	118	27.1%

Table 3. Characteristics of Audit and Feedback Interventions *(Continued)*

Variable	35	8.0%
Once only	117	26.9%
Format (can be more than one)		
Verbal (didactic)	109	25.1%
Verbal (interactive)	157	36.1%
Written	346	79.5%
Both verbal and written	152	34.9%
Electronic dashboard	31	7.1%
Feedback deliverer		
Local champion	77	17.7%
External champion	107	24.6%
External organisation	73	16.8%
Research team	115	26.4%
Other	18	4.1%
Unclear	3	0.6%
Nature of the Comparator (can be more than one)		
Historical data	123	28.3%
Performance of top peers	39	9.0%
Performance of similar peers	185	42.5%
Average performance of all audited	85	19.5%
Target not based on local data	66	15.2%
Other	8	1.8%
Action		
Plans for improvement provided		
General	327	75.2%
Specific	101	23.2%
Co-developed plans	91	20.9%
Target of Change		

Table 3. Characteristics of Audit and Feedback Interventions *(Continued)*

Organisation or team	250	57.5%
Provider	331	76.1%
Patient	62	14.2%

*Measured in 277 trials and 417 arms (stepped-wedge trials not included)

Table 4. Stepped-Wedge Trials: Topics and Effects

Study	Topic	Reported findings regarding effectiveness of A&F intervention
Amanyire 2016	antiretroviral therapy in Africa	80% in the intervention group had started ART by 2 weeks after eligibility compared with 38% in the control group (risk difference 41.9%, 95% CI 40.1 to 43.8)
Brown 2018	care for high-risk prostate cancer	The proportion of patients discussed at a MDT meeting increased from 17% in the control-phase to 59% in the intervention-phase (adjusted RR = 4.32; 95% CI [2.40 to 7.75]; P < 0·001), but no significant difference in referral to radiation oncology (intervention 32% vs control 30%; adjusted RR = 1.06; 95% CI [0.74 to 1.51]; P = 0.879).
Diamantouros 2017	appropriate thromboprophylaxis	Intervention groups increased compared to controls (85% vs. 76% in hip fracture wards; 67% vs. 54% in general surgery wards; 64% vs. 62% in medicine wards), but the difference was statistically significant only in the medicine wards.
Dreischulte 2016	high-risk prescribing	Targeted high-risk prescribing was significantly reduced, from a rate of 3.7% (1102 of 29,537 patients at risk) immediately before the intervention to 2.2% (674 of 30,187) at the end of the intervention (adjusted odds ratio, 0.63; 95% confidence interval [CI] 0.57 to 0.68; P < 0.001).
Fuller 2012	hand hygiene	OR for hand hygiene compliance rose post-randomisation (1.44; 95% CI 1.18, 1.76; P = 0.001) in intensive therapy units, but not in elderly wards.
Huffman 2018	acute MI patients in India	Mixed findings across many process outcomes
Morrison 2015	out of hospital cardiac arrest	Active quality improvement interventions conferred no additional improvements in rates of successful targeted temperature management (26.9% active vs 25.7% passive; odds ratio, 0.96; 95% CI 0.63 to 1.45; P = 0.84).
Ralph 2018	prophylaxis for rheumatic heart disease	No change
Rodriguez 2015	hand hygiene in ICUs	Compliance with hand hygiene in the control group was 66.0% (2354/3565) vs. 75.6% (5190/6864) in the intervention group (OR 1.08; 95% CI 1.03 to 1.14).
Scales 2016	out of hospital cardiac arrest	Appropriate neurologic prognostication increased after the intervention (68% vs. 74% patients; odds ratio [OR], 1.79; 95% confidence interval [CI] 1.01 to 3.19; P = 0.05
Trent 2019	pneumonia and sepsis management	The odds of providing guideline adherent care were 1.8 (95% CI 1.01 to 3.2) after the introduction of feedback with blinded peer comparison.
Wu 2019	acute MI care in China	The intervention showed a significant improvement in the composite KPI score (mean [SD], 0.69 [0.22] vs 0.61 [0.23]; P < 0.01) and in 7 individual KPIs,

Table 4. Stepped-Wedge Trials: Topics and Effects *(Continued)*

		including the early use of antiplatelet therapy and the use of appropriate secondary prevention medicines at discharge.

A&F: Audit and feedback
ART: Antiretroviral therapy
CI: Confidence interval
ICU: Intensive Care Unit
KPI: Key performance indicator
MDT: Multidisciplinary team
SD: Standard deviation

Table 5. Studies with only Continuous Outcome Data: Topics and Effects

Study	Topic	Reported findings regarding effectiveness of A&F intervention
Bregnhøj 2009	polypharmacy in elderly	Reduced number of medications by 1.03 (95% CI 0.3, 1.7)
Dormuth 2012	statin prescribing	Significant beneficial effect for primary prevention but not secondary
Everett 1983	laboratory tests	Mixed findings
Hemkens 2017	antibiotic prescribing	No change
Holm 1990	sedative medications	No change
Houston 2015	referrals for smoking cessation	More referrals: 31% of all smokers referred versus 11 %, $P < 0.001$
Kerry 2000	x-ray referrals	In practices receiving the guidelines, there was a 20% reduction in requests for spinal examinations compared with control practices ($P < 0.05$).
Martin 1980	laboratory test-ordering and radiology requests	Reduction in test-ordering but not radiology
Marton 1985	laboratory test-ordering	Reduction in number of tests ordered approached statistical significance ($P = 0.07$).
McConnell 1982	antibiotic prescribing	Reduction in intervention group significantly greater than control ($P < 0.05$)
Navarro 2012	prescribing for alcohol dependence	Prescribing rates in the intervention communities significantly increased.
Nejad 2016	parenteral steroids	Being in the paper and text feedback arms resulted in 36.1 and 41.7 units of decrease in DDD respectively, compared to the control arm ($P < 0.02$ and $P < 0.005$).
Ruangkanchanasetr 1993a; Ruangkanchanasetr 1993b; Ruangkanchanasetr 1993c	laboratory testing	No changes
Ryskina 2018	laboratory test-ordering in hospital	No change except for those in the highest-ordering group
Smith 1998	sedative medications	Reduced number of prescriptions ($P = 0.004$)

Table 5. Studies with only Continuous Outcome Data: Topics and Effects *(Continued)*

Socolar 1998	chart documentation for child sexual abuse	No improvement in most aspects of documentation
Van der Weijden 1999	cholesterol testing	No change
Verstappen 2003a; Verstappen 2003b	test-ordering	Mixed findings
Vingerhoets 2001	patient reported processes of primary care	No changes in patient evaluation of care scores
Wadland 2007	referrals for smoking cessation	More referrals from the intervention than from the control practices (484 vs 220; P < 0.001)
Wahlström 2003	infectious disease management in hospitals in Laos	Improvement in quality of care score
Wattal 2015	antibiotic prescribing in a hospital in Delhi	No change
Wones 1987	laboratory testing	No change

A&F: Audit and feedback
CI: Confidence interval
DDD: Defined Daily Dose

APPENDICES

Appendix 1. Search strategies for studies included in analysis

MEDLINE (Ovid)

MEDLINE database used: MEDLINE (Ovid MEDLINE® Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE® Daily and Ovid MEDLINE®) 1946 to present

Date searched: June 11, 2020

No.	Search terms	Results
1	(audit* adj3 (feedback or fed back)).ti,ab,kf.	3327
2	clinical audit/	1708
3	medical audit/	16997
4	nursing audit/	3007
5	dental audit/	424
6	management audit/	2506
7	benchmarking/	13335
8	"commission on professional and hospital activities"/	279

(Continued)

9	feedback/	29725
10	feedback, psychological/	3475
11	utilization review/	8072
12	drug utilization review/	3756
13	concurrent review/	380
14	peer review, health care/	1436
15	(audit or audits or auditing).ti,ab,kf.	40126
16	(feedback or fed back).ti,ab,kf.	137282
17	(review adj4 record?).ti,ab,kf.	20998
18	chart review?.ti,ab,kf.	41342
19	(practice data or hospital* data).ti,ab,kf.	6226
20	(benchmark* or bench mark*).ti,ab,kf.	41466
21	(scorecard? or score card? or reportcard? or report card?).ti,ab,kf.	2645
22	(dashboard* or dash board*).ti,ab,kf.	1224
23	patient registry.ti,ab,kf.	2257
24	(business intelligence or process control).ti,ab,kf.	3486
25	(panel adj (management or support or view*)).ti,ab,kf.	85
26	(performance adj2 (scor* or measur* or monitor* or improv*)).ti,ab,kf.	90608
27	or/2-26	416306
28	exp health personnel/	511000
29	(health* professional? or health care professional? or health* team? or health care team? or physician? or clinician? or nurs* or provider* or doctor? or intern? or caregiver? or gp? or therapist? or health* personnel or health care personnel or practitioner? or resident? or pharmacist?).ti,ab,kf.	1681805
30	quality assurance, health care/	55796
31	quality indicators, health care/	15547
32	quality improvement/	24291
33	quality of health care/	72477
34	(quality adj (assurance or improvement or control or indicator?)).ti,ab,kf.	112388

(Continued)

35	(health care quality or healthcare quality or quality of healthcare or quality of health care or quality of care).ti,ab,kf.	62639
36	((influenc* or chang*) adj3 (behaviour* or behavior* or practice)).ti,ab,kf.	103606
37	or/28-36	2167338
38	exp randomized controlled trial/	508044
39	controlled clinical trial.pt.	93710
40	randomi#ed.ti,ab,kf.	623767
41	randomly.ti,ab,kf.	335785
42	random allocation/	102942
43	clinical trials as topic.sh.	191512
44	trial.ti.	219805
45	or/38-44	1353026
46	exp animals/ not humans/	4705997
47	45 not 46	1223313
48	1 or (27 and 37)	100302
49	47 and 48	8972
50	(2019* or 2020*).dt,dp,ed,ep,yr.	2746772
51	49 and 50	1544

Embase (Ovid)

Embase database used: Embase 1974 to present

Date searched: June 11, 2020

No.	Search terms	Results
1	(audit* adj3 (feedback or fed back)).ti,ab,kw.	4176
2	clinical audit/	4536
3	nursing audit/	31
4	benchmarking/	5030
5	feedback system/	80081

(Continued)

6	psychological feedback/	311
7	utilization review/	59653
8	drug utilization review/	506
9	concurrent review/	59653
10	(audit or audits or auditing).ti,ab,kw.	83365
11	(feedback or fed back).ti,ab,kw.	177255
12	(review adj4 record?).ti,ab,kw.	34335
13	chart review?.ti,ab,kw.	84615
14	(practice data or hospital* data).ti,ab,kw.	10556
15	(benchmark* or bench mark*).ti,ab,kw.	50234
16	(scorecard? or score card? or reportcard? or report card?).ti,ab,kw.	3504
17	(dashboard* or dash board*).ti,ab,kw.	2368
18	patient registry.ti,ab,kw.	4458
19	(business intelligence or process control).ti,ab,kw.	5525
20	(panel adj (management or support or view*)).ti,ab,kw.	173
21	(performance adj2 (scor* or measur* or monitor* or improv*)).ti,ab,kw.	111848
22	or/2-21	628510
23	exp health care personnel/	1535674
24	(health* professional? or health care professional? or health* team? or health care team? or physician? or clinician? or nurs* or provider* or doctor? or intern? or caregiver? or gp? or therapist? or health* personnel or health care personnel or practitioner? or resident? or pharmacist?).ti,ab,kw.	2175764
25	health care quality/	239315
26	(quality adj (assurance or improvement or control or indicator?)).ti,ab,kw.	174366
27	(health care quality or healthcare quality or quality of healthcare or quality of health care or quality of care).ti,ab,kw.	85654
28	((influenc* or chang*) adj3 (behaviour* or behavior* or practice)).ti,ab,kw.	131947
29	or/23-28	3299067
30	1 or (22 and 29)	197876
31	random*.ti,ab.	1542714

(Continued)

32	factorial*.ti,ab.	38147
33	(crossover* or cross over*).ti,ab.	106994
34	((doubl* or singl*) adj blind*).ti,ab.	233222
35	(assign* or allocat* or volunteer* or placebo*).ti,ab.	1038976
36	crossover procedure/	63483
37	single blind procedure/	39201
38	randomized controlled trial/	607598
39	double blind procedure/	173561
40	or/31-39	2334969
41	exp animal/ not human/	4780999
42	40 not 41	2102033
43	30 and 42	26066
44	limit 43 to embase	9830
45	limit 44 to yr="2019 -Current"	1101

CENTRAL (The Cochrane Library)

Database: CENTRAL Issue/year: 6/2020

Date searched: June 11, 2020

No.	Search terms	Results
#1	(audit* near/3 (feedback or (fed next back))):ti,ab	467
#2	[mh "clinical audit"]	289
#3	[mh “medical audit”]	223
#4	[mh “nursing audit”]	46
#5	[mh “dental audit”]	4
#6	[mh “management audit”]	98
#7	[mh “benchmarking”]	96
#8	[mh “feedback”]	1225
#9	[mh “feedback, psychological”]	1720

(Continued)

#10	[mh "utilization review"]	214
#11	[mh "drug utilization review"]	103
#12	[mh "concurrent review"]	3
#13	[mh "peer review, health care"]	34
#14	(audit or audits or auditing):ti,ab	2019
#15	(feedback or (fed next back)):ti,ab	9931
#16	(review near/4 record?):ti,ab	813
#17	(chart next review?):ti,ab	1217
#18	((practice next data) or (hospital* next data)):ti,ab	374
#19	(benchmark* or (bench next mark*)):ti,ab	783
#20	((score next card?) or (report next card?)):ti,ab	239
#21	(dashboard* or (dash next board*)):ti,ab	88
#22	(patient next registry):ti,ab	132
#23	((business next intelligence) or (process next control)):ti,ab	67
#24	(panel next (management or support or view*)):ti,ab	13
#25	(performance near/2 (scor* or measur* or monitor* or improv*)):ti,ab	9446
#26	{or #2-#25}	25681
#27	[mh "health personnel"]	7742
#28	((health* next professional?) or (health next care next professional?) or (health* next team?) or (health next care next team?) or physician? or clinician? or nurs* or provider* or doctor? or intern? or caregiver? or gp? or therapist? or (health* next personnel) or (health next care next personnel) or practitioner? or resident? or pharmacist?):ti,ab	92385
#29	[mh ^"quality assurance, health care"]	610
#30	[mh ^"quality indicators, health care"]	204
#31	[mh ^"quality improvement"]	549
#32	[mh ^"quality of health care"]	794
#33	(quality next (assurance or improvement or control or indicator?)):ti,ab	3303
#34	((health next care next quality) or (healthcare next quality) or ("quality of healthcare" or "quality of health care" or "quality of care")):ti,ab	1827
#35	((influenc* or chang*) near/3 (behaviour* or behavior* or practice)):ti,ab	8789

(Continued)

#36	{or #27-#35}	103592
#37	#1 or (#26 and #36)	7760
#38	#1 or (#26 and #36) with Cochrane Library publication date Between Dec 2010 and Jun 2020	5563

CINAHL (EBSCOhost)

Date searched: June 11, 2020

No.	Search terms	Results
S1	TI (audit* N3 (feedback or fed back)) or AB (audit* N3 (feedback or fed back))	1,503
S2	MH “audit”	18,863
S3	MH “nursing audit”	805
S4	MH “benchmarking”	7,766
S5	MH “feedback”	16,471
S6	MH “utilization review”	2,057
S7	MH “drug utilization”	8,446
S8	MH “record review”	70,764
S9	MH “peer review”	7,758
S10	TI (audit or audits or auditing) or AB (audit or audits or auditing)	24,408
S11	TI (feedback or (fed next back)) OR AB (feedback or (fed next back))	37,307
S12	TI (review N4 record#) OR AB (review N4 record#)	8,111
S13	TI (chart next review#) OR AB (chart next review#)	8
S14	TI (practice data or hospital* data) OR AB (practice data or hospital* data)	34,801
S15	TI (benchmark* or (bench mark*)) OR AB (benchmark* or (bench mark*))	10,291
S16	TI (scorecard# or score card# or reportcard# or report card#) OR AB (scorecard# or score card# or reportcard# or report card#)	1,775
S17	TI (dashboard* or dash board*) OR AB (dashboard* or dash board*)	880
S18	TI (patient registry) OR AB (patient registry)	7,267
S19	TI ((business intelligence) or (process control)) OR AB ((business intelligence) or (process control))	4,104

(Continued)

S20	TI (panel N0 (management or support or view*)) OR AB (panel N0 (management or support or view*))	91
S21	TI (performance N2 (scor* or measur* or monitor* or improv*)) OR AB (performance N2 (scor* or measur* or monitor* or improv*))	38,348
S22	S2 OR S3 OR S4 OR S5 OR S6 OR S7 OR S8 OR S9 OR S10 OR S11 OR S12 OR S13 OR S14 OR S15 OR S16 OR S17 OR S18 OR S19 OR S20 OR S21	256,396
S23	MH “health personnel+”	591,984
S24	TI (health* professional# or health care professional# or health* team# or health care team# or physician# or clinician# or nurs* or provider* or doctor# or intern# or caregiver# or gp# or therapist# or health* personnel or health care personnel or practitioner# or resident# or pharmacist#) OR AB (health* professional# or health care professional# or health* team# or health care team# or physician# or clinician# or nurs* or provider* or doctor# or intern# or caregiver# or gp# or therapist# or health* personnel or health care personnel or practitioner# or resident# or pharmacist#)	976,762
S25	MH “quality assurance”	21,664
S26	MH “quality indicators”	366
S27	MH “quality improvement”	58,695
S28	MH “quality of health care”	78,424
S29	MH “quality of nursing care”	14,725
S30	TI (quality N0 (assurance or improvement or control or indicator#)) OR AB (quality N0 (assurance or improvement or control or indicator#))	44,812
S31	TI (health care quality or healthcare quality or quality of healthcare or quality of health care or quality of care) OR AB (health care quality or healthcare quality or quality of healthcare or quality of health care or quality of care)	91,534
S32	TI ((influenc* or chang*) N3 (behaviour* or behavior* or practice)) OR AB ((influenc* or chang*) N3 (behaviour* or behavior* or practice))	55,273
S33	S24 OR S25 OR S26 OR S27 OR S28 OR S29 OR S30 OR S31 OR S32	1,268,092
S34	S1 OR (S22 AND S33)	94,438
S35	PT randomized controlled trial	131,417
S36	PT clinical trial	109,255
S37	TI (randomis* or randomiz* or randomly) OR AB (randomis* or randomiz* or randomly)	322,817
S38	(MH "Clinical Trials+")	318,711
S39	(MH "Random Assignment")	68,322
S40	S35 OR S36 OR S37 OR S38 OR S39	498,188

(Continued)

S41	S34 AND S40	7,542
S42	S41 Limiters - Published Date: 20190101-20201231; Exclude MEDLINE records	482

ClinicalTrials.gov

Date searched: June 11, 2020

Field	Search terms	Results
Other terms	audit AND feedback	199

WHO International Clinical Trials Registry Platform (ICTRP)

Date searched: February 28, 2019

No.	Search terms	Results
	audit AND feedback	140

Appendix 2. Search strategies for 2022 update search (studies not included in analysis)

MEDLINE (Ovid)

MEDLINE database used: MEDLINE (Ovid MEDLINE® Epub Ahead of Print, In-Process & Other Non-Indexed Citations, Ovid MEDLINE® Daily and Ovid MEDLINE®) 1946 to present

Date searched: February 14, 2022

No.	Search terms	Results
1	(audit* adj3 (feedback or fed back)).ti,ab,kf.	3869
2	clinical audit/	1978
3	medical audit/	17366
4	nursing audit/	3013
5	dental audit/	425
6	management audit/	2529
7	benchmarking/	15287
8	"commission on professional and hospital activities"/	289

(Continued)

9	feedback/	31725
10	feedback, psychological/	3696
11	utilization review/	8142
12	drug utilization review/	3869
13	concurrent review/	383
14	peer review, health care/	1455
15	(audit or audits or auditing).ti,ab,kf.	44621
16	(feedback or fed back).ti,ab,kf.	156746
17	(review adj4 record?).ti,ab,kf.	23652
18	chart review?.ti,ab,kf.	47992
19	(practice data or hospital* data).ti,ab,kf.	7511
20	(benchmark* or bench mark*).ti,ab,kf.	52173
21	(scorecard? or score card? or reportcard? or report card?).ti,ab,kf.	2862
22	(dashboard* or dash board*).ti,ab,kf.	1965
23	patient registry.ti,ab,kf.	2737
24	(business intelligence or process control).ti,ab,kf.	4191
25	(panel adj (management or support or view*)).ti,ab,kf.	98
26	(performance adj2 (scor* or measur* or monitor* or improv*)).ti,ab,kf.	107877
27	or/2-26	479644
28	exp health personnel/	572761
29	(health* professional? or health care professional? or health* team? or health care team? or physician? or clinician? or nurs* or provider* or doctor? or intern? or caregiver? or gp? or therapist? or health* personnel or health care personnel or practitioner? or resident? or pharmacist?).ti,ab,kf.	1895041
30	quality assurance, health care/	56757
31	quality indicators, health care/	16920
32	quality improvement/	30286
33	quality of health care/	76287
34	(quality adj (assurance or improvement or control or indicator?)).ti,ab,kf.	130577

(Continued)

35	(health care quality or healthcare quality or quality of healthcare or quality of health care or quality of care).ti,ab,kf.	72326
36	((influenc* or chang*) adj3 (behaviour* or behavior* or practice)).ti,ab,kf.	119764
37	or/28-36	2430618
38	exp randomized controlled trial/	559931
39	controlled clinical trial.pt.	94699
40	randomi#ed.ti,ab,kf.	713687
41	randomly.ti,ab,kf.	376869
42	random allocation/	106591
43	clinical trials as topic.sh.	199165
44	trial.ti.	256641
45	or/38-44	1492298
46	exp animals/ not humans/	4957999
47	45 not 46	1350417
48	1 or (27 and 37)	115115
49	47 and 48	10335
50	202*.dt,dp,ed,ep,yr. B178	3978940
51	49 and 50	2202

Embase (Ovid)

Embase database used: Embase 1974 to present

Date searched: February 14, 2022

No.	Search terms	Results
1	(audit* adj3 (feedback or fed back)).ti,ab,kw.	4835
2	clinical audit/	6553
3	nursing audit/	39
4	benchmarking/	7797

(Continued)

5	feedback system/	85140
6	psychological feedback/	441
7	utilization review/	59617
8	drug utilization review/	788
9	concurrent review/	59617
10	(audit or audits or auditing).ti,ab,kw.	93372
11	(feedback or fed back).ti,ab,kw.	201273
12	(review adj4 record?).ti,ab,kw.	39015
13	chart review?.ti,ab,kw.	99908
14	(practice data or hospital* data).ti,ab,kw.	12647
15	(benchmark* or bench mark*).ti,ab,kw.	62769
16	(scorecard? or score card? or reportcard? or report card?).ti,ab,kw.	3764
17	(dashboard* or dash board*).ti,ab,kw.	3450
18	patient registry.ti,ab,kw.	5375
19	(business intelligence or process control).ti,ab,kw.	6222
20	(panel adj (management or support or view*)).ti,ab,kw.	202
21	(performance adj2 (scor* or measur* or monitor* or improv*)).ti,ab,kw.	130920
22	or/2-21	718263
23	exp health care personnel/	1748935
24	(health* professional? or health care professional? or health* team? or health care team? or physician? or clinician? or nurs* or provider* or doctor? or intern? or caregiver? or gp? or therapist? or health* personnel or health care personnel or practitioner? or resident? or pharmacist?).ti,ab,kw.	2460959
25	health care quality/	253694
26	(quality adj (assurance or improvement or control or indicator?)).ti,ab,kw.	187451
27	(health care quality or healthcare quality or quality of healthcare or quality of health care or quality of care).ti,ab,kw.	97332
28	((influenc* or chang*) adj3 (behaviour* or behavior* or practice)).ti,ab,kw.	149869
29	or/23-28	3716928
30	1 or (22 and 29)	225634

(Continued)

31	random*.ti,ab.	1754399
32	factorial*.ti,ab.	43034
33	(crossover* or cross over*).ti,ab.	116476
34	((doubl* or singl*) adj blind*).ti,ab.	253713
35	(assign* or allocat* or volunteer* or placebo*).ti,ab.	1151479
36	crossover procedure/	69415
37	single blind procedure/	45134
38	randomized controlled trial/	695218
39	double blind procedure/	192161
40	or/31-39	2621232
41	exp animal/ not human/	5044913
42	40 not 41	2364134
43	30 and 42	29761
44	limit 43 to embase	11208
45	limit 44 to yr="2020 -Current"	1919

CENTRAL (The Cochrane Library)

Database: CENTRAL Issue/year: 1/2022

Date searched: February 14, 2022

No.	Search terms	Results
#1	(audit* near/3 (feedback or (fed next back))):ti,ab	851
#2	[mh "clinical audit"]	303
#3	[mh “medical audit”]	230
#4	[mh “nursing audit”]	47
#5	[mh “dental audit”]	4
#6	[mh “management audit”]	122
#7	[mh “benchmarking”]	120
#8	[mh “feedback”]	1440

(Continued)

#9	[mh "feedback, psychological"]	2022
#10	[mh "utilization review"]	261
#11	[mh "drug utilization review"]	151
#12	[mh "concurrent review"]	3
#13	[mh "peer review, health care"]	33
#14	(audit or audits or auditing):ti,ab	3928
#15	(feedback or (fed next back)):ti,ab	18408
#16	(review near/4 record?):ti,ab	1662
#17	(chart next review?):ti,ab	2248
#18	((practice next data) or (hospital* next data)):ti,ab	651
#19	(benchmark* or (bench next mark*)):ti,ab	1498
#20	(scorecard? or score card? or reportcard? or report card?):ti,ab	1508
#21	(dashboard* or (dash next board*)):ti,ab	304
#22	(patient next registry):ti,ab	230
#23	((business next intelligence) or (process next control)):ti,ab	104
#24	(panel next (management or support or view*)):ti,ab	22
#25	(performance near/2 (scor* or measur* or monitor* or improv*)):ti,ab	15641
#26	{or #2-#25}	45127
#27	[mh "health personnel"]	10052
#28	((health* next professional?) or (health next care next professional?) or (health* next team?) or (health next care next team?) or physician? or clinician? or nurs* or provider* or doctor? or intern? or caregiver? or gp? or therapist? or (health* next personnel) or (health next care next personnel) or practitioner? or resident? or pharmacist?):ti,ab	170854
#29	[mh ^"quality assurance, health care"]	620
#30	[mh ^"quality indicators, health care"]	236
#31	[mh ^"quality improvement"]	797
#32	[mh ^"quality of health care"]	918
#33	(quality next (assurance or improvement or control or indicator?)):ti,ab	5394
#34	((health next care next quality) or (healthcare next quality) or ("quality of healthcare" or "quality of health care" or "quality of care")):ti,ab	2943

(Continued)

#35	((influenc* or chang*) near/3 (behaviour* or behavior* or practice)):ti,ab	15754
#36	{or #27-#35}	188709
#37	#1 or (#26 and #36)	15274
#38	#1 or (#26 and #36) with Cochrane Library publication date Between Jan 2020 and Dec 2022	3926 C108

CINAHL (EBSCOhost)

Date searched: February 14, 2022

No.	Search terms	Results
S1	TI (audit* N3 (feedback or fed back)) or AB (audit* N3 (feedback or fed back))	1,621
S2	MH “audit”	18,422
S3	MH “nursing audit”	782
S4	MH “benchmarking”	7,820
S5	MH “utilization review”	2,042
S6	MH “drug utilization”	8,570
S7	MH “feedback”	17,108
S8	MH “record review”	65,065
S9	MH “peer review”	7,869
S10	TI (audit or audits or auditing) or AB (audit or audits or auditing)	25,101
S11	TI (feedback or (fed next back)) OR AB (feedback or (fed next back))	39,989
S12	TI (review N4 record*) OR AB (review N4 record*)	10,264
S13	TI (chart next review*) OR AB (chart next review*)	8
S14	TI (practice data or hospital* data) OR AB (practice data or hospital* data)	38,364
S15	TI (benchmark* or (bench mark*)) OR AB (benchmark* or (bench mark*))	11,170
S16	TI (scorecard* or score card* or reportcard* or report card*) OR AB (scorecard* or score card* or reportcard* or report card*)	9,387
S17	TI (dashboard* or dash board*) OR AB (dashboard* or dash board*)	1,140
S18	TI (patient registry) OR AB (patient registry)	9,285

(Continued)

S19	TI ((business intelligence) or (process control)) OR AB ((business intelligence) or (process control))	4,512
S20	TI (panel N0 (management or support or view*)) OR AB (panel N0 (management or support or view*))	103
S21	TI (performance N2 (scor* or measur* or monitor* or improv*)) OR AB (performance N2 (scor* or measur* or monitor* or improv*))	40,977
S22	S2 OR S3 OR S4 OR S5 OR S6 OR S7 OR S8 OR S9 OR S10 OR S11 OR S12 OR S13 OR S14 OR S15 OR S16 OR S17 OR S18 OR S19 OR S20 OR S21	272,024
S23	MH "health personnel+"	611,035
S24	TI (health* professional* or health care professional* or health* team* or health care team* or physician* or clinician* or nurs* or provider* or doctor* or intern* or caregiver* or gp* or therapist* or health* personnel or health care personnel or practitioner* or resident* or pharmacist*) OR AB (health* professional* or health care professional* or health* team* or health care team* or physician* or clinician* or nurs* or provider* or doctor* or intern* or caregiver* or gp* or therapist* or health* personnel or health care personnel or practitioner* or resident* or pharmacist*)	1,427,265
S25	MH "quality assurance"	21,377
S26	MH "quality indicators"	0
S27	MH "quality improvement"	61,896
S28	MH "quality of health care"	80,172
S29	MH "quality of nursing care"	15,130
S30	TI (quality N0 (assurance or improvement or control or indicator*)) OR AB (quality N0 (assurance or improvement or control or indicator*))	41,903
S31	TI (health care quality or healthcare quality or quality of healthcare or quality of health care or quality of care) OR AB (health care quality or healthcare quality or quality of healthcare or quality of health care or quality of care)	97,613
S32	TI ((influenc* or chang*) N3 (behaviour* or behavior* or practice)) OR AB ((influenc* or chang*) N3 (behaviour* or behavior* or practice))	58,895
S33	S23 OR S24 OR S25 OR S26 OR S27 OR S28 OR S29 OR S30 OR S31 OR S32	1,887,875
S34	S1 OR (S22 AND S33)	117,486
S35	PT randomized controlled trial	139,117
S36	PT clinical trial	110,329
S37	TI (randomis* or randomiz* or randomly) OR AB (randomis* or randomiz* or randomly)	351,879
S38	(MH "Clinical Trials+")	332,456
S39	(MH "Random Assignment")	72,648

(Continued)

S40	S35 OR S36 OR S37 OR S38 OR S39	527,675
S41	S34 AND S40	9,626
S42	S41 Limiters - Published Date: 20200101-20221231; Exclude MEDLINE records	881

ClinicalTrials.gov

Date searched: February 14, 2022

Field	Search terms	Results
Other terms	audit AND feedback	294

WHO International Clinical Trials Registry Platform (ICTRP)

Date searched: February 14, 2022

No.	Search terms	Results
	audit AND feedback	164

WHAT'S NEW

Date	Event	Description
22 June 2026	New citation required but conclusions have not changed	Corrected figures 4, 5 and 6. Corrected confidence intervals reported in text.
22 June 2026	Amended	Minor correction to address figure scaling and a typographical error.

HISTORY

Protocol first published: Issue 3, 1996
Review first published: Issue 1, 1998

Date	Event	Description
25 March 2025	New citation required and conclusions have changed	This updated version has led to new conclusions
25 March 2025	New search has been performed	Update completed and peer review responses incorporated; 10 years of new studies included in this update.

Date	Event	Description
16 May 2012	New search has been performed	New search; 32 additional studies included
16 May 2012	New citation required and conclusions have changed	32 new studies; new authors on team
30 September 2011	New search has been performed	Identified studies awaiting assessment
10 December 2010	Amended	Updated search applied for revised protocol
8 November 2010	Amended	Edits to protocol
8 November 2010	Amended	Further edits to protocol
29 April 2008	Amended	Converted to new review format

CONTRIBUTIONS OF AUTHORS

NI led the previous version of this review (Ivers 2012). JMG and NI conceived the overall plan for this update. NI, JMG, KJK and KAB conceived the methodological approach used in the review. MS, AJ and SY provided project management, co-ordination and quality control for the review. ML, AXTX, KW, NMS, TAW, SL, MG, HC, MDC, GR, GF, MH, CS, AB, MS, BJM, E-AB, TC, SA, CR screened search results and completed data abstraction. SY, SL, and CR classified interventions according to the EPOC taxonomy (EPOC 2002). JP, FL, CM, VA, and JC extracted behaviour change techniques (BCTs) as components of the interventions. DOC, AL, AF, EG, HJ, JW, JCH, LG, LB, MDC, SC, TR, SY, ML, AXTX, and NI performed quality assessment of the included studies. NI, ML and SY conducted descriptive analyses of the papers. KAB conducted the quantitative analyses and contributed to the interpretation of the review results. NI, KAB, SY and JMG drafted the review. All the authors provided feedback on drafts.

DECLARATIONS OF INTEREST

Sarah Alderson: none known.

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Liesl Grobler: no relevant interests. South African Medical Research Council (Employment), NHMRC (Employment).

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- Department of Family and Community Medicine at Women's College Hospital and the University of Toronto, Canada

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- Australian Commission on Safety and Quality in Health Care , Australia

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DIFFERENCES BETWEEN PROTOCOL AND REVIEW

Differences between the previous review version (Ivers 2012) and this current update

Numerous changes in methods compared to the prior version of the review are listed below and described in more detail throughout the text:

- We focused on the performance of health professionals compared to the previous version which included patient outcomes; therefore, for this update, we excluded studies that provided data only on patient outcomes. Acknowledging that, ultimately, the goal of all quality improvement efforts is to improve patient outcomes, the rationale for this change was three-fold. First, effects on patient outcomes of A&F depend entirely on the strength of association between the health professional behaviour targeted by the A&F and the patient outcome of interest. This will naturally vary widely across behaviours and settings. We argue this represents an issue with the selection of indicators to focus upon and sometimes an issue with health professional guidelines rather than the effects of A&F itself. Second, there is no way to quantitatively synthesise patient outcomes, as there is no latent shared concept across the wide range of potential patient outcomes. Third, the effects on patient outcomes usually take longer to achieve, since they depend on a sequence of events from health professional change in behaviour to patient change in behaviour and most trials are not long enough nor adequately powered for these outcomes. Since A&F is an intervention that aims to change health professional behaviour to be more aligned with standards, evaluating A&F requires focusing on these direct, proximal, attributable effects.
- We included studies in which any intervention used audit and feedback (A&F) rather than studies in which the A&F was used alone or was considered a 'core' aspect of a multi-faceted intervention as in the previous review. We reasoned that the prior approach was subjective and that we could use analytical approaches in the updated review to address multi-faceted interventions.
- We extracted behaviour change techniques as components of quality improvement strategies using novel methods developed by our team (Presseau 2015). We hope this will offer future researchers an opportunity to conduct further evaluations to understand mechanisms of action and advance theory.
- The search strategy was revised and updated to reflect changes in search terminology and methodological filters, and we searched trial registries because this was not done in previous versions of this review.
- We included cluster trials only if they had at least four clusters per arm versus previous versions of the review in which clusters were included with three clusters per arm. Given the number of trials in the review, we were able to exclude smaller cluster trials where the estimates of effect are difficult to disentangle from the clustering.
- We described continuous outcomes narratively rather than quantitatively as in the previous version. The vast majority of trials did report results in a manner that could be defined as dichotomous (i.e. proportion of events completed in accordance with the standard). For studies that reported both dichotomous and continuous outcomes, we extracted the dichotomous outcomes only. For the relatively few studies with only continuous outcomes, there was concern about misinterpretation of a meta-analytical result because there was often not a clear 'standard' for the outcome. For example, an intervention might aim to reduce the count of tests, but it is often not clear what target this is aimed at.
- For risk of bias, we added 'recruitment bias' and 'incorrect analysis' as domains specific to cluster trial designs, and we did not assess 'blinding of participants' as A&F interventions typically cannot allow for this, although in many trials the participants may not have been aware of the trial outcomes.

- We extracted outcomes at only the first follow-up time point, and we did not extract data from further follow-up time points when available as the sustainability of effects was beyond the scope of this review.
- Meta-analyses could not be conducted in the manner proposed in the protocol given the substantial heterogeneity as explained in the text.